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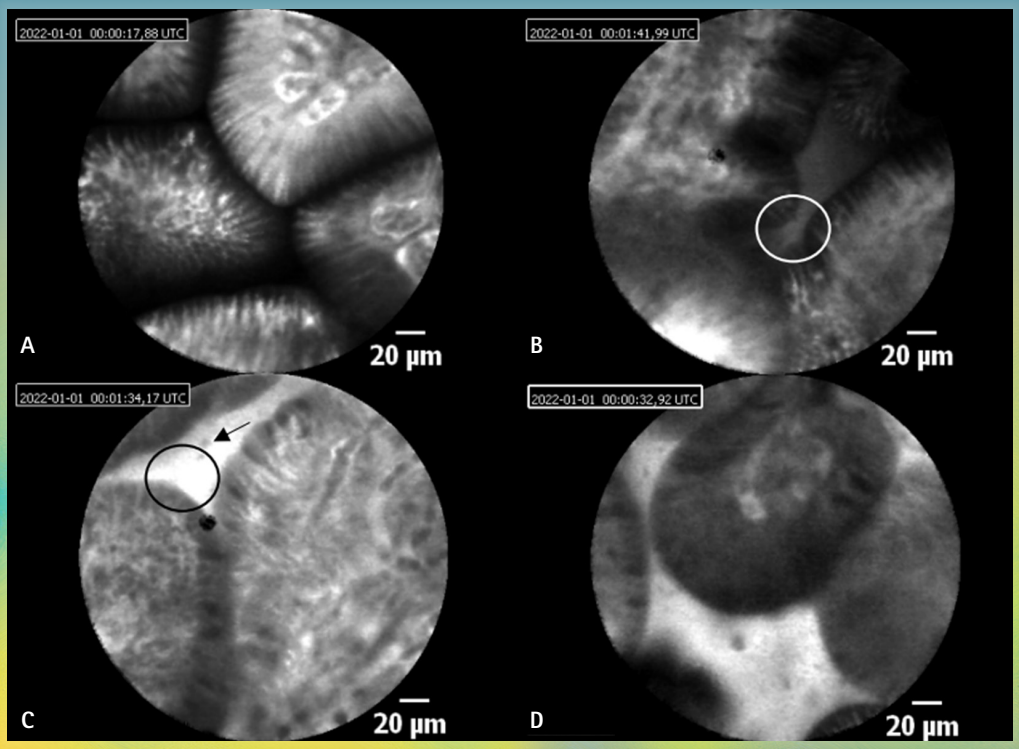

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Confocal laser endomicroscopy assessment before (A, B) and after (C, D) nutrient exposure in the duodenum. See Raymenants et al on page 350.

Editorial
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Physical Activity as a Therapeutic Strategy for Functional Dyspepsia

Yeon Ji Kim

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Article: The effects of physical activity on functional dyspepsia: a mendelian randomization and randomized controlled study
Wei Z, Tao Z, Hao Y, Liu N, Wang J, Xing X
(J Neurogastroenterol Motil 2026;32:405-416)

Functional dyspepsia (FD) is a disorder of gut-brain interaction affecting approximately 10-20% of adults worldwide and one of the most frequent reasons for gastroenterology consultation.[1,2] Despite its high prevalence, pharmacotherapy for FD remains limited in efficacy: proton pump inhibitors, prokinetics, and neuromodulators each provide modest and often transient benefit, and no single agent reliably achieves sustained symptom control.[3] In this context, growing attention has been directed toward lifestyle and other non-pharmacological approaches as adjunctive strategies in the management of FD. The 2022 British Society of Gastroenterology guidelines discussed gut–brain behavioral interventions such as cognitive behavioral therapy and hypnotherapy, together with lifestyle measures, as part of FD management.[4] The 2022 Japanese Society of Gastroenterology guidelines also acknowledged the potential role of lifestyle-related factors in the management of FD.[5] However, practical recommendations regarding the optimal exercise regimen for FD have remained limited because of the lack of supporting clinical trial data. The study by Wei et al[6] helps strengthen the limited evidence base for non-pharmacological management of FD.

The association between physical inactivity and FD has been documented in observational studies. A population-based study of 3160 Australians reported an association between FD and lower levels of physical activity, particularly among patients with the postprandial distress syndrome subtype.[7] However, these observational studies cannot exclude the possibility of reverse causation, as patients with more severe symptoms may be less physically active because of symptom-related limitations. To further explore causality, Wei et al[6] performed a Mendelian randomization analysis using genome-wide association study data from the UK Biobank, including 7662 FD cases and 455 348 controls. Six physical activity phenotypes were evaluated as genetically predicted exposures. Participation in strenuous sports was associated with a significant reduction in FD risk (inverse variance weighted: OR = 0.924; 95% CI, 0.875-0.976; $P = 0.005$). A suggestive inverse association was also observed for activities including swimming, cycling, and fitness training, which broadly represented moderate-intensity exercise (OR = 0.964; 95% CI, 0.937-0.993; $P = 0.013$). Light domestic activity and recreational walking did not reach statistical significance after sensitivity analyses. These findings suggest that the protective association may be more closely related to structured aerobic exercise than to general physical activity alone. Wei et al[6] subsequently conducted a single-center randomized controlled trial involving 260 patients with FD diagnosed according to the Rome IV criteria. Participants were assigned either to a 12-week moderate-intensity exercise program consisting of jogging or cycling at least 5 times weekly for 30-60 minutes per session, or to a control group receiving usual lifestyle guidance with weekly telephone follow-up. Of 260 enrolled participants, 227 completed the study (114 exercise, 113 control). Adequate overall symptom relief, defined as complete relief during at least half of the assessed weeks between weeks 3 and 12, was significantly more frequent in the exercise group than in the control group (46.9% vs 30.8%, $P = 0.008$). Exercise participants also showed significantly greater reductions in total 8-item dyspepsia symptom scores, Short-form Leeds Dyspepsia Questionnaire (SF-LDQ) total score, most SF-LDQ subscale scores, and all 5 Short-form Nepean Dyspepsia Index quality-of-life domains. Among individual symptoms, epigastric pain,

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epigastric burning sensation, postprandial fullness, abdominal distension, and nausea showed greater improvement in the exercise group, whereas early satiety, belching, and vomiting did not differ significantly between groups. Anxiety and depression scale improved similarly in both groups. However, these findings should not be interpreted as evidence that the benefits of exercise are unrelated to gut–brain mechanisms, as the effects of physical activity may not be fully reflected by conventional assessments of anxiety and depression. Exercise may influence symptoms through pathways related to gut–brain interaction, in addition to its effects on gastrointestinal function. Further studies are needed to clarify the mechanisms underlying the benefits of physical activity in FD. The randomized controlled trial represents one of the largest exercise intervention studies in FD conducted to date, exceeding the size of a previous trial by Rane et al[8] (n = 72) by more than threefold, and is the first to consistently apply Rome IV diagnostic criteria. That said, several limitations deserve acknowledgment. In the Mendelian randomization analysis, the causal estimate for strenuous sports was sensitive to a single genetic variant (rs12571549): its exclusion in leave-one-out analysis nullified the association, introducing uncertainty about the robustness of that specific estimate. The randomized controlled trial was conducted at a single tertiary center in China, limiting generalizability to primary care and to other ethnic groups. Exercise adherence was assessed by self-report during telephone follow-up rather than by objective measures such as accelerometry or heart rate monitoring, which may have resulted in variability in compliance. The use of concomitant medications was not controlled and may have influenced the observed treatment effect. Finally, the 12-week follow-up period may be relatively short for a chronic and relapsing condition such as FD, and the long-term durability of exercise-related benefits remains uncertain.

Recent years have seen growing interest in non-pharmacological and lifestyle-based approaches to FD. This study adds clinical trial data to the limited evidence supporting exercise as a potential non-pharmacological approach for FD. Moderate-intensity aerobic exercise, such as jogging or cycling for 30-60 minutes at least 5 times per week, was associated with adequate symptom relief in nearly 50% of patients with FD. For clinicians managing a condition that is common, chronic, and often difficult to treat, this degree of symptom improvement is noteworthy. However, the findings should be interpreted with caution, given the single-center design, self-reported exercise adherence, and lack of long-term follow-up. Future multicenter trials with objective adherence monitoring, longer follow-up, and subgroup analyses by FD subtype will be needed to refine these recommendations further. The present findings provide a useful starting point for future studies aimed at clarifying the role of physical activity in FD, including the effects of different exercise types, intensities, and durations, as well as potential differences in treatment response according to FD subtype.

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None.

Author contributions
Yeon-Ji Kim conceptualized the editorial, reviewed the literature, interpreted the data, and drafted and revised the manuscript.

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Beyond Dysbiosis Restoration: Does the Success of Fecal Microbiota Transplantation for Irritable Bowel Syndrome Lie in Repairing the Intestinal Stem Cell Niche?

Yong Sung Kim[1,2] **and Suck Chei Choi**[1,3]*

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Article: Possible mechanisms underlying the effects of fecal microbiota transplantation in patients with irritable bowel syndrome
El-Salhy M, Skarvatun T, Hoff DAL, et al
(J Neurogastroenterol Motil 2026;32:449-463)

Fecal microbiota transplantation (FMT) has established a clear and definitive role in the treatment of recurrent *Clostridioides difficile* infection, where therapeutic success is directly linked to restoration of a profoundly depleted microbial ecosystem.[1,2] In contrast, the application of FMT in irritable bowel syndrome (IBS) remains highly controversial. Clinical trials have yielded remarkably heterogeneous outcomes, ranging from robust long-term efficacy to complete therapeutic failure.[3-5] Such inconsistency is likely due to the complex pathogenesis of IBS, which involves not only altered gut microbiota but also various other factors such as motility disturbance, visceral hypersensitivity, altered mucosal and immune function, and altered central nervous system processing.[6,7] Accordingly, the simplistic concept of merely "correcting dysbiosis" appears insufficient to fully explain or predict clinical responses to FMT in IBS. To date, the cellular and molecular cascades initiated within the host gastrointestinal milieu by donor-derived microbiota largely remain a mechanistic black box.

In this issue of the *Journal of Neurogastroenterology and Motility*, El-Salhy et al[8] provide novel mechanistic insights by providing direct histopathological evidence of how FMT structurally reshapes the host intestinal architecture. Utilizing mucosal biopsy specimens obtained from a rigorous randomized controlled trial, they demonstrated that donor FMT significantly increased the densities of Musashi-1–expressing intestinal stem cells (ISCs) and neurogenin-3–positive early endocrine progenitor cells. These findings strongly suggest that healthy donor-derived microbiota deliver key luminal signals that reactivate the host ISC niche, thereby restoring mucosal proliferative capacity and regenerative homeostasis. Importantly, this regenerative process was accompanied by a profound restoration of the enteroendocrine milieu. The significant increase in glucagon-like peptide-1- and peptide YY-producing cells following FMT is particularly compelling, given that these hormones are known to modulate gut motility and visceral sensation. Thus, the FMT-induced restoration of this endocrine balance likely contributes to the observed alleviation of IBS symptoms. Low-grade mucosal inflammation, characterized by an increased number and activation of mast cells, is a well-recognized mechanism underlying IBS. The investigators demonstrated a significant reduction in lamina propria CD45+ cells and submucosal mast cells, suggesting a resolution of this low-grade mucosal inflammation. Notably, these cellular changes were directly linked to clinical outcomes, including both IBS symptom severity and systemic fatigue. This suggests that by mitigating chronic immune activation, FMT may effectively dampen the neuro-immune signaling pathways that drive visceral hypersensitivity and IBS-related discomfort. Taken together, these findings suggest that FMT alleviates IBS symptoms not merely by restoring microbial composition, but by rebuilding the cellular networks governing gastrointestinal physiology.

This study aligns with an emerging paradigm that the gut microbiota does not merely modulate the function of mature epithelial cells, but fundamentally resets the host tissue developmental trajectory, ISC fate, and lineage commitment topology. This dif-

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ferentiation pathway is highly sensitive to the microbial context; commensal microbiota promote ISC differentiation toward enterocytes, whereas pathogens drive lineage commitment toward enteroendocrine cells, inducing their long-term accumulation.[9] Furthermore, during early postnatal development, the microbiota actively shapes the ISC niche and epithelial composition via mesenchymal and macrophage signaling.[10] Several studies revealed microbiota-metabolite-stem cell regulatory axis. For instance, *Blautia coccoides* produces indole-3-lactic acid, which is converted by other commensal bacteria to indole-3-propionic acid, stimulating β-hydroxybutyrate synthesis in intestinal epithelial cells. β-hydroxybutyrate activates $HOPX^+$ reserve ISCs and promotes regeneration of the $LGR5^+$ ISC pool, accelerating epithelial repair.[11] Similarly, microbial metabolites like valeric acid promote tryptophan hydroxylase 2 expression in serotonergic neurons, triggering a cooperative cascade involving macrophage-derived prostaglandin E2 and Wnt/β-catenin signaling to drive ISC self-renewal.[12] These findings demonstrate that the microbiota not only modulate mature cell function but also reset stem cell pool regeneration and differentiation programs.

Direct evidence that transplanted microbiota promote epithelial development and functional maturation also exists, albeit primarily demonstrated in germ-free animal models. For example, the transplantation of microbiota enriched with beneficial microbes capable of modulating bile acid metabolism has been shown to upregulate epithelial development genes and enhance intestinal barrier function.[13] Given this scientific background, the work by El-Salhy et al[8] provides a compelling histopathological framework that connects clinical FMT efficacy to the molecular regulation of the ISC niche in IBS.

While these findings are promising, they warrant cautious interpretation. The study's placebo-controlled design strengthens evidence linking FMT to clinical and histological improvements. However, the simultaneous observation of these changes only suggests a potential mechanistic pathway. Claiming a direct causal link between ISC density and symptom relief or establishing ISC restoration as the sole mediator requires further validation. Although Musashi-1 is a widely recognized ISC marker, its expression is not exclusively restricted to intestinal stem cells and can be observed in other proliferating cell populations. Nonetheless, it serves as a valuable indicator of mucosal proliferative capacity within this study. Methodological caveats also merit attention. The study is limited by its modest sample size, a lack of explicit information regarding the standardization of biopsy sites, potential bias from non-blinded histopathological assessment, and the unknown long-term durability of ISC niche restoration. Furthermore, as studies on reduced ISC density in IBS patients have been conducted almost exclusively by this single research group,[14] validation in independent, larger, multicenter cohorts across diverse ethnicities is essential before this hypothesis can be accepted as one of the contributing factors in the pathophysiology of IBS.

The key contribution of El-Salhy's study is that it provides a mechanistic framework for previously poorly understood microbial intervention.[8] Ultimately, the intrinsic risks, lack of standardization, and donor-dependent variability of conventional FMT limit its long-term utility as a therapeutic option for chronic conditions like IBS. However, by mapping the precise cellular receptors, enteroendocrine hormones, and immune cell populations altered by microbial engraftment, this study paves the way for standardized, next-generation microbiome-based therapeutics. A critical next step in translational research will be to verify whether these microbial metabolites can modulate stem cell differentiation and immune suppression in the absence of a raw fecal matrix. Future studies should evaluate the administration of targeted butyrate-producing bacterial consortia or specific beneficial strains in germ-free animal models. Moreover, since bacterial dysbiosis does not fully encapsulate the pathogenesis of IBS, alternative pathways of symptom relief via FMT must be explored. Given the bidirectional nature of the microbiota-gut-brain axis, FMT may partially exert its therapeutic effects by modulating CNS processing or alleviating psychiatric comorbidities, which secondarily reduces gastrointestinal symptoms.[15] In the future, gut microbiota-based therapeutics for IBS may gradually shift from crude fecal suspensions toward target-engineered synthetic microbial consortia, precision postbiotics, or potentially intestinal epithelial organoid supplementations,[16] aiming to directly restore the intestinal stem cell niche once clinically validated.

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None.

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Advances in Understanding Calcium Sensitization in Gastrointestinal Smooth Muscles: Pathways, Interstitial Cells, and Disease

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Contractile activity of gastrointestinal smooth muscles is triggered by increased intracellular Ca^{2+}, which activates myosin light chain kinase (MLCK) and leads to phosphorylation of myosin regulatory light chain (RLC), enabling contraction. Myosin light chain phosphatase (MLCP) dephosphorylates RLC for relaxation. The balance between MLCK and MLCP determines the relationship between calcium levels and contractile force. Shifting the balance between MLCK and MLCP in favor of MLCK by inhibiting MLCP allows MLCK to phosphorylate more RLC and generate increased force at any given Ca^{2+} concentration. Thus, when MLCP is inhibited, the myofilaments are more sensitive to calcium, allowing greater force at any given Ca^{2+} concentration. Major mechanisms for calcium sensitization in GI smooth muscle include protein kinase C-mediated phosphorylation of protein phosphatase 1 regulatory subunit 14A (CPI-17) and Rho-associated coiled-coil kinase 2 (ROCK2)-mediated phosphorylation of myosin phosphatase targeting subunit 1. Calcium sensitization is crucial for sphincter tone, sustained contraction, peristalsis, energy conservation, and coordinated motility. Smooth muscle cells, interstitial cells of Cajal, and platelet-derived growth factor alpha positive (PDGFRα$^+$) cells form the SIP syncytium, which regulates excitability and calcium sensitization. This review updates our current understanding of the regulatory mechanisms of calcium sensitization with the recent findings from genetically engineered mice with deficiencies in protein kinase C and ROCK2 calcium sensitization pathways, and the involvement of interstitial cells in regulating GI smooth muscle cell excitability. Dysregulated calcium sensitization pathways in motility disorders are summarized, to potentially aid in the development of new pharmacological strategies to prevent and treat GI motility disorders.

Keywords: Interstitial cells of Cajal; Knockout mice; Phosphorylation; Signal transduction; Smooth muscle

INTRODUCTION

Gastrointestinal (GI) motility depends on the coordinated contraction and relaxation of smooth muscle cells (SMCs) within the tunica muscularis, a process regulated by complex interactions among SMCs, enteric motor neurons, and interstitial cell populations. Contraction is initiated through excitation–contraction coupling, whereby increases in intracellular Ca^{2+} activate calmodulin-dependent myosin light chain kinase (MLCK), resulting in phosphorylation of the regulatory light chain (RLC) of myosin and the initiation of cross-bridge cycling.[1-3] However, it is now well established that the magnitude and duration of GI smooth muscle contraction cannot be explained solely by changes in intracellular Ca^{2+} concentration. Instead, mechanisms that regulate the Ca^{2+} sensitivity of the contractile apparatus are critical for modulating contractile force as necessary for proper motor function.[4,5]

Calcium sensitization refers to the enhancement of force generation at a given intracellular Ca^{2+} concentration, primarily through preservation of myosin RLC phosphorylation by inhibition of myosin light chain phosphatase (MLCP). MLCP plays a central role in determining the phosphorylation state of myosin RLC, thus the balance between MLCK and MLCP activities is a key determinant of smooth muscle contractility.[6,7] Regulatory pathways that inhibit MLCP, including protein kinase C (PKC)-dependent phosphorylation of protein phosphatase 1 regulatory subunit 14A (CPI-17) and RhoA/Rho-associated coiled-coil kinase 2 (ROCK)-mediated phosphorylation of the myosin phosphatase targeting subunit (MYPT1), have emerged as major mechanisms underlying Ca^{2+} sensitization in GI smooth muscles.[8-11] These pathways enable sustained contractile responses, contribute to sphincter tone, and support the generation of coordinated motility patterns.[12,13]

Although the biochemical mechanisms underlying MLCP regulation have been extensively studied, their physiological roles in intact GI tissues have proven to be more complex than initially

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appreciated. Emerging evidence suggests that the relative contributions of PKC/CPI-17 and ROCK/MYPT1 pathways differ depending on muscle phenotype, anatomical location, and mode of stimulation.[9,11,14] Studies using pharmacological approaches have traditionally supported a prominent role for ROCK-mediated MYPT1 phosphorylation in Ca^{2+} sensitization.[15,16] However, more recent investigations, including those employing genetically engineered mouse models, indicate that constitutive MLCP inhibition and basal tone may be more strongly associated with MYPT1 phosphorylation at Thr694, while physiological activation of Ca^{2+} sensitization during neurotransmission may preferentially involve CPI-17 phosphorylation.[14,17] These findings highlight the need to distinguish between mechanisms derived from pharmacological stimulation and those that operate under physiological conditions.

A key factor contributing to this complexity is the organization of GI smooth muscle within the SIP syncytium, a functional network composed of SMCs, interstitial cells of Cajal (ICC), and platelet-derived growth factor receptor-α–positive (PDGFRα$^+$) cells.[18,19] These cell types are electrically coupled and act in concert to regulate excitability, integrate enteric neural inputs, and coordinate contractile responses. ICC serve as pacemaker cells and intermediaries for excitatory neurotransmission, whereas PDGFRα$^+$ cells mediate inhibitory signaling.[18,19] The anatomical and functional relationships within the SIP syncytium restrict neurotransmitter access to specific cellular targets and shape downstream signaling pathways. As a result, the activation of Ca^{2+} sensitization mechanisms in SMCs depends on the mode of stimulation and the cellular context in which signaling occurs. Recent studies have begun to reveal that physiological activation of GI smooth muscle via enteric neurotransmission may differ from responses elicited by exogenous agonists. Neural stimulation appears to preferentially activate PKC-dependent CPI-17 phosphorylation, whereas bath-applied agonists can engage both PKC- and ROCK-dependent pathways, reflecting broader receptor activation across multiple cell types within the SIP syncytium.[14,20] These observations underscore the importance of interpreting experimental findings within the context of physiological signaling and highlight potential limitations of pharmacological approaches in defining the mechanisms of Ca^{2+} sensitization.

Dysregulation of Ca^{2+} sensitization pathways has been implicated in a variety of GI motility disorders, including gastroparesis, achalasia, and inflammatory bowel disease.[21-24] Alterations in the expression or phosphorylation of CPI-17, MYPT1, and associated signaling proteins have been linked to impaired contractility and abnormal motility patterns, suggesting that targeting components of the Ca^{2+} sensitization machinery may represent a promising therapeutic strategy.[23-25] Despite its central importance, relatively few recent reviews have focused specifically on calcium sensitization in GI smooth muscle, instead focusing on how these incorporate sensitization mechanisms within broader discussions of Ca^{2+} signaling, electromechanical coupling, and integrated motor control.[26,27] While this reflects the increasingly comprehensive view of GI motility, it also underscores the need for a focused reappraisal of calcium sensitization within this expanded conceptual framework.

In this review, we seek to provide an updated synthesis of the mechanisms regulating Ca^{2+} sensitization in GI smooth muscle, with particular emphasis on the roles of MLCP regulation, interstitial cells within the SIP syncytium, cytoskeletal dynamics, and insights gained from genetically engineered mouse models. We summarize the current understanding of calcium sensitization mechanisms in GI smooth muscles and examine their role in the regulation of motility under physiological and pathophysiological conditions. Emphasis is placed on the integration of classical MLCP regulatory pathways with emerging concepts in ion channel function, intercellular signaling, and neuromuscular coordination. We also examine the distinctions between physiological and pharmacological activation of sensitization pathways and discuss how dysregulation of these processes contributes to GI motility disorders. By integrating molecular, cellular, and systems-level perspectives, this review aims to emphasize Ca^{2+} sensitization as an integral component of the broader regulatory networks governing gastrointestinal motor function.

ROLE OF MYOSIN LIGHT CHAIN KINASE IN SMOOTH MUSCLE CONTRACTION

Normal GI smooth muscle motility patterns are essential for maintaining homeostasis and the adaptive responses to stresses imposed by pathological disorders. A network of signaling pathways centered on the molecular motor myosin as well as membrane proteins associated with calcium handling and cell adhesion regulate myosin phosphorylation for cross-bridge cycling and GI smooth muscle contractile responses.[5,28-30] G protein-coupled receptor (GPCR) signaling pathways, voltage-dependent Ca^{2+} influx, or release from the sarcoplasmic reticulum, increase the smooth muscle cell cytoplasmic Ca^{2+} concentration to activate the Ca^{2+}/calmodulin-dependent MLCK (Fig. 1).[1,3] Activated MLCK phosphorylates the RLC of smooth muscle myosin, allowing myosin to form cross bridges and bind to actin filaments to generate force.[2,3,31,32] MLCK is the only kinase that phosphorylates RLC Ser19 (S19) physiologically in different kinds of smooth muscles.[1,33,34] MLCK activation is also modulated by phosphorylation that reduces its activity, to fine tune the RLC phosphorylation response.[35,36] MLCK

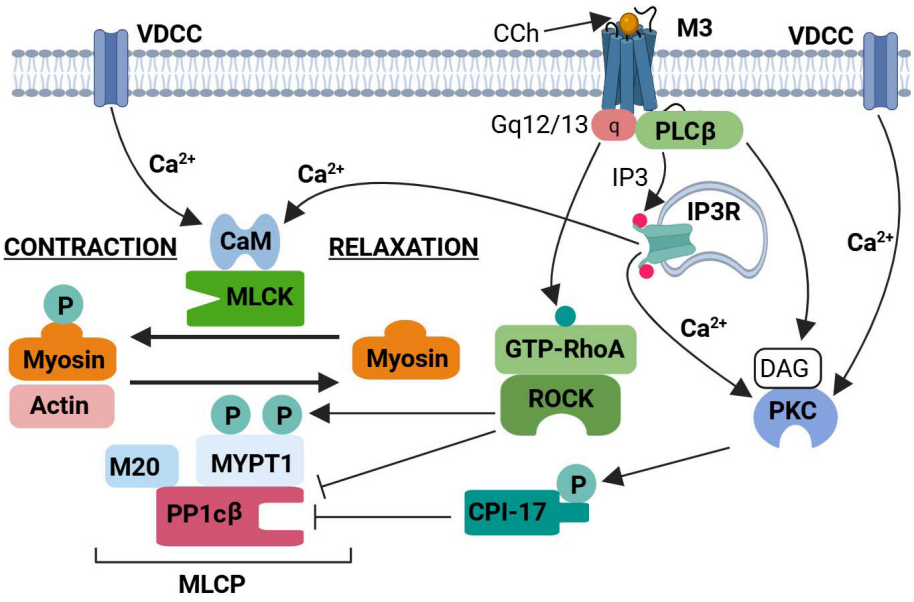

Figure 1. Schematic diagram of calcium sensitization pathways in smooth muscle cells (SMCs). Depolarization of the SMC plasma membrane opens dihydropyridine-sensitive voltage-dependent Ca^{2+} channels (VDCC), increasing the concentration of Ca^{2+} in the cytoplasm. Ca^{2+} binds to calmodulin (CaM), which binds to and activates myosin light chain kinase (MLCK), which phosphorylates the regulatory light chain of thick filament myosin to enable myosin binding to thin filament actin for cross-bridge cycling and muscle contraction. Simultaneously, voltage-dependent Ca^{2+} influx activates protein kinase C (PKC), which phosphorylates protein phosphatase 1 regulatory subunit 14A (CPI-17). Phosphorylated CPI-17 binds to and inhibits myosin light chain phosphatase (MLCP), inhibiting the dephosphorylation of myosin regulatory light chain. Thus, during voltage-dependent Ca^{2+} influx, the ratio of MLCK activity to MLCP activity is higher and contraction is augmented because the PKC-mediated calcium sensitization pathway is activated. In receptor-mediated smooth muscle contraction, cholinergic agonists bind to the muscarinic M3 receptor and activate Gq12/13. Gq12/13 stimulates guanosine 5′-triphosphate (GTP) binding to RhoA, which binds to and activates Rho-associated coiled-coil kinase (ROCK). ROCK phosphorylates myosin phosphatase targeting subunit 1 (MYPT1), and other proteins, leading to reduced MLCP activity. Gq12/13 also activates phospholipase Cβ (PLCβ), generating inositol 1,4,5-trisphosphate (IP3) and diacylglycerol (DAG). IP3 releases Ca^{2+} from intracellular stores into the cytoplasm. This Ca^{2+} binds to CaM which activates MLCK to phosphorylate and enable myosin to bind to thin filament actin for cross-bridge cycling. Ca^{2+} and DAG also bind to and activate PKC, which phosphorylates CPI-17 to further inhibit MLCP. Thus, during receptor-mediated contraction, the ratio of MLCK activity to MLCP activity is even higher, and contraction is further augmented because both the PKC- and ROCK-mediated calcium sensitization pathways are activated. CCh, carbachol; PP1cβ, catalytic type 1 phosphatase subunit; M20, 20 kDa regulatory subunit of MLCP.

is inactivated by the decrease in cytosolic Ca^{2+} concentrations that dissociates calmodulin from MLCK. RLC S19 is dephosphorylated by MLCP, initiating relaxation by returning myosin to an inhibited state (Fig. 1).[6] Thus, the ratio of MLCK/MLCP activities is a major determinant of the extent of RLC phosphorylation and contractile force.[4] In contrast to striated muscle, the force and shortening velocities of smooth muscles are regulated functions of cross-bridge phosphorylation.[12]

ROLE OF MYOSIN LIGHT CHAIN PHOSPHATASE IN SMOOTH MUSCLE CONTRACTION

Although both Ca^{2+} influx and myosin RLC phosphorylation are required for smooth muscle contraction, it is now well established that regulation of MLCP activity, independent of the Ca^{2+}-dependent activation of MLCK, is an important factor in determining the extent of RLC phosphorylation and contractile force.[5,13] Shifting the ratio of MLCK/MLCP activities in favor of MLCK activity by inhibiting MLCP allows MLCK to phosphorylate more myosin RLC at any given Ca^{2+} concentration. This means that the contractile proteins are essentially more sensitive to calcium, leading to increased muscle contraction for a given calcium level. Thus, myofilament calcium sensitization in smooth muscles refers to the phenomenon that greater force is generated when MLCP activity is inhibited at any given concentration of Ca^{2+} which activates MLCK, or the muscle can generate greater force at lower intracellular calcium concentrations.[6,37,38]

The MLCP holoenzyme is composed of 3 subunits: the catalytic type 1 phosphatase subunit (PP1cβ; Human gene, PPP1CB; Mouse gene, Ppp1cβ), a regulatory subunit (myosin phosphatase targeting subunit 1 [MYPT1]; Protein phosphatase 1 regulatory subunit 12A; Human gene, PPP1R12A; Mouse gene, Ppp1r12a) and a 20

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kDa subunit of unknown function (Fig. 1).[39,40] MLCP dephosphorylation of RLC (Myosin-RLC; Human gene MYL9; Mouse gene, Myl9) is involved in various biological processes involving different myosin II isoforms, including smooth muscle contraction, cell division, cell migration, morphogenesis and other developmental processes.[7] PP1cβ activity is increased by binding to MYPT1, and inhibited by MYPT1 phosphorylation.[41-43] MYPT1 also binds myosin and/or other proteins, thus serving as a scaffolding protein.[41,42]

Two different signaling pathways primarily inhibit MLCP activity and increase RLC phosphorylation in the absence of additional cytoplasmic Ca^{2+} (Fig. 1).[8,13,44] Phosphorylation of MYPT1 inhibits MLCP activity and contributes to smooth muscle Ca^{2+} sensitization.[17,41-43] In most smooth muscles, MYPT1 exhibits high constitutive phosphorylation of MYPT1 at Thr694 (T694, mouse sequence numbering; Thr696, T696 in human MYPT1) which inhibits MLCP activity, but is not extensively increased in response to contractile agonists.[9-11,15,17,20,45-47] Activation of GPCRs by contractile agonists has clearly been shown to increase MYPT1 phosphorylation at Thr852 (T852, mouse sequence numbering; Thr853, T853 in human MYPT1) by ROCK2, and increase the contractile responses of smooth muscles, suggesting that MYPT1 phosphorylation at T852 inhibits MLCP activity.[14,48-52] Ca^{2+} sensitization in smooth muscles is also mediated by CPI-17 (protein phosphatase 1 regulatory subunit 14A, human gene PPP1R14A; mouse gene Ppp1r14a), which is converted into a potent inhibitor of MLCP by phosphorylation of Thr38 (T38) by PKC.[53-58]

ROLE OF MYOSIN PHOSPHATASE TARGETING SUBUNIT 1 PHOSPHORYLATION IN SMOOTH MUSCLE CONTRACTION

A large number of studies over the past several decades have been conducted to improve our understanding of the physiological functions of MYPT1 phosphorylation at T694 and T852 in smooth muscles. The constitutive phosphorylation of MYPT1 T694 now appears to be a primary mechanism for regulating force maintenance of smooth muscle, rather than phosphorylation of MYPT1 T852.[17,20] Studies using a reconstituted recombinant MLCP enzyme show that MLCP activity is almost completely blocked when MYPT1 is phosphorylated at T694.[59,60] The phospho-T694 residue docks directly at the active site of PP1, causing autoinhibition.[59,60] The majority of reports demonstrate that T694 phosphorylation is constitutively high in resting smooth muscle tissues and is barely elevated by any contractile agonist stimulus that activates ROCK2, or reduced by any relaxant stimuli.[8,14,44,49,61-63] MYPT1 T694 phosphorylation is not reduced by ROCK2 or PKC inhibitors that affect MYPT1 T852 or CPI-17 phosphorylation.[14,17,48,49,64,65] Thus, it appears that MYPT1 phosphorylation at T694 is not responsible for ROCK2- or PKC-mediated Ca^{2+} sensitization of smooth muscle. The kinase(s) involved in the constitutive phosphorylation of MYPT1 T694 are still unidentified. AMPK, ZIPK, ILK or p21-activated kinase can phosphorylate MYPT1 T694 in vitro, but whether they phosphorylate MYPT1 T694 in vivo is still unknown.[66-68]

The phosphorylation of MYPT1 at T852 has been a major focus of studies investigating the regulation of smooth muscle calcium sensitization because it has been consistently observed that T852 phosphorylation is increased by contractile agonists that increase ROCK2 activity.[5,6,14,17,37,38,49,64,69] However, most of the studies investigating the effects of MYPT1 phosphorylation at T852 in smooth muscles in vivo has been determined primarily with pharmacological approaches using ROCK2 inhibitors. These pharmacological perturbations affecting MYPT1 T852 phosphorylation associated with diminution of force responses have formed a fundamental model of signaling mechanisms for MLCP inhibition and force maintenance by MYPT1 T852 phosphorylation in a variety of smooth muscles.[8,9,44,51,62] Despite these observations, the experimental evidence about MYPT1 T852 phosphorylation and its functional roles are conflicting.[10,59,60,70] Some reports propose that the regulatory mechanism of phosphorylated T852 is via direct inhibition of PP1c activity, while others suggest that T852 phosphorylation triggers dissociation of the MLCP complex from myosin filaments.[10,43,70-72] Using chicken gizzard MYPT1 mutants containing T850 with nonphosphorylatable T695A, or T695 with nonphosphorylatable T850A (chicken isoform numbering), Feng et al[59] showed that the T850A mutant almost completely inhibited MLCP activity in vitro, suggesting that phospho-T695, not phospho-T850, is the inhibitory site for MLCP activity. Similarly, Khasnis et al[60] developed a novel procedure to express MLCP reconstituted in mammalian cells and showed MYPT1 T694 phosphorylation inhibited MLCP activity whereas T852 phosphorylation did not. Khromov et al[31] demonstrated in vitro that T852 phosphorylation displayed 30-fold less inhibitory activity towards MLCP than T694; while Murányi et al[70] concluded that T852 phosphorylation regulated MLCP activity equivalently to T694.

Investigations of ROCK2-mediated Ca^{2+} sensitization were significantly advanced by transgenic approaches to rule out off-target effects where pharmacological inhibitors may act non-selectively on other protein kinases or proteins in different signaling pathways that regulate contractile responses. To further investigate the regulation of MLCP activity in vivo, Chen et al[17] established 2 mouse lines with Ala substitution of MYPT1 T694 or T852. Isolated ileum and bladder smooth muscles from T852A mice displayed no significant changes in RLC phosphorylation or force re-

sponses, but force was still inhibited by a ROCK2 inhibitor. In contrast, smooth muscles containing the T694A mutation showed a significant reduction in force along with reduced RLC phosphorylation. The contractile responses of T694A mutant smooth muscles were also independent of ROCK2 activation. These findings suggest that phosphorylation of MYPT1 T694, but not T852, appears to be the primary mechanism contributing to the constitutive inhibition of MLCP activity, preservation of basal RLC phosphorylation levels, and force maintenance in vivo in smooth muscles.

Evidence supporting a role for MYPT1 T852 phosphorylation in contributing to enhanced RLC phosphorylation in intact tissues depends mostly on results obtained with pharmacological inhibitors of ROCK2. ROCK2 inhibitors clearly reduce MYPT1 T852 phosphorylation, and inhibit GI smooth muscle contractions and reduce basal tone. However, ROCK2 is a multifunctional Ser/Thr protein kinase that phosphorylates several substrates, and other potential physiological substrates and processes affecting smooth muscle contraction have been identified. Thus, pharmacological inhibition of ROCK2 would inhibit the phosphorylation of other proteins by ROCK2 that promote contractile responses. For example, ROCK2 appears to be involved in agonist activation of Ca^{2+} entry distinct from voltage or store operated channels.[73,74] ROCK2 also affects actin dynamics in multiple signaling modules, including in particular the phosphorylation of LIMK, which phosphorylates and inactivates cofilin's ability to sever F-actin.[75,76] The resulting increase in actin polymerization is associated with enhanced force development through membrane adhesions linking the actomyosin motor complex with the extracellular matrix.[77] Treating smooth muscles with ROCK2 inhibitors inhibits F-actin polymerization and contraction.[78,79] Thus, even selective pharmacological inhibition of ROCK2 is not selective for inhibiting MYPT1 T852 phosphorylation, as the phosphorylation of multiple proteins affecting smooth muscle contraction are also inhibited. Finally, the secreted protein Mfge8 was described as a novel gastric braking protein in 2016 by Khalifeh et al[80] MFG-E8 decreases the force of antral contractions and curtails gastric emptying. MFG-E8 binds to α8integrin on gastric SMCs and inhibits ROCK2 phosphorylation of MYPT1 at T852, resulting in RLC dephosphorylation and muscle relaxation. Lack of MFG-E8 results in a 50% increase in the rate of gastric emptying, demonstrating the importance of this pathway in regulating gastric contractions, but is probably due to inhibition of phosphorylation of other proteins by ROCK2 that would otherwise promote contractile responses.

ROLE OF PROTEIN PHOSPHATASE 1 REGULATORY SUBUNIT 14A IN SMOOTH MUSCLE CONTRACTION

In GI smooth muscles, inhibition of MLCP in response to pharmacological agonist-induced Ca^{2+} sensitization, and Ca^{2+} sensitization induced by membrane depolarization-induced Ca^{2+} influx may primarily be due to phosphorylation of CPI-17 at T38 by PKC.[11,56,81] CPI-17 was first identified in pig aorta smooth muscle homogenates as an endogenous MLCP inhibitor protein that exhibits elevated potency upon phosphorylation at T38 by PKCα and PKCδ.[56,63,65,81-85] Unlike MYPT1, CPI-17 expression is highly restricted to SMCs and neurons.[86-88] When CPI-17 is phosphorylated, the phospho-T38 selectively docks at the active site of PP1c associated with MYPT1, with a Kd value in the sub nM range, causing a potent inhibition.[55] Accumulated evidence shows that CPI-17 regulates MLCP activity in smooth muscles.[9] The extent of Ca^{2+} sensitization is directly correlated with CPI-17 expression in smooth muscle tissues.[88,89] CPI-17 expression is also linked to individual smooth muscle contractile characteristics. CPI-17 expression is higher in tonic muscles and lower in phasic muscles, such as urinary bladder and vas deferens.[88] CPI-17 expression also positively correlates with the extent of phorbol ester-induced Ca^{2+} sensitization of force.[88] CPI-17 is not expressed in farm chicken smooth muscles, but the levels of RhoA, ROCK2, PKC, and MYPT1 are comparable to those in rabbit and pigeon smooth muscles.[89] Stimulation with endothelin-1, phenylephrine, or phorbol ester, induces a minimal agonist-induced Ca^{2+} sensitization of contraction in chicken arterial smooth muscle, but in rabbit and pigeon arteries that do express CPI-17, these agonists induce a robust contraction.[89] The recent development of CPI-17 knockout (KO) and T38A mice has confirmed the importance of MLCP inhibition by CPI-17 phosphorylation at T38 by PKC for the regulation of vascular tone and blood pressure, and that CPI-17 expression is positively correlated with the extent of phorbol ester-induced Ca^{2+} sensitization of force.[58,90] Phorbol ester-induced aorta contraction is completely abolished in CPI-17 KO mice, and severely diminished in T38A KO mice, demonstrating that CPI-17 phosphorylation at T38 mediates PKC-mediated Ca^{2+} sensitization.[58] However, studies using CPI-17 KO mice and T38A mice have not been carried out in GI smooth muscles.

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BASAL VS CA^{2+}-STIMULATED PROTEIN PHOSPHATASE 1 REGULATORY SUBUNIT 14A PHOSPHORYLATION

In GI smooth muscles, CPI-17 is basally phosphorylated in unstimulated muscles, although the levels vary between sphincter, tonic, or phasic smooth muscles.[91] CPI-17 phosphorylation is increased by K^+ depolarization or muscarinic receptor stimulation in GI smooth muscles.[16,49,92-95] In contrast to MYPT1 phosphorylation at T852, the use of pharmacological inhibitors or activators of PKC has clearly demonstrated the functional role of CPI-17 phosphorylation at T38 in GI smooth muscle Ca^{2+} sensitization. Although PKC phosphorylates pT38, the roles of the different PKC isozymes involved in T38 phosphorylation is still unclear. PKC is a family of conventional Ca^{2+}-dependent α, β, and γ cPKCs, novel Ca^{2+}-independent δ, ε, θ, and η nPKCs, and atypical ξ, and ι/λ aPKCs.[96] The cPKCs are activated by Ca^{2+}, diacylglycerol, and phosphatidylserine.[96] The nPKCs do not bind Ca^{2+}, and are activated by diacylglycerol and phosphatidylserine, but not Ca^{2+}.[96] The cPKCs α, βI, βII, and γ; and the nPKC δ are all highly expressed in GI smooth muscle tissues, suggesting that CPI-17 can be phosphorylated by Ca^{2+}-dependent and Ca^{2+}-independent PKCs.[94,95,97-101] This could explain some of the conflicting reports describing the effects of pharmacological inhibition of PKC on basal CPI-17 phosphorylation and basal tone.[102] Efforts aimed at developing specific nPKC inhibitors, along with the specific cPKC inhibitors, may make it possible to determine the contributions and roles of cPKCs and nPKCS to basal and agonist-mediated CPI-17 phosphorylation in different GI smooth muscles with different functions.[103] Sphincters are generally characterized by high, sustained tone, in contrast to the stomach and intestines which are characterized by rhythmic, propulsive contractions to provide mixing and motility of material within their lumens. To specifically target agonist-or Ca^{2+}-induced CPI-17 phosphorylation without inhibiting basal CPI-17 phosphorylation could potentially provide novel therapeutic strategies for improving GI motility without adversely affecting GI sphincter function.

CA^{2+} SENSITIZATION IN GASTROINTESTINAL SPHINCTERS

Several groups have extensively studied calcium sensitization pathways in feline and human lower esophageal sphincter (LES).[16,104-108] However, the data regarding the regulation of LES spontaneous tone and agonist-evoked contractions by calcium sensitization mechanisms are conflicting, and the relative contributions of ROCK2 and PKC-mediated pathways are still unclear. A study by Hillemeier et al[109] showed that the broad-spectrum cPKC and nPKC inhibitors H-7 and calphostin c reduce spontaneous resting tone in feline LES muscle strips. In addition, U73122, an inhibitor of phosphatidylinositol-4,5-bisphosphate-specific phospholipase C, and D609, an inhibitor of phosphatidylcholine-specific phospholipase C, reduced diacylglycerol formation and LES tone in a concentration-dependent manner. These data suggest that resting LES tone is associated with elevated diacylglycerol levels and PKC activity. However, it should be noted that the effects of Rho kinase inhibitors on the cat LES were not examined. In contrast, a report by Sims et al[16] showed that in human LES muscles, the Rho-kinase blockers Y-27632, and HA-1077, caused concentration-dependent reductions of tension in spontaneously contracted LES muscles. These ROCK2 inhibitors also reduced intrinsic nerve-evoked and carbachol (CCh)-evoked contraction. Calphostin C and chelerythrine reduced CCh-evoked contractions to a lesser extent. Neither PKC inhibitor tested caused a statistically significant change in the resting tension of LES smooth muscle. These findings suggest that in human LES muscles, calcium sensitization by ROCK2 is more important for maintaining resting tone and agonist-induced increases in tone than PKC.[16] Neither MYPT1 or CPI-17 phosphorylation were investigated in these studies. In murine gastric fundus and proximal colon muscles, but not gastric antrum muscles, we found that the broad-spectrum PKC inhibitor, Gf109203x, reduces CPI-17 phosphorylation below the basal level.[49] We subsequently found that either Gf109203x and Go6976, a selective inhibitor of cPKCs, inhibited CCh- or cholinergic nerve-stimulated increase in T38 phosphorylation, but only Gf109203x reduced T38 phosphorylation below basal levels in the murine gastric fundus.[14] Similarly, basal CPI-17 phosphorylation and tone in the monkey LES is unaffected by the cPKC inhibitor Go6976, but is reduced by the broad-spectrum PKC inhibitor Gf109273X.[110] These findings are consistent with a nPKC contributing to LES tone. These findings suggest that a nPKC maintains basal CPI-17 phosphorylation, and that a cPKC mediates agonist-evoked increases in CPI-17 T38 phosphorylation.

Calcium sensitization pathways in rat and human internal anal sphincter (IAS) have been extensively studied by Rattan and coworkers.[111,112] These studies revealed significantly greater expression levels of RhoA, and ROCK2, and greater T694 phosphorylation levels in the tone-generating IAS than in predominantly phasic rectum smooth muscle. ROCK2 inhibitors abolished MYPT1 T694 phosphorylation and were more potent antagonists of contraction in the IAS than in other phasic muscles. However, the effects of ROCK2 inhibitors on MYPT1 T853 were not examined. Rat and human IAS and rectum CPI-17 levels, and constitutive T38 phosphorylation were also found to be greater in the IAS than in rec-

tum. However, due to the limited ability of PKC inhibitors to reduce IAS tone, the authors concluded that tone in the IAS is primarily due to calcium sensitization via the ROCK2/MYPT1 pathway.[112] The ROCK2/MYPT1 calcium sensitization pathway in the monkey IAS and rectum were examined by Keef and co-workers.[110] Their results mostly agree with those of Rattan et al[110] in that T694 and T852 phosphorylation and ROCK2 expression levels are greater in the IAS than in rectum. However, little difference in the potency of ROCK2 inhibitors between muscles was found. Likewise, little difference between the potency of ROCK2 inhibitors in the phasically active mouse colon versus the tone-generating fundus has been found.[49] Together, these data suggest that some degree of constitutive myofilament sensitization is likely to be present in both phasic and tone-generating muscles.[102]

PHYSIOLOGICAL VS PHARMACOLOGICAL STIMULATION OF CA^{2+} SENSITIZATION VIA MYOSIN LIGHT CHAIN PHOSPHATASE INHIBITION

Physiological activation, in contrast to pharmacological stimulation, of smooth muscle contraction elicits different phosphorylation responses for MYPT1 and CPI-17. Cholinergic neurostimulation results in increased CPI-17 phosphorylation, but not MYPT1, in gastric fundus or urinary bladder smooth muscles (Fig. 2).[14,113] However, pharmacological stimulation of muscarinic receptors with bath applied muscarinic agonists increased MYPT1 and CPI-17 phosphorylation in both types of smooth muscles (Fig. 3).[14,113] The agonist-evoked increase in T852 phosphorylation observed in these 2 smooth muscle tissues may be due to pharmacologically intense activation of signaling modules not observed physiologically, due to exposure of the agonist to a larger population of muscarinic receptors. In contrast, the myogenic response in arterial resistance vessels is associated with increased MYPT1 T852 phosphorylation and actin polymerization, but no CPI-17 phosphorylation.[114,115] These responses are diminished with ROCK2 inhibition, implicating a ROCK2-dependent contribution to the myogenic response by phosphorylation of MYPT1 T852 and/or cofilin. A subsequent study confirmed the importance of ROCK2 in cofilin phosphorylation and enhanced actin polymerization in the myogenic response.[115]

Many studies investigating the role of Ca^{2+} sensitization in response of GI smooth muscles to muscarinic stimulation have not acknowledged the physiological fact that acetylcholine (ACh) released from neurons does not necessarily elicit responses in all post-junctional cells in the SIP syncytium (Fig. 2).[18,19] Ca^{2+} sensitization pathways of GI smooth muscle contraction have typically been studied by immersing muscles in solutions containing contractile agonists, such as CCh (Fig. 3). However, stimulating muscles by bath-applied agonists is not equivalent to neurotransmitter release because different post-junctional receptors are activated in response to these different modes of stimulation (Fig. 3).[14,64,113] We previously found that a bath-applied cholinergic agonist activates Ca^{2+} sensitization mechanisms in murine gastric fundus smooth muscles that are different than those of cholinergic neurotransmission.[14] Electrical field stimulation (EFS) of cholinergic enteric motor neurons only increased CPI-17 phosphorylation, while bath-applied CCh increased both CPI-17 and MYPT1 phosphorylation. We then used the cholinesterase inhibitor neostigmine to strengthen the cholinergic signal during EFS of cholinergic enteric motor neurons. With neostigmine present, both CPI-17 and MYPT1 phosphorylation were increased by EFS. These results support the conclusion that the increase in MYPT1 T852 phosphorylation is the result of a stronger cholinergic signal due to exposure of the agonist to a wider population of muscarinic receptors, that may not typically occur physiologically. Further support for this conclusion is provided from our results using fundus muscles of *W/W^v* mice, which lack intramuscular ICC (ICC-IM). In fundus muscles of *W/W^v* mice, EFS alone of cholinergic enteric motor neurons was able to increase both CPI-17 and MYPT1 phosphorylation. These data suggest that ICC-IM and cholinesterase activity restrict ACh accessibility to a select population of muscarinic receptors, possibly only those expressed by ICC-IM. Cholinergic responsiveness was altered by cholinesterase inhibition or by the absence of ICC-IM, by allowing ACh released from nerve terminals to reach muscarinic M3 receptors on SMCs and enable the activation of ROCK2-dependent Ca^{2+} sensitization pathways and MYPT1 phosphorylation that are not normally activated when cholinesterases are active and ICC are present. Thus, it appears neural stimulation of contraction evokes GI smooth muscle Ca^{2+} sensitization primarily by increasing CPI-17 phosphorylation. The anatomical arrangement of GI enteric motor neurons, ICC-IM, and SMCs within the SIP syncytium may account for these physiological responses. However, these findings have only been reported for murine gastric fundus muscles.

THE SIP SYNCYTIUM: GENERATION OF CA^{2+} ACTION POTENTIALS IN SMOOTH MUSCLE CELLS AND CA^{2+} SENSITIZATION

SMCs provide the forces necessary for GI motility, but interstitial cells, electrically coupled to SMCs, generate the pacemaker activity that underlies major motor patterns, tune SMC excitabil-

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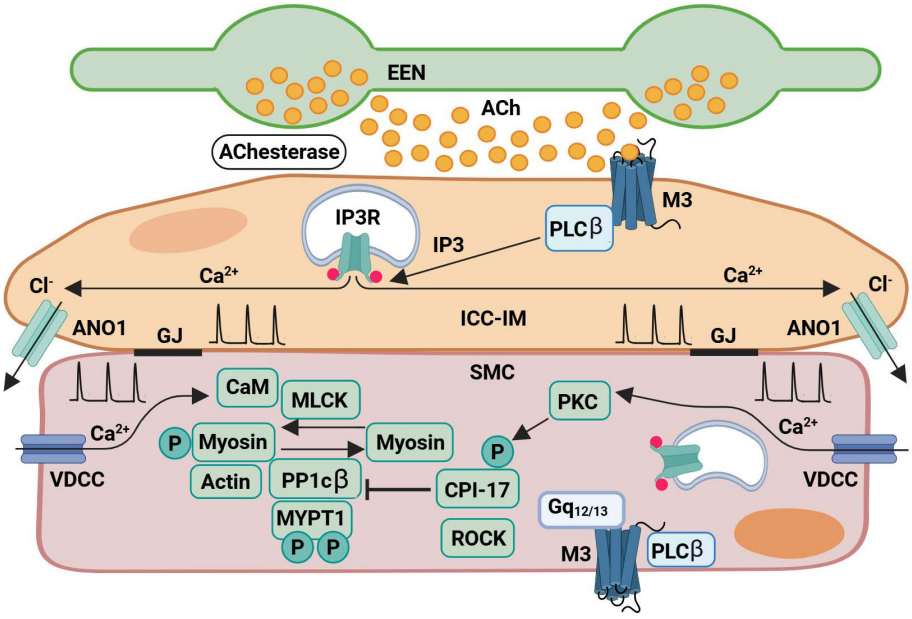

Figure 2. Schematic diagram of physiological activation of calcium sensitization pathways in smooth muscle cells (SMCs). Cholinergic excitatory neurotransmission activates muscarinic M3 receptors on intramuscular interstitial cells of Carjal (ICC-IM) via specialized synapses that exist between enteric nerve terminals and ICC-IM. The acetylcholine (ACh) released is restricted to the post-junctional region between the enteric excitatory neuron (EEN) and the ICC-IM by acetylcholinesterase (AChesterase) activity. ACh binding to muscarinic M3 receptors activates Gqα and phospholipase Cβ (PLCβ), generating inositol 1,4,5-trisphosphate (IP3). IP3 releases Ca^{2+} from intracellular stores into the cytoplasm, causing a transient increase in intracellular Ca^{2+}. These Ca^{2+} transients activate anoctamin-1 (ANO1) Cl– channels, generating inward currents that elicit a transient depolarization of the ICC-IM that conducts to SMCs via gap junctions (GJ), causing depolarization of the SMCs. Depolarization of SMCs opens voltage-dependent Ca^{2+} channels (VDCC), increasing the Ca^{2+} concentration in the cytoplasm. Ca^{2+}/Ca^{2+} binds to calmodulin (CaM) activates myosin light chain kinase (MLCK), which phosphorylates myosin to enable myosin binding to actin for muscle contraction. Voltage-dependent Ca^{2+} influx also activates protein kinase C (PKC), which phosphorylates protein phosphatase 1 regulatory subunit 14A (CPI-17). Phosphorylated CPI-17 binds to and inhibits myosin light chain phosphatase (MLCP), inhibiting the dephosphorylation of myosin regulatory light chain. Thus, physiological cholinergic neurotransmission depolarizes ICC-IM, which is conducted to SMCs and elicits voltage-dependent Ca^{2+} influx into SMCs, activating MLCK, and partially inhibiting MLCP by activating the PKC-mediated calcium sensitization pathway. PP1cβ, catalytic type 1 phosphatase subunit.

ity, and transduce inputs from enteric motor neurons that relax or contract GI smooth muscles.[18] The interstitial cells that regulate SMCs are ICCs and PDGFRα$^+$ cells.[19] Together these cells form the SIP syncytium.[13,19] ICC and PDGFRα$^+$ cells express signature Ca^{2+}-dependent conductances: ICC express Ca^{2+}-activated Cl^- channels, encoded by Ano1, that generate inward current, and PDGFRα$^+$ cells express Ca^{2+}-activated K^+ channels, encoded by Kcnn3, that generate outward current.[26] ICC and PDGFRα$^+$ cells are intermixed with smooth muscle fibers in networks within the circular muscle layer (and in some regions in the longitudinal muscle layer also). ICC-IM and PDGFRα$^+$ cells are closely associated with the varicosities of enteric motor neurons and express receptors, second-messenger pathways, and ion channels that facilitate the functional responses to enteric motor neurotransmitters.[116,117] ICC and PDGFRα$^+$ cells are electrically coupled to SMCs via gap junctions, so responses elicited in the interstitial cells can conduct to SMCs and regulate the excitability of the musculature.

Cholinergic excitatory neurotransmission activates muscarinic M3receptors on ICC-IM via specialized synapses between the enteric nerve varicosities and ICC-IM, activating Gq and phospholipase Cβ and generating inositol 1,4,5-trisphosphate (IP3).[118-120] IP3 releases Ca^{2+} from intracellular stores into the cytoplasm, causing a transient rise in intracellular Ca^{2+}. These Ca^{2+} transients couple to activation of ANO1 Cl^- channels, generating inward currents that elicit a transient depolarization of the ICC-IM that conducts to SMCs via gap junctions, causing depolarization of the SMCs.[18,19,26] Depolarization of SMCs elicits action potentials that occur by activation of L-type voltage-dependent Ca^{2+} channels.[120,121] Ca^{2+} entry into SMCs through L-type (dihydropyridine-sensitive) Ca^{2+} channels during action potentials is substantial, as indicated by the rapid upstroke velocities of action potentials (up to approximately 20 V/s).[119] The opening of L-type Ca^{2+} channels causes an immediate and considerable increase in the cytoplasmic concentration of Ca^{2+} and Ca^{2+} binding to calmodulin, which then binds to and activates MLCK, and is the major mechanism through which GI SMCs achieve excitation-contraction coupling by enabling my-

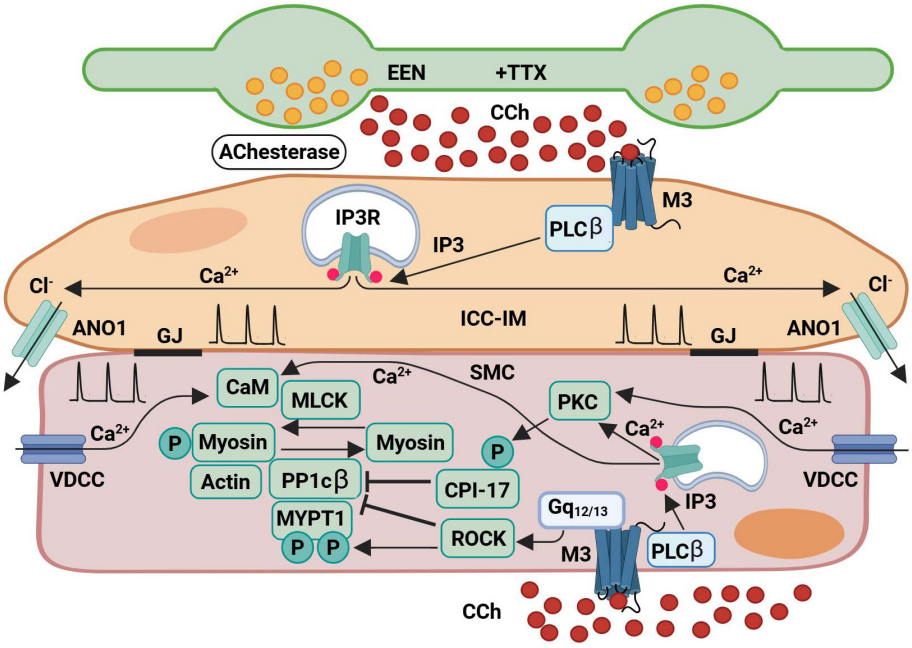

Figure 3. Schematic diagram of pharmacological activation of calcium sensitization pathways in smooth muscle cells. In receptor-mediated smooth muscle contraction, bath-applied cholinergic agonists (CCh) bind to the muscarinic M3 receptors on both intramuscular interstitial cells of Carjal (ICC-IM) and smooth muscle cells (SMCs). CCh binding to muscarinic M3 receptors on ICC-IM activates Gq and phospholipase Cβ (PLCβ), generating inositol 1,4,5-trisphosphate (IP3) and Ca^{2+} release from intracellular stores, causing a transient increase in cytoplasmic Ca^{2+}. These Ca^{2+} transients activate anoctamin-1 (ANO1), generating inward currents that elicit a depolarization of the ICC-IM that conducts to SMCs via gap junctions (GJ), causing depolarization of the SMCs. Depolarization of SMCs opens voltage-dependent Ca^{2+} channels (VDCCs), increasing the Ca^{2+} concentration in the cytoplasm. Ca^{2+}/Ca^{2+} binds to calmodulin (CaM) activates myosin light chain kinase (MLCK), which phosphorylates myosin to enable myosin binding to actin for muscle contraction. Voltage-dependent Ca^{2+} influx into SMCs also activates protein kinase C (PKC), which phosphorylates protein phosphatase 1 regulatory subunit 14A (CPI-17). Phosphorylated CPI-17 binds to and inhibits myosin light chain phosphatase (MLCP), inhibiting the dephosphorylation of myosin regulatory light chain. CCh binding to muscarinic M3 receptors on SMCs activates Gq12/13, stimulating guanosine 5′-triphosphate (GTP) binding to RhoA, which then binds to and activates Rho-associated coiled-coil kinase (ROCK). ROCK then phosphorylates myosin phosphatase targeting subunit 1 (MYPT1), and other proteins, leading to reduced MLCP activity. Gq12/13 also activates PLCβ, generating IP3 and diacylglycerol (DAG). IP3 releases Ca^{2+} from intracellular stores into the cytoplasm. This Ca^{2+} binds to CaM which activates MLCK to phosphorylate and enable myosin to bind to thin filament actin for cross-bridge cycling. Ca^{2+} and DAG also bind to and activate PKC, which phosphorylates CPI-17 to further inhibit MLCP. Thus, during pharmacologically-mediated contraction, the ratio of MLCK activity to MLCP activity is even higher, and contraction is further augmented because both the PKC- and ROCK-mediated calcium sensitization pathways are activated. EEN, enteric excitatory neuron; TTX, tetrodotoxin; PP1cβ, catalytic type 1 phosphatase subunit.

osin RLC phosphorylation and myosin binding to actin myofilaments for cross-bridge cycling.[119] Thus, the increase in the cytoplasmic concentration of Ca^{2+} that occurs by the depolarization of L-type Ca^{2+} channels appears to be the major mechanism by which GI SMCs achieve excitation-contraction coupling, and appears to be the primary physiological mechanism by which GI SMCs contract. The increase in T852 phosphorylation observed in GI smooth muscle tissues by bath applied CCh, as well as the increase in T852 phosphorylation by EFS in the presence of a cholinesterase inhibitor likely represents pharmacologically-evoked intense activation of Ca^{2+} sensitization pathways not observed physiologically. The data showing that excitatory motor neuron neurotransmission only increases PKC-dependent CPI-17 phosphorylation supports this mechanism of excitation-contraction coupling in GI smooth muscles and supports the conclusion that neural stimulation of GI smooth muscle contraction evokes Ca^{2+} sensitization primarily by increasing CPI-17 phosphorylation (Fig. 2).

ACCURATE QUANTITATION OF PROTEIN PHOSPHATASE 1 REGULATORY SUBUNIT 12A, PROTEIN PHOSPHATASE 1 REGULATORY SUBUNIT 14A, AND MYOSIN REGULATORY LIGHT CHAIN PHOSPHORYLATION

The primary requirement for measuring in vivo levels of protein phosphorylation in cells and tissues is to preserve protein phos-

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phorylation during homogenization of the cells or tissues for analysis. However, a common problem in assessing in vivo protein phosphorylation is the occurrence of nonspecific dephosphorylation of phosphorylated proteins during sample preparation.[122,123] Studies of the regulation of MLCP activity in vivo by protein kinases require accurate quantitative measurements of the phosphorylated substrates; in particular MYPT1, CPI-17, as well as the substrate for MLCK and MLCP, myosin RLC. There are important considerations for measuring the phosphorylation of proteins, including tissue fixation methods, the extraction of phosphoproteins, and the methods utilized for measuring the extent of phosphorylation of specific phosphorylated sites.[124-126] Important considerations for preparing tissue for analysis include: minimizing dissection trauma or other disturbances; freezing or fixing the tissue rapidly enough to prevent changes from occurring during transfer of the tissue to the freezing or fixing medium, preventing alterations during storage and during manipulation before tissue lysis; and avoiding alteration of phosphoproteins by kinases and phosphatases in the lysis buffer and during tissue lysis.[127] Freezing tissue rapidly enough is critical to prevent artificial changes due to the uncontrolled activities of protein kinases and phosphatases. However, smooth muscle tissues are particularly challenging, due to the high expression levels of both MLCK and MLCP. Tissue sample handling and processing, storage, as well as buffer formulations are all significant variables in the measurement of MYPT1, CPI-17, and RLC phosphorylation. Rapidly placing the frozen tissue into ice-cold lysis buffer before it can thaw and allow enzymatic activities to recover is just as important as rapidly freezing the tissues to stop enzymatic activity. In tissues homogenized directly in SDS sample buffer without phosphatase inhibitors and sufficient tissue dilution, RLC can be dephosphorylated, resulting in low baseline tissue RLC phosphorylation values.[128,129] Therefore, in addition to containing protease inhibitors, smooth muscle tissues must be homogenized in buffers containing sufficient concentrations of phosphatase inhibitors.[122,123] A typical lysis buffer designed to inhibit phosphatase activities contains sodium fluoride, sodium orthovanadate, sodium pyrophosphate, and β-glycerophosphate.[122,123] In addition, prior to freezing, the smooth muscle tissue should be submerged into an ice-cold solution of 10% trichloroacetic acid (TCA) dissolved in acetone.[67,125,129,130] The acetone permeabilizes the tissue, allowing TCA to rapidly penetrate the tissue. TCA is a strong protein precipitant and denaturant, which rapidly inactivates cellular phosphatases, proteases, and kinases. Utilization of trichloroacetic acid to fix proteins in a phosphorylated state is a powerful tool for accurate quantification of protein phosphorylation. Fixing cells or tissues in 10% TCA in acetone has also been shown to preserve protein phosphorylation for detection by immunofluorescence microscopy.[131,132] Recently, RNA later has been shown to preserve proteins and protein phosphorylation for proteomic studies, but smooth muscle tissues have not been tested.[133]

DISEASE

Alterations to Ca^{2+} sensitization mechanisms have been reported in certain models of common GI motility disorders. Gastroparesis is characterized by abnormal gastric function resulting in delayed gastric emptying in the absence of mechanical or physical obstruction.[134] Gastroparesis is a common complication of diabetes, adversely impacting quality of life with symptoms of abdominal discomfort, nausea, and vomiting.[22] The pathogenesis of this disorder is not well understood, but an innate immune dysregulation and injury to the ICC and other components of the enteric nervous system through paracrine and oxidative stress mediators are likely central to the pathogenesis of gastroparesis.[135-137] Smooth muscle atrophy, degeneration, and fibrosis have been observed in severe diabetes.[136] Investigations have reported alterations to Ca^{2+} sensitization mechanisms in some animal models of gastroparesis. The *ob/ob* leptin-deficient mouse model of obesity and type 2 diabetes develops delayed gastric emptying, providing an animal model for investigating how gastric smooth muscle dysfunction may contribute to the pathophysiology of diabetic gastroparesis.[138] Using this model of obesity and diabetes, we found that ROCK2 expression in gastric antrum smooth muscle was decreased compared to age-matched controls.[48] Basal and agonist-evoked MYPT1 and RLC phosphorylation, but not CPI-17 phosphorylation, were also reduced compared to age-matched controls.[48] These findings suggest that decreased ROCK2 phosphorylation in gastric antrum smooth muscles could potentially contribute to the antral dysmotility of diabetic gastroparesis, but gastric emptying was not measured.

Streptozotocin induced type 1 diabetes leads to gastric motor abnormalities and altered gastric emptying in rats.[139] Reports show that streptozotocin induced diabetes results in reduced activities of phospholipase C, PKC, and MLCK activities, and a decrease in the contractility of gastric smooth muscles, due to reduced Ca^{2+} signaling, although CPI-17 phosphorylation was not investigated.[140,141] Diabetes occurs spontaneously in 90% of partially inbred BB/W diabetic rats.[142] This animal model resembles human insulin-dependent diabetes and has been used to investigate the mechanism of diabetic gastroparesis.[143-145] Six months after induction of diabetes, CCh-induced responses of gastric smooth muscles were significantly reduced in diabetic rats compared to controls.[146] In addition, there was also impaired PKC activation and translo-

cation from the cytosol to the plasma membrane in gastric muscles from diabetic BB/W rats. These observations indicate that functional myogenic impairment occurred in diabetic BB/W rats, resulting from altered intracellular signal transduction involving abnormal IP3 production and PKC translocation, although CPI-17 phosphorylation was not investigated.[146]

Ulcerative colitis and Crohn's disease are chronic inflammatory bowel diseases (IBDs). It is well known that IBD is associated with dysmotility of the gastrointestinal tract that can lead to diarrhea, constipation or both.[147] In patients with IBD and in experimental colitis animal models, it has been demonstrated that spontaneous contractions and agonist induced contractions of isolated colonic smooth muscles are decreased by functional alternations to the smooth muscle, such as impaired Ca^{2+} mobilization, enhancement of ATP-sensitive K^+ channels, and/or dysfunction of the enteric nervous system.[148,149] However, there are also a few reports about the influence of intestinal inflammation on smooth muscle contractile systems. In the mouse model of dextran sodium sulphate (DSS)-induced ulcerative colitis it was found that CCh-induced contractions of colon smooth muscles were reduced, while the expression of muscarinic M3 receptors, RhoA, ROCK1, ROCK2, and MYPT1 in smooth muscles was not changed.[24] However, CPI-17 expression was significantly decreased in DSS-treated mouse colon. These results suggest that the suppression of CCh-induced contraction in mice with colitis may be at least partially attributable to the increased activity of myosin phosphatase by the downregulation of CPI-17 expression. In the TNBS-induced colitis model in rats, CCh-induced contractions of colon smooth muscles were reduced, and PKC expression and CPI-17 phosphorylation were reduced.[21] These findings suggest that Ca^{2+}-sensitization in rat colon involves a PKC-dependent pathway, and that the reduced CCh-induced contraction in colitis was due to inflammation-induced changes in Ca^{2+}-sensitization involving a defect in the PKC pathway. IL-10 KO mice develop spontaneous and chronic colitis accompanied by immune cell infiltration, submucosal fibrosis, and thickening of the muscularis externa.[23] α-smooth muscle actin protein expression in colon smooth muscles did not change, but CPI-17 protein was significantly decreased compared with healthy wild-type controls. Colon smooth muscle contractile force and RLC phosphorylation induced by a muscarinic agonist were reduced in the KO mice. In addition, in colonic smooth muscle tissues from patients with ulcerative colitis, CPI-17 protein expression was decreased compared with healthy control subjects.[23] These findings suggest that CPI-17 downregulation may contribute to the decreased motor function in chronic inflammatory bowel diseases. Th1 and Th2 cytokines cause hypocontractility and hypercontractility of inflamed intestinal smooth muscle.[150-152] Th1 cytokines downregulate CPI-17 and L-type Ca^{2+} channels and upregulate regulators of G protein signaling 4, which contributes to the hypocontractility of inflamed intestinal smooth muscle.[152]

CONCLUSIONS AND FUTURE DIRECTIONS

This review emphasizes the importance of smooth muscle calcium sensitization to GI motility, and highlights the value of genetically engineered mouse models with targeted deficiencies in the PKC and ROCK2 pathways for more clearly defining the physiological roles of CPI-17 and MYPT1 phosphorylation in regulating MLCP activity and smooth muscle contractile responses, especially in physiological versus pharmacological contexts. The evidence supporting the role of ICCs, as part of the SIP syncytium, in regulating the activation of calcium sensitization pathways within SMCs is discussed. This review summarizes how dysregulation of calcium sensitization pathways may contribute to GI motility disorders. Future studies should expand on these models to dissect the specific roles of these proteins in sphincter function and GI motility, and the impact of their dysregulation in GI motility and sphincter disorders. Translating findings from animal models and cellular studies into clinical research to identify new drug targets and treatment strategies for GI motility disorders could yield potentially beneficial results.

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Author contributions
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Gastroesophageal Reflux Disease and Reflux-like Phenotypes: Using Lyon Consensus 2.0 to Guide Clinical Approach

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Gastroesophageal reflux disease (GERD) is common worldwide, but troublesome symptoms alone are not sufficient for a conclusive diagnosis. The Montreal Consensus introduced a symptom-based definition of GERD and defined mucosal injury phenotypes of the disease. The Lyon Consensus adapted mucosal injury findings as well as defined abnormal reflux burden criteria for the diagnosis of conclusive GERD. The diagnosis of esophageal disorders of gut–brain interaction (E-DGBI) using Rome V criteria requires exclusion of GERD, which is also described in the Lyon Consensus 2.0 criteria. Collectively utilizing these various criteria, distinct phenotypes can be identified in patients with troublesome reflux-like symptoms. The mucosal injury phenotype, initially defined by the Montreal Consensus, requires reflux management for healing of esophagitis to improve symptoms, and surveillance to identify and prevent GERD complications including Barrett's esophagus and esophageal adenocarcinoma. The abnormal reflux burden phenotype requires reflux management for symptom improvement. The E-DGBI phenotype requires specific management of each of the functional disorders for symptom relief. Hypervigilance and symptom-specific anxiety can overlap with all the phenotypes, and E-DGBI can coexist with conclusive GERD. Therefore, there are intricate interrelationships between Montreal, Lyon and Rome criteria that aid evaluation of troublesome symptoms, and concepts from each of these contribute to accurate phenotyping and optimal management of patients.

Keywords: Esophageal hypervigilance; Esophagitis; Functional esophageal disorders; Gastroesophageal reflux disease; Rumination

INTRODUCTION

Gastroesophageal reflux disease (GERD) is common worldwide, with prevalence of 8%-33%.[1] Two decades ago, the Montreal Consensus introduced a symptom-based definition of GERD, where the reflux of gastric content causes troublesome symptoms, disease-specific complications, or both,[2] but troublesome symptoms are not always indicative of GERD.[3] The Lyon Consensus was conceived to identify features from esophageal testing that define GERD, as distinct from absence of GERD.[4] When findings are inconclusive, the Lyon Consensus recommends further differentiation towards or away from conclusive GERD using adjunctive metrics from esophageal testing.[4,5]

Concurrent with these advances, Rome diagnostic criteria have evolved to non-erosive reflux disease (NERD) requiring symptoms, a normal endoscopy and abnormal reflux burden (Table 1).[6,7] Esophageal disorders of gut–brain interaction (E-DGBI) require exclusion of GERD, making the Lyon Consensus directly relevant to Rome diagnoses.[5] Hypervigilance and symptom-specific anxiety are prevalent across all reflux phenotypes,[8] and E-DGBI can overlap with conclusive GERD.[9] Thus, GERD and related symptom-based phenotypes constitute a family of disorders with distinct but related pathophysiology that includes disruption of the esophagogastric junction (EGJ), mucosal injury and complications, esophageal clearance, esophageal perception, and esophageal hypervigilance.[10] This review addresses the interrelationships between Montreal, Rome and Lyon criteria in the clinical application of Lyon Consensus 2.0.

TROUBLESOME SYMPTOMS

Troublesome typical symptoms of heartburn, acid regurgitation and esophageal chest pain introduced by the Montreal Consensus are the starting point for empiric GERD treatment trials.[2,5,9] Qualification and quantification of symptoms remain central to the approach to GERD, which directly impact health-related quality of life.[2,11,12] Validated patient reported outcome instruments offer

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valuable insights into the likelihood of GERD,[13] but are not reliable to diagnose GERD.[14,15] Resolution of symptoms is an important therapeutic target, although prevention and treatment of mucosal inflammation, esophageal strictures, Barrett's esophagus (BE), and esophageal adenocarcinoma are important considerations in both symptomatic and asymptomatic patients (Figure).[16]

Typical symptoms have a higher likelihood of response to antisecretory therapy compared to atypical symptoms,[17] hence empiric treatment trials are offered in the absence of alarm symptoms. In randomized controlled trials, over 70% of patients with heartburn respond to short-term proton pump inhibitor (PPI) therapy, but placebo response can be as high as 25%.[18] Approximately 50% of patients with regurgitation also report improvement with PPIs, but regurgitation can persist despite improvement of concurrent heartburn,[19,20] and similar placebo response is reported.[21] After ruling out cardiac etiologies, GERD is identified in up to 60% of patients with esophageal chest pain (Table 2), where PPI response can match that of heartburn (56-85% with at least 50% improvement) in contrast to patients without objective GERD where PPI response is not significantly better than with placebo.[22] Using this data, PPI responsiveness has a sensitivity of 84% and specificity of 76% in discriminating GERD from non-GERD presentations in esophageal chest pain.[23] However, not all PPI responders have underlying conclusive GERD. A study of 312 endoscopy negative patients with heartburn found that of 144 patients who experienced ≥ 75% symptom improvement with PPIs, only 55% had abnormal reflux burden on pH-impedance monitoring, while the remainder had functional syndromes including reflux hypersensitivity (RH, 26%) and functional heartburn (FH, 19%).[3] Therefore, troublesome typical symptoms are a starting point for empiric therapy where response predicts presence of GERD with modest sensitivity (78%) albeit with low specificity (54%).[17] The Lyon Consensus 2.0 requires conclusive GERD evidence on objective testing prior to escalation of management.

Laryngopharyngeal symptoms (LPS) encompass several atypical symptoms that are poorly responsive to PPI, including chronic cough, hoarse voice, throat clearing, sore throat, and excessive phlegm in the throat.[24,25] Retrograde movement of gastric contents is rarely demonstrated, making laryngopharyngeal reflux disease infrequent in LPS presentations (Table 2).[26] and alternative mechanisms are more likely.[27] In randomized controlled trials, PPI therapy has shown no significant benefit over placebo in undifferentiated LPS as well as uninvestigated chronic cough,[28,29] and abnormal reflux is more likely when typical reflux symptoms coexist with LPS.[30] Both Lyon Consensus 2.0 and the San Diego Consensus for LPS/laryngopharyngeal reflux disease recommend objective upfront esophageal testing prior to anti-reflux therapy for isolated LPS,[5,25] which is further supported by cost benefit analysis.[31]

In contrast to other LPS, globus is negatively associated with conclusive GERD (OR, 0.5; 95% CI, 0.30-0.86) and has been vali-

Table 1. Stratification of Gastroesophageal Reflux Disease Phenotypes by Lyon 2.0, Montreal, and Rome IV Consensus Classification Definitions

Phenotype	Lyon 2.0 (2024)	Montreal (2006)	Rome V (2026)
No GERD	Normal endoscopy AET < 4% (each day of study) Supported by reflux episodes < 40 and MNBI > 2500 Ω	-	Normal endoscopy Normal biopsies for FCP Physiologic AET Absence of motor disorders Positive RSA for RH
Inconclusive GERD	LA grade A esophagitis AET 4-6% (24-hour study or ≥ 2 days on prolonged wireless study)	Typical reflux syndrome Reflux chest pain syndrome Extraesophageal (atypical) syndromes	-
Conclusive GERD with mucosal injury	LA grades B, C, and D esophagitis Peptic esophageal stricture Biopsy proven Barrett's mucosa	Reflux esophagitis Reflux stricture Barrett's esophagus Esophageal adenocarcinoma	-
Conclusive GERD with abnormal reflux burden	AET ≥ 6% on 24-hour study or on ≥ 2 days on prolonged wireless study OR Inconclusive GERD with supportive evidence: > 80 reflux episodes, MNBI < 1500 Ω, RSA	-	Pathologic AET distinguishes NERD from FCP, RH, and FH FCP, RH, and FH can overlap with conclusive GERD
Behavioral phenotypes	No GERD in pure behavioral phenotypes; Behavioral phenotypes can overlap with GERD	-	Supragastric belching Rumination syndrome

GERD, gastroesophageal reflux disease; AET, acid exposure time; MNBI, mean nocturnal baseline impedance; LA, Los Angeles classification; RSA, reflux symptom association; FCP, functional chest pain; RH, reflux hypersensitivity; NERD, non-erosive reflux disease; FH, functional heartburn.

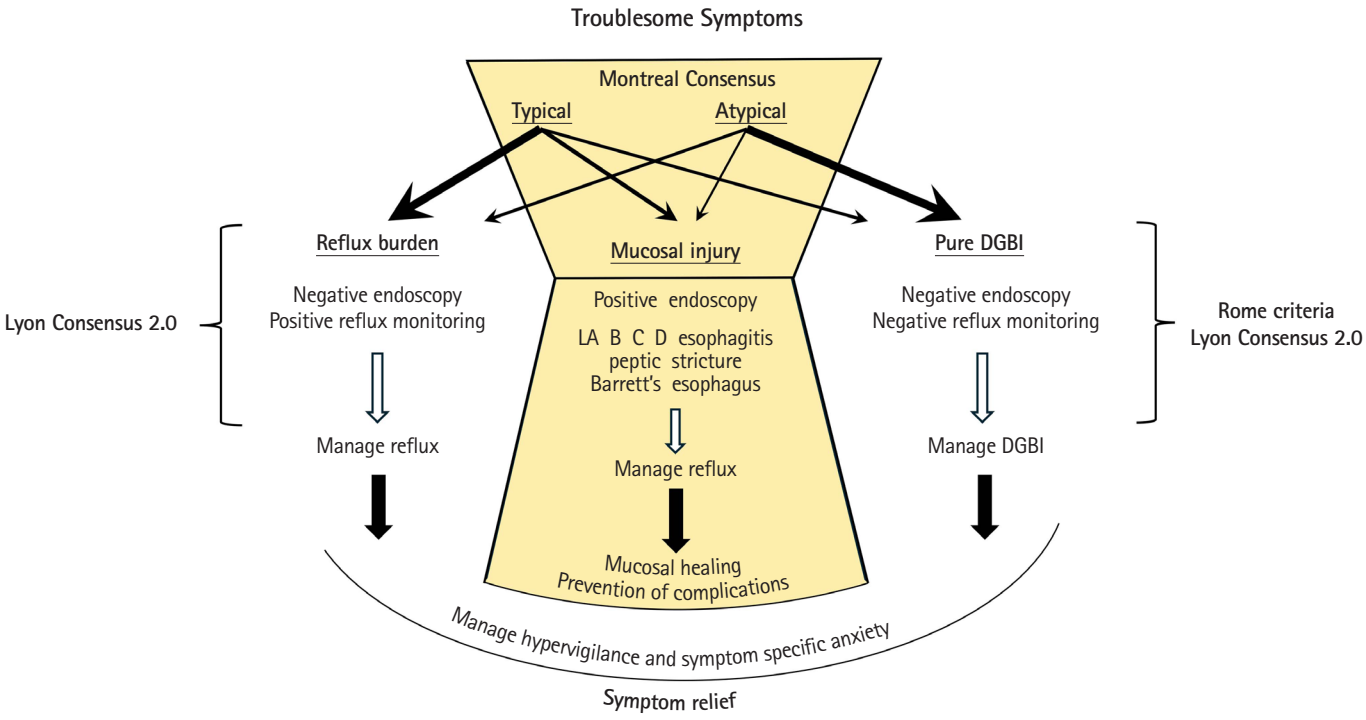

Figure. Interrelationships between Montreal Consensus, Lyon Consensus 2.0 and Rome Criteria. Montreal Consensus starts with patients with troublesome symptoms, but only a third of these patients will have the mucosal injury phenotype. Lyon Consensus 2.0 adapted findings of esophageal mucosal injury into conclusive evidence for gastroesophageal reflux disease (GERD) and defined the abnormal reflux burden phenotype. Rome Criteria, on the other hand, requires exclusion of both mucosal injury and abnormal reflux burden phenotypes to identify pure disorders of gut brain interaction (DGBI), and the updated version utilizes Lyon Consensus 2.0 criteria for this purpose. Hypervigilance and symptom-specific anxiety can be present in all the symptom phenotypes, and DGBI can overlap with conclusive GERD diagnosed using Lyon Consensus 2.0 criteria. The ultimate goal is symptom relief, achieved primary through management of GERD in the mucosal injury and abnormal reflux burden phenotypes, and through management of DGBI when evidence of GERD is not found.

dated as a negative GERD predictor in a weighted scoring system.[32] Belching may associate with reflux events, but can also be part of behavioral syndromes, making upfront pH-impedance monitoring a useful initial approach (Table 2).[33] Other isolated foregut symptoms such as nausea, dyspepsia, abdominal pain, and dysphagia have very low likelihood of objective GERD, and reflux testing serves primarily to rule out a reflux based mechanism.[5] The more atypical the symptom, the higher the need for documentation of GERD prior to management, as non-GERD etiologies including DGBI and behavioral syndromes are likely to contribute to symptom burden more often than reflux mechanisms.

Therefore, given competing mechanisms and substantial overlap with non-reflux etiologies, troublesome symptoms alone do not define phenotypes with mucosal injury or abnormal acid burden.[34]

DIAGNOSTIC EVALUATION

Diagnostic evaluation becomes necessary in patients with typical symptoms who fail to respond to empiric treatment trials, in patients with LPS or other atypical symptoms, and in the presence of alarm symptoms, where a high-quality upper endoscopy is typically the first test performed (Table 2).[5,24,35]

Although endoscopy is an important tool for the evaluation of esophageal changes from reflux, erosive esophagitis can be identified in only a third of treatment naïve patients with typical reflux symptoms, and the diagnostic yield decreases to < 10% in patients on PPIs despite persistent symptoms.[36] Since erosive esophagitis is expected to heal in > 70% treated with PPIs and > 90% with potassium competitive acid blockers,[37] endoscopic evaluation is recommended off therapy for 2-4 weeks if GERD diagnosis is uncertain.[38,39] Screening for BE is recommended in high-risk populations including patients with chronic GERD symptoms and risk factors including older age, Caucasian race, obesity, family history of esophageal cancer, and tobacco/alcohol use.[40-43]

Reflux monitoring technology has advanced beyond 24-hour pH measurements alone to include impedance monitoring that can quantify mucosal integrity, and prolonged wireless pH monitoring that records acid burden for up to 96 hours. In patients with unproven GERD and negative endoscopy, ambulatory reflux monitor-

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Table 2. Evaluation and Potential Mechanisms of Common Symptoms

Symptom	Work-up	Likely	Less likely	Rare	Exclusions
Heartburn	EGD, reflux monitoring (wireless preferred) ± HRM/HRIM	ERD, NERD	RH, FH EoE	Achalasia, motor disorders	Cardiac disorders
Chest pain	EGD with biopsies, reflux monitoring (wireless preferred) ± HRM/HRIM	ERD, NERD	RH, FCP EoE	Achalasia, motor disorders	Cardiac disorders
Regurgitation	EGD, reflux monitoring (pH impedance preferred) + HRIM	ERD, NERD	Achalasia, Motor disorders	Rumination, EoE	Vomiting disorders, Gastric disorders
Cough/throat clearing/ throat phlegm	EGD, reflux monitoring (pH impedance preferred) + HRM/HRIM	Hypersensitivity and hypervigilance	LPRD	ERD, Motor disorders	Allergic conditions, post-nasal drip, Pulmonary and ENT disorders
Hoarseness/throat pain/ voice changes	EGD, reflux monitoring (pH impedance preferred) + HRM/HRIM	Hypersensitivity and hypervigilance	LPRD	ERD, Motor disorders	Allergic conditions, post-nasal drip, ENT and pulmonary disorders
Globus	EGD, reflux monitoring (pH impedance or wireless) ± HRM/HRIM	Hypersensitivity and hypervigilance	Inlet patch	ERD, NERD, motor disorders	ENT disorders
Belching	EGD, reflux monitoring (pH impedance preferred) + HRIM	ERD, NERD, gastric belching	Supragastric belching	Motor disorders	

EGD, esophagogastroduodenoscopy; HRM, high-resolution manometry; HRIM, high-resolution impedance manometry; ERD, erosive reflux disease; NERD, non-erosive reflux disease; RH, reflux hypersensitivity; FH, functional heartburn; EoE, eosinophilic esophagitis; LPRD, laryngopharyngeal reflux disease; FCP, functional chest pain; ENT, ear nose throat.

ing off antisecretory therapy differentiates NERD from non-GERD syndromes including RH and FH (Figure), using either prolonged wireless pH monitoring or 24-hour catheter based pH/pH-impedance monitoring.[5] The wireless pH probe can be placed during initial diagnostic endoscopy, overcomes day-to-day variability in reflux burden and does not interfere with the patient's daily activities that might confound 24-hour studies.[44] However, wireless pH monitoring may not be universally available, comes with higher cost, and lacks impedance-based metrics that improve sensitivity and specificity of reflux monitoring.[5] Catheter based pH/pH-impedance monitoring requires high-resolution manometry (HRM) for identification of the proximal margin of the LES for proper placement. This has higher sensitivity in detection of not just acidic but also weakly/non-acidic reflux events, adding to the value of reflux-symptom association. Baseline impedance, a longitudinal marker of esophageal mucosal integrity, can be measured as an adjunctive metric of reflux burden.[5] Further, pH-impedance monitoring on optimized antisecretory therapy can help determine the mechanism of persisting symptoms despite therapy in patients with conclusive GERD.[45]

Although insufficient for a conclusive GERD diagnosis by itself, HRM can assess esophageal peristaltic performance, motility patterns associated with reflux and EGJ morphology.[46,47] Intact primary esophageal peristalsis helps promptly clear refluxate from the esophagus after reflux events, and brings neutralizing saliva for recovery of mucosal acidification.[16] Hypomotility disorders such as ineffective esophageal motility and absent contractility can prolong contact of the noxious refluxate with esophageal mucosa, potentially causing reflux symptoms and mucosal complications.[48] HRM is an essential component of pre-operative evaluation prior to anti-reflux intervention,[46] when achalasia and obstructive motor disorders mimicking GERD were found in approximately 3% of > 1000 patients being evaluated for anti-reflux surgery.[49]

Barium esophagography lacks diagnostic accuracy in identifying conclusive GERD and does not replace ambulatory reflux monitoring in GERD diagnosis. A recent retrospective evaluation of 182 patients who underwent both barium esophagram and ambulatory pH-impedance monitoring found low sensitivity (37.6%) and specificity (71.6%) of barium radiography in diagnosing conclusive GERD according to Lyon Consensus 2.0 criteria.[50] The role of barium esophagography in GERD presentations is primarily in characterizing EGJ morphology, and as an adjunctive study in defining achalasia or other obstructive processes.[46,47]

MUCOSAL INJURY PHENOTYPES

The Montreal Consensus first outlined syndromes of esophageal injury including reflux esophagitis, peptic stricture, and BE to define end organ complications from GERD (Figure).[2] The Lyon Consensus adopted these mucosal injury phenotypes as endoscopic criteria for conclusive GERD (Table 1).[4]

Visible erosions in the distal esophageal mucosa remain the most reliable endoscopic findings of acid injury. The Los Angeles (LA) classification assigns higher grades to more profound mucosal injury, which in turn associate with higher acid exposure time (AET) on reflux monitoring, worse symptom severity at baseline, and greater symptom improvement with antisecretory therapy.[51,52] While endoscopic healing with antisecretory therapy occurs more often in lower LA grades of esophagitis,[38,53] recurrence of esophagitis after withdrawal of therapy is more likely in higher LA grades.[54] The updated Lyon Consensus 2.0 elevated LA grade B esophagitis to conclusive GERD evidence, in light of studies demonstrating similar reflux evidence on pH-impedance testing and response to PPI compared to LA grade C esophagitis, but markedly higher reflux burden than LA grade A esophagitis.[52,55] However, interobserver variation in assigning LA grades can confound Lyon Consensus designation of conclusive GERD,[56,57] and borderline LA grade B with limited erosions may need to be interpreted similar to LA grade A esophagitis.[58]

Peptic esophageal strictures remain an important mucosal injury phenotype, although prevalence has decreased with widespread PPI use.[59,60] Peptic strictures are short (< 2 cm), manifest with or without concurrent esophagitis, and typically present with troublesome dysphagia.[61] Biopsies can to rule out alternative mechanisms such as eosinophilic esophagitis or neoplasia. Acid suppression alters the natural history of peptic strictures, and PPIs have been shown to be superior to histamine-2 receptor blockers for resolution of dysphagia, prevention of recurrence, and need for repeated dilation.[62]

Salmon colored mucosa in the distal esophagus raises suspicion of BE, and biopsies showing intestinal metaplasia with goblet cells is required to confirm BE in most but not all international guidelines.[41,63] Patients with BE have higher reflux burden compared to healthy volunteers, as well as patients with NERD and erosive esophagitis.[64,65] Long-segment BE (> 3 cm) associates with higher AET, greater number of total reflux episodes, more proximal reflux, and longer acid clearance time.[64,66,67] Professional societies vary on inclusion of ultrashort segment (< 1 cm) BE and irregular Z line in the BE diagnostic pool, since large retrospective studies show negligible risk of progression to high grade dysplasia or esophageal adenocarcinoma.[68,69] However, a recent retrospective study of 167 US veterans with ultrashort segment BE on index endoscopy found persistence of BE in over half of patients, and definitive dysplasia in 8.4% on follow up[70]; thus exclusion from diagnostic criteria is controversial.[71] Given these discrepancies, the Lyon Consensus 2.0 requires biopsy proven BE for conclusive GERD diagnosis, but makes no specifications regarding BE length, and leaves this designation to the treating physician or surgeon.[5]

ABNORMAL REFLUX BURDEN PHENOTYPES

When endoscopic evidence of mucosal injury is not evident, further investigation using ambulatory reflux monitoring can segregate NERD from non-GERD phenotypes (Table 1).[6] Meta-analysis of patients with heartburn treated with PPI found similar rates of complete symptom response between erosive esophagitis (72.0%) and NERD with abnormal reflux burden (73.5%), while negative endoscopy without further evaluation had a lower heartburn response (50.5%).[70]

Rome V criteria and Lyon Consensus 2.0 both require abnormal AET on either wireless or catheter-based pH/pH impedance monitoring to define NERD.[5,9] Cohort studies have consistently demonstrated total AET as the most reproducible metric from ambulatory reflux monitoring, with high sensitivity (77-100%) and specificity (85-100%) for differentiation of patients with esophagitis from healthy controls.[72-75] and for predicting symptom improvement after medical or surgical management.[76] The 95th percentile AET threshold lies between 5.3% and 6.3% in healthy controls.[77-79] Based on these and other studies comparing acid burden across the GERD spectrum to healthy controls, the AET threshold for pathologic GERD was set at 6% for Western populations, while AET < 4% is considered physiologic, and values 4-6% are inconclusive and require additional supportive data.[4,72] In Asian countries, a lower AET threshold of 4% is used to define conclusive GERD.[80]

The Lyon Consensus 2.0 suggests prolonged wireless pH testing as the preferred ambulatory modality for evaluation of typical symptoms and negative endoscopy, especially perceptive symptoms (heartburn, esophageal chest pain) (Table 2). In a retrospective study of PPI refractory patients, 85% of patients with discordant results on days 1 and 2 had a predominant physiologic, pathologic or borderline pattern evident with additional days of analysis.[81] Utilizing an AET threshold of < 4% on all days of a 96-hour wireless study, a negative predictive value of 99% was demonstrated in discriminating healthy controls from patients with erosive esophagitis.[52] Additionally, the cumulative number of days with total AET < 4% has added prognostic value in predicting PPI discontinuation.[82]

In patients with borderline total AET (4-6%), presence of adjunctive criteria can increase confidence in the likelihood of GERD (Ta-

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ble 1). While various reflux episode thresholds have been reported,[83] ≥ 80 total reflux episodes off acid suppression on pH-impedance monitoring provides the strongest evidence for escalating reflux management.[84] High numbers of total reflux episodes can differentiate PPI dependent GERD patients from healthy controls and a reduction of reflux episodes to below physiologic thresholds post intervention is associated with satisfaction with therapy.[84,85] Post hoc analysis of 123 patients randomized to twice daily PPI versus magnetic sphincter augmentation in patients with troublesome regurgitation found that > 80 total reflux episodes predicted satisfaction (≥ 50% reduction of symptoms) with intervention in 31 PPI non-responders who crossed over to magnetic sphincter augmentation.[84]

Baseline impedance on pH impedance monitoring is an indicator of esophageal mucosal integrity that can segregate GERD from healthy controls and provide discriminative value in patients with borderline AET.[86-88] Mean nocturnal baseline impedance (MNBI) is the average at the 3 cm or 5 cm impedance channel over three 10-minute overnight intervals (1 AM, 2 AM, and 3 AM). Low MNBI (< 1500 Ω) is found in many patients with pathologic or borderline AET but is rare in healthy controls.[81] In a retrospective study of 371 patients who underwent pH impedance monitoring off therapy, symptomatic improvement was similar between patients with pathologic AET and those with borderline AET with low MNBI (73% vs 76%) but similar to physiologic AET when MNBI was normal (33.3% vs 38.6%).[89]

Reflux symptom association (RSA) using symptom association probability (> 95%) and symptom index (> 50%) assesses temporal relationships between reported symptoms and reflux events.[76] Rome V utilizes RSA in the diagnosis of RH irrespective of response to anti-secretory therapy (Table 1). RSA may provide insights into response to anti-reflux therapy,90 however, interpretation requires caution and should be used in conjunction with other supporting evidence prior to escalation of anti-reflux therapy.[91,92]

Mucosal biopsies have a limited role in the diagnosis of GERD. Small studies have shown that microscopic esophagitis is less common in healthy controls and patients with FH (13% and 15% respectively) compared to patients with erosive esophagitis, NERD, and esophageal hypersensitivity (95%, 77%, and 65%, respectively).[93] Dilated intracellular spaces on electron microscopy are more frequent in patients with increased reflux burden (60%) compared to patients with FH (9%), and correlate negatively with MNBI (r = -0.230), however, this technology is not readily available in most clinical settings.[94] Overall, diagnostic performance of histopathology to differentiate conclusive GERD from non-GERD phenotypes is suboptimal and not recommended.[95]

NORMAL REFLUX BURDEN PHENOTYPES

Rome Diagnoses

When AET is physiologic (< 4%) in symptomatic patients, E-DGBI, particularly functional chest pain, FH and RH, are diagnosed using ROME criteria (Figure).[9] Patients with heartburn or esophageal chest pain for at least 6 months, without a structural, mucosal or motor disorder on esophageal testing and physiologic acid exposure with no RSA fulfil criteria for FH and functional chest pain. If RSA is present, RH is diagnosed.

Diagnosis of E-DGBI could be confounded by day-to-day variation in reflux burden if reflux monitoring is performed using 24-hour pH-impedance, and with lack of impedance-based markers if wireless pH monitoring is utilized. Some patients with physiologic AET could have adjunctive pH-impedance based metrics that are not within physiologic range. For instance, very low MNBI and/or high reflux episode counts in the context of physiologic AET could raise suspicion of GERD, and indeed, patients fulfilling these criteria demonstrate PPI response.[96] In contrast, patients with AET < 4% and no adjunctive GERD feature (normal baseline impedance, normal reflux episode counts) have psychological profiles and PPI response similar to true functional disorders, which are treated with neuromodulators and adjunctive measures.

Behavioral Phenotypes

Patients with regurgitation and belching should be carefully evaluated to discriminate between reflux-related mechanisms, behavioral phenotypes, and overlap between the 2 mechanisms.

Distinction between supragastric belching (SGB), gastric belching, and aerophagia can be made with a combination of HRM with impedance and pH impedance monitoring.[97] Gastric belches result from transient relaxation of the LES, in contrast to SGB where air sucked or pushed into the esophagus and rapidly expelled, never reaching the stomach.[98] Aerophagia describes air swallowing and typically presents with bloating and increased flatulence rather than belching as the primary symptom.[97] A single center retrospective cohort study determined > 13 SGB over 24 hours on pH impedance monitoring as abnormal,[99] with patients reporting both bothersome belching and other upper GI symptoms, rather than insolated SGB. Reflux events associated with SGB are more likely non-acidic and are often seen in PPI non-responders.[100,101] Diaphragmatic breathing, cognitive behavioral therapy, and neuromodulators reduce symptoms and improve quality of life, and antireflux surgery should not be offered for isolated SGB.[102]

Rumination syndrome refers to the effortless regurgitation of recently ingested food from the stomach to the oral cavity followed by expectoration or re-mastication and re-swallowing, in the ab-

sence of nausea or retching.[103] Rumination episodes, indistinguishable from reflux events, occur in the immediate post prandial period but stop when the regurgitate becomes acidic. While rare in the general population, rumination can be identified in PPI-non responders using post prandial HRM with impedance monitoring, or pH impedance monitoring.[101,104] Management consists of patient education, behavioral therapy, and diaphragmatic breathing rather than anti-reflux therapies.[105,106]

Behavioral disorders can also overlap with GERD, where reflux metrics may be pathologic despite symptom presentations and test findings suggestive of SGB or rumination syndrome.[107]

ROLE OF HYPERVIGILANCE

Perception of esophageal stimuli occurs from the complex interplay between visceral afferent nerves in the luminal gut and peripheral nervous system as well as central cognitive and psychological processes.[100] Disturbances in mood or emotional arousal can impair and amplify interpretation of esophageal stimuli.[108,109] Hyperawareness of esophageal sensation leads to the perception of stimuli as frightening, and patients can develop a disproportionate behavioral response from catastrophic thinking about potential consequences. The periodic improvement of symptoms may be inappropriately attributed to close selective attention and avoidance of perceived triggers, further reinforcing the behavior pattern. This cognitive-affective pattern is collectively referred to as "esophageal hypervigilance" and can add to PPI refractory symptom burden and significantly impaired quality of life.[13]

While understanding of esophageal hypervigilance has evolved over the last decade, the concept of hypervigilance is well established in both mental health and pain disorders.[110] The esophageal hypervigilance and anxiety scale (EHAS) is a 15-item patient reported outcome adapted to the esophagus from validated hypervigilance questionnaires.[111] In the initial validation cohort of 982 patients, EHAS demonstrated excellent internal consistency, reliability, and both total and sub-scores for symptom-specific anxiety and hypervigilance positively correlated with symptom severity and negatively correlated with quality of life.[111] Subsequent studies evaluating patients with reflux symptoms found that EHAS predicts symptom severity regardless of acid burden on ambulatory testing with either pH impedance or prolonged wireless pH monitoring across different reflux phenotypes.[8,112] Higher EHAS scores are associated with more severe LPS despite no difference in esophageal reflux metrics.[113] In a study of 102 patients who underwent anti-reflux surgery for GERD, pre-operative EHAS was an independent predictor of both symptom severity before surgery and symptom improvement after surgery.[114]

Esophageal hypervigilance offers a novel mechanism to explain discordance between reflux symptoms and physiologic testing and provides additional targets for therapy including cognitive behavioral therapy and gut-directed hypnotherapy.[115,116]

CONCLUSIONS

While GERD evaluation starts with characterization of troublesome symptoms suggested by the Montreal consensus, symptoms alone do not define actionable GERD, although typical symptoms without alarm features may warrant empiric acid suppressive therapy. Mucosal injury phenotypes are identified in a minority of symptomatic patients on upper endoscopy, especially when studied off therapy. When endoscopy is normal or demonstrates LA grade A esophagitis, GERD remains unproven, and ambulatory reflux monitoring can identify or rule out NERD, and potentially diagnose RH, FH and behavioral disorders. Diagnosis of E-DGBI requires exclusion of mucosal injury and abnormal reflux burden phenotypes, identified using Lyon Consensus criteria. Hypervigilance and symptom-specific anxiety, as well as E-DGBI diagnosed using Rome criteria can coexist across all the symptomatic reflux phenotypes. Therefore, Montreal, Lyon and Rome are intricately linked together in patient evaluation, and concepts from each of these contribute to accurate phenotyping and optimal management.[10]

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Potassium-competitive Acid Blockers in Gastroesophageal Reflux Disease and Functional Dyspepsia: A Korean Expert Review With Original Meta-analyses

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Potassium-competitive acid blockers (P-CABs) have emerged as a potent alternative to proton pump inhibitors (PPIs) for the treatment of gastroesophageal reflux disease and functional dyspepsia (FD). This narrative review incorporating original meta-analyses summarizes the latest evidence from randomized controlled trials and meta-analyses, emphasizing the efficacy, safety, and clinical applications of P-CABs such as vonoprazan, tegoprazan, fexuprazan, keverprazan, and zastaprazan. P-CABs demonstrate rapid onset, sustained acid suppression, and superior healing rates in erosive esophagitis (EE), particularly in severe EE (grade C/D). Maintenance therapy in healed EE with P-CABs also yields higher endoscopic remission rates compared to PPIs. In non-erosive reflux disease, P-CABs offer rapid and sustained symptom control compared with placebo. Their pharmacologic profiles, including flexible dosing and rapid action, support on-demand therapy, potentially enhancing patient satisfaction. In addition, P-CABs may help overcome unmet needs from suboptimal efficacy of PPIs in nocturnal heartburn and extraesophageal symptoms. Emerging economic evaluation suggests P-CABs may be cost-effective in Asian healthcare systems. In FD, tegoprazan offers effective symptom relief without delaying gastric emptying, distinguishing it from PPIs. Limited clinical and real-world data indicate that second-generation P-CABs do not exhibit significant hepatotoxicity, however, long-term hypergastrinemia and mucosal changes warrant further surveillance. This review highlights the evolving role of P-CABs in gastroesophageal reflux disease and FD and emphasizes the need for further research to refine their clinical positioning, optimize therapeutic strategies, and evaluate long-term safety outcomes.

Keywords: Dyspepsia; Gastroesophageal reflux; Potassium-competitive acid blockers

INTRODUCTION

Potassium-competitive acid blockers (P-CABs) represent a potent class of acid suppressant that reversibly inhibits the hydrogen-potassium ATPase (H+/K+-ATPase) in gastric mucosal cells, thereby providing faster, stronger, and more sustained gastric acid suppression than traditional proton pump inhibitors (PPIs). Following the clinical use of vonoprazan in Japan, other P-CABs such as tegoprazan, fexuprazan, and zastaprazan are now clinically used in Korea, while keverprazan is used in China. Beyond their pharmacological advantages of acid stability and minimized impact from CYP2C19 polymorphism, their convenience—allowing administration regardless of meal timing while still suppressing gastric acid secretion—makes them suitable for acid-related disorders such as gastroesophageal reflux disease (GERD), *Helicobacter pylori* infection, and peptic ulcer disease. Despite these advantages, the adoption of P-CAB varies significantly by country, due to differences in drug supply situations, insurance reimbursement policies, and cost-effectiveness considerations. In the United States, where drug prices tend to be higher than in other countries, expert guidelines

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recommend P-CAB for patients who have failed standard or double-dose PPI therapy.[1] Conversely, P-CABs have gained significant traction in actual clinical settings in Korea. This is because Korea has 3 P-CAB options, which are comparable in cost to PPIs, and offer relative advantages over PPIs for acid-related diseases. Consequently, P-CABs are frequently prescribed for *H. pylori* eradication therapy and acid-related diseases, which are different patterns than in other regions. Korean physicians are among the highest prescribers of P-CABs globally, making it important to review the clinical experience with P-CABs in Korea. Given the financial implications, the availability of various P-CABs, and the ease of access to specialists, reviewing research and sharing clinical experience in Korea is imperative. This review will provide a synopsis and comparative analysis of the efficacy and safety of P-CABs for GERD and functional dyspepsia (FD) in Korea, incorporating clinical experience compared with Western expert review.

POTASSIUM-COMPETITIVE ACID BLOCKERS IN GASTROESOPHAGEAL REFLUX DISEASE

Erosive Esophagitis

To date, a total of 11 randomized controlled trials (RCTs) comparing the healing effects of P-CABs and PPIs on erosive esophagitis (EE) have been published, including 4 vonoprazan studies, 3 tegoprazan studies, 2 fexuprazan studies, 1 keverprazan study, and 1 zastaprazan study (Table 1). Previous studies revealed that the healing rate of EE after 4-week and 8-week therapy with P-CAB was very high. In RCTs for the treatment of EE, all types of P-CABs are non-inferior to PPIs.[2-12] Furthermore, studies investigating the 2-week healing rate showed that P-CABs demonstrated significantly higher healing rates than PPIs, and this difference was greater than the differences observed in the 4-week or 8-week healing rates.[2,3,5,10,11] These results suggest that the healing effect of P-CABs is more potent and faster than that of PPIs. There was no difference in healing efficacy based on the timing of P-CAB administration. Regarding the timing of administration, specific clinical evidence for fexuprazan has shown that its efficacy in healing EE is not significantly affected by meal timing. A recent RCT demonstrated that post-meal administration of fexuprazan was non-inferior to pre-meal administration in terms of EE healing rates.[13] While the pharmacological profile of P-CABs—characterized by rapid absorption and meal-independent inhibition of H+/K+-ATPase—suggests potential flexibility in dosing across the class, dedicated clinical trials for other P-CAB agents are still needed to confirm this generalizability.

A large amount of clinical data has been accumulated on this, leading to systematic reviews and meta-analyses. A meta-analysis published in 2023 comparing vonoprazan to lansoprazole was conducted with 4 published non-inferiority RCTs mentioned above.[14] At 8 weeks post-treatment, vonoprazan resulted in significantly higher rates of endoscopic healing as compared to lansoprazole (risk ratios [RR], 1.041; 95% confidence interval [CI], 1.003-1.081); however the difference was very slight. Another meta-analysis compared tegoprazan and PPI, including 4 RCTs and 963 patients. The meta-analysis showed that the healing rates between tegoprazan and PPIs were similar, with no significant differences observed at both 4 weeks and 8 weeks.[15] RCTs on keverprazan, fexuprazan, and zastaprazan have also shown similar results.[6,7,10,11]

Maintenance therapy in healed erosive esophagitis

As of now, 3 RCTs have been conducted to evaluate endoscopic remission after maintenance therapy (2 with vonoprazan and 1 with tegoprazan) (Table 2). Vonoprazan was non-inferior to lansoprazole in maintenance of healing of EE at 24 weeks, and superior to lansoprazole in maintenance of healing grade C/D esophagitis.[8] Another study also showed that vonoprazan was not inferior to lansoprazole for maintaining endoscopic remission at 24 weeks and 12 weeks.[16] Meanwhile, tegoprazan maintenance therapy also demonstrated non-inferiority to lansoprazole in terms of preventing EE recurrence at 24 weeks and 12 weeks.[17]

Comparison of Healing Effect for Erosive Esophagitis Between Potassium-competitive Acid Blockers

Meta-analyses compared the healing effects of EE according to the types of P-CABs and PPIs. One meta-analysis compared only severe esophagitis (grade C/D), and vonoprazan was most effective in the initial and maintained healing of severe esophagitis (surface under the cumulative ranking curve [SUCRA] 0.89 and 0.87, respectively).[18] Another meta-analysis revealed that the maximum SUCRA value of vonoprazan 40 mg for the healing rate in EE was 90.7 in the first subgroup (vonoprazan, keverprazan, and lansoprazole) and the maximum SUCRA value of tegoprazan 50 mg was 72.1 in the second subgroup (tegoprazan, fexuprazan, and esomeprazole).[19] This result suggests that standard doses of vonoprazan and tegoprazan can be very effective drugs for the healing of EE. In another meta-analysis, keverprazan showed the best results in healing rates after 8 weeks, with a SUCRA value of 84.7, and this was also true for the subgroup with severe EE at baseline.[20]

Original Meta-analysis of Erosive Esophagitis: Potassium-competitive Acid Blocker vs Proton Pump Inhibitor

In this review, we conducted independent meta-analyses on

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Table 1. Summary of Randomized Controlled Trials on the Healing Effects of Potassium-competitive Acid Blockers on Erosive Esophagitis

Year	Nation[a]	First author	Drugs	8 Weeks healing rate			4 Weeks healing rate			8 Weeks healing rate, LA-C/D		4 weeks healing rate, LA-C/D	
				ITT	FAS	PPS	ITT	FAS	PPS	FAS	PPS	FAS	PPS
2015	Japan (66)	K. Ashida	VPZ 20 mg		139/144 (96.5)			136/144 (94.4)		50/50 (100.0)		50/50 (100.0)	
			LPZ 30 mg		126/132 (95.5)			123/132 (93.2)		43/46 (93.5)		40/46 (87.0)	
2016	Japan (39)	K. Ashida	VPZ 20 mg		203/205 (99.0)			198/205 (96.6)		75/75 (100)		72/75 (96.0)	
			LPZ 30 mg		190/199 (95.5)			184/199 (92.5)		63/72 (87.5)		58/72 (80.6)	
2020	Asia[b] (56)	Y. Xiao	VPZ 20 mg		220/238 (92.4)			203/238 (85.3)		63/75 (84.0)		55/75 (73.3)	
			LPZ 30 mg		210/230 (91.3)			192/230 (83.5)		54/67 (80.6)		45/67 (67.2)	
2023	USA (77), Europe (34)[c]	L. Laine	VPZ 20 mg	478/514 (92.9)		462/488 (94.7)					162/177 (91.7)		
			LPZ 30 mg	431/510 (84.6)		410/474 (86.6)					125/174 (72.0)		
2019	South Korea (27)	KJ Lee	TPZ 50 mg	95/99 (96.0)		91/92 (98.9)	87/99 (87.9)		84/92 (91.3)		3/3 (100.0)		3/3 (100.0)
			EPZ 40 mg	92/99 (92.9)		87/88 (98.9)	87/99 (87.9)		83/88 (94.3)		3/3 (100.0)		2/3 (66.7)
2024	China (32)	H. Zhu	TPZ 50 mg		112/123 (91.1)	108/117 (92.3)				11/11 (100.0)			
			EPZ 40 mg		116/125 (92.8)	112/118 (94.9)				8/10 (80)			
2025	South Korea (20)	CM Shin	TPZ 50 mg					98/103 (95.2)	95/98 (96.9)			8/9 (88.9)	
			LPZ 30 mg					94/109 (86.2)	92/95 (96.8)			7/10 (70.0)	
2022	South Korea (25)	KN Lee	FPZ 40 mg		112/116 (96.6)	106/107 (99.1)		98/111 (88.3)	93/103 (90.3)		7/8 (87.5)		5/8 (62.5)
			EPZ 40 mg		112/115 (97.4)	110/111 (99.1)		94/108 (87.0)	92/104 (88.5)		8/8 (100.0)		6/7 (85.7)
2024	China (48)	Q. Zhuang	FPZ 40 mg		146/165 (88.5)	145/149 (97.3)		126/165 (76.4)	126/149 (84.6)	41/51 (80.4)	40/43 (93.0)	29/51 (56.9)	29/43 (67.4)
			EPZ 40 mg		145/163 (89.0)	143/146 (97.9)		120/163 (73.6)	118/146 (80.8)	43/47 (91.5)	43/45 (95.6)	33/47 (70.2)	33/45 (73.3)
2022	China (44)	S. Chen	KPZ 20 mg		114/119 (95.8)	110/111 (99.1)		98/119 (82.4)	97/111 (87.4)	22/24 (91.7)	20/21 (95.2)	14/24 (58.3)	14/21 (66.7)
			LPZ 30 mg		107/119 (89.9)	102/110 (92.7)		97/119 (81.5)	93/110 (84.6)	20/25 (80.0)	18/23 (78.3)	16/25 (64.0)	14/23 (60.9)
2025	South Korea (29)	JH Oh	ZPZ 20 mg		141/144 (97.9)	136/136 (100.0)		137/144 (95.1)	132/136 (97.1)	6/6 (100.0)	6/6 (100.0)	6/6 (100.0)	6/6 (100.0)
			EPZ 40 mg		131/138 (94.9)	118/121 (97.5)		121/138 (87.7)	112/121 (92.6)	5/6 (83.3)	4/5 (80.0)	4/6 (66.7)	4/5 (80.0)

[a]The number in parentheses indicates the number of hospitals where the study was conducted.
[b]The Asian countries included are China (27 sites), South Korea (12 sites), Taiwan (11 sites), and Malaysia (6 sites).
[c]The European countries included are Poland, Czech Republic, Hungary, Bulgaria, and United Kingdom.
LA-C/D, Los Angeles classification-C/D; ITT, intention-to-treat; FAS, full analysis set; PPS, per protocol set; VPZ, vonoprazan; LPZ, lansoprazole; EPZ, esomeprazole; TPZ, tegoprazan; FPZ, fexuprazan; KPZ, keverprazan; ZPZ, zastaprazan.
Data are presented as n (%).

Table 2. Summary of Randomized Controlled Trials on Endoscopic Remission After Maintenance Therapy With Potassium-competitive Acid Blockers on Erosive Esophagitis

Year	Nation[a]	First author	Drugs	24 Weeks endoscopic remission rate		12 Weeks endoscopic remission rate		24 Weeks endoscopic remission rate, LA-C/D
				FAS	PPS	FAS	PPS	FAS
2024	China, South Korea, and Malaysia (64)	Y. Xiao	VPZ 10 mg	157/181 (86.7)		13/18 (72.2)		28/39 (71.8)
			VPZ 20 mg	150/171 (87.7)		18/20 (90.0)		28/32 (87.5)
			LPZ 15 mg	137/184 (74.5)		13/20 (65.0)		21/36 (58.3)
2023	USA (77), Europe (34)[b]	L. Laine	VPZ 10 mg	232/293 (79.2)	231/259 (89.2)			71/95 (74.7)
			VPZ 20 mg	235/291 (80.7)	223/246 (90.7)			71/92 (77.2)
			LPZ 15 mg	211/294 (72.0)	200/251 (79.7)			59/96 (61.5)
2023	South Korea (33)	YK Cho	TPZ 25 mg	133/154 (86.4)	125/138 (90.6)	143/154 (92.9)	128/138 (92.8)	6/8 (75.0)
			LPZ 15 mg	127/151 (84.1)	111/124 (89.5)	145/151 (96.0)	119/124 (96.0)	6/10 (60.0)

[a]The number in parentheses indicates the number of hospitals where the study was conducted.
[b]The European countries included are Poland, Czech Republic, Hungary, Bulgaria, and United Kingdom.
LA-C/D, Los Angeles classification-C/D; FAS, full analysis set; PPS, per protocol set; VPZ, vonoprazan; LPZ, lansoprazole; TPZ, tegoprazan.

the healing and maintenance for EE based on RCTs published to date. We searched the PubMed database using the keywords "vonoprazan," "tegoprazan," "fexuprazan," "keverprazan," and "zastaprazan" to identify relevant studies published until June 2025. All RCTs evaluating the clinical outcomes of P-CAB vs PPI for EE treatment were considered eligible for inclusion in the first meta-analysis. The second analysis included RCTs that compared the efficacy of P-CAB and PPI in the maintenance therapy in EE. This analysis was also conducted independently by the authors. Nine RCTs were used in the analysis of 4-week treatment for EE, 10 RCTs were used in the analysis of 8-week treatment, and 3 RCTs were used in the analysis of 24-week maintenance treatment. Statistical analyses were performed using Review Manager 5.4, employing a random-effects model to calculate pooled RRs and 95% CIs.

At 4 weeks, P-CABs demonstrated a significantly higher endoscopic healing rate than PPIs, although the difference was minimal (RR, 1.04; 95% CI, 1.01-1.06), while no statistically significant difference was observed at 8 weeks (RR, 1.02; 95% CI, 1.00-1.05) (Fig. 1). In cases of severe esophagitis (Los Angeles classification-C/D [LA-C/D]), P-CABs did not show significantly superior results compared to PPIs in endoscopic healing rates at 4 weeks after treatment (RR, 1.10; 95% CI, 0.99-1.23); however, at 8 weeks after treatment, P-CABs showed significantly higher endoscopic healing rates than PPIs (RR, 1.09; 95% CI, 1.01-1.18) (Fig. 2). The success rate of 24-week maintenance therapy with P-CABs was significantly higher than that with PPIs, with a RR of 1.09 (95% CI, 1.02-1.18), and the difference was even greater for patients with severe esophagitis (LA-C/D) (RR, 1.22; 95% CI, 1.04-1.44) (Fig. 3).

Non-erosive Reflux Disease

Non-erosive reflux disease (NERD) is a subcategory of GERD, which is characterized by troublesome reflux symptoms with abnormally increased gastroesophageal reflux observed on 24-hour ambulatory pH-impedance monitoring in the absence of esophageal mucosal injury confirmed on endoscopy.[21] NERD is the most common manifestation among GERD, which accounts for 60–70% in westernized countries.[22] It is a persistent medical condition that markedly influences the overall quality of life of affected individuals,[23,24] thereby requiring personalized therapeutic strategies to regulate the manifestation of symptoms. Despite its prevalence, management of NERD with conventional PPI therapy has been challenging, previous studies showed that the PPI efficacy for symptom control is much less for NERD than erosive esophagitis.[25,26]

In a Japanese RCT with NERD in 2016, vonoprazan did not demonstrate superiority over placebo, however, the study design excluded patients whose heartburn symptoms improved with antacids during a 1-week run-in phase.[27] As a result, the study may not have included suitable patients with acid-induced symptoms, potentially limiting the assessment of treatment efficacy.[27] A study conducted in South Korea showed that once-daily tegoprazan (50 mg

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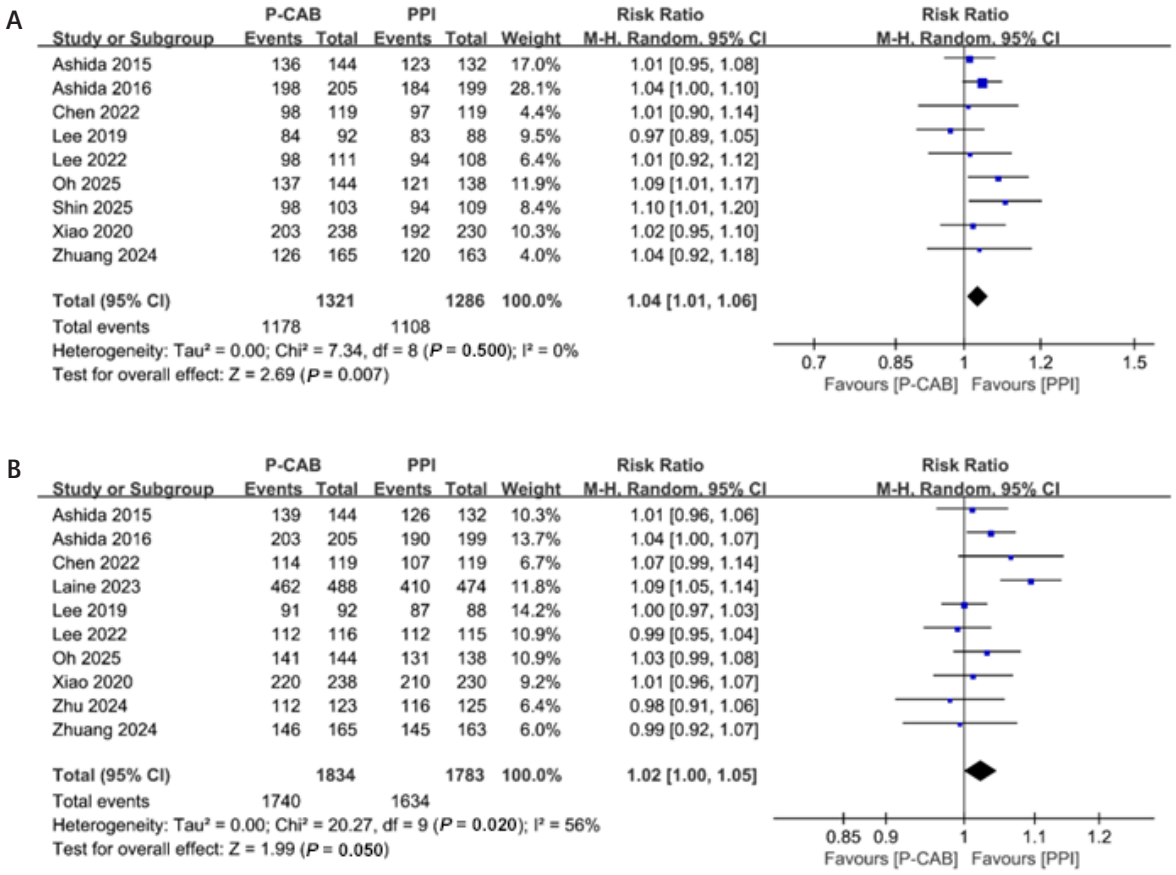

Figure 1. Forest plots of endoscopic healing rates of erosive esophagitis for potassium-competitive acid blocker (P-CAB) versus proton pump inhibitor (PPI) in 4-week (A) and 8-week (B) after treatment. M-H, Mantel-Haenszel; CI, confidence interval.

or 100 mg) was significantly more effective than placebo in resolving heartburn symptoms and, notably, regurgitation as well.[28] In the meta-analysis of 3 RCTs evaluating the efficacy of P-CABs—specifically vonoprazan (10 mg and 20 mg) and tegoprazan (50 mg and 100 mg)—in patients with NERD, the number of symptom-free days over a 4-week period was compared between P-CABs and placebo.[29] The pooled analysis showed that P-CABs resulted in a mean increase of 7.94 symptom-free days compared to placebo (95% CI, 3.80-12.08).[29] Sensitivity analyses, which combined various P-CAB types and doses across the 3 NERD studies, confirmed a statistically significant benefit of P-CABs over placebo, with no evidence of heterogeneity.[29] A recent trial in the United States showed that vonoprazan demonstrated a rapid onset of action, alleviating heartburn symptoms as early as day 1.[30] This treatment effect was maintained consistently over the 20-week extension period.[30] Therefore, in NERD patients with clearly acid-related symptoms, P-CABs may offer advantages over PPIs in terms of more rapid and sustained symptom relief.

On-demand Therapy

In the long-term management of GERD, effective symptom control is the primary goal. Both the US and Korean guidelines recommend on-demand or intermittent therapy as appropriate treatment strategies in patients with NERD.[21,31] On-demand therapy, in which medication is taken at the onset of symptoms, is particularly effective when using agents with a rapid onset and sustained duration of action. However, PPIs are less appropriate for on-demand therapy because of their relatively slow onset of action and the need to be converted into an active form in an acidic environment, which may delay symptom relief.[32] These limitations of PPI have been recognized as an unmet need in the treatment of GERD,[32] and P-CABs have emerged as a promising alternative treatment option in on-demand therapy due to their rapid onset of action and flexible dosing that are not dependent on meal timing.

A North American randomized controlled trial in patients with NERD showed that on-demand vonoprazan significantly improved heartburn relief compared to placebo, with a higher proportion of episodes completely resolved within 3 hours and sustained relief lasting 24 hours.[33] Notably, a statistically significant difference was observed as early as 1 hour after administration.[33] A recent RCT compared the efficacy of on-demand tegoprazan 50 mg versus esomeprazole 20 mg in GERD patients who had previously respond-

ed to acid-suppressive therapy.[34] Among 69 patients completing 8 weeks of on-demand treatment, patient satisfaction scores were comparable between groups, however, tegoprazan led to significantly faster symptom relief, with 26.2% of episodes improving within 30 minutes versus 16.1% in the esomeprazole group.[34] Another single-center prospective cohort study involving 31 GERD

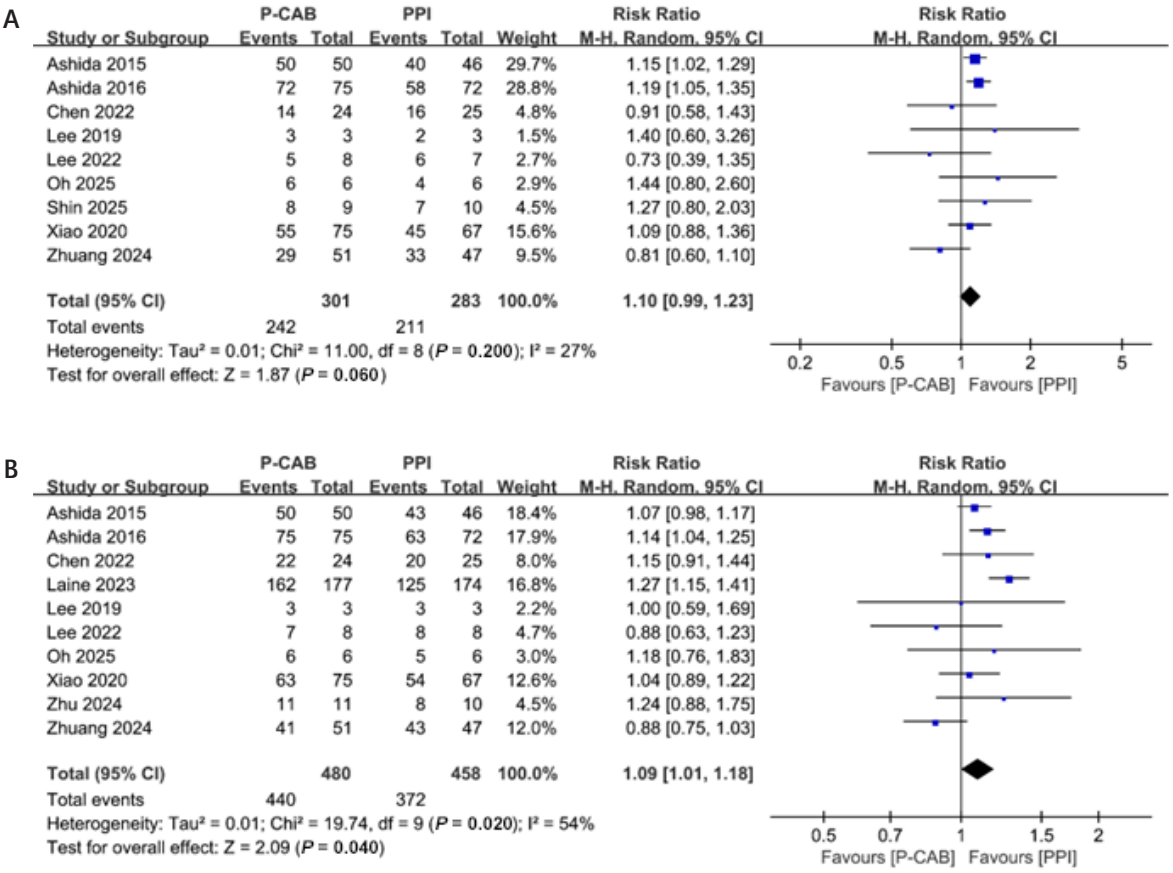

Figure 2. Forest plots of endoscopic healing rates of severe erosive esophagitis (Los Angeles classification-C/D [LA-C/D]) for potassium-competitive acid blocker (P-CAB) versus proton pump inhibitor (PPI) in 4- (A) and 8-week (B) after treatment. M-H, Mantel-Haenszel; CI, confidence interval.

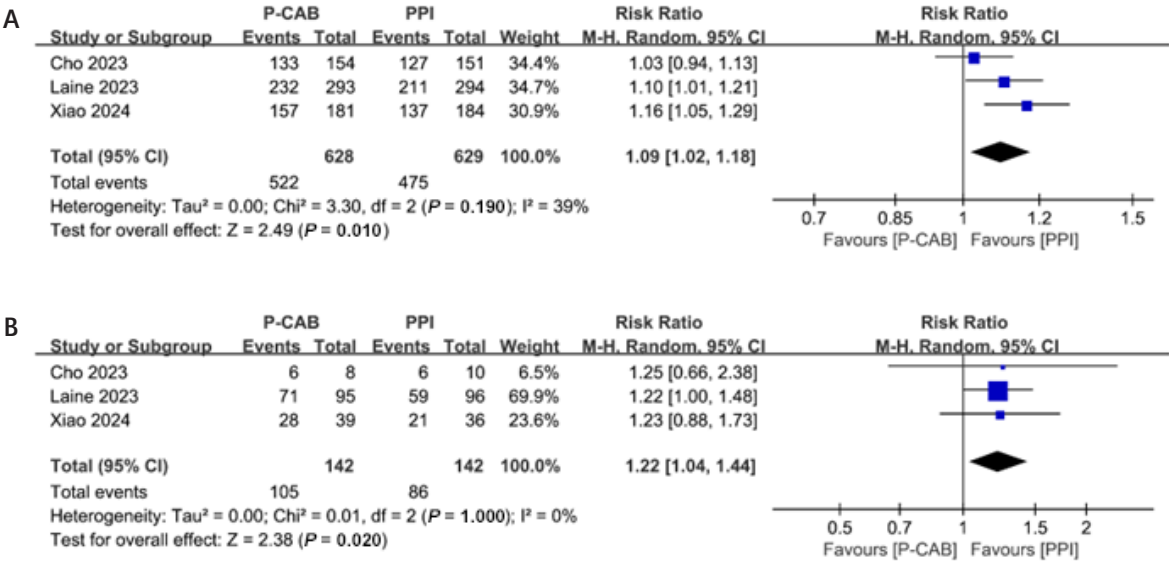

Figure 3. Forest plots of success rates of 24-week maintenance therapy for potassium-competitive acid blocker (P-CAB) versus proton pump inhibitor (PPI) in erosive esophagitis (A) and severe erosive esophagitis (Los Angeles classification-C/D [LA-C/D]) (B). M-H, Mantel-Haenszel; CI, confidence interval.

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patients demonstrated that fexuprazan was effective for both the initial 4-week treatment and subsequent 4-week maintenance phase.[35] Approximately 68% of patients preferred on-demand therapy, which provided symptom control comparable to that of continuous therapy.[35] These findings suggest that P-CABs, compared to PPIs, may serve as a more effective, faster-acting, and patient-preferred option for on-demand management of GERD.

Economic perspectives on Potassium-competitive Acid Blockers vs Proton Pump Inhibitors

In the United States, the use of P-CABs in first-line treatment of uninvestigated heartburn symptoms or NERD is not recommended by cost-effectiveness concerns due to their higher price compared to PPIs, which are widely available as over-the-counter medications.[1] P-CABs are currently available only as branded prescription medications, whereas several PPIs are widely accessible as low-cost generic or over-the-counter formulations. As a result, the acquisition cost of P-CABs is substantially higher than that of PPIs in the US healthcare setting.[36] Reflecting these considerations, a recent AGA Clinical Practice Update noted that, at current US pricing, the modest clinical advantages of P-CABs over standard- or double-dose PPIs may not be sufficient to justify their routine use as first-line therapy from a cost-effectiveness perspective.[1]

In contrast, the economic landscape differs considerably in East Asian countries. PPIs are not readily available over the counter in Asian countries, and the cost gap between P-CABs and PPIs is relatively modest compared to that observed in the United States. In East Asian countries including Japan, Korea, and China, various P-CABs are commercially available with broader insurance coverage, and the narrower price differential between P-CABs and PPIs creates a more favorable economic environment for their clinical adoption.[37,38]

Recent economic evaluations have demonstrated that P-CABs, particularly vonoprazan, are a cost-effective option for GERD management. Intermittent P-CAB therapy showed superior cost-effectiveness compared to maintenance PPI therapy, with fewer medication days and clinic visits, and high patient acceptability.[39] A vonoprazan-first strategy also provided greater QALYs at lower costs than PPI-based step-up approaches.[40] These findings were consistent across GERD severity levels, supporting the early use of P-CABs as a clinically effective and economically efficient option for long-term management.[40] A cost-effectiveness analysis in China similarly concluded that vonoprazan generates incremental QALYs at lower costs compared with PPIs, demonstrating its dominant cost-effectiveness profile in Asian healthcare settings.[41]

Accordingly, in Asian countries, P-CABs—despite being slightly more expensive than PPIs—may be preferred from an economic perspective due to their superior efficacy and cost-effectiveness profile. However, these favorable economic profiles in Asian countries may not be generalizable to Western settings. Future economic evaluations will be needed to assess the cost-effectiveness of individual P-CAB considering current market prices and regional healthcare system characteristics.

Extraesophageal Symptoms

P-CABs have emerged as potential therapeutic alternatives to PPIs for the management of GERD, particularly in patients with extraesophageal symptoms (EES) such as chronic cough, hoarseness, and globus sensation. These symptoms are often refractory to PPI therapy, likely due to limitations such as delayed onset, short plasma half-life, and inadequate nocturnal acid suppression.[42] In contrast, P-CABs offer rapid onset, potent and sustained acid inhibition, and acid-independent activation, making them theoretically more effective in EES management. However, it should be noted that EES are often weakly associated with esophageal acid exposure and are frequently driven by non-acid reflux, hypersensitivity, or functional and central mechanisms, which may limit the clinical relevance of intensified acid suppression in this population.[43]

Four randomized controlled trials have been conducted to investigate the efficacy of P-CABs in patients with EES. A preliminary, double-blind, randomized, placebo-controlled trial involving 35 patients with laryngopharyngeal reflux disease (LPRD) demonstrated that tegoprazan (50 mg daily) administration improved LPRD symptoms and signs, however, did not show superiority over placebo.[44] Kim et al[45] performed an 8-week randomized, double-blind study that compared fexuprazan with esomeprazole in treating LPRD patients, and both fexuprazan and esomeprazole significantly reduced the total reflux symptom index (RSI) and reflux finding score. Notably, in patients with severe LPRD (RSI > 18), fexuprazan led to significantly greater RSI improvement than esomeprazole after 4 weeks, especially in chronic cough and hoarseness symptoms. In the GERD-related cough, both vonoprazan and esomeprazole significantly reduced cough severity scores; however, vonoprazan was not superior to esomeprazole.[46] A recent 8-week randomized, double-blind study comparing fexuprazan with esomeprazole in GERD patients with chronic cough demonstrated that both treatments provided comparable efficacy in improving cough-related quality of life, however, fexuprazan led to significantly greater improvement in typical reflux symptoms based on RDQ scores.[47]

The current evidence on the role of P-CABs in EES remains limited by small sample sizes and short durations. Given the multifactorial etiology of EES and the uncertain role of acid suppression

in their pathogenesis, P-CABs cannot be routinely recommended over PPIs for EES. However, P-CABs may be considered on an individual basis in selected patients with documented acid-related EES refractory to optimized PPI therapy. Thus, large-scale, placebo-controlled randomized trials with validated EES assessment tools are warranted to determine the definitive role of P-CABs in the treatment of extraesophageal GERD.

Nighttime Heartburn

Nighttime heartburn symptoms are commonly observed in individuals with frequent GERD complaints and are associated with greater impairment in health-related quality of life.[48] These symptoms are often more severe and less responsive to PPIs due to impaired acid clearance during sleep.[32] As a result, management of nocturnal GERD remains an important unmet need, with growing interest in alternative therapies with a faster onset and longer duration of acid suppression, such as P-CABs. Two randomized controlled trials have evaluated the effect of P-CABs on night-time heartburn. In one study, tegoprazan 50 mg demonstrated a shorter time to first night-time heartburn-free day (1.5 days vs 3.0 days with PPI), although the difference was not statistically significant.[49] In another trial, vonoprazan 20 mg provided significantly higher complete relief of night-time heartburn compared to lansoprazole during the first 7 days of treatment ($P < 0.01$), suggesting a more rapid symptom resolution in patients with nocturnal reflux.[50] Based on these findings, P-CABs may be preferred in GERD patients with night-time symptoms to improve sleep quality and overall quality of life.

POTASSIUM-COMPETITIVE ACID BLOCKERS IN FUNCTIONAL DYSPEPSIA

FD is a common gastrointestinal disorder whose pathophysiology is known to involve gastric acid. As a result, most clinical guidelines recommend the use of antisecretory agents, particularly PPIs. Consequently, P-CABs may have a mechanistic rationale for alleviating FD symptoms. However, evidence on P-CABs in FD remains limited. A recent prospective, multicenter, single-arm study evaluated tegoprazan (50 mg once daily for 8 weeks) in patients with FD according to Rome IV criteria. Results demonstrated satisfactory symptom relief rates of 74.6% at 4 weeks and 86.7% at 8 weeks, with significant improvement in Nepean Dyspepsia Index–Korean and GERD-Health-Related Quality of Life scores, irrespective of FD subtype or *H. pylori* status.[51] If future RCTs comparing P-CABs with PPIs or placebo in FD demonstrate efficacy, P-CABs may be considered for inclusion in future FD guidelines.

PPIs have been associated with delayed gastric emptying. A systematic review reported consistent delays in gastric emptying of solid meals due to impaired peptic hydrolysis under acid suppression.[52] Clinical studies further support this finding. For example, omeprazole increased gastric half-emptying time by approximately 17 minutes,[53] and Benini et al[54] observed an increase from ~199.6 to 230.9 minutes after four days of omeprazole pretreatment. In contrast, in an in vivo canine model, tegoprazan administration at pharmacologically active doses induced phase III–like contractions of the migrating motor complex in the stomach. These contractile patterns are associated with enhanced gastric clearance, suggesting a potential prokinetic effect beyond acid suppression.[55] Recently, a randomized, double-blind, placebo-controlled study in healthy adults showed that tegoprazan (50 mg) did not significantly alter gastric half-emptying time or gastric retention at 4 hours compared to placebo.[56] A contemporary commentary emphasizes that, unlike PPIs, tegoprazan offers potent, rapid acid suppression without delaying gastric emptying.[57]

Taken together, P-CABs like tegoprazan exhibit promising symptom relief in FD without adversely affecting gastric emptying, unlike PPIs. This suggests that P-CABs may offer therapeutic advantages in FD management. Nonetheless, larger RCTs comparing P-CABs with PPIs and placebo in FD populations are needed. Further studies should also explore long-term effects on gastric motility, gut microbiota, and overall symptomatology.

SHORT-TERM AND LONG-TERM SAFETY ISSUES IN POTASSIUM-COMPETITIVE ACID BLOCKERS

PPIs have been widely prescribed for the treatment and prevention of a broad spectrum of acid-related disorders (GERD, peptic ulcer disease, nonsteroidal anti-inflammatory drug-related ulcer prevention, *H. pylori* eradication, etc), and a substantial proportion of patients receive these agents on a long-term basis. Population-based studies and narrative reviews have repeatedly documented high rates of chronic and often inappropriate PPI use.[58] Concerns about the safety of prolonged PPI therapy have accumulated over the last decade and include associations—typically observed in observational studies and meta-analyses—with pneumonia, *Clostridioides difficile* infection, acute and chronic kidney disease, hypomagnesaemia, vitamin B12 deficiency, bone fracture risk, and other outcomes; these signals have driven calls for regular re-evaluation and deprescribing when appropriate.[59]

P-CABs are a newer class of antisecretory agents that achieve rapid, potent and sustained acid suppression by reversible

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K^+-competitive inhibition of the gastric H^+/K^+ ATPase. Early developmental compounds in the P-CAB class (historical prototypes such as SCH28080 and other imidazopyridine derivatives) were associated with hepatotoxicity in preclinical and early clinical work, which slowed their development and raised safety concerns for the class.[60] Subsequent, clinically used P-CABs—including vonoprazan, tegoprazan, fexuprazan, keverprazan, and zastaprazan—have undergone more extensive clinical testing and postmarketing surveillance. Overall, the contemporary clinical and real-world evidence indicates that these later-generation P-CABs have not demonstrated a consistently higher risk of clinically significant hepatotoxicity compared with PPIs. For example, a nationwide Korean claims-database cohort study reported no increase in hepatotoxicity with tegoprazan relative to 6 common PPIs (and in some analyses lower hepatotoxicity risk with tegoprazan).[61] Clinical trials and early postmarketing evaluations of fexuprazan and zastaprazan have likewise reported no clinically meaningful liver-safety signal to date, although vigilance continues.[62]

A biologically plausible adverse effect of very potent and sustained acid suppression is hypergastrinemia. Vonoprazan—the most extensively used P-CAB worldwide—has been repeatedly shown to raise fasting serum gastrin to a greater extent than conventional PPIs (in a dose- and duration-dependent manner). Long-term vonoprazan therapy has been associated with significant gastrin increases and gastric mucosal changes, including parietal-cell and foveolar hyperplasia as well as fundic gland polyps in some cohorts.[63] Observational data have raised further concern: recent epidemiologic analyses have reported an association between P-CAB exposure (primarily vonoprazan in Japan) and an increased risk of gastric cancer among populations previously treated for *H. pylori*.[64] However, the observational nature of these studies precludes causal inference, and independent replication is warranted.

By contrast, available data for other P-CABs are more heterogeneous. Tegoprazan and fexuprazan have generally produced smaller gastrin elevations that are similar to those seen with PPIs in short- to medium-term studies, suggesting that not all P-CABs produce the same degree of hypergastrinemia as vonoprazan. Nevertheless, long-term comparative data on gastrin dynamics and related histologic change remain limited.[56]

Evidence on systemic long-term harms associated specifically with chronic P-CAB use (eg, osteoporosis/fracture risk, community-acquired pneumonia, renal disease, or other PPI-linked outcomes) is currently sparse. The VISION long-term maintenance program for vonoprazan (a randomized, multicenter open-label trial with maintenance follow-up) and its interim reports (3-year and 5-year data) found no clear excess of clinically important systemic complications compared with lansoprazole over the trial period, although they did confirm higher gastrin levels and gastric mucosal changes with vonoprazan and emphasize the need for ongoing surveillance and longer follow-up.[65] Larger, population-based studies with longer follow-up are needed to determine whether class-level differences exist for these systemic endpoints.[38]

Until longer-term, well-powered comparative safety studies are available, clinicians should apply the same stewardship principles to P-CAB therapy as are recommended for PPIs: regularly re-assess the indication for ongoing antisecretory treatment, limit duration to the minimum necessary, down-titrate to the lowest effective dose when possible and avoid unnecessary chronic use. Recent expert guidance (AGA clinical practice updates and AGA technical reviews on integrating P-CABs into practice) explicitly recommends routine re-evaluation of ongoing antisecretory therapy and suggests that deprescribing strategies and "lowest-effective-dose" approaches that were developed for PPIs are applicable to P-CABs as well.[66]

In summary, early P-CAB prototypes raised hepatotoxicity concerns, but second-generation P-CABs (vonoprazan, tegoprazan, fexuprazan, keverprazan, zastaprazan) have not shown a consistent hepatic safety signal compared with PPIs in available clinical trials and real-world cohorts; nevertheless, potent acid suppression by some agents (notably vonoprazan) leads to greater hypergastrinemia and mucosal changes, and isolated observational reports have suggested a possible association with gastric neoplasia in specific contexts (eg, after *H. pylori* eradication), although causality remains unestablished. Given the current evidence gaps regarding long-term systemic risks, clinicians should minimize unnecessary P-CAB exposure by adopting evidence-based deprescribing and monitoring strategies, as established for PPIs. Continued long-term comparative safety studies and post-marketing surveillance are mandatory to define the true risk–benefit profile of each P-CAB. A summary of evidence-based recommendations for P-CAB use in GERD and FD is presented in Table 3.

CONCLUSION

P-CABs represent a significant advancement in acid suppression therapy, offering clinical benefits that address limitations of conventional PPIs. Their rapid onset, potent acid inhibition, and flexible dosing make them particularly suitable for diverse GERD phenotypes, including EE, NERD, and extraesophageal manifestations. Evidence supports their superior efficacy in healing and maintenance of EE, and their utility in on-demand therapy enhances patient-centered care. In FD, P-CABs like tegoprazan show encouraging results without impairing gastric motility, suggesting

Table 3. Summary Recommendation of Potassium-competitive Acid Blockers in Gastroesophageal Reflux Disease and Functional Dyspepsia

Clinical phenotype	Evidence	Suggestion
GERD		
Initial treatment of EE	RCTs show non-inferiority or superiority of P-CABs over PPIs in the healing of EE, more rapid healing of EE, and superiority in the healing of severe EE.	P-CABs can be considered as a treatment option for healing patients with EE, especially those with severe EE.
Maintenance of healed EE	RCTs demonstrate that P-CABs (vonoprazan and tegoprazan) are superior to PPIs in the maintenance of healed EE.	P-CABs are recommended for use in maintaining the healing of EE.
Initial treatment of NERD	RCTs show that P-CABs (vonoprazan and tegoprazan) provide significantly more symptom-free days than placebo in NERD.	P-CABs can be used as an initial treatment for NERD in patients with clearly acid-related symptoms.
On-demand therapy	RCT evidence directly comparing P-CABs with PPIs is limited.	P-CABs, with their rapid onset of action and flexible dosing, may be more suitable than PPIs for on-demand therapy.
Extraesophageal symptoms	Limited evidence shows tegoprazan and fexuprazan improve RSI/RFS and chronic cough scores; vonoprazan comparable to PPI.	P-CABs may be considered over PPIs in patients with severe extraesophageal symptoms refractory to PPIs, however, routine use over PPIs is not recommended without objective evidence of pathologic reflux.
Nighttime heartburn	Small RCTs show P-CABs (vonoprazan and tegoprazan) provide faster relief of nocturnal reflux compared to PPIs.	P-CABs may be preferred in GERD patients with nighttime symptoms to improve sleep quality and overall quality of life.
FD		
Functional dyspepsia	Evidence is sparse. Confirmation from large RCTs is essential.	P-CABs such as tegoprazan may be considered for FD, as they offer effective symptom relief without impairing gastric emptying.
Safety issue		
Long-term safety	Limited clinical and real-world data suggest no consistent increase in hepatotoxicity with second-generation P-CABs compared with PPIs.	Long-term comparative safety studies are still needed.

GERD, gastroesophageal reflux disease; EE, erosive esophagitis; NERD, non-erosive reflux disease; FD, functional dyspepsia; RCT, randomized controlled trial; P-CABs, potassium-competitive acid blockers; PPIs, proton pump inhibitors; RSI, reflux symptom index; RFS, reflux finding score.

broader therapeutic potential. Safety profiles of newer P-CABs appear favorable, though vigilance is necessary regarding hypergastrinemia and long-term mucosal changes. Economic analyses further support their adoption in Asian contexts, where cost-effectiveness aligns with clinical performance. With the increasing use of P-CABs, future studies should focus on comparative effectiveness, long-term safety, and personalized treatment algorithms. In summary, P-CABs represent a significant therapeutic advancement for GERD and FD, overcoming efficacy gaps of PPIs and expected to play a key role in future management.

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Conflicts of interest
None.

Author contributions
Study concept and design: Seung In Seo, Sang Pyo Lee, Cheol Min Shin, and Moo In Park; literature search and study selection: Seung In Seo, Sang Pyo Lee, and Cheol Min Shin; data extraction and meta-analysis: Sang Pyo Lee; drafting of the manuscript: Seung In Seo, Sang Pyo Lee, and Cheol Min Shin; critical revision of the manuscript for important intellectual content: Yu Kyung Cho, Jeong Hwan Kim, Jung-Hwan Oh, and Moo In Park; and supervision: Moo In Park. All authors reviewed and approved the final version of the manuscript.

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Confocal Laser Endomicroscopy With Duodenal Food Challenge Does Not Predict Triggering Food Items in Eosinophilic Esophagitis: Results From a Pilot Trial

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Background/Aims: Targeted elimination of triggering food items is an attractive treatment option in eosinophilic esophagitis (EoE). However, no reliable biomarkers exist to identify culprit foods. In a previous uncontrolled study, 6/9 responded to duodenal food challenge (DFC), visualized by confocal laser endomicroscopy (CLE). This study aims to assess efficacy of a targeted elimination diet based on identification of nutrients that trigger acute mucosal alterations in the duodenum compared to a sham diet.

Methods: Randomized double-blind cross-over study in EoE. After baseline endoscopy, patients with peak eosinophil count of > 15 eosinophils/high power field underwent CLE. Following baseline assessment, a solution of foods (grains/milk/egg/soy/fish/nuts) was sprayed on the duodenal mucosa, with repeat imaging thereafter. Mucosa was judged as altered in case of fluorescein leakage and particle shedding. Based on the results, patients underwent a blinded randomized 6-week cross-over elimination diet with follow-up endoscopy. Exploratory measurements were done for small intestinal permeability, eosinophil- and mast cell activation.

Results: Seven patients completed the study. At CLE, 1 patient displayed baseline alterations, 5 reacted to 1 to 2 nutrients in 2 sessions (milk = 4, grains = 2, nuts = 2, fish = 1). There was no association between CLE results, and presence of food-specific IgE ($P > 0.999$). Both dietary arms resulted in a similar histological response ($P > 0.999$), with histological remission in only 1 patient (excluding gluten-containing grains). Permeability, eosinophil or mast cell activation were not different between arms.

Conclusions: In this pilot blinded cross-over trial, a targeted elimination diet based on the results of CLE with DFC did not generate a superior response compared to a sham diet.

Keywords: Diet; Duodenum; Eosinophilic esophagitis; Permeability

INTRODUCTION

Since its first description in the early nineties, the number of patients diagnosed with eosinophilic esophagitis (EoE) has seen an enormous increase.[1] This has been paralleled by an impressive growth in knowledge regarding clinical presentation, underlying pathophysiology, and management strategies.[1-4] Nevertheless, an adequate diagnostic tool or biomarker to identify the causing allergens—typically food antigens—remains to be discovered. Up to now, the gold standard for dietary management of EoE—and the only way to eliminate the cause of the inflammation—remains an exclusion diet, empirically excluding the 6-, 4-, 2-, or even the single most common food antigen(s) from the diet,[5-7] followed by a reassessment with esophagogastroduodenoscopy (EGD) and esophageal biopsies. These diets can be effective in 34% to 70% percent of patients but are cumbersome and require repeat endoscopies with biopsies after each elimination and reintroduction step.[8] Less invasive methods such as cystosponge or string test are being developed, however not yet available for routine clinical practice.[1]

The initial hypothesis that immunoglobulin E (IgE) could play a

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key role in pathogenesis, was invalidated as treatment with an anti-IgE antibody resulted in a remission rate of only 33% in a non-controlled setting[9,10] and no differences were found in a randomized controlled trial.[11] Also targeted elimination diets based on IgE sensitization to food allergens only led to improved symptoms and histology in a minority of adult patients.[9,12] The same study that tested an anti-IgE antibody suggested a role for IgG4.[11] However in a recent study, the fall in median peak esophageal eosinophil count (PEC) on a targeted exclusion diet based on food-specific IgG4 was not statistically significant, and the methodology needs further evaluation.[13]

Because of the limitations of current test methods and the intensity of an empiric food elimination diet, dietary treatments are not frequently used in adult patients with EoE. Therefore, developing reliable diagnostic tests to identify causative foods remains an unmet need in EoE.[1]

In patients with active EoE, the esophageal mucosal integrity is impaired and intercellular spaces are dilated.[14-16] In other atopic diseases, barrier dysfunction not only affects the target organ, but also the small intestine, the most immunogenic organ in the gut. It has been postulated that duodenal permeability might also be increased in EoE patients.[14-17] An increase in small intestinal permeability was shown in a study measuring the lactulose to mannitol ratio in urine,[14] however not confirmed in another.[15] It remains however plausible that increased permeability and sensitization for antigens (nutrients, microbiota etc) in the duodenum is present, but that we lack the right tools to measure it.[17]

Confocal laser endomicroscopy (CLE) is a technique that allows for real-time microscopic visualization of the mucosa during upper endoscopy.[18] In patients with irritable bowel syndrome (IBS), assessment of the duodenal mucosa using CLE has shown acute food-induced mucosal alterations within minutes after intraluminal food administration.[19] Acute mucosal alterations were defined as fluorescein leakage and the appearance of particles in the lumen.[18] This definition has shown substantial intra- and inter-observer agreement rates in a blinded assessment of experienced investigators.[20]

This technique, including duodenal food challenge (DFC), has been used in a proof-of-concept study in 9 patients with histologically proven EoE. One patient showed baseline mucosal alterations and of the remaining 8, 61% showed alterations upon DFC. These were, in order of frequency: wheat, milk, soy, and egg. The patients were treated with established medical treatment options for EoE (including proton pump inhibitor [PPI], swallowed topical corticosteroids [STC] or endoscopic dilation) and an additional open-label food exclusion of the triggering nutrient. All patients reported a reduction in symptoms, however the study did not report endoscopic/histological responses.[21] Assessment of the epithelial barrier using CLE in the esophagus has been undertaken,[22] but consists of a tight barrier and is not subject to acute alterations after food application as seen in the duodenum.

The aim of our study is first to assess nutrient-triggered acute mucosal alterations in the duodenum of EoE patients, using CLE. Secondly, in patients in whom these alterations were observed, the aim is to assess the efficacy of a targeted elimination diet based on the identification of nutrients triggering acute mucosal reactions in the duodenum in a blinded, randomized controlled trial. In addition, we aim to explore permeability and markers of inflammation in the duodenum.

MATERIALS AND METHODS

Trial Design and Study Population

We performed a double-blind randomized controlled trial in patients diagnosed with EoE based on symptoms of esophageal dysfunction, endoscopic features and histological confirmation,[2] with non-response or intolerance to PPIs, and motivated to start an elimination diet. An EGD with duodenal and esophageal biopsies was performed at baseline, along with blood samples, urine samples, and baseline questionnaires. In patients with active EoE based on symptoms and histology, this was followed by a second EGD with probe-based CLE at least 2 weeks later. Three days prior to the procedure, patients followed the six-food elimination diet (SFED), to avoid interactions during the measurement. If acute mucosal alterations were visualized during the CLE procedure (see criteria below), patients were randomized to receive a blinded cross-over diet for 6 weeks; excluding either the nutrient(s) triggering alterations during CLE, or a sham diet excluding (a) nutrient(s) that did not cause acute alterations. In patients where no alterations were visualized, or in case of no adequate negative comparator for the sham diet, a control diet was started using consecutively gluten-containing grains and milk in random order. At the end of each diet period, baseline measurements were repeated (Fig. 1, study flowchart). Primary outcome was the difference in histological response rate, defined as a PEC of < 15 eosinophils per high power field (PEC < 15/HPF) in patients undergoing duodenal CLE-targeted elimination diet compared to the sham diet. Secondary outcomes included difference in deep histological remission (PEC < 6/HPF); in PEC before and after the targeted elimination diet and symptomatic and endoscopic changes in patients undergoing duodenal CLE-targeted elimination diet compared to the sham diet. Measures of permeability and duodenal inflammation were assessed as exploratory outcomes. The study was conducted according to the

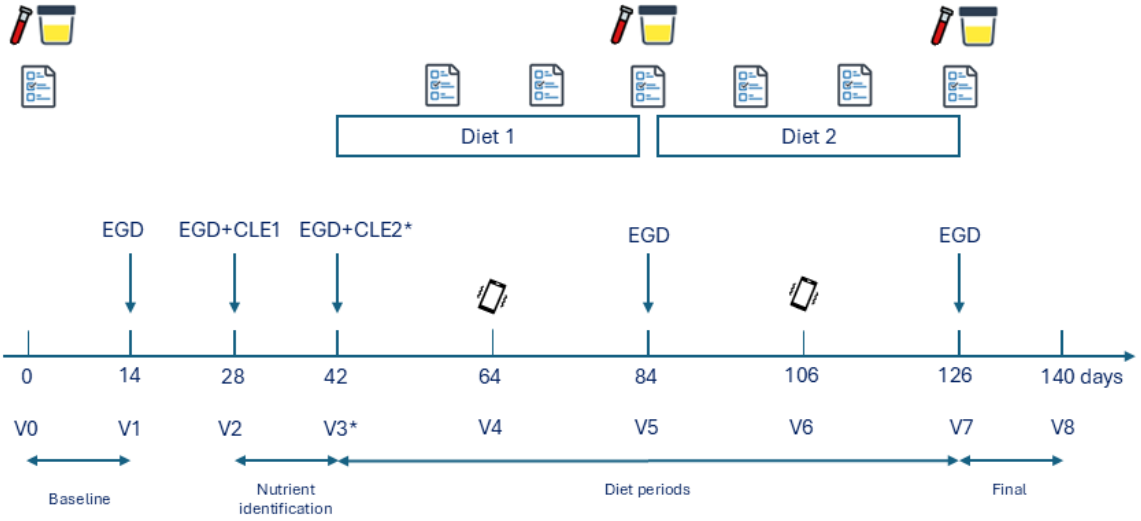

Figure 1. Study flowchart. *V3 with second confocal laser endomicroscopy (CLE) was performed in case of early alterations seen at first CLE (at baseline, or after administration of 1 of the first 3 foods). Blood samples and urine samples were taken at baseline, and after each elimination diet. Esophagogastroduodenoscopy (EGD) was repeated after each elimination diet. Questionnaires were filled out at baseline, and at regular timepoints during the diet periods. A phone-call visit was scheduled half-way each diet period.

Declaration of Helsinki and Good Clinical Practice regulations after approval by an Ethics Committee (S66223, NCT NCT05695456). Written informed consent was obtained from all study participants before inclusion in the study.

Study Procedures

Probe-based confocal laser endomicroscopy

The procedure was performed according to guidelines[18]: During EGD with propofol sedation, a CLE probe was passed through the working channel of the endoscope and gently placed against the mucosa in the second part of the duodenum (D2). After administration of IV fluorescein (2.5 mL, 10% solution) as a contrast agent, baseline assessment of the mucosa was performed for 2 minutes (Fig. 2). Subsequently, the 6 foods of the SFED (10 mL solutions of milk/egg/soy/lyophilized fish and shellfish/nuts/a mixture of grains containing wheat, barley, and rye) were applied sequentially in a semi-random order. The endoscopist (J.T.) was blinded for the applied nutrient. The order was semi-randomized starting with either soy or egg, followed by either grains or milk, and ending with fish and nuts. This order was chosen to preferentially test the nutrients that are known to be the more common culprits in EoE. After application of the nutrient, evaluation of the mucosa was repeated for 4 minutes, and the mucosa was judged as altered in case of presence of both intraluminal fluorescein and luminal particles at least 2 minutes after food application (Fig. 2).[18,20] In case alterations were observed, the exam was terminated as changes can no longer reliably be assessed after acute alterations. When this occurred after the application of one of the first 3 foods, the procedure was repeated at least 2 weeks later to allow for restoration of the mucosal wall, and the remaining foods were tested. Biopsies from D2 were collected at the end of the examination.

Biological Material

Blood samples

Baseline blood samples were used to assess full blood counts, high-sensitivity C-reactive protein (hsCRP), serum tryptase, total IgE, and food-specific IgE (FS-IgE) to each of the tested nutrients (measured using ImmunoCAP on Phadia 1000; ThermoFisher Scientific, Waltham, MA, USA: egg white, cow's milk, soy, a mixture of grains, a mixture of nuts, and a mixture of fish and shellfish). Repeat serum tryptase level was measured at least 30 minutes after a reaction occurred during CLE. HsCRP was repeated after each exclusion diet (real/sham).

Urinary lactulose/mannitol ratio

Participants ingested a fixed dose of 5g lactulose and 1g mannitol dissolved in 250 mL of water, and urine was collected for 2 hours after ingestion in a dry container containing 250 mg neomycin in powder form. The containers were brought to the laboratory and after filtration through a 0.45 µm filter, 1 mL aliquots were frozen until analysis. Urinary lactulose and mannitol concentrations were quantified with gas chromatography-mass spectrometry (GC-MS).[23] Limits of quantification were 15 mg/L and 5 mg/L for mannitol and lactulose, respectively. The lactulose-mannitol ratio (LMR) was calculated as the fractional lactulose concentration divided by the fractional mannitol concentration.

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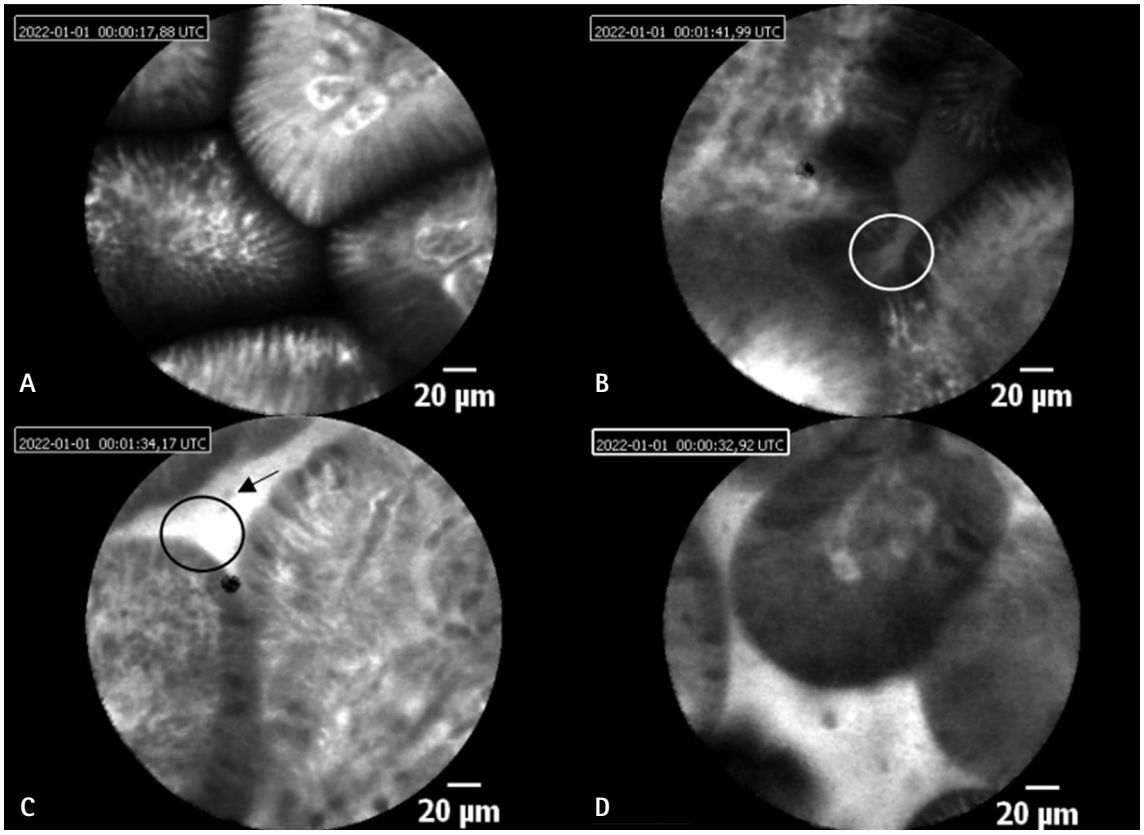

Figure 2. Confocal laser endomicroscopy (CLE) assessment before (A, B) and after (C, D) nutrient exposure in the duodenum. (A) Baseline CLE image with presence of epithelial cells in the villi made visible through intravenously administered fluorescein. At baseline the lumen is dark. (B) Some fluorescein in the lumen (circle) but no shedding of particles. (C) Positive CLE image with extravasation of fluorescein into the lumen and particles (arrow) previously termed "cell-shedding." (D) Positive CLE image: contrast inversion with the lumen appearing white, and the epithelium dark.

Ussing chamber experiments

The protocol used for Ussing chamber experiments was reported previously.[24] In summary, biopsies were mounted in modified 3-mL Ussing chambers (Mussler Scientific Instruments) within 30 minutes of collection in open-circuit conditions. The surface of exposed biopsies was 0.017 cm^2. For each chamber, transepithelial electrical resistance (TEER) was calculated from voltage deflections induced by bipolar constant-current pulses of 16 µA every 60 seconds with duration of 200 msec and was recorded every 30 minutes over 2 hours. The average TEER of timepoints 60-90-120 minutes was calculated and presented as Ω x cm^2. Paracellular flux was measured using fluorescein (molecular mass of 400 Da, 20 mg/200 µL; Sigma-Aldrich, St Louis, Missouri, USA), which was added to the mucosal compartment of the 3 Ussing chambers containing esophageal biopsies. A fluorescent labeled dextran (FD4 molecular mass of 4 kDa; Sigma-Aldrich) was used for the 3 duodenal biopsies. Serosal samples were collected every 30 minutes for 2 hours and the passage of fluorescein or FD4 was measured using a fluorescence reader (FLUOstar Omega; BMG Labtech, Ortenberg, Germany). The average was calculated for each chamber at 90 and 120 minutes presented as pmol/cm^2. Paracellular flux could only be measured at baseline, and after each diet period, as the IV administration of fluorescein during CLE procedure interfered with the measurement.

Histology

A total of 6 esophageal biopsies were taken at 3 levels (proximal, mid- and distal esophagus) at baseline, and after each exclusion diet of 6 weeks. All biopsies were assessed for PEC by one and the same expert pathologist (G.D.H.). The pathologist was blinded to the timepoint of the study (ie, baseline or during dietary intervention) and to the nature of the diet (ie, targeted diet or sham diet).

Duodenal eosinophil and mast cell counting of baseline biopsies was performed blinded after H&E and c-kit staining, respectively. Stained duodenal biopsy sections were scanned with a ZEISS Axioscan 7 slide scanner in a random and blinded fashion and visualized using QuPath software (version 0.5.1, UK).[25] Duodenal eosinophils (bilobar nucleus and eosinophilic granules on H&E staining) and mast cells (visible nucleus and cytoplasmic c-kit staining) were counted according to the Leuven Intestinal Counting Proto-

col (LICP).[26] The threshold was set at 400 eosinophils/mm^2 and 700 mast cells/mm^2 for abnormal duodenal infiltration.[26]

Biopsy supernatants

Two duodenal biopsies were obtained using standard size biopsy forceps and placed in culture medium (Roswell Park Memorial Institute containing glutamine, supplemented with 10% fetal bovine serum and 1% penicillin/streptomycin 10,000 U/mL, (all from Gibco, Life Technologies). After 1 hour, biopsies were transferred to a 48-well plate containing the same culture medium and incubated for 24 hours at 37°C. Supernatants were obtained after 24 hours, snap frozen and kept at –80°C until analysis.

Beta-2 tryptase and eosinophil derived neurotoxin (EDN) in biopsy supernatants were measured using commercially available ELISA kit (TPSB2 ELISA, ThermoFisher Scientific, Life Technologies; EDN ELISA, Diagnostics Development, Uppsala, Sweden, respectively). Resulting beta-tryptase and EDN content was corrected for biopsy weight and reported as µg/mg tissue.

Mucosal protein extracts

Eosinophilic cationic peptide (ECP), alpha tryptase and IgE were measured on mucosal protein extracts with Immunocaps, using Phadia 250 Instrument (ThermoFisher Scientific) and corrected for biopsy protein content measured using BC Assay (Interchim, Montluçon, France). Results for IgE are reported in kU/g, results for ECP and alpha tryptase are reported as ng/mg protein.[27]

Comparison with healthy volunteers

We used available data from a cohort of healthy volunteers to compare baseline measurements (TEER, mucosal IgE, ECP, and alpha-tryptase). The cohort of healthy volunteers was recruited in our center in the context of a similar study protocol (NCT05097872), with main exclusion criteria being acute or chronic gastrointestinal symptoms, any gastrointestinal comorbidity such as coeliac disease, EoE or inflammatory bowel disorder, or history of gastro-intestinal surgery. All healthy volunteers were negative for serum IgE specific for any of the administered foods.

Questionnaires

At baseline, and at regular timepoints during the diet periods, symptoms were assessed using dysphagia symptom questionnaire (DSQ), Eosinophilic Esophagitis Activity Index (EEsAI), and eosinophilic esophagitis quality of life questionnaire for adults (EoE-QoL-A) (Fig. 1). The DSQ is a short, validated, 4-question assessment of dysphagia symptoms over the last 24 hours, which was used as a daily diary for 2 weeks at baseline and during the entire diet periods.[28] The EEsAI is a symptom-based severity index designed and validated for EoE; and evaluates symptoms related to food intake over a recall period of 7 days.[29] The EoE-QoL-A is a reliable disease-specific health-related quality of life measure for adult EoE patients.[30] The latter 2 questionnaires were filled out at baseline, and every 2 weeks during the diet. All questionnaires were endorsed by the international multidisciplinary consensus as part of the core outcome set for therapeutic studies in Eosinophilic Esophagitis (Coreos).[28] The Monash University Comprehensive Nutrition Assessment Questionnaire was filled in at baseline, a short food frequency questionnaire was filled in by participants weekly during both diet periods to assess compliance with the exclusion diet. FIRE questionnaire (food induced immediate response), was filled in at baseline and at the end of each diet period.

Statistical Analysis

This is a pilot study using a new technique (CLE) in EoE. Since no studies were published on the use of CLE in EoE patients, the sample size calculation was based on data available from dietary studies not using CLE. Assuming that the CLE directed diet will be effective in 75% of patients (similar to the 6FED), and the sham diet in 10% of patients, using a 2-tailed McNemar test with an α of 0.05, 1-β of 0.8, we calculated a sample size of 15. Adjusting for one drop-out, we aimed to include 16 CLE positive patients. A futility analysis was performed after 50% of planned participants were included (ie, 8 patients) due to the invasive nature of the study. As we did not reach our anticipated sample size, the power of the study did not reach 0.8.

Descriptive statistics were used for baseline characteristics. Data are represented as median (interquartile range). A McNemar test was used to assess the primary outcome. Non-parametric tests (Wilcoxon signed rank test, Kruskal Wallis rank sum test) were used for secondary and exploratory outcomes, as well as Spearman/Pearson correlations where applicable. A logistic regression model was used to test for carry-over effect. *P*-value of < 0.05 was considered statistically significant.

RESULTS

Baseline Characteristics and Measurements

We included 8 patients, of which 1 was later excluded due to inactive EoE at baseline (PEC < 15/HPF). Median age at inclusion was 41.5 (28.0-47.0) years, 5 (62.5%) participants were male, with a median body mass index of 27.9 (26.3-29.8) kg/m^2.

Of the 7 patients who had active EoE at baseline (PEC > 15/HPF), median PEC was 80/HPF (50-95), median endoscopic reference score (EREFS) score was 3 (3-4), with moderate symptoms and sig-

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nificant impact on quality of life (Table 1).

All patients had previously taken PPI as first line treatment, which was ineffective in six, and was not tolerated by the seventh patient (skin rash). Five patients were previously treated with STC, with no response in 1 patient, only short-term and suboptimal use of a flixotide inhaler in another patient, and response in the 3 other patients. These latter patients wanted to stop medication and switch to diet-based therapy, for which STC was interrupted at least 2 months before inclusion.

A personal history of atopy was present in 5 of 7 patients; most common were hay fever and allergic rhinitis. One patient was treated with anti-TNF therapy (certolizumab subcutaneously) for severe psoriasis. Another patient had severe atopic dermatitis, food allergy to soy, and oral allergy syndrome for stone fruits, kiwi and banana. None of the patients however, had a history of anaphylaxis. Two patients took antihistamines at baseline, which was continued on a stable dose throughout the study.

Four out of seven patients experienced FIRE (food-induced immediate response of the esophagus) symptoms at baseline, with an acute reaction to wheat mentioned by all 4 of them (specifically beer, bread, and pasta), raw fruits or vegetables mentioned by 3 (raw fruit and vegetables, fresh fruit juice, banana), and dairy products by 1 (cheese, yoghurt).

Table 1. Demographics and Baseline Measurements

Variable	value
Demographics	
N	7
Age (yr)	41.5 (28-47)
Sex (%)	62.5% male
BMI (kg/m^2)	27.9 (26.3-29.8)
PPI use	7 (100%)
STC use	5 (71%)
History of atopy	5 (71%)
Antihistamine use	2 (29%)
Baseline EoE severity and impact	
PEC (max eosinophils/HPF)	80 (50-95)
EREFS	3 (3-4)
EEsAI	42 (33-59.5)
EoE-QoI-A (/96)	52 (37.5-74)
DSQ (/84)	12 (0-25.5)

EoE, eosinophilic esophagitis; BMI, body mass index; PPI, proton pump inhibitor; STC, Swallowed topical corticosteroids; PEC, peak eosinophil count; HPF, high-power field; EREFS, Endoscopic reference score; EEsAI, Eosinophilic Esophagitis Activity Index; EoE-QoI-A, eosinophilic esophagitis quality of life score for adults; DSQ, dysphagia symptom questionnaire.

Laboratory measurements

Total serum IgE was elevated in all but 1 patient, with a median of 346 (148-505) kU/L. FS-IgE were also elevated in these 6 patients, with FS-IgE for milk elevated in 5 patients, for nuts in 4, for grains, soy, egg and fish in 3 patients (Supplementary Table 1).

Baseline white blood cell counts were normal in all patients, with a slightly increased peripheral eosinophil count in 2 patients (0.4×10^9/L and 1.0×10^9/L, respectively). Baseline serum tryptase was low in all patients with a median of 6.8 (4.6-7.2) μg/L.

Confocal Laser Endomicroscopy

At least 2 weeks after baseline EGD, 7 patients underwent a second EGD with CLE. During the first CLE examination, 1 patient displayed baseline mucosal alterations, all other patients reacted to one of the foods applied in the duodenum. In 3 patients, a second CLE examination was performed with mucosal alterations after food application in all 3. We were able to test at least 3 nutrients in all patients (3 in 1 patient, 4 in 3 patients, and 5 nutrients in 3 patients). As a result, 5 patients reacted to 1 nutrient, and 2 patients reacted to 2 nutrients in 2 different sessions (milk = 4, grains = 2, nuts = 2, fish = 1) (Table 2).

There was no association between reaction on CLE, and the presence of serum specific IgE for the tested nutrients ($P > 0.999$). Moreover, we saw no association between reaction on CLE, and baseline intake of the tested nutrients ($P = 0.230$) (Supplementary Table 2). Also, there was no significant correlation between baseline intake and FS-IgE, except for fish ($\rho = 0.87$; 95% CI, 0.33-0.98; $P = 0.010$), although FS-IgE for fish were only elevated in 2 patients.

Response to Targeted Exclusion Diet

Five patients started with a blinded diet for the trigger nutrients (real diet), 2 patients started with a sham diet (Supplementary Table 3).

The primary endpoint of PEC < 15/HPF, was only reached by 1 patient after the real diet (elimination of gluten-containing grains), the response rate was not different between diet arms ($P > 0.999$). One other patient had a peak eosinophil count of 15/HPF after the real diet (grains), and of 18/HPF after the sham diet (soy). A third patient had symptomatic improvement and a reduction in PEC from 100/HPF at baseline to 25/HPF after the real diet (milk and nuts), and a resurgence after the sham diet (grains and fish) to 230/HPF. None of the other patients showed histological response to either of the diets. Adherence to the diet was excellent except for 2 patients, who had consumed lactose-free milk where they had to exclude all dairy products containing animal-protein (Supplementary Table 2). A logistic regression model was built to test

Table 2. Overview of Alterations as Assessed With Confocal Laser Endomicroscopy

Patient	Baseline	Grains	Milk	Soy	Eggs	Fish	Nuts	N CLE	N applied nutrients
1	neg	neg	neg	neg	neg	pos		1	5
2	neg	neg	pos	neg	neg		pos	2	2-3
4	neg	neg	pos	neg	neg			1	4
5	neg	pos	neg	neg	neg			1	4
6	neg	neg	pos	neg	neg			1	4
7	neg	neg	pos	neg	neg		pos	2	2-2
8	pos	pos		neg	neg			2	0-3
pos/tested	1/10	2/7	4/6	0/7	0/7	1/1	2/2		

N CLE, number of confocal laser endomicroscopy (CLE) examinations performed per subject; N applied nutrients, number of applied nutrients per examination; neg, negative; pos, positive; pos/tested, total amount of alterations seen divided by the times each nutrient was applied in all examinations. In subjects where 2 CLE procedures were performed, the numbers for each procedure are separated by -.

for carry-over effect, however this could not be formally tested because no responses were observed in the second diet phase.

Regarding the secondary endpoints, PEC did not change (baseline: 80.0 (50.0-95.0), real diet: 35.0 (21.5-88.0), sham diet: 115.0 (26.5-176.0), P = 0.450, Fig. 3A). Median EREFS did not change after either of the diets (baseline: 3.0 (3.0-4.0), real diet: 3.0 (2.5-4.0), sham diet: 3.0 (3.0-4.5), Fig. 3B). DSQ over a period of 2 weeks (maximal total score of 84) was compared at baseline versus the last 2 weeks of each diet period. There was no significant change in DSQ score from baseline after any of the diet periods (P = 0.180). Median EEsAI was 58.0 (21.0-68.5) after the real diet, and 42.0 (13.5-68.5) after the sham diet. Both were not different from baseline (P = 0.400, Fig. 3C). Also median EoE-QoI-A was not different from baseline (real: 50.0 (41.0-82.5), sham: 69.0 (47.0-85.5), P = 0.740, Fig. 3D).

Exploratory Measurements of Intestinal Permeability and Inflammation

Exploratory results can be found in Table 3.

Healthy volunteers

Data from 8 healthy volunteers were used (6/8 female), with a median age of 22.0 years (20.8-24.0), and median body mass index of 23.4 (21.8-25.3).

Baseline

As a marker for duodenal permeability, TEER was compared between included EoE patients with data from healthy volunteers who underwent the same protocol, and was not significantly different (P = 0.650). Paracellular flux of FD4 was not performed in our cohort of healthy volunteers, however when comparing with a previous cohort, results were similar.[31] As a marker of small intestinal permeability, baseline LMR was assessed, which was again not different from healthy volunteers (P = 0.480).

At baseline, duodenal mast cells and eosinophils were counted using the LICP.[26] Well-oriented slides were available for all but 1 patient. Median duodenal eosinophil count was 254/mm^2 (207-350), which falls within the normal range as previously defined (< 400 eosinophils/mm^2).[26] Median mast cell count at baseline was 627/mm^2 (569-713), with the highest value being 826/mm^2, which is on the high end as compared with data from healthy volunteers (normal < 700 mast cells/mm^2).[26]

When comparing local total IgE levels in duodenal biopsies with healthy volunteers, we found a trend towards higher total IgE levels in active EoE (P = 0.050). Also local alpha-tryptase levels in duodenal biopsies tended to be higher in EoE patients than controls (P = 0.050). We found no difference in presence of ECP (P = 0.610).

Before vs after confocal laser endomicroscopy

After an acute mucosal reaction during CLE, TEER was not different from baseline (P = 0.840, Fig. 4A). We saw no difference in secretion of EDN (P = 0.640) or beta-tryptase (P > 0.999) in duodenal biopsy supernatant after CLE, as compared with baseline. Also systemically measured tryptase in serum, taken 30-60 minutes after acute alterations were observed during CLE, was not higher than baseline, and was even lower than baseline (P = 0.040, Fig. 4B).

Comparison with exclusion diets

LMR was not significantly different between conditions (P = 0.340). Duodenal TEER (P = 0.290), as well as paracellular flux of FD4 (P = 0.870) did not significantly change from baseline after any of the diets. Esophageal TEER was low at baseline, with a median of 62.8 Ω × cm^2 (30.3-115.0), and did not improve after any of the diet periods (P = 0.460). Trans- and paracellular flux of fluorescein in esophageal biopsies could not reliably be assessed be-

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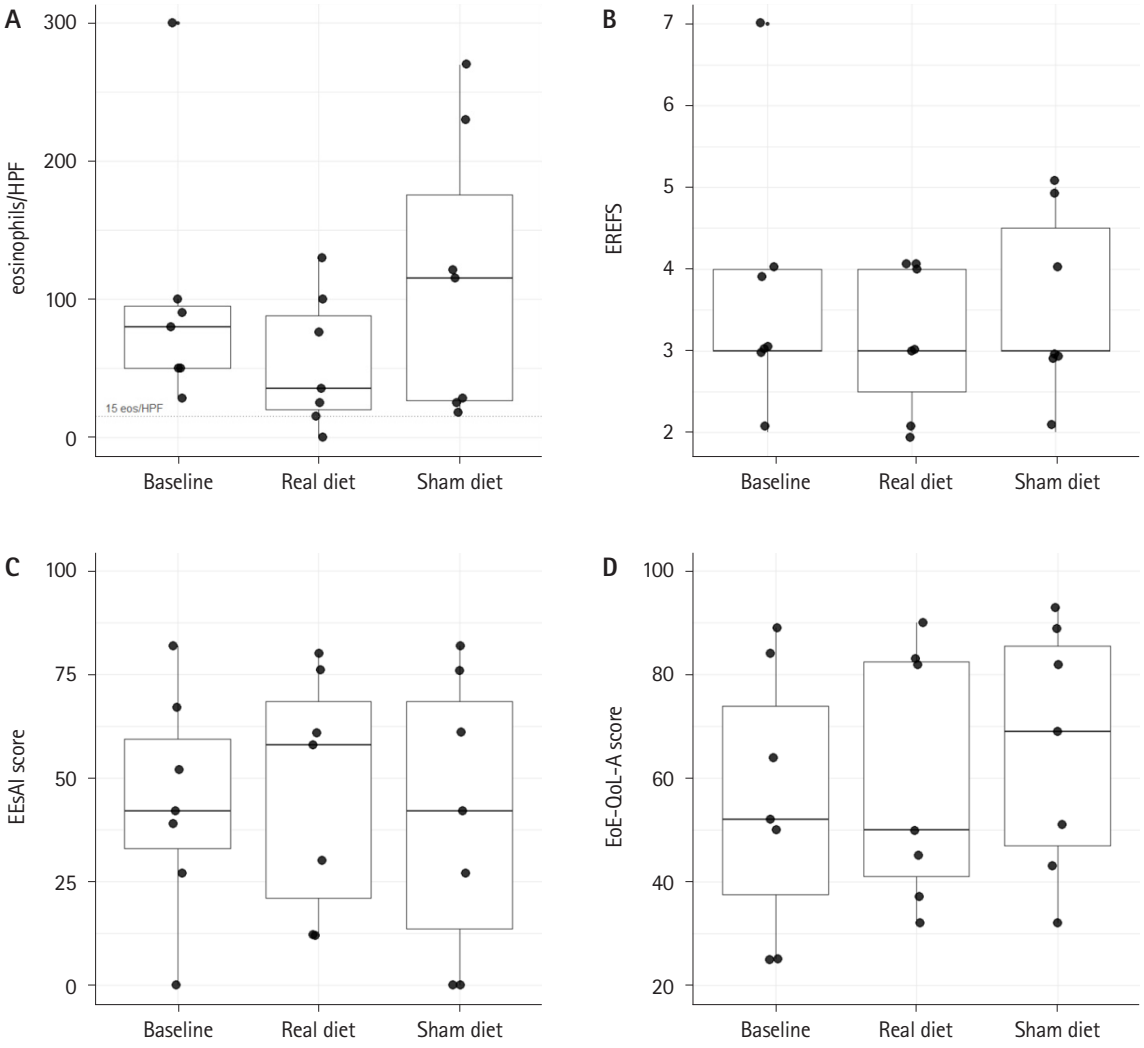

Figure 3. Evolution of peak eosinophil count (PEC), endoscopic reference score (EREFS), Eosinophilic Esophagitis Activity Index (EEsAI), and eosinophilic esophagitis quality of life score for adults (EoE-QoL-A) from baseline and after cross-over diet periods. (A) PEC at baseline, after the real diet and sham diet. Median PEC remained above the threshold for histological remission of 15/HPF, 1 patient was in histological remission after the real diet. There was no significant change from baseline after the diet periods. (B) EREFS at baseline, after the real diet and sham diet. There was no significant change from baseline after the diet periods. (C) EEsAI score at baseline, after the real diet and sham diet. There was no significant change from baseline after the diet periods. (D) EoE-QoL-A score at baseline, after the real diet and sham diet. There was no significant change from baseline after the diet periods.

cause of the severely impaired barrier function in most patients.

HsCRP remained stable during follow-up (P = 0.420). There was no change in any of the markers of inflammation tested: alpha-tryptase (P = 0.810), ECP (P = 0.440), and total IgE (P = 0.600) in mucosal protein extracts; EDN (P = 0.800) and beta-tryptase (P = 0.680) in duodenal biopsy supernatants.

DISCUSSION

In this pilot randomized controlled trial, a targeted exclusion diet was evaluated, using the results of acute mucosal alterations to food exposure in the duodenum, as assessed by confocal laser endomicroscopy. This technique allows for live, in vivo assessment of the intestinal mucosa during endoscopy on a microscopic level, and has shown acute mucosal alterations to food exposure in the duodenum in IBS patients.[19,32]

In our study, we found that all 7 participants reacted to at least 1 food applied to the duodenal mucosa, and when the examination was repeated at least 2 weeks later, 2 patients reacted to a second food item (1 patient showed baseline alterations in 1 of the 2 exams). In the landmark paper in IBS patients, an acute reaction occurred in 70% of patients after duodenal food challenge,[19] and a similar observation was reported for 6 out of 8 in a pilot study in patients with EoE (with a ninth patient showing baseline alterations).[21] It is unclear why the response rate to CLE was as

Table 3. Exploratory Measurements

Measurement	Eosinophilic Esophagitis	Healthy volunteers		*P*-value
Total IgE in duodenal biopsy (kU/g)	5.99 (3.18-9.11)	1.31 (1.10-2.75)		0.050
ECP in duodenal biopsy (ng/mg)	41.6 (29.6-52.7)	37.5 (22.7-51.3)		0.610
Alpha tryptase in duodenal biopsy (ng/mg)	492 (249-662)	177 (133-322)		0.050
TEER (Ω × cm²)	23.6 (22.2-24.1)	23.3 (22.6-27.5)		0.650
Lactulose mannitol ratio	1.16 (1.07-1.27)	0.96 (0.63-1.29)		0.480
	Baseline	CLE		*P*-value
Duodenum				
Eosinophil count (/mm²)	254 (207-350)			
Mast cell count (/mm²)	627 (569-713)			
TEER (Ω × cm²)	23.6 (22.2-24.1)	23.7 (22.5-25.3)		0.840
Beta-tryptase from biopsy supernatant (μg/mg)	0.64 (0.44-0.81)	0.67 (0.27-1.25)		0.640
EDN from biopsy supernatant (μg/mg)	5.15 (3.33-5.98)	5.62 (4.77-6.64)		> 0.999
Serum tryptase (μg/L)	6.8 (4.6-7.2)	4.7 (3.35-5.3)		0.040
	Baseline	Real diet	Sham diet	*P*-value
Esophagus				
TEER (Ω × cm²)	62.8 (39.4-102)	55.3 (36.0-89.8)	44.3 (25.5-72.5)	0.460
Duodenum				
TEER (Ω × cm²)	23.6 (22.2-24.1)	26.2 (24.5-34.1)	25 (22.7-26.4)	0.290
FD4 flux (pmol/cm²)	1788 (1286-2356)	1884 (1866-2440)	1884 (1798-2233)	0.870
Total IgE in duodenal biopsy (kU/g)	5.99 (3.18-9.11)	4.30 (2.30-7.87)	5.99 (4.43-15.2)	0.600
ECP in duodenal biopsy (ng/mg)	41.6 (29.6-52.7)	31 (21.6-46.4)	59.1 (27.0-95.1)	0.440
Alpha tryptase in duodenal biopsy (ng/mg)	492 (249-662)	263 (182-544)	504 (212-676)	0.810
Lactulose mannitol ratio	1.16 (1.07-1.27)	0.913 (0.88-1.46)	0.946 (0.73-1.10)	0.340
High-sensitivity CRP	1.02 (0.49-1.92)	0.98 (0.83-1.08)	0.42 (0.28-0.75)	0.420

IgE, immunoglobulin E; ECP, eosinophil cationic peptide; TEER, transepithelial electrical resistance; CLE, confocal laser endomicroscopy; EDN, eosinophil derived neurotoxin; FD4, fluorescent labeled dextran.

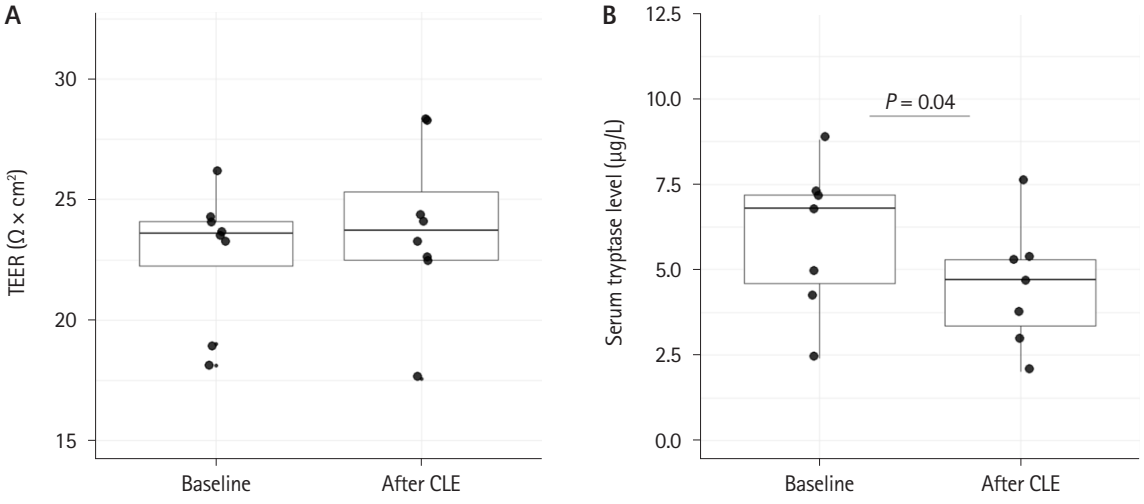

Figure 4. Transepithelial electrical resistance (TEER) and serum tryptase at baseline versus after confocal laser endomicroscopy (CLE). (A) Comparison of TEER (Ω × cm²) in ussing chambers at baseline, versus after CLE. There was no significant difference between conditions. (B) Comparison of serum tryptase levels (μg/L) at baseline versus after CLE. There was no increase in serum tryptase after CLE, instead there was a significant decrease in serum tryptase levels taken 30-60 minutes after mucosal alterations were seen during CLE examination ($P = 0.040$).

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high as 100% in our study. Similar to previous studies, our patients followed a hypo-allergenic diet 3 days prior to the CLE examination. It is unlikely that this was too short, as we noted first unaltered mucosa during baseline assessment in all but 1 patient, and an acute change after application of a certain nutrient. In addition, most of our patients were exposed to several nutrients triggering no CLE reaction, before alterations were observed. Other than in previous studies, most of our patients had positive serum FS-IgE to at least 1 of the tested nutrients, however, we did not find an association between the presence of serum IgE antibodies, and a reaction to the same nutrient, applied in the duodenum. The nutrients tested in this study were slightly different from those in previous studies, where wheat, milk, soy, yeast, and egg white were tested. However, we chose to test the 6 foods of the SFED, including nuts (positive in 2 patients) and fish and shellfish (positive in 1 patient). Yeast was not tested in this study. In addition, we used probe-based CLE, whereas most previous studies used endoscopy-based CLE (which is no longer available), making direct comparisons difficult. For probe-based CLE, a recent blinded assessment of CLE images by experienced investigators from 3 European expert centers showed substantial intra- and interobserver agreement rate.[20] Lastly, we cannot exclude other mechanisms for the alterations we see, such as duration of the exam or physiological changes after application of nutrients to the intestinal mucosa, and therefore results of the application in healthy volunteers should be awaited.

When putting the patients on a blinded cross-over exclusion diet, excluding either the trigger nutrient(s), or a sham diet for 6 weeks, only 1 patient achieved histological remission, as defined by PEC < 15/HPF. There was no difference in response between the real diet (1 patient, excluding gluten-containing grains) and the sham diet (none) (primary endpoint). Two other patients had a partial response, one to the real diet (milk & nuts), and the other to both the real diet (grains) and the sham diet (soy). When looking at the evolution of PEC over time, we saw no difference between the real and the sham diet. Also, endoscopic findings (EREFS) and symptom scores (DSQ, EEsAI, and EoE-QoL-A) did not change over the course of the study. In the previous study mentioned above, where 6 out of 8 EoE patients reacted to DFC as assessed by CLE, the effect of a targeted exclusion diet was not assessed in a blinded, controlled manner as was the case in this study. The patients followed an open-label exclusion diet, alongside established treatment options for EoE. Furthermore, there is no mentioning of a histological response after the exclusion diet.[21] This is thus the first study evaluating the results of DFC during CLE in a blinded, controlled targeted elimination diet, which was not more effective than a sham diet, excluding nutrients that did not trigger acute alterations. When comparing with empirical elimination diets, a one-food elimination diet was shown to be effective in 34% of adults,[5] and a 2-food elimination diet (excluding wheat and milk) in 43% of adults,[33] and even more in a systematic review and meta-analysis also including children.[8] The 1 participant that achieved histological remission in this study, excluded gluten-containing grains, one of the most common triggering food antigens in EoE.[8] In summary, from the 7 patients included in this study, only 1 responded to a targeted diet based on CLE results, and there was no difference in primary and secondary outcomes between real and sham diet. To be able to generalize these results, they should be confirmed in a larger sample.

We explored several markers of permeability and of inflammation in the duodenum. As previously documented, we saw a highly increased permeability ex-vivo of the esophageal mucosal wall in patients with active EoE.[15,34] As we had only one patient in histological remission after treatment, we could not compare with EoE in remission.

Katzka et al[14] showed an increase in LMR in urine, a measurement correlating to small bowel permeability, of active EoE patients as opposed to the ones in remission. However, Warners et al[15] found no difference in LMR. Furthermore, the latter also used duodenal impedance measurement in vivo, and TEER measurement and transepithelial flux in vitro in Ussing chambers, and found no difference in duodenal integrity between EoE patients and controls. Also in our study, we could not confirm an increased permeability in the small intestine of EoE patients with active inflammation.[31] In addition, we found normal eosinophil counts and ECP values, as compared with healthy volunteers. As a conclusion, in this small cohort of patients with active EoE at baseline, we did not find signs of small intestinal increased permeability or eosinophil activation, however the sample size is too small to exclude involvement of the small intestine in EoE.

We did find a trend towards an increased level of local IgE in mucosal protein extracts of duodenal biopsies, as compared with healthy volunteers. Even though only exploratory, higher levels of local total IgE have previously been demonstrated in patients with confirmed food allergy[27]; also in our cohort, the majority had positive FS-IgE in serum for the 6 tested nutrients, and possibly others which were not tested. There was a trend towards higher alpha-tryptase concentrations in mucosal biopsies, and also duodenal mast cell counts were on the high end, as compared with a previous cohort of healthy volunteers using the same counting protocol.[26] This suggests an increased duodenal mast cell infiltration and/or activation in the duodenum of EoE patients, similarly to patients with atopic dermatitis,[35] but should be confirmed. With only seven patients concluding the study, it is underpowered to

draw firm conclusions. In addition, in this cross-over study, more patients started with the real diet than the sham diet, inducing a possible order effect. However, we did not see any difference between both diet arms, and we relied on the objective measurement of PEC, therefore it is unlikely that a more balanced distribution would have changed our results. A formal test for carry-over effect could not be used because of no responses in the second diet phase. Three of our patients were in histological remission on STC before the start of the study, and discontinued medication at least 2 months prior to baseline measurements. Even though all patients had a PEC > 15, these patients might not have been fully relapsed at the start of the study, inducing a possible bias in further evaluations. Only 1 of these patients responded to 1 of the diets, suggesting that this did not affect our results. Lastly, most of the ex vivo measurements are exploratory and should be interpreted with caution.

In conclusion, in this blinded cross-over pilot trial, the results of a duodenal food challenge during CLE were not associated with response to a targeted exclusion diet. Targeted elimination diets using CLE are unlikely to provide a benefit over established empirical exclusion diets in EoE. In addition, in this study, no small intestinal increased permeability was found.

SUPPLEMENTARY MATERIALS

Note: To access the supplementary tables mentioned in this article, visit the online version of Journal of Neurogastroenterology and Motility at http://www.jnmjournal.org/, and at http://doi.org/10.5056/jnm25132.

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SUPPLEMENTARY MATERIALS

Supplementary Table 1. Total Peripheral Blood Immunoglobulin E Levels and Specific IgE for the Tested Foods

Patient	Total IgE	Nuts	Fish	Grains	Eggs	Milk	Soy
1	580	1.16	0	28.10	0	0	0
2	346	3.45	0.1	0.31	0	0.25	0.21
4	31	0	0	0	0	0	0
5	380	0.72	0.92	0.47	1.58	0.17	0
6	505	26.00	0	0.48	0	0.15	0.15
7	148	0.12	0	0.23	0	4.10	0.13
8	132	0.13	0.58	0	0.12	0.17	0

IgE, immunoglobulin E.
All values are given in kU/L. Values above the clinical threshold for abnormality are indicated in red.

Supplementary Table 2. Baseline Food Intake of the 6 Foods of the 6-Food Elimination Diet

Patient	Nuts	Fish	Grains	Eggs	Milk	Soy
1	4	4	34	4	20	8
2	0	4	46	4	16	0
4	0	0	8	0	4	0
5	0	12	154	30	12	8
6	0	0	8	8	30	0
7	8	0	36	8	8	0
8	34	4	70	34	42	0

Baseline food intake based on The Monash University Comprehensive Nutrition Assessment Questionnaire. For grains, all gluten-containing grains were included. Intake is represented as the monthly consumption of a certain food item.

Supplementary Table 3. Overview of Exclusion Diets ("Real" vs "Sham") and Compliance to the Diet

Patient	Diet 1	Compliance	Diet 2	Compliance
1	Fish (real)	0	Nuts (sham)	0
2	Fish, grains (sham)	1	Milk, nuts (real)	0
4	Milk (real)	13	Grains (sham)	0
5	Grains (real)	0	Milk (sham)	0
6	Grains (real)	0	Milk (sham)	0
7	Milk, nuts (real)	7	Grains, fish (sham)	0
8	Grains (real)	4	Soy (sham)	0

Compliance is expressed in number of days of reported intake of one of the food items the patient had to exclude over the entire diet period of 6 weeks.

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The Utility of the Functional Sphincter Area: A Novel Functional Lumen Imaging Probe Parameter for the Assessment of Anal Sphincter Structure and Function in Fecal Incontinence

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Background/Aims: The functional lumen imaging probe (FLIP) is a unique tool to measure the anal sphincter during distension. Using detailed geometric data generated by FLIP, we present the functional sphincter area (FSA), a novel measurement of sphincter morphology with potential clinical utility in the investigation of fecal incontinence (FI).

Methods: Parous female patients with FI presenting to a tertiary pelvic floor unit were prospectively recruited. FLIP measurements were obtained alongside endoanal ultrasound and high-resolution anorectal manometry.

Results: Thirty-one parous female patients with FI (median age: 62) were recruited. On endoanal ultrasound, 14 (45.2%) were found to have a sphincter defect in either the external sphincter (n = 10), or in both the external and internal sphincter (n = 4). The FSA demonstrated a consistent pattern of change during distension of the FLIP bag in all patients. At rest, the FSA was greater in those without anal sphincter defects at 30 mL (72.6 vs 53.2 mm^2; $P = 0.029$) and 40 mL (71.5 vs 45.5 mm^2; $P = 0.048$) bag volumes. The same pattern was observed during voluntary squeeze at 30 mL (67.8 vs 42.8 mm^2; $P = 0.044$) and 40 mL (61.2 vs 38.6 mm^2; $P = 0.040$) bag volumes. By contrast, there were no differences in symptom severity or manometry measurements between the 2 groups.

Conclusions: The FSA may offer a clinically useful parameter and, in combination with other FLIP measurements, could become a useful tool to assess the structure and function of the anal canal in FI.

Keywords: Anal canal; Fecal incontinence; Manometry; Pelvic floor disorders

INTRODUCTION

Fecal continence is maintained by coordinated interactions between the anorectum, the anal sphincter, and the pelvic floor orchestrated by central and peripheral neural activities.[1] Deficiencies in these components can lead to fecal incontinence (FI), a devastating condition affecting 8% of the population.[2,3] However, anal sphincter dysfunction is considered a principal cause, particularly following obstetric or iatrogenic injury.[4,5] High-resolution anorectal manometry (HRAM) is the established test of sphincter function.[6] However, HRAM uses fixed-diameter catheters to measure the pressure generated by the anal sphincter muscle under isometric conditions. This may explain why assessing sphincter function with HRAM only demonstrates a limited association with the severity of anal sphincter defects in patients with FI.[7]

The functional lumen imaging probe (FLIP) is a unique tool to assess the resistance of the anal sphincter during distension,[6,8-10] which has been proposed as the main determinant of its function.[11-13] It generates a dynamic geometric profile of the whole anal canal at different distension volumes with the ability to measure the cross-sectional area (CSA) over an 8 cm length in combination with pressure measurements.[8] In doing so, FLIP can examine the length-tension relationship of the sphincter muscle to describe its function and has therefore been proposed as an alternative tool to HRAM.[9,14] Furthermore, new metrics, such as the functional sphincter area (FSA), can now be extracted from the traditional FLIP measurements. We therefore hypothesised that this advanced FLIP measurement of the anal canal could be used

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to identify differences in sphincter function between patients with anal sphincter defects and those without. If so, this could be used to support the use of FLIP in the investigation of FI and address the limitations currently experienced with HRAM.

This study of FI patients describes the FSA as a novel FLIP parameter to assess the anal sphincter in those with and without evidence of anal sphincter defects. In addition, conventional FLIP and HRAM parameters were explored in the same patients. The primary aim is to investigate if differences exist in measurements with these devices between parous female patients with and without anal sphincter defects.

MATERIALS AND METHODS

Study Design

This prospective observational study received approval from the local research ethics committee (Greater Manchester West, United Kingdom; Ref: 19/NW/0633). Patients with FI attending the pelvic floor unit were identified prospectively and were invited to take part in this study. Patients were excluded if they were under the age of 18, lacked cognitive capacity to provide informed consent, were pregnant, or had active inflammatory bowel disease or a history of anorectal cancer. All patients provided written informed consent.

The severity of FI was assessed using the Vaizey incontinence score.[15] A score of 0 indicates no incontinence symptoms, whilst the maximum score of 24 suggests severe FI symptoms. The impact of FI on quality of life was assessed with the Manchester Health Questionnaire.[16] Scores range from 0, indicating no impact, to 900 indicating the most severe impact on an individual.

Endoanal ultrasound (EAUS) investigations (3D 9052 Transducer, BK Medical, MA, USA) were performed by a consultant radiologist with an interest in pelvic floor dysfunction. Patients were considered to have an anal sphincter defect if there was evidence of previous sphincter trauma on EAUS including the presence of a scar, or sphincter disruption in either the external anal sphincter (EAS) or internal anal sphincter (IAS) or both (Fig. 1). The standardized London protocol was used for HRAM measurements[17] with a 14 Fr water-perfused catheter containing 10 circumferential pressure sensors spaced at 8 mm intervals (Mui Scientific, ON, Canada). Mean resting pressure, maximum squeeze pressure, and incremental squeeze pressure were recorded in mmHg. In addition, the functional anal canal length was also recorded, defined as the length of the anal canal in which pressure exceeded rectal pressure by > 5 mmHg, recorded in cm.[18]

Functional Lumen Imaging Probe Measurement

Analysis of the functional lumen imaging probe data

Data from the FLIP measurements were analyzed using a customized MATLAB subroutine (R2022a; Mathworks, Natick, MA, USA). The data can be used to construct profiles of the anal canal at any bag volume.[22] A smoothed diameter distribution along the anal canal was obtained using one-dimensional interpolation at any time point (Fig. 2A). Parameters were calculated at the area of minimum CSA, including the yield pressure, closing pressure, pressure-strain elastic modulus, and distensibility index (DI) during rest and voluntary squeeze, which have been described previously.[9,20,21,23] In addition, the bag pressures at each bag volume during

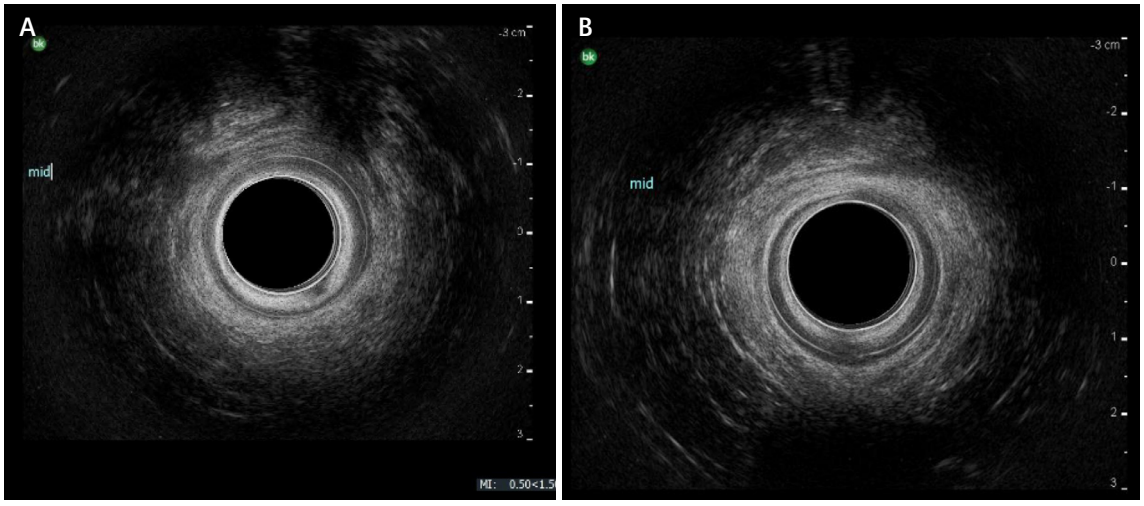

Figure 1. Endoanal ultrasound images of anal sphincter defects. (A) An endoanal ultrasound scan of the mid anal canal in a patient demonstrating intact anal sphincter muscles. (B) An endoanal ultrasound scan of the mid anal canal in a patient, who had previously suffered obstetric trauma, demonstrating a scar throughout the length of the external anal sphincter at 1 o'clock.

rest and maximum squeeze were used to calculate the increase in (incremental) bag pressure. Resting measurements were calculated as an averaged measurement from the final 10 seconds of the 30 second resting period, and the squeeze function was during the 10 second squeeze.

Measuring the Functional Sphincter Area

The pressure and diameters along the anal canal were analyzed during the initial state at the second distension to 10 mL, and at each bag volume of 20, 30, 40, and 50 mL (Fig. 2). At the initial state, the proximal and distal limits were defined at the points where the diameters clearly started to increase.[19,22] Using these defined limits, the configuration of the anal canal can be determined using a technique that has previously been described with the proximal and distal anal canal defined as the points 75% of the distance from the middle to the proximal and distal limits.[22] During every distension after the initial state, the middle of the anal canal was redefined as the middle of the narrow zone of the anal canal at half of the distension volume. Proximal and distal ends of the narrow zone were defined as the locations with CSA equal to 110% of the narrowest CSA (Fig. 2B). The proximal and distal anal canal points were also recalculated by assuming the proximal and distal anal canal lengths during distension are the same as that at the initial state (proximal anal canal length: length from the middle to the proximal limit of the anal canal; distal anal canal length: length from the middle to the distal limit of the anal canal). The FSA, measured in mm^2, represents the area contained by the outside of the FLIP bag wall and the line drawn between the proximal and distal anal canal, termed the functional anal canal length (Fig. 2B). The FSA was considered to represent the area of tissue contained by the outside of the FLIP bag wall as it increases in volume. The FSA was measured at rest and during voluntary squeeze. The CSA was calculated from the radius of the FLIP bag measured from the center of the catheter to the wall of the bag.

Statistical Methods

Statistical analysis was performed using SPSS Version 29.0 (IBM, NY, USA). Data are presented as median (interquartile range) unless otherwise stated. The Kruskal-Wallis test and the Mann-Whitney *U* test were used for comparisons between independent samples of non-parametric data. Statistical significance was considered at the $P < 0.05$ level. A formal sample size calculation was not

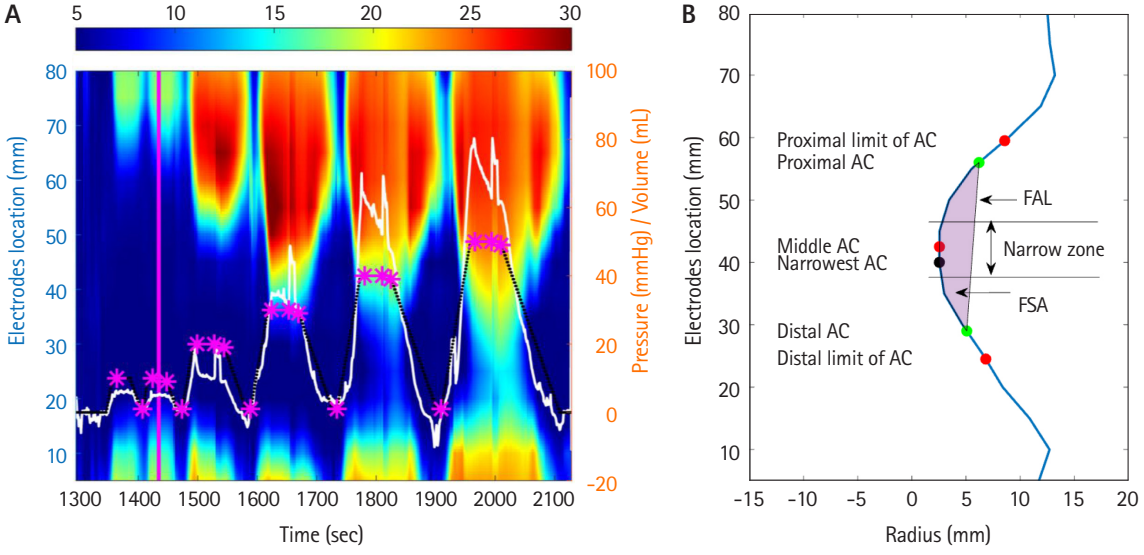

Figure 2. A spatio-temporal diameter map of a single functional lumen imaging probe (FLIP) measurement. (A) A representative spatio-temporal diameter map of the anal canal from a single patient with fecal incontinence. The changing colors from blue to red indicate an increasing diameter. Each column represents the configuration of the anal canal at a particular time point, and each row the change in diameter over time, along the 8 cm measurable length of the FLIP catheter. The vertical axis on the right represents pressure in mmHg (white line) and bag volume (heavy black dotted line). The magenta line is the selected initial state time point. The magenta stars are the detected starting time points of inflation, resting, squeezing, and deflation states. (B) The configuration of the anal canal in the same patient. The center of the FLIP catheter is indicated along the horizontal axis at radius = 0 mm. The vertical axis indicates the position of the 16 pairs of detection electrodes spaces along a length of 8 cm. The blue line indicates the geometric profile of the FLIP bag in the anal canal. This configuration demonstrates the proximal and distal anal canal points, the narrow zone of the anal canal, the functional anal canal length (FAL) and the functional sphincter area (FSA). The narrow zone was the region with cross-sectional area (CSA) less than 110% of the narrowest CSA along the anal canal, the FAL is the distance between the proximal and distal anal canal (AC), and the FSA is the area of the anal canal contained outside of the FLIP bag wall.

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possible given the exploratory nature of this first study to investigate a novel FLIP parameter.

RESULTS

Thirty-one parous female patients with FI (median age 62 [range 38-84]) were recruited. Table 1 presents baseline clinical data from the included patients. None of the included patients had previously undergone a secondary anal sphincter repair.

The FLIP measurements were easy for the investigator to perform and were well tolerated by patients. However, in 1 patient, distension to 50 mL bag volume was not tolerated due to discomfort. There were no other adverse events related to the study investigation.

The Functional Sphincter Area

As the FLIP bag filled with conductive solution and the volume increased, the FSA changed in size. A consistent finding in all patients was an FSA that began to increase in size as the FLIP bag inflated from 10 to 50 mL bag volume. This increase in FSA continued until a point was reached where the minimum CSA started to increase in size, as the anal canal opened. The pressure observed at this point, as the narrowest part of the anal canal began to open, is commonly described as the 'yield pressure'. As the anal canal continued to open, the FSA then decreased in size until the maximum bag volume was reached (50 mL) (Fig. 3). There was no difference in the median rate of change of the FSA between patients with and without an anal sphincter defect (2.02 [1.21-2.67] mm^2/s vs 2.00 [1.82-2.73] mm^2/s; $P = 0.518$). In all measurements, the yield pressure, or pressure at which the FSA began to decrease in size occurred between 30 to 50 mL bag volume.

Functional Sphincter Area Measurements in Anal Sphincter Defects

Of the 31 included patients, 17 (54.8%) had no evidence of a sphincter defect on EAUS, whilst 14 (45.2%) were found to have a sphincter defect in either the EAS in isolation (n = 10), or in both the EAS and IAS (n = 4). There were no differences between the groups with and without anal sphincter defects in median age (60 vs 63; $P = 0.922$), weight (69.5 vs 71.0 kg; $P = 0.399$), Vaizey score (14 vs 16; $P = 0.173$), Manchester Health Questionnaire score (459.17 vs 505.00; $P = 0.830$), parity (2 vs 2; $P = 0.830$), or vaginal deliveries (2 vs 2; $P = 0.710$).

The differences in FLIP measurements between the 2 groups are presented in Table 2. At rest, the FSA at 30 mL (72.6 vs 53.2 mm^2; $P = 0.029$) and 40 mL (71.5 vs 45.5 mm^2; $P = 0.048$) bag volumes were greater in those with an intact anal sphincter complex. At 50 mL bag volume, there was a trend towards a reduced FSA in patients with evidence of anal sphincter defects (40.5 vs 22.0 mm^2; $P = 0.101$). Fig. 4 demonstrates an example of FSA recordings in two patients with and without an anal sphincter defect. With the exception of yield pressure (26.4 vs 17.6 mmHg; $P = 0.040$), there were no other differences in FLIP measurements at rest between the 2 groups ($P > 0.05$).

During voluntary squeeze, only squeeze FSA at 30 mL (67.8 vs 42.8 mm^2; $P = 0.044$) and 40 mL (61.2 vs 38.6 mm^2; $P = 0.040$) bag volumes were different between those with intact anal sphincters and those with evidence of an anal sphincter defect.

There were no differences in HRAM results between the 2 groups (Table 3).

Table 1. Included Patient Demographics

Parameter	Patients (n = 31)
Age (yr, median [range])	62 (38-84)
Weight (kg, median [range])	70 (46-162)
Bowel symptom history (n [%])	
ODS	12 (38.7)
Fecal incontinence	31 (100.0)
Urge fecal incontinence	20 (64.5)
Passive fecal incontinence	5 (16.1)
Mixed fecal incontinence	6 (19.4)
Fecal leakage	16 (51.6)
Vaizey score (median [IQR])	14 (12-17)
MHQ score (median [IQR])	477.50 (354.17-615.00)
Urinary incontinence (n [%])	12 (38.7)
Obstetric history (n [%])	
Parous	31 (100.0)
Parity [median (IQR)]	2 (2-3)
Vaginal delivery	31 (100.0)
Cesarean section	5 (16.1)
Forceps use	11 (35.5)
Ventouse use	4 (12.9)
Episiotomies or perineal tear	30 (96.8)
Previous anorectal surgery (n [%])	4 (12.9)
Hemorrhoidectomy	2
Lord's procedure	1
Delorme's procedure	1

ODS, obstructed defecation syndrome; IQR, interquartile range; MHQ, Manchester Health Questionnaire.

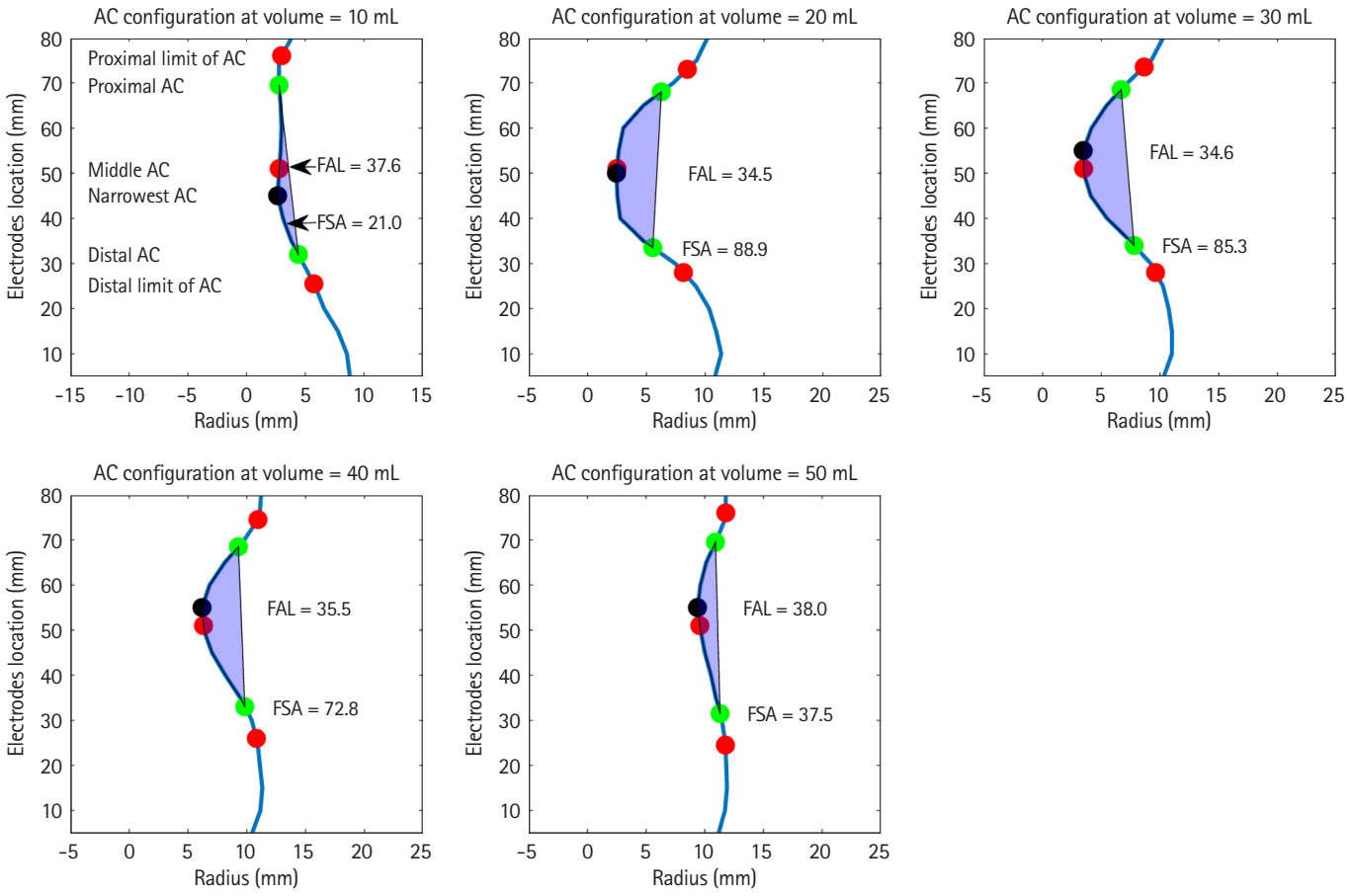

Figure 3. The functional sphincter area and functional anal canal length at rest in a patient without an anal sphincter defect at 10, 20, 30, 40, and 50 mL. AC, anal canal; FAL, functional anal canal length; FSA, functional sphincter area.

The Severity of Anal Sphincter Defects and the Functional Sphincter Area

Table 4 presents FSA measurements at rest and during voluntary squeeze stratified according to the severity of anal sphincter trauma. The FSA measurements in patients with combined EAS and IAS sphincter defects tended to be lower compared to those with isolated EAS defects and those with intact anal sphincters (Fig. 5), although this did not reach statistical significance when comparing differences ($P > 0.05$ for all; Kruskal-Wallis test).

DISCUSSION

The anal sphincter complex is an important component of the continence mechanism,[1] and its dysfunction from iatrogenic or obstetric injury is often implicated in the aetiology of FI in female patients.[24] This prospective observational study of parous females describes a new dynamic parameter of the whole anal canal that may be of interest in the assessment of anal sphincter morphology, the functional sphincter area. This area was lower in patients with sphincter defects, indicating a reduced sphincter volume. In addition, the yield pressure is reduced in this group, indicating an anal canal that opens more easily at a reduced pressure. To date, most studies with FLIP have reported measurements of the anal canal obtained at a single point along its length; at the minimum measurable diameter (or cross-sectional area) at a particular static bag volume using the distensibility index.[21,25] By contrast, the FSA combines geometric measurements from the whole anal sphincter to quantify the sphincter area and has identified reduced measurements in patients with anal sphincter defects. These findings may help to explain the mechanism behind FI in this group with a small volume anal sphincter, that opens at a lower pressure. No differences in HRAM parameters, including the functional anal canal length, were seen between patients with anal sphincter defects and those without.

FLIP provides a detailed geometric and pressure profile of the whole anal canal, and it has previously been used in the investigation of FI.[21,22,26] The DI (minimum CSA/bag pressure; $mm^2/mmHg$)

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Table 2. Functional Lumen Imaging Probe Parameters in Patients With and Without Anal Sphincter Defects

FLIP parameter	Intact anal sphincters (n = 17)	Anal sphincter defect (n = 14)	*P*-value[a]
Resting parameters			
Yield pressure (mmHg)	26.4 (16.5-48.1)	17.6 (11.8-32.1)	0.040
Ep opening (kPa)	4.0 (2.3-4.9)	2.9 (2.0-3.7)	0.215
Closing pressure (mmHg)	18.0 (10.5-33.0)	12.2 (9.3-19.5)	0.064
Ep closing (kPa)	2.6 (1.8-3.9)	1.8 (1.2-2.9)	0.077
DI at 40 mL (mm^2/mmHg)	1.1 (0.6-3.2)	2.0 (0.9-3.8)	0.230
[b]DI at 50 mL (mm^2/mmHg)	2.2 (1.3-4.5)	4.4 (2.3-5.3)	0.101
FSA at 30 mL (mm^2)	72.6 (59.9-87.4)	53.2 (30.4-69.4)	0.029
FSA at 40 mL (mm^2)	71.5 (44.8-84.3)	45.5 (27.8-65.5)	0.048
[b]FSA at 50 mL (mm^2)	40.5 (20.9-73.5)	22.0 (7.1-41.1)	0.101
Squeeze parameters			
Squeeze DI at 40 mL, mm^2/mmHg	0.5 (0.3-1.3)	0.8 (0.5-1.5)	0.570
[b]Squeeze DI at 50 mL, mm^2/mmHg	1.6 (0.9-3.1)	1.8 (1.2-2.4)	0.637
Incremental squeeze bag pressure at 40 mL, mmHg	17.1 (11.2-24.7)	18.6 (9.2-31.2)	0.739
[b]Incremental squeeze bag pressure at 50 mL, mmHg	13.0 (4.2-24.3)	21.1 (10.4-28.9)	0.208
Squeeze FSA at 30 mL, mm^2	67.8 (57.3-82.6)	42.8 (26.9-65.5)	0.044
Squeeze FSA at 40 mL, mm^2	61.2 (45.3-84.2)	38.6 (30.2-58.9)	0.040
[b]Squeeze FSA at 50 mL, mm^2	29.0 (17.7-69.0)	16.2 (9.2-36.3)	0.093

[a]Mann-Whitney *U* test.
[b]Denotes measurements from 16 participants with intact anal sphincters used for analysis.
FLIP, functional lumen imaging probe; Ep, pressure-strain elastic modulus; DI, distensibility index; FSA, functional sphincter area.
Significance at $P < 0.05$ level.

is typically used to describe the resistance of the anal sphincter to distension and is calculated at the point of minimum CSA at a particular bag volume.[10,27] Using this parameter alone, authors have described FLIP to be a complimentary tool to HRAM.[26] This measurement taken at rest, or during voluntary squeeze, is unable to differentiate between patients with intact sphincters and varying degrees of sphincter injury.[6] Indeed, the DI does not always correlate with the symptoms of FI.[6] In this present study, no differences were seen in the DI between patients with and without sphincter defects. However, data can be downloaded from the FLIP device to generate additional parameters in post-acquisition analysis.[19] These additional parameters seek to quantify the function of the anal canal during opening and closing. During inflation of a FLIP bag to 50 mL, the yield pressure has been described as the pressure at which the narrowest part of the anal canal just starts to open.[19] This is lower in patients with FI,[20] and those treated with neoadjuvant therapy for anal or rectal cancer.[23] In this study, the yield pressure was reduced in patients with anal sphincter defects. It suggests that this parameter, measuring the resistance to distension of the anal canal, may offer additional clinical utility above that provided by DI alone at a static bag volume, and may help to explain the mechanism of FI in this group.

EAUS is the gold standard technique to assess the anatomy of the anal sphincter complex and, whilst it is a well-tolerated technique, it is invasive and requires the use of specialist equipment and experienced technicians.[28] Its role in the routine investigation of FI is unclear and it is only recommended on a case-by-case basis, particularly where a sphincter repair is being considered, or to plan subsequent deliveries following obstetric trauma.[29,30] Further work with FSA could see FLIP proposed as a first-line test prior to EAUS to provide both a functional and structural assessment of the anal sphincter complex in FI. In select patients, without suspicion of an anal sphincter defect, an additional EAUS investigation could therefore be avoided. Further work is now required to generate a normal dataset to inform the clinical utility and reliability of the FSA and other FLIP measurements in the assessment of FI. This should include whether the FSA can correlate with the severity of sphincter injury, including combined IAS and EAS injuries, and if it corresponds with patient reported symptoms of FI. It is acknowledged that the intra- and inter-rater reliability and repeatability of the novel parameters presented here have not currently been explored and should be the subject of future work in

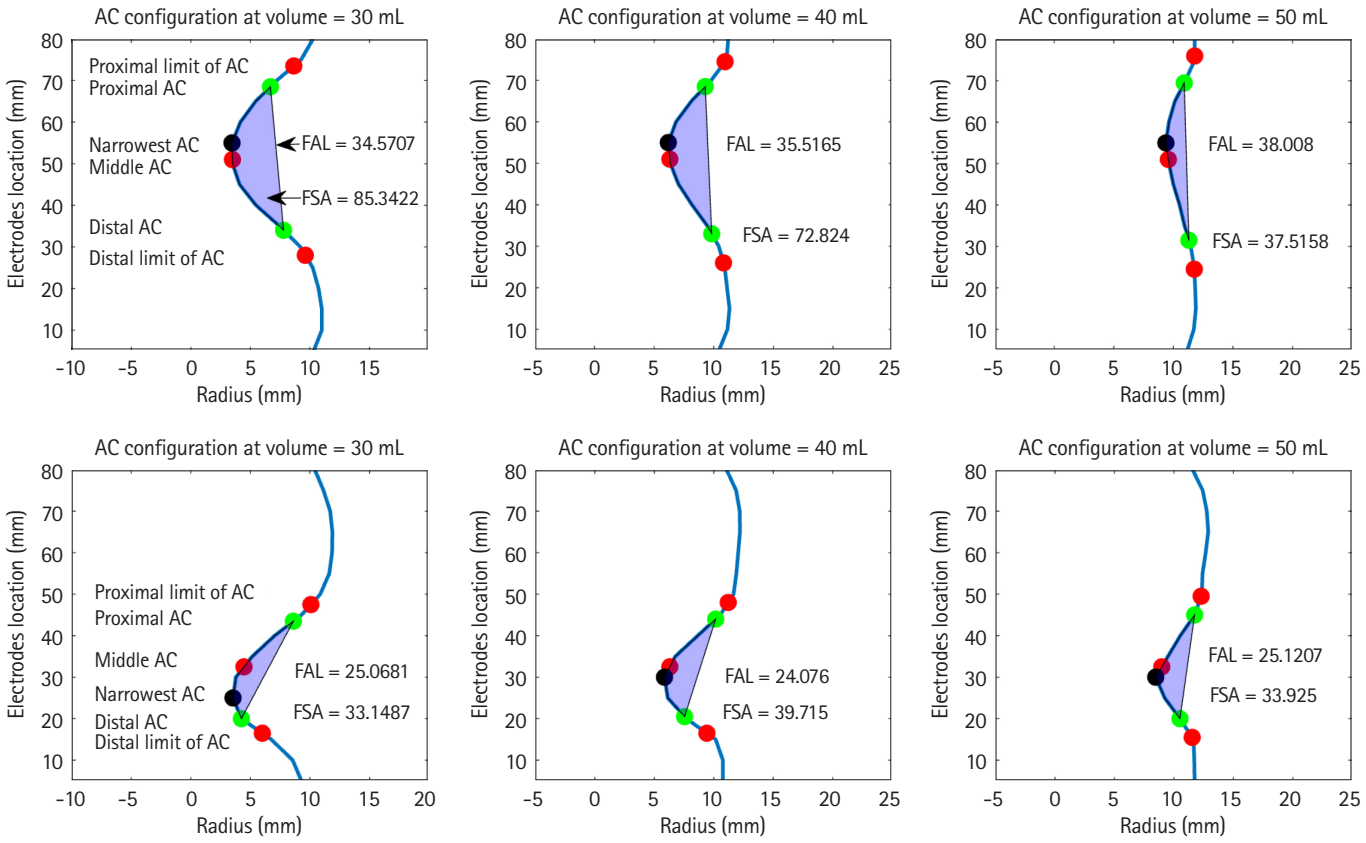

Figure 4. A representative functional sphincter area (FSA) and functional anal canal length (FAL) at 30, 40, and 50 mL at rest from patients without (top panels) and with (bottom panels) an anal sphincter defect (FAL, unit in mm; FSA, unit in mm^2). AC, anal canal.

Table 3. High-resolution Anorectal Manometry Results

HRAM parameter	Intact anal sphincters (n = 17)	Anal sphincter defect (n = 14)	*P*-value[a]
Functional anal canal length (cm)	2.9 (2.4–3.2)	2.5 (2.0–3.3)	0.421
Mean resting pressure (mmHg)	49.0 (33.5–62.5)	31.3 (26.8–61.5)	0.493
Maximum squeeze pressure (mmHg)	106.0 (66.0–133.1)	85.5 (71.5–154.5)	0.279
Incremental squeeze pressure (mmHg)	51.2 (24.0–83.0)	43.5 (21.3–86.0)	0.739

[a]Mann-Whitney *U* test.
HRAM, high resolution anorectal manometry.
Significance at $P < 0.05$ level.

healthy controls and symptomatic patients. Indeed, when the reliability of HRAM was explored, the variation seen in repeated measurements of pelvic floor function affected the precision of the test, which could have implications in the clinical utility of these tools.[31]

This is the first study to use FLIP to quantify the ability of the whole anal sphincter to resist distension. However, in calculating the FSA, and indeed other geometric measurements with FLIP, it is assumed that the FLIP bag is cylindrical inside the anal canal. This has been demonstrated in healthy volunteers using a customized magnetic resonance compatible FLIP catheter,[32] but remains to be demonstrated in symptomatic patients with a sphincter defect, thus representing a limitation of this study. In addition, this study is limited by its single center design with small overall numbers with 1 patient unable to tolerate distension to 50 mL bag volume. The use of a water-perfused HRAM device rather than a 3D solid-state catheter may also limit the ability of HRAM to detect difference between the 2 groups. EAUS remains operator dependent and despite the experienced radiologists in our institution there may be instances of missed external sphincter injuries in our study. Finally, the FLIP parameters described can currently only be generated in detailed post-acquisition analysis limiting the generalisability of the findings. Future work therefore is required to

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Table 4. A Comparison of Functional Lumen Imaging Probe Measurements Between Patients With Varying Severity of Anal Sphincter Defects

Parameter	Anal sphincter defect			*P*-value[a]
	Intact (n = 17)	EAS Only (n = 10)	EAS and IAS (n = 4)	
Resting Parameters				
FSA at 30 mL (mm^2)	72.6 (59.9-87.4)	55.1 (41.5-69.4)	37.6 (15.0-93.2)	0.090
FSA at 40 mL (mm^2)	71.5 (44.8-84.3)	50.4 (38.0-65.6)	27.3 (12.3-85.9)	0.114
FSA at 50 mL (mm^2)[b]	40.5 (20.9-73.5)	28.4 (11.9-41.1)	14.3 (2.8-73.9)	0.239
Squeeze Parameters				
Squeeze FSA at 30 mL (mm^2)	67.8 (57.3-82.6)	42.8 (34.4-65.5)	37.1 (15.5-94.2)	0.126
Squeeze FSA at 40 mL (mm^2)	61.2 (45.3-84.2)	40.4 (36.7-58.9)	17.6 (6.6-83.2)	0.087
Squeeze FSA at 50 mL (mm^2)[b]	29.0 (17.7-69.0)	17.1 (13.0-36.3)	5.0 (2.7-67.3)	0.176

[a]Kruskal-Wallis test.
[b]Denotes measurements from 16 participants with intact anal sphincters used for analysis.
EAS, external anal sphincter; IAS, internal anal sphincter; FSA, functional sphincter area.
Significance at $P < 0.05$ level.

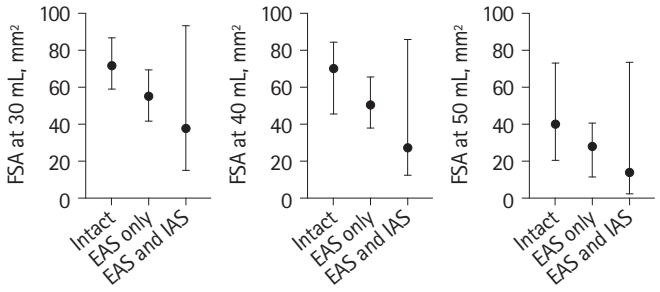

Figure 5. Median and interquartile range bar charts of functional sphincter area measurements at different bag volumes at rest in patients with varying severity of anal sphincter defects. FSA, functional sphincter area; EAS, external anal sphincter; IAS, internal anal sphincter.

explore the accuracy of FLIP in identifying anal sphincter defects, with the generation of 'normal' data from healthy volunteers and in parous women without FI symptoms. If proven to be clinically useful, these additional parameters should be incorporated into the FLIP device to provide 'real-time' measurements as the bag inflates. This would avoid the need for detailed post-acquisition analysis which currently limits the routine clinical use of these measurements.

In conclusion, in patients with anal sphincter defects, the FSA measured with FLIP may offer a parameter of clinical utility. In combination with other measurements previously described, FLIP could be a valuable tool to assess the structure and function of the anal canal. Future work is now required to define the diagnostic utility of FLIP and its measured parameters in the assessment of FI and anal sphincter trauma.

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Conflicts of interest

None.

Author contributions

Alexander O'Connor, Donghua Liao, Edward Kiff, Abhiram Sharma, and Karen Telford designed the study; Alexander O'Connor and Donghua Liao undertook data acquisition; Alexander O'Connor, Donghua Liao, Matthew Davenport, Abhiram Sharma, and Karen Telford were involved in data analysis and interpretation; Alexander O'Connor, Edward Kiff, Matthew Davenport, Abhiram Sharma, and Dipesh H Vasant assisted in drafting the initial manuscript; Asbjørn M Drewes, Abhiram Sharma, Dipesh H Vasant, John McLaughlin, Edward Kiff, and Karen Telford supervised the project and the preparation of the manuscript; and all authors provided critical revision of the manuscript, were involved in intellectual discussion, approve the final version of the manuscript, and agree to be accountable for all aspects of the work.

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Disorders of Gut–Brain Interaction and Psychological Comorbidity in Singapore: Insights From the Rome IV Global Epidemiology Study

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Background/Aims: Disorders of gut–brain interaction (DGBI) affect ~40% of the global population, yet data from Asia, including urbanized Singapore, remain limited. We aim to estimate prevalence, characterize psychological comorbidities, and assess burden using Rome IV criteria.

Methods: As part of the Rome Foundation Global Epidemiology Study, we conducted a cross-sectional, internet-based survey of 2047 nationally representative Singaporean adults (January-March 2018). The Rome IV Diagnostic Questionnaire identified 22 DGBI; participants with physician-diagnosed organic disease were excluded. Psychological comorbidities were assessed with Patient Health Questionnaire-4 (PHQ-4 [anxiety/depression]) and Patient Health Questionnaire-12 (PHQ-12 [somatization]), and quality of life with Patient-Reported Outcomes Measurement Information System Global-10. Multivariable logistic regression identified factors associated with DGBI and psychological comorbidity.

Results: DGBI prevalence was 31.1% (95% CI, 29.1-33.2), lower than the pooled global prevalence (~40%). Functional bowel disorders predominated (25.3%), with constipation-related conditions (irritable bowel syndrome with constipation and functional constipation) affecting 9.9%. DGBI were more common in younger adults and women, but not education-related. Participants with multi-region DGBI reported higher somatization and anxiety/depression scores, poorer physical and mental health, and greater healthcare use. Regression showed younger age (per-year OR, 0.978; 95% CI, 0.970-0.986), female sex (OR, 0.667; 95% CI, 0.546-0.813), and higher somatization and anxiety scores independently associated with DGBI. Among those with DGBI, psychological comorbidity was independently associated with female sex, polypharmacy, and multi-region involvement.

Conclusions: In Singapore, DGBI are common but less prevalent than global estimates. Constipation-related disorders and psychological comorbidities contribute substantially to symptom burden and healthcare use, particularly in women and younger adults. Findings support integrated, culturally tailored care pathways to guide regional practice and public health policy.

Keywords: Brain-gut axis; Constipation; Irritable bowel syndrome; Prevalence; Surveys and questionnaires

INTRODUCTION

Disorders of gut–brain interaction (DGBI), previously known as functional gastrointestinal disorders, include conditions such as irritable bowel syndrome (IBS), functional dyspepsia (FD), and functional constipation (FC). Collectively, they affect an estimated 40% of the global population, representing a major public health challenge due to their substantial impact on quality of life, healthcare utilization, and economic burden.[1,2]

DGBI are characterized by chronic gastrointestinal symptoms without structural abnormalities, arising from dysregulation of the gut-brain axis—a complex, bidirectional communication pathway linking the gastrointestinal tract and the central nervous system.[3] This dysregulation helps explain the frequent co-occurrence of psychological conditions such as anxiety, depression, and somatization, which exacerbate symptom severity and complicate treatment.[4,5]

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The interplay between gastrointestinal symptoms and psychological distress creates a vicious cycle, in which emotional stress amplifies gut symptoms, while chronic gastrointestinal dysfunction contributes to psychological burden.[6] Epidemiological studies suggest that up to 60% of individuals with DGBI exhibit psychological comorbidities, which adversely affects treatment outcomes and drives increased healthcare costs.[7] The Rome IV diagnostic criteria, a globally validated framework, offers a standardized approach to diagnosing DGBI, enabling consistent assessments across diverse populations.[8]

Despite extensive research in Western cohorts, data from Asian countries remain limited, particularly in highly urbanized, multicultural settings such as Singapore.[9] Singapore's unique demographic profile—comprising predominantly Chinese (74.3%), with sizeable Malay (13.5%), Indian (9.0%), and other ethnic populations—together with a high-stress urban environment and advanced healthcare system, offers a unique context for investigating DGBI prevalence and psychological comorbidities.[10] Cultural factors, including dietary habits, healthcare-seeking behaviors, and stigma surrounding mental health, may influence DGBI presentation and management in Singapore, highlighting the need for region-specific research.[11]

Using the Rome IV criteria, this study estimated the prevalence of DGBI and associated psychological distress in Singaporean adults, addressing critical gaps in Asian epidemiological knowledge. The primary aims of the study were to estimate Rome-IV DGBI prevalence, to characterize psychological comorbidities among individuals with DGBI, and to assess their burden in Singapore using the dataset of the Rome Foundation Global Epidemiology Study (RFGES).

MATERIALS AND METHODS

Study Design

This study represents the Singapore arm of the RFGES, a cross-sectional, population-based survey conducted across 33 countries that was designed to evaluate the prevalence and burden of DGBI using Rome IV criteria.[2] The Singapore arm estimated DGBI prevalence, characterized gastrointestinal tract anatomical distribution, assessed psychological comorbidities (anxiety, depression, somatization), and evaluated their impact on healthcare utilization and quality of life in a multi-ethnic urban population. This descriptive epidemiological approach followed the RFGES protocol and was designed without a priori hypotheses.

Population and Sampling

The Singapore study population consisted of a nationally representative sample of 2,047 adults aged 18 years and older. Participants were recruited via a professional survey company (Qualtrics, LLC, Provo, UT, USA) using stratified sampling to endeavor to obtain a balanced sex (50% male, 50% female) and age distribution (40% aged 18-39, 40% aged 40-64, 20% aged ≥ 65 years), consistent with the global protocol.[2] The sample size was determined to provide adequate statistical power for prevalence estimates, aligning with the global study's minimum of 2000 participants per country, with adjustments for Singapore's population.

Data Collection

Data were collected through a secure, anonymous Internet survey administered between January and March 2018, leveraging Singapore's high Internet penetration (> 80%) for feasibility. The survey, hosted by Qualtrics LLC. (Provo, UT, USA), was accessible only to preselected, invited participants. Quality control measures included 2 attention-check questions, a completion-speed check, and repeat questions to detect inconsistent responses. Automated skip patterns eliminated missing data for mandatory questions, yielding complete symptom datasets.

The survey instruments included:

- Rome IV diagnostic questionnaire for adults: used to diagnose 22 DGBI (eg, IBS, FD, and FC) based on standardized criteria. Participants self-reporting physician-diagnosed organic diseases (eg, celiac disease and inflammatory bowel disease) were excluded from analyses to minimize misclassification.[2]
- 80-Item supplemental questionnaire: captured sociodemographic data (age, sex, education, relationship status, community size, and religion), medical history, psychosocial factors, diet, and quality of life. Embedded validated instruments included:
  - Patient Health Questionnaire-4 (PHQ-4) for anxiety and depression screening[12]
  - PHQ-12 for somatization, adapted from the PHQ-5[13]
  - Patient-Reported Outcomes Measurement Information System (PROMIS) Global-10 for physical and mental health-related quality of life[14]
  - Questions on healthcare utilization (medical visits, medication use, and gastrointestinal interventions)

The questionnaire was available in English, Chinese, and Bahasa Malay. The original English version was translated in simplified Chinese for China and Bahasa Malay for Malaysia, with linguistic validation and cultural adaptation by the professional translation company TransPerfect, Inc. (New York, NY, USA). The original English questionnaire was then localized for Singapore by the local

principal investigator (R.K.W.)[2]

Statistical Methods

The RFGES dataset facilitated analyses by individual country, so the data for Singapore were analyzed independently of the other 25 countries surveyed through the Internet and compared with the other 25.[2] Prevalence rates for DGBI were reported as percentages with 95% CIs, stratified by sex, age group, years of education, and gastrointestinal (GI) anatomical region (esophageal, gastroduodenal, bowel, anorectal). Prevalence rates were pooled across the 25 other countries using Yang's meta-prevalence method.[15] Descriptive statistics summarized single vs multi-region DGBI, psychological comorbidity scores (PHQ-4 and PHQ-12), quality of life (PROMIS-10 physical and mental health scores), and healthcare utilization metrics.

Associations among DGBI, psychological comorbidities (PHQ-4 anxiety/depression diagnosis and PHQ-12 somatization), healthcare utilization, and quality of life were examined using chi-square tests for categorical variables and Mann-Whitney *U* test or Kruskal-Wallis H-test for continuous variables.

Two multivariable logistic regression models were conducted. The first aimed to identify independent associations with DGBI cases (yes/no) in the overall sample. Covariates included age, sex, PROMIS-10 physical and mental health scores, and PHQ-12 somatization scores.

The second model examined associations of psychological comorbidities among participants with DGBI. Participants were categorized as having psychological comorbidities if they had a PHQ-4 score ≥ 3 or a PHQ-12 score ≥ 8. Independent variables included age (continuous), gender (male vs female), history of visiting a doctor for bowel-related issues (yes/no), mean number of medications (continuous), mean number of abdominal surgeries (continuous), and DGBI region involvement (categorized as 1, 2, or 3/4 regions among esophageal, gastroduodenal, bowel, and anorectal). All variables were entered simultaneously using the enter method.

For both models, model fit was assessed using the Hosmer–Lemeshow goodness-of-fit test, and statistical significance was defined as $P < 0.05$. Results are presented as ORs with 95% CIs. The model's explanatory power was estimated using Nagelkerke's R^2.

Ethical Considerations

This study was exempt from a full institutional review board ethics review as an anonymous internet survey without any possibility of identifying participants during the course of the study or at any subsequent time.

RESULTS

Study Population Characteristics (Table 1)

The sample included 2,047 participants (mean age = 40.4 ± 12.7 years; 50% female). Most had 12 or more years of education (91.5%). Over half were married (54.0%), and 39.8% were single. The most common religious affiliation was Buddhism (29.3%), followed by spiritual but unaffiliated (14.1%), Protestantism (11.5%), and Catholicism (9.9%).

Prevalence of Disorders of Gut–Brain Interaction in Singapore (Table 2)

Among 2047 participants, the overall prevalence of Rome IV DGBI was 31.1% (95% CI: 29.1-33.2). Functional bowel disorders were the most common GI anatomic group, affecting 25.3% of respondents. Within this category, constipation-related disorders were especially prominent, with 9.9% of individuals affected by either IBS with constipation (IBS-C) or FC. This burden increased further when anorectal conditions such as proctalgia fugax and levator ani syndrome were included, highlighting a substantial population affected by constipation-related symptoms in Singapore.

Compared to the pooled Rome IV data from the other 25 RFGES Internet countries, Singapore showed a lower overall prevalence of DGBI (31.1% vs 40.6%), but a similar female predominance, and markedly lower rates of esophageal disorders (2.0% vs 6.1%).

Gastrointestinal Anatomical Distribution and Multi-region Involvement

A total of 636 participants (31.1%) reported one or more DGBI. GI anatomical region analysis revealed that 76.9% of participants had a DGBI diagnosis in a single region; 19.0% reported involvement of 2 regions, and 4.1% reported diagnoses involving 3 or more regions. Bowel disorders accounted for the vast majority of single-region DGBI. Among multi-region presentations, the most common combinations were bowel with gastroduodenal (n = 48) and anorectal (n = 52). Only 5 individuals had confirmed diagnoses in all 4 anatomical regions (Fig. 1).

Participants with a higher number of DGBI were more likely to be female and of younger age (18-39), with no significant differences in education level. As the number of involved GI regions increased, there was an incremental increase in somatization (PHQ-12) and anxiety/depression (PHQ-4) (Fig. 2) and an incremental decrease in PROMIS-10 physical and mental scores ($P < 0.001$) (Table 3).

Gender, Age, and Education Disparities

Among 2047 participants, 2.0% reported any esophageal dis-

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Table 1. Socio-demographic Characteristics of Study Population (N = 2047)

Characteristic	N	(%)
Age (yr)		
18-39	976	47.7%
40-64	995	48.6%
65+	76	3.7%
Mean ± SD	40.4 ± 12.7	
median	40	
Range	18-81	
	2047	
Gender		
Male	1023	50.0%
Female	1024	50.0%
	2047	
Education (yr)		
0-11	174	8.6%
12-15	904	44.6%
16+	951	46.9%
Mean ± SD	15.4 ± 3.2	
median	15	
Range	0-30	
	2029 (mis = 18)	
Current relationship status		
Single	815	39.8%
Married	1105	54.0%
Divorced	75	3.7%
Widowed	17	0.8%
Co-habiting	35	1.7%
	2047	
Religious identity		
Buddhism	599	29.3%
No particular religion, but I believe in God or a higher power	288	14.1%
Protestantism	235	11.5%
Catholicism	203	9.9%
No religion, and I do not believe in God or a higher power	182	8.9%
Islam	155	7.6%
Taoism or traditional Chinese religion	122	6.0%
Other religion not listed above	105	5.1%
I do not wish to answer this question	98	4.8%
Hinduism	43	2.1%
Other (Sikhism/Judaism/Jainism/Shinto/Mormonism	17	0.8%
	2047	

mis, missing data (n = 18) were excluded from percentage calculations.

order, 8.1% any gastroduodenal disorder, 25.3% any bowel disorder, and 4.4% any anorectal disorder (Table 4). Overall, 31.1% of participants reported at least 1 disorder of gut-brain interaction (DGBI), with significantly higher prevalence among females (36.9% vs 25.2%, $P < 0.001$) and younger adults aged 18-39 (36.3% vs 26.3%, $P < 0.001$). Esophageal disorders were more common in males and younger participants, while FD and its subtypes (postprandial distress syndrome and epigastric pain syndrome) were more prevalent among younger individuals ($P < 0.0001$). FC showed a strong gender difference, being more frequent in females ($P < 0.001$). Functional bloating/distention was also more common in females ($P = 0.017$) and younger individuals ($P = 0.040$). Bowel disorders overall were more prevalent in females ($P < 0.001$) and in younger participants ($P = 0.008$). There were no significant differences in the overall prevalence of DGBI by education level, though a few individual conditions (eg, functional dysphagia and rumination syndrome) showed associations with lower education.

Psychological Comorbidities and Impact

Psychological distress was strongly associated with DGBI presence and extent of GI anatomic region involvement. Individuals with multi-region DGBI had progressively higher PHQ-12 somatization scores, with those affected in 4 anatomical regions reaching a mean score of 16.6, compared to 4.5 in those without DGBI ($P < 0.001$, Table 5A). Among single-region cases, anorectal disorders were particularly linked to elevated anxiety and depression scores (PHQ-4: 5.9 ± 3.1, Table 5B).

Patients with DGBI and psychological comorbidities were more likely to utilize healthcare services, with higher rates of medical consultations (43.5% vs 32.4%, $P = 0.003$, Table 6) and medication use (mean 0.99 ± 1.75 vs 0.43 ± 0.89, $P < 0.001$, Table 6).

Independent Associations (Regression Models)

Two multivariable logistic regression models were built (Table 7). The first was used to identify factors associated with the presence of DGBI. The model was statistically significant ($P < 0.001$), showed good fit (Hosmer–Lemeshow $P = 0.220$), and explained 18.3% of the variance (Nagelkerke $R^2 = 0.183$).

Older age (OR = 0.988, $P = 0.001$) and male gender (OR = 0.696, $P < 0.001$) were associated with lower odds of DGBI presence. In contrast, visiting a doctor for a bowel issue (OR = 1.705, $P < 0.001$), poorer physical health (OR = 0.874, $P < 0.001$), and higher PHQ-4 (OR = 1.104, $P < 0.001$) and PHQ-12 scores (OR = 1.077, $P < 0.001$) were linked to increased odds. Relationship status was not significantly associated ($P = 0.212$), and other non-significant variables were excluded from the model.

The second model was conducted to identify factors associated

Table 2. Pooled Prevalence rates (% and 95% CI) for the 22 Disorders of Gut–Brain Interaction, Assessed by Rome IV, in a Combined Population-based Internet Survey Sample in 25 Countries (N = 52 080) and in Singapore Alone (N = 2047)

DGBI	Overall	Sex		Age group (yr)			Education group (yr)		
		Female	Male	18-39	40-64	65+	0-11	12-15	16+
Any DGBI									
Singapore	31.1 (29.1-33.2)	36.9 (34.0-39.8)	25.2 (22.6-28.0)	36.3 (33.7-39.4)	26.8 (24.2-29.6)	19.7 (11.8-28.9)	27.6 (20.7-34.5)	29.0 (26.0-31.7)	33.4 (30.4-36.5)
All 25 Internet countries	40.6 (40.2-41.0)	46.9 (46.3-47.5)	34.6 (34.0-35.2)	44.6 (44.0-45.3)	40.0 (39.4-40.7)	32.0 (31.0-32.9)	40.9 (32.7-49.1)	39.4 (33.4-45.2)	41.7 (35.9-47.4)
Esophageal disorders									
Functional chest pain									
Singapore	0.5 (0.2-0.8)	0.4 (0.0-0.8)	0.6 (0.2-1.2)	0.6 (0.2-1.1)	0.4 (0.1-0.8)	-	1.7 (0.0-4.0)	0.4 (0.0-0.9)	0.3 (0.0-0.7)
All 25 Internet countries	1.4 (1.3-1.5)	1.5 (1.4-1.7)	1.3 (1.2-1.4)	1.4 (1.3-1.6)	1.5 (1.3-1.7)	1.0 (0.8-1.3)	1.5 (1.3-1.7)	1.4 (1.3-1.6)	1.3 (1.1-1.5)
Functional heartburn									
Singapore	0.3 (0.1-0.6)	0.1 (0.0-0.3)	0.6 (0.2-1.1)	0.5 (0.1-0.9)	0.2 (0.0-0.5)	-	1.1 (0.0-2.9)	0.1 (0.0-0.3)	0.4 (0.1-0.8)
All 25 Internet countries	1.2 (1.1-1.3)	1.3 (1.2-1.4)	1.0 (0.9-1.2)	1.3 (1.2-1.5)	1.2 (1.1-1.3)	0.7 (0.5-0.8)	1.3 (1.1-1.5)	1.1 (1.0-1.3)	1.1 (0.9-1.2)
Reflux hypersensitivity									
Singapore	0.3 (0.1-0.5)	-	0.6 (0.2-1.1)	0.4 (0.1-0.8)	0.2 (0.0-0.5)	-	-	0.2 (0.0-0.6)	0.4 (0.1-0.8)
All 25 Internet countries	0.9 (0.8-0.9)	0.9 (0.8-1.0)	0.8 (0.7-0.9)	0.9 (0.8-1.0)	1.0 (0.9-1.1)	0.5 (0.4-0.6)	0.9 (0.7-1.1)	0.9 (0.8-1.0)	0.8 (0.6-0.9)
Globus									
Singapore	0.3 (0.1-0.5)	0.2 (0.0-0.5)	0.4 (0.1-0.8)	0.4 (0.1-0.8)	0.2 (0.1-0.8)	-	1.1 (0.0-2.9)	0.2 (0.0-0.6)	0.2 (0.0-0.5)
All 25 Internet countries	0.8 (0.7-0.8)	0.9 (0.8-1.0)	0.7 (0.6-0.8)	0.8 (0.8-1.0)	0.9 (0.8-1.0)	0.5 (0.4-0.7)	0.8 (0.6-1.0)	0.8 (0.7-0.9)	0.7 (0.6-0.8)
Functional dysphagia									
Singapore	1.0 (0.6-1.5)	0.6 (0.2-1.1)	1.5 (0.8-2.2)	1.8 (1.0-2.8)	0.3 (0.0-0.7)	-	-	0.6 (0.1-1.1)	1.7 (0.9-2.5)
All 25 Internet countries	3.2 (3.1-3.4)	3.6 (3.4-3.8)	2.9 (2.7-3.1)	3.4 (3.1-3.6)	3.3 (3.0-3.5)	2.7 (2.4-3.1)	3.4 (3.0-3.7)	3.2 (3.0-3.5)	3.2 (2.9-3.4)
Any esophageal disorder									
Singapore	2.0 (1.4-2.6)	1.2 (0.5-1.9)	2.8 (2.0-3.9)	3.2 (2.1-4.3)	1.0 (0.4-1.7)	-	3.4 (1.1-6.3)	1.2 (0.6-2.0)	2.5 (1.6-3.6)
All 25 Internet countries	6.1 (5.9-6.3)	6.8 (6.5-7.1)	5.5 (5.2-5.8)	6.3 (6.0-6.6)	6.5 (6.2-6.9)	4.6 (4.2-5.1)	6.4 (6.0-6.9)	6.1 (5.8-6.5)	5.8 (5.5-6.1)
Gastroduodenal disorders									
Functional dyspepsia									
Singapore	5.9 (4.9-6.9)	6.6 (5.2-8.2)	5.2 (3.7-6.6)	9.1 (7.4-11.1)	3.2 (2.2-4.3)	-	3.4 (1.1-6.3)	6.2 (4.6-7.7)	6.0 (4.5-7.6)
All 25 Internet countries	7.3 (7.1-7.5)	8.8 (8.5-9.2)	5.8 (5.5-6.1)	9.2 (8.8-9.5)	6.7 (6.4-7.1)	3.9 (3.5-4.3)	8.2 (7.7-8.7)	7.0 (6.7-7.4)	7.0 (6.6-7.3)
Belching disorder									
Singapore	0.6 (0.3-1.0)	0.9 (0.4-1.6)	0.4 (0.1-0.8)	0.8 (0.3-1.4)	0.4 (0.1-0.8)	1.3 (0.0-3.9)	-	0.9 (0.3-1.5)	0.5 (0.1-0.9)
All 25 Internet countries	1.0 (0.9-1.1)	1.1 (1.0-1.2)	0.9 (0.8-1.0)	1.1 (1.0-1.2)	1.0 (0.8-1.1)	0.7 (0.5-0.8)	1.0 (0.8-1.2)	1.0 (0.8-1.1)	0.9 (0.8-1.1)

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Table 2. Continued

DGBI	Overall	Sex		Age group (yr)			Education group (yr)		
		Female	Male	18-39	40-64	65+	0-11	12-15	16+
Rumination syndrome									
Singapore	1.5 (1.0-2.1)	1.1 (0.5-1.8)	2.0 (1.2-2.8)	1.7 (0.9-2.6)	1.3 (0.6-2.1)	1.3 (0.0-3.9)	2.3 (0.6-4.6)	0.6 (0.1-1.1)	2.3 (1.5-3.3)
All 25 Internet countries	2.8 (2.7-3.0)	3.2 (3.0-3.4)	2.5 (2.3-2.7)	2.8 (2.6-3.0)	3.1 (2.9-3.3)	2.4 (2.1-2.7)	2.6 (2.2-2.9)	3.0 (2.7-3.2)	2.8 (2.6-3.0)
Chronic nausea vomiting syndrome									
Singapore	0.3 (0.1-0.6)	0.5 (0.1-1.0)	0.1 (0.0-0.3)	0.3 (0.0-0.7)	0.3 (0.0-0.7)	-	0.6 (0.0-1.7)	0.1 (0.0-0.3)	0.4 (0.1-0.8)
All 25 Internet countries	1.0 (0.9-1.0)	1.2 (1.1-1.3)	0.7 (0.6-0.8)	1.4 (1.2-1.5)	0.8 (0.6-0.9)	0.4 (0.3-0.5)	1.1 (0.9-1.3)	0.9 (0.8-1.0)	0.9 (0.8-1.1)
Cyclic vomiting syndrome									
Singapore	0.9 (0.5-1.3)	0.5 (0.1-1.0)	1.3 (0.6-2.1)	1.2 (0.5-2.0)	0.6 (0.2-1.1)	-	0.6 (0.0-1.7)	0.6 (0.1-1.1)	1.3 (0.6-2.0)
All 25 Internet countries	1.2 (1.1-1.3)	1.2 (1.1-1.4)	1.1 (1.0-1.2)	1.6 (1.5-1.8)	0.9 (0.8-1.0)	0.6 (0.5-0.8)	1.5 (1.3-1.8)	1.0 (0.8-1.1)	1.1 (1.0-1.3)
Cannabinoid hyperemesis syndrome									
Singapore	0.0 (0.0-0.1)	-	0.1 (0.0-0.3)	-	0.1 (0.0-0.3)	-	0.6 (0.0-1.7)	-	-
All 25 Internet countries	0.05 (0.03-0.07)	0.02 (0.00-0.04)	0.08 (0.05-0.11)	0.11 (0.07-0.16)	0.00 (0.00-0.01)	0.01 (-0.01-0.03)	0.1 (0.0-0.1)	0.04 (0.01-0.07)	0.04 (0.01-0.06)
Any gastroduodenal disorder									
Singapore	8.1 (6.9-9.2)	8.8 (7.0-10.4)	7.4 (5.8-9.1)	11.2 (9.2-13.2)	5.5 (4.1-7.0)	2.6 (0.0-6.6)	6.9 (3.4-10.9)	7.7 (6.1-9.4)	8.6 (6.8-10.3)
All 25 Internet countries	10.7 (10.5-11.0)	12.5 (12.1-12.9)	9.0 (8.6-9.3)	13.0 (12.6-13.5)	10.0 (9.6-10.4)	6.7 (6.2-7.2)	11.7 (11.0-12.3)	10.4 (10.0-10.9)	10.3 (9.9-10.7)
Bowel disorders									
Rome-IV IBS									
Singapore	1.3 (0.8-1.9)	1.4 (0.8-2.1)	1.3 (0.6-2.1)	2.4 (1.4-3.4)	0.4 (0.1-0.9)	-	0.6 (0.0-1.7)	1.0 (0.4-1.7)	1.5 (0.7-2.4)
All 25 Internet countries	4.2 (4.0-4.3)	5.4 (5.1-5.7)	4.2 (4.0-4.3)	5.4 (5.1-5.7)	3.9 (3.6-4.1)	1.7 (1.4-1.9)	4.1 (3.7-4.5)	3.9 (3.7-4.2)	4.4 (4.1-4.7)
Functional Constipation									
Singapore	9.5 (8.2-10.8)	14.2 (11.9-16.5)	4.8 (3.5-6.2)	10.1 (8.3-12.0)	9.1 (7.3-10.9)	5.3 (1.3-10.5)	6.9 (3.4-10.9)	9.6 (7.6-11.5)	9.9 (8.0-11.8)
All 25 Internet countries	11.8 (11.5-12.1)	15.3 (14.8-15.7)	8.4 (8.1-8.8)	13.4 (12.9-13.8)	11.1 (10.7-11.5)	9.5 (8.8-10.1)	11.3 (10.7-11.9)	11.5 (11.1-11.9)	12.4 (11.9-12.8)
Opioid-induced constipation									
Singapore	0.4 (0.2-0.7)	0.6 (0.2-1.1)	0.3 (0.0-0.7)	0.5 (0.1-1.0)	0.4 (0.1-0.8)	-	-	0.6 (0.1-1.1)	0.4 (0.1-0.8)
All 25 Internet countries	1.6 (1.5-1.7)	1.8 (1.6-2.0)	1.4 (1.3-1.6)	1.5 (1.4-1.7)	1.7 (1.5-1.9)	1.6 (1.3-1.8)	1.8 (1.5-2.0)	1.4 (1.3-1.6)	1.7 (1.5-1.8)
Functional diarrhea									
Singapore	4.3 (3.4-5.1)	3.5 (2.4-4.7)	5.0 (3.6-6.4)	4.4 (3.1-5.7)	4.1 (2.9-5.3)	3.9 (0.0-7.9)	5.7 (2.9-9.8)	3.4 (2.3-4.6)	4.7 (3.5-6.1)
All 25 Internet countries	4.7 (4.5-4.9)	4.1 (3.9-4.3)	5.3 (5.1-5.6)	4.6 (4.3-4.9)	5.1 (4.8-5.4)	4.1 (3.7-4.5)	5.2 (4.7-5.6)	4.4 (4.2-4.7)	4.8 (4.5-5.1)
Functional bloating/distention									
Singapore	3.6 (2.8-4.4)	4.6 (3.3-6.0)	2.5 (1.7-3.4)	4.5 (3.2-5.8)	2.7 (1.7-3.7)	2.6 (0.0-6.6)	1.7 (0.0-4.0)	3.8 (2.7-5.1)	3.8 (2.6-5.3)
All 25 Internet countries	3.4 (3.3-3.6)	4.6 (4.3-4.8)	2.4 (2.2-2.5)	3.4 (3.2-3.6)	3.9 (3.7-4.2)	2.4 (2.1-2.7)	3.4 (3.3-3.6)	3.2 (3.0-3.5)	3.6 (3.3-3.8)

Table 2. Continued

DGBI	Overall	Sex		Age group (yr)			Education group (yr)		
		Female	Male	18-39	40-64	65+	0-11	12-15	16+
Unspecified functional bowel disorder									
Singapore	6.8 (5.7-7.9)	7.9 (6.3-9.5)	5.7 (4.2-7.2)	6.9 (5.2-8.4)	6.8 (5.3-8.5)	5.3 (1.3-10.5)	7.5 (3.4-11.5)	5.6 (4.3-7.2)	7.6 (5.9-9.1)
All 25 Internet countries	8.9 (8.6-9.1)	9.5 (9.2-9.9)	8.2 (7.9-8.6)	9.6 (9.2-10.0)	8.8 (8.4-9.2)	7.2 (6.7-7.8)	8.7 (8.1-9.2)	8.8 (8.4-9.2)	9.2 (8.8-9.6)
Any bowel disorder									
Singapore	25.3 (23.4-27.2)	31.3 (28.4-34.1)	19.3 (16.9-21.7)	28.0 (25.2-31.0)	23.2 (20.8-25.9)	17.1 (9.2-26.3)	22.4 (16.7-28.7)	23.5 (20.7-26.1)	27.1 (24.2-30.1)
All 25 Internet countries	33.7 (33.3-34.1)	39.2 (38.6-39.8)	28.4 (27.9-29.0)	37.1 (36.4-37.7)	33.4 (32.8-34.1)	25.9 (25.0-26.8)	33.6 (32.7-34.5)	32.6 (31.9-33.2)	35.0 (34.3-35.6)
Central nervous system disorders of GI pain									
Centrally mediated abdominal pain syndrome									
Singapore	-	-	-	-	-	-	-	-	-
All 25 Internet countries	0.02 (0.01-0.03)	0.04 (0.01-0.06)	-	0.03 (0.01-0.05)	0.01 (0.00-0.02)	0.01 (-0.01-0.03)	-	0.02 (0.00-0.04)	0.02 (0.00-0.04)
Biliary disorders									
Functional biliary pain									
Singapore	0.1 (0.0-0.2)	0.2 (0.0-0.5)	-	0.1 (0.0-0.3)	0.1 (0.0-0.3)	-	-	0.1 (0.0-0.3)	0.1 (0.0-0.3)
All 25 Internet countries	0.07 (0.04-0.09)	0.11 (0.07-0.15)	0.02 (0.00-0.04)	0.11 (0.07-0.15)	0.04 (0.01-0.07)	0.02 (-0.01-0.05)	0.09 (0.03-0.15)	0.08 (0.04-0.12)	0.08 (0.04-0.11)
Anorectal disorders									
Fecal incontinence									
Singapore	1.0 (0.6-1.5)	1.1 (0.5-1.7)	1.0 (0.4-1.7)	0.8 (0.3-1.4)	1.2 (0.6-2.0)	1.3 (0.0-3.9)	1.1 (0.0-2.9)	1.2 (0.6-2.0)	0.8 (0.3-1.5)
All 25 Internet countries	1.6 (1.5-1.7)	1.5 (1.4-1.7)	1.7 (1.5-1.8)	1.1 (1.0-1.3)	1.8 (1.6-1.9)	2.3 (2.0-2.7)	1.5 (1.3-1.7)	1.6 (1.5-1.8)	1.6 (1.4-1.7)
Levator ani syndrome									
Singapore	0.6 (0.3-1.0)	0.8 (0.3-1.4)	0.5 (0.1-1.0)	0.7 (0.2-1.3)	0.6 (0.2-1.1)	-	1.1 (0.0-2.9)	0.7 (0.2-1.3)	0.5 (0.1-1.1)
All 25 Internet countries	1.2 (1.1-1.3)	1.4 (1.3-1.5)	0.9 (0.8-1.1)	1.3 (1.2-1.5)	1.2 (1.1-1.4)	0.6 (0.4-0.8)	1.2 (1.0-1.4)	1.1 (1.0-1.2)	1.2 (1.1-1.4)
Proctalgia fugax									
Singapore	3.0 (2.3-3.8)	3.4 (2.3-4.6)	2.6 (1.7-3.7)	3.7 (2.6-5.0)	2.3 (1.4-3.3)	3.9 (0.0-7.9)	1.1 (0.0-2.9)	2.4 (1.5-3.5)	3.8 (2.6-4.9)
All 25 Internet countries	5.7 (5.5-5.9)	6.6 (6.2-6.9)	4.8 (4.6-5.1)	6.2 (5.9-6.5)	5.9 (5.5-6.2)	3.9 (3.5-4.3)	5.8 (5.3-6.3)	5.2 (4.9-5.5)	6.1 (5.7-6.4)
Any anorectal disorder									
Singapore	4.4 (3.6-5.3)	5.2 (3.9-6.5)	3.6 (2.4-4.8)	4.9 (3.6-6.5)	3.9 (2.8-5.2)	3.9 (0.0-7.9)	2.9 (0.6-5.7)	4.2 (3.1-5.5)	4.7 (3.5-6.1)
All 25 Internet countries	7.9 (7.6-8.1)	9.0 (8.6-9.3)	6.8 (6.5-7.1)	8.1 (7.7-8.5)	8.2 (7.8-8.6)	6.4 (5.9-7.0)	7.9 (7.6-8.1)	7.3 (6.9-7.7)	8.3 (7.9-8.7)

DGBI, disorders of gut–brain interaction.

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with psychological comorbidities among patients with DGBI. The model was statistically significant overall ($P < 0.001$, Omnibus test) and demonstrated good fit (Hosmer–Lemeshow $P = 0.837$). The Nagelkerke R^2 was 0.170, indicating that the model explained approximately 17.0% of the variance in psychological comorbidities.

Older age (OR = 0.965; 95% CI, 0.951-0.980; $P < 0.001$) and male gender (OR = 0.528; 95% CI, 0.372-0.751; $P < 0.001$) were associated with significantly lower odds of psychological comorbidities. In contrast, a higher number of medications was associated with increased odds (OR = 1.389; 95% CI, 1.174-1.645; $P < 0.001$).

Patients with DGBI involving 2 regions (OR = 1.938; $P = 0.007$) or 3/4 regions (OR = 6.770; $P = 0.012$) had significantly higher odds of psychological comorbidities compared to those with DGBI in one region. Other variables, such as having visited a doctor for bowel issues or number of abdominal surgeries, were not statistically significant.

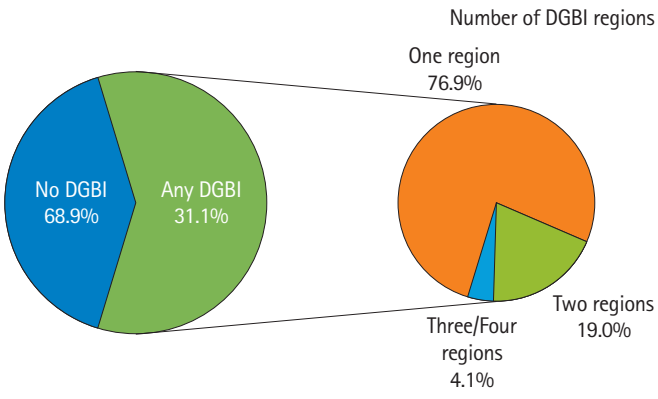

Figure 1. Distribution of disorders of gut–brain interaction (DGBI) regions.

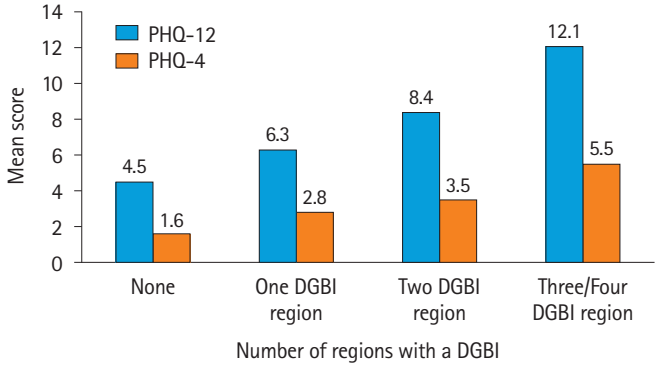

Figure 2. Psychological burden by disorders of gut–brain interaction (DGBI). PHQ, Patient Health Questionnaire.

Table 3. Sociodemographic and Psychosocial Characteristics According to Number of Disorders of Gut–Brain Interaction

Characteristic	No. of DGBIs								*P*-value
	0 (n = 1411)		1 (n = 489)		2 (n = 121)		3-4 (n = 26)		
	n	%	n	%	n	%	n	%	
Gender									
Male	765	54.2%	194	39.7%	50	41.3%	14	53.8%	
Female	646	45.8%	295	60.3%	71	58.7%	12	46.2%	< 0.001
	1411		489		121		26		
Age group (yr)									
18-39	622	44.1%	267	54.6%	70	57.9%	17	65.4%	
40-64	728	51.6%	210	42.9%	48	39.7%	9	34.6%	< 0.001
65+	61	4.3%	12	2.5%	3	2.5%	0	0.0%	
	1411		489		121		26		
Education group (yr)									
0-11	126	9.0%	34	7.0%	14	11.8%	0	0.0%	
12-15	642	45.8%	207	42.8%	44	37.0%	11	44.0%	0.104
16+	633	45.2%	243	50.2%	61	51.3%	14	56.0%	
	1401	(mis = 10)	484	(mis = 5)	119	(mis = 2)	25	(mis = 1)	
PHQ-12 somatization score (mean ± SD)	4.5 ± 3.7		6.3 ± 4.0		8.4 ± 4.2		12.1 ± 5.7		< 0.001
PHQ-4 anxiety/depression score (mean ± SD)	1.6 ± 2.2		2.8 ± 2.8		3.5 ± 2.8		5.5 ± 3.9		< 0.001
PROMIS-10 physical score (mean ± SD)	15.5 ± 2.2		14.5 ± 2.2		13.7 ± 2.3		12.0 ± 2.7		< 0.001
PROMIS-10 mental score (mean ± SD)	13.9 ± 2.9		12.7 ± 3.0		11.8 ± 2.8		10.7 ± 4.1		< 0.001

DGBIs, disorders of gut–brain interaction; PHQ, Patient Health Questionnaire; PROMIS, Patient-Reported Outcomes Measurement Information System.

Table 4. Prevalence rate N (%) for 22 Rome IV functional gastrointestinal diagnoses by sex, age and education

Rome IV diagnosis	Overall (N = 2047)		Sex					Age group (yr)					Education group (yr) (mis = 18)						
			Male (n = 1023)		Female (n = 1024)		*P*-value	18-39 (n = 976)		40+ (n = 1071)		*P*-value	0-11 (n = 174)		12-15 (n = 904)		16+ (n = 951)		*P*-value
	n	%	n	%	n	%		n	%	n	%		n	%	n	%	n	%	
Any DGBI	636	31.1%	258	25.2%	378	36.9%	< 0.001	354	36.3%	282	26.3%	< 0.001	48	27.6%	262	29.0%	318	33.4%	0.070
A. Esophageal disorders																			
Functional chest pain	10	0.5%	6	0.6%	4	0.4%	0.752	6	0.6%	4	0.4%	0.642	3	1.7%	4	0.4%	3	0.3%	0.049
Functional heartburn	7	0.3%	6	0.6%	1	0.1%	0.124	5	0.5%	2	0.2%	0.378	2	1.1%	1	0.1%	4	0.4%	0.088
Reflux hypersensitivity	6	0.3%	6	0.6%	0	0.0%	0.039	4	0.4%	2	0.2%	0.602	0	0.0%	2	0.2%	4	0.4%	0.552
Globus	6	0.3%	4	0.4%	2	0.2%	0.686	4	0.4%	2	0.2%	0.602	2	1.1%	2	0.2%	2	0.2%	0.095
Functional dysphagia	21	1.0%	15	1.5%	6	0.6%	0.076	18	1.8%	3	0.3%	< 0.001	0	0.0%	5	0.6%	16	1.7%	0.021
Any esophageal disorder	41	2.0%	29	2.8%	12	1.2%	0.011	31	3.2%	10	0.9%	< 0.001	6	3.4%	11	1.2%	24	2.5%	0.051
B. Gastroduodenal disorders																			
Functional dyspepsia	121	5.9%	53	5.2%	68	6.6%	0.191	89	9.1%	32	3.0%	< 0.0001	6	3.4%	56	6.2%	57	6.0%	0.359
PDS	114	5.6%	50	4.9%	64	6.3%	0.212	81	8.3%	33	3.1%	< 0.0001	6	3.4%	53	5.9%	53	5.6%	0.440
EPS	20	1.0%	12	1.2%	8	0.8%	0.499	19	1.9%	1	0.1%	< 0.0001	0	0.0%	8	0.9%	11	1.2%	0.333
Belching disorder	13	0.6%	4	0.4%	9	0.9%	0.266	8	0.8%	5	0.5%	0.469	0	0.0%	8	0.9%	5	0.5%	0.338
Rumination syndrome	31	1.5%	20	2.0%	11	1.1%	0.146	17	1.7%	14	1.3%	0.533	4	2.3%	5	0.6%	22	2.3%	0.006
Chronic nausea vomiting syndrome	6	0.3%	1	0.1%	5	0.5%	0.218	3	0.3%	3	0.3%	0.999	1	0.6%	1	0.1%	4	0.4%	0.365
Cyclic vomiting syndrome	18	0.9%	13	1.3%	5	0.5%	0.094	12	1.2%	6	0.6%	0.166	1	0.6%	5	0.6%	12	1.3%	0.239
Cannabinoid hyperemesis syndrome	1	0.0%	1	0.1%	0	0.0%	0.999	0	0.0%	1	0.1%	0.999	1	0.6%	0	0.0%	0	0.0%	-
Any gastroduodenal disorder	166	8.1%	76	7.4%	90	8.8%	0.295	109	11.2%	57	5.3%	< 0.0001	12	6.9%	70	7.7%	82	8.6%	0.656
C. Bowel disorders																			
Rome-IV IBS	27	1.3%	13	1.3%	14	1.4%	0.999	23	2.4%	4	0.4%	< 0.0001	1	0.6%	9	1.0%	14	1.5%	0.472
IBS-C	9	0.4%	6	0.6%	3	0.3%	0.506	8	0.8%	1	0.1%	0.030	0	0.0%	3	0.3%	4	0.4%	0.682
IBS-D	8	0.4%	3	0.3%	5	0.5%	0.727	6	0.6%	2	0.2%	0.231	1	0.6%	3	0.3%	4	0.4%	0.882
IBS-U	3	0.1%	2	0.2%	1	0.1%	0.999	2	0.2%	1	0.1%	0.930	0	0.0%	0	0.0%	3	0.3%	-
IBS-M	7	0.3%	2	0.2%	5	0.5%	0.453	7	0.7%	0	0.0%	0.010	0	0.0%	3	0.3%	3	0.3%	-
Functional Constipation	194	9.5%	49	4.8%	145	14.2%	< 0.001	99	10.1%	95	8.9%	0.364	12	6.9%	87	9.6%	94	9.9%	0.461
Opioid-induced constipation	9	0.4%	3	0.3%	6	0.6%	0.508	5	0.5%	4	0.4%	0.886	0	0.0%	5	0.6%	4	0.4%	0.597
Functional diarrhea	87	4.3%	51	5.0%	36	3.5%	0.123	43	4.4%	44	4.1%	0.822	10	5.7%	31	3.4%	45	4.7%	0.222
Functional bloating/distention	73	3.6%	26	2.5%	47	4.6%	0.017	44	4.5%	29	2.7%	0.040	3	1.7%	34	3.8%	36	3.8%	0.382
Unspecified functional bowel disorder	139	6.8%	58	5.7%	81	7.9%	0.054	67	6.9%	72	6.7%	0.967	13	7.5%	51	5.6%	72	7.6%	0.230
Any bowel disorder	517	25.3%	197	19.3%	320	31.3%	< 0.001	273	28.0%	244	22.8%	0.008	39	22.4%	212	23.5%	258	27.1%	0.131

Table 4. Continued

Rome IV diagnosis	Overall (N = 2047)		Sex					Age group (yr)					Education group (yr) (mis = 18)							
			Male (n = 1023)		Female (n = 1024)		*P*-value	18–39 (n = 976)		40+ (n = 1071)		*P*-value	0–11 (n = 174)		12–15 (n = 904)		16+ (n = 951)		*P*-value
	n	%	n	%	n	%		n	%	n	%		n	%	n	%	n	%	
D. Central nervous system disorders of GI pain																			
Centrally mediated abdominal pain syndrome	0	0.0%	0	0.0%	0	0.0%	-	0	0.0%	0	0.0%	-	0	0.0%	0	0.0%	0	0.0%	-
E. Biliary disorders																			
Functional biliary pain	2	0.1%	0	0.0%	2	0.2%	0.999	1	0.1%	1	0.1%	0.999	0	0.0%	1	0.1%	1	0.1%	-
F. Anorectal disorders																			
Fecal incontinence	21	1.0%	10	1.0%	11	1.1%	0.999	8	0.8%	13	1.2%	0.509	2	1.1%	11	1.2%	8	0.8%	0.663
Levator ani syndrome	13	0.6%	5	0.5%	8	0.8%	0.405	7	0.7%	6	0.6%	0.864	2	1.1%	6	0.7%	5	0.5%	0.634
Proctalgia fugax	62	3.0%	27	2.6%	35	3.4%	0.369	36	3.7%	26	2.4%	0.125	2	1.1%	22	2.4%	36	3.8%	0.077
Any anorectal disorder	90	4.4%	37	3.6%	53	5.2%	0.106	48	4.9%	42	3.9%	0.322	5	2.9%	38	4.2%	45	4.7%	0.523

DGBI, disorder of gut–brain interaction; PDS, postprandial distress syndrome; EPS, epigastric pain syndrome; IBS, irritable bowel syndrome; IBS-C, IBS with constipation; IBS-D, IBS with diarrhea; IBS-U, unsubtyped IBS; IBS-M, mixed IBS; GI, gastrointestinal.

DISCUSSION

This study, part of the RFGES, utilized Rome IV criteria to provide the first nationally representative assessment of DGBI in Singapore. The observed prevalence of 31.1% highlights a substantial population-level burden that is comparable to other Asian cohorts but lower than global averages.[2] Functional bowel disorders, particularly constipation-related conditions, emerged as the most prevalent subgroup, affecting 25.3% of respondents. Constipation-related conditions, including IBS-C and FC, affected 9.9% of the population.

Significant sex and age disparities were observed, with females showing a higher overall prevalence (36.9% vs 25.2%, $P < 0.001$) and greater burden of constipation-related disorders (14.2% vs 4.8%). Younger adults (18-39 years) also exhibited higher prevalence rates (36.3%), emphasizing the need for age-tailored approaches.

A strong association was found between DGBI and psychological comorbidities. Multi-region DGBI involvement was linked to markedly elevated PHQ-12 somatization scores (mean 16.6 vs 4.5, $P < 0.001$) and PHQ-4 anxiety and depression scores (mean 5.8 vs 1.6, $P < 0.001$). These findings highlight the complex interplay of physiological, psychological, and cultural factors in Singapore's multi-ethnic population, and underscore the importance of integrated, culturally sensitive models of care.[4,16]

The exceptionally low IBS prevalence in Singapore (1.3%) compared with prior local Rome III estimates, such as 20.9% as reported by Siah et al (2016),[11] likely reflects multiple factors. The shift from Rome III to Rome IV criteria, which require more stringent symptom–pain correlations, has been shown globally to reduce IBS prevalence estimates by up to half.[17] Earlier online surveys also differed in sampling strategy and exclusion of organic disease. Indeed, meta-analyses by Lovell and Ford (2012)[18] and Sperber et al (2017)[19] reported pooled IBS prevalence of 11.2 % and 8.8 %, respectively—making the 2016 Singapore rate unusually high by international standards. Thus, methodological evolution rather than abrupt epidemiologic change best explains the difference.

Comparison With Australian Findings (Table 8)

The Australian arm of the Rome Foundation Global Epidemiology Study included analyses of DGBI and psychological health, providing a detailed assessment of their relationship; we therefore compared our findings with this study to contextualize the psychological comorbidities observed in Singapore.

Singapore's DGBI prevalence of 31.1% is lower than Australia's reported 37.7% in the RFGES. While unspecified functional bowel disorder (8.0%) and FC (7.7%) were the most prevalent DGBI in

Table 5. Psychological Comorbidity by Regional Involvement of Disorders of Gut–Brain Interaction

A. Number of Disorders of Gut–Brain Interaction Regions and Mental Health Conditions

No. of DGBI regions	N	PHQ-12 (somatization) mean score	*P*-value	PHQ-4 (anxiety/depression) mean score	*P*-value
0	1411	4.5 ± 3.7	< 0.001	1.6 ± 2.2	< 0.001
1	489	6.3 ± 4.0		2.8 ± 2.8	
2	121	8.4 ± 4.2		3.5 ± 2.8	
3	21	11.0 ± 5.2		5.4 ± 4.2	
4	5	16.6 ± 6.4		5.8 ± 3.0	

DGBI, disorder of gut–brain interaction; PHQ, Patient Health Questionnaire.
Data are presented as mean ± SD.

B. Single Disorders of Gut–Brain Interaction Region and Mental Health Conditions

DGBI region	N	PHQ-12 (somatization) mean score	*P*-value	PHQ-4 (anxiety/depression) mean score	*P*-value
Esophageal	9	5.7 ± 5.2	0.107	3.0 ± 2.3	0.003
Gastroduodenal	79	7.2 ± 4.5		3.4 ± 3.1	
Bowel	443	6.2 ± 3.8		2.6 ± 2.6	
Anorectal	11	8.1 ± 4.2		5.9 ± 3.1	

DGBI, disorder of gut–brain interaction; PHQ, Patient Health Questionnaire.
Data are presented as mean ± SD.

Table 6. Comparison of Healthcare Utilization Among Disorder of Gut–Brain Interaction Patients With and Without Psychological Comorbidities

Healthcare utilization measure	Psychological comorbidities				*P*-value
	Yes (n = 395)		No (n = 241)		
	n	%	n	%	
Have you ever visited a doctor because of a bowel problem?					
Yes	172	43.5%	78	32.4%	
No	223	56.5%	163	67.6%	0.003
	395		241		
How often do you go to a doctor for your health?					
Once a month or more	41	10.4%	10	4.1%	
A few times a year	231	58.5%	146	60.6%	
Once a year	58	14.7%	35	14.5%	0.029
Less than once a year	56	14.2%	47	19.5%	
Never	9	2.3%	3	1.2%	
	395		241		
Mean number of medications					
Mean ± SD	0.99 ± 1.75		0.43 ± 0.89		< 0.001
Mean number of abdominal surgeries					
Mean ± SD	0.10 ± 0.31		0.08 ± 0.27		0.394

Australia, Singapore's burden was dominated by constipation-related subtypes (FC and IBS-C) at 9.9%.

Both countries demonstrated higher prevalence of DGBI among females (Singapore: 36.9% vs 25.2%; Australia: 43.4% vs 32.0%), though Singapore showed a more pronounced gender disparity in constipation prevalence (14.2% vs 4.8% for FC in females vs males) compared to Australia (10.1% vs 7.2% for FC in females vs males). This may reflect differences in dietary fiber intake, cultural barriers to discussing bowel symptoms, and greater prevalence of anxiety among women, which is known to exacerbate constipation via gut–brain axis mechanisms.[11,20]

Psychological comorbidity patterns also differed between the 2

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Table 7. Logistic Regression Models

Variable	OR	95% CI		*P*-value
		Lower	Upper	
Model 1- Any DGBI				
Age (yr)	0.988	0.979	0.997	0.001
Gender (male)	0.696	0.567	0.854	< 0.001
Current relationship status (married/co-habiting)	0.869	0.697	1.083	0.212
Visited a doctor because of a bowel problem (yes)	1.705	1.374	2.115	< 0.001
Global Physical Health component score - PROMIS Global-10	0.874	0.829	0.921	< 0.001
PHQ-4 score (anxiety and depression questionnaire)	1.104	1.056	1.155	< 0.001
PHQ-12 Somatic Symptom Scale score	1.077	1.047	1.107	< 0.001
Model 2- Psychological comorbidities among DGBI patients				
Age (yr)	0.965	0.951	0.980	< 0.001
Gender (male)	0.528	0.372	0.751	< 0.001
Visited a doctor because of a bowel problem (yes)	0.767	0.532	1.105	0.155
Mean number of medications	1.389	1.174	1.645	< 0.001
Mean number of abdominal surgeries	1.401	0.761	2.581	0.279
DGBI region numbers				
1 region	1			
2 regions	1.938	1.202	3.125	0.007
3/4 regions	6.770	1.511	30.335	0.012

DGBI, disorders of gut–brain interaction; PROMIS, Patient-Reported Outcomes Measurement Information System; PHQ, Patient Health Questionnaire.

Table 8. Demographic and Clinical Differences Between Singapore and Australia

Parameter	Singapore	Australia
Sample size (n)	2047	2024
Overall DGBI prevalence	31.1%	37.7%
Leading DGBI subtypes (top 2)	Functional constipation: 9.5% IBS-C: 0.4%	Unspecified functional bowel disorder: 8.0% Functional constipation: 7.7%
DGBI by gender	Female: 36.9% Male: 25.2%	Female: 43.4% Male: 32.0%
Significant associations with DGBI	Age: OR = 0.988, 95% CI (0.979-0.997) Male gender OR = 0.696, 95% CI (0.567-0.854) PHQ-4 OR = 1.104, 95% CI (1.056-1.155) PHQ-12 OR = 1.077, 95% CI (1.047-1.107)	Age, gender, anxiety not significantly associated Depression OR = 1.17, 95% CI (1.04-1.32) Somatization OR = 1.17, 95% CI (1.14- 1.20)
Anxiety/depression	PHQ-4 score (anorectal cohort): 5.9 ± 3.1	Anxiety prevalence: 26.6% Depression prevalence: 29.0%
Multi-region DGBI	7.2%	9.1%
Older adults (≥ 65 yr)	3.7%	19%

DGBI, disorders of gut–brain interaction; IBS-C, irritable bowel syndrome with constipation.
Data are presented as n, %, or mean ± SD.

countries. Australia reports an overall prevalence of 26.6% for anxiety and 29.0% for depression among DGBI patients. In Singapore, anorectal disorders, including constipation-related conditions, were strongly associated with elevated PHQ-4 anxiety/depression scores (5.9 ± 3.1). By contrast, in Australia, functional chest pain exhibited the highest anxiety (55.0%) and depression (57.5%) rates, while IBS-M showed the highest somatization rate (75.9%).[21] However, comparisons of psychological burden should be interpreted cautiously because the Singapore dataset used continuous PHQ-4 scores (eg, for anorectal disorders, mean 5.9 ± 3.1), whereas the

Australian analysis reported categorical caseness rates (% with anxiety or depression). These non-equivalent metrics highlight an inherent limitation in cross-country comparisons within the Rome IV framework.

These differences between Singapore and Australia likely reflect variations in cultural attitudes towards mental health, health system accessibility, and psychological stressors such as occupational stress in a high-pressure urban environment.[2,9,22] For instance, Australia's greater access to mental health services, including subsidized psychological care through the Better Access initiative, and higher rates of antidepressant use (17.8% of DGBI patients) may mitigate psychological burden.[21,23]

Burden of Constipation in Singapore

Constipation-related disorders (IBS-C and FC) affected 9.9% of Singapore's population, highlighting a substantial clinical burden, particularly among females and older adults. This rate is comparable to Australia's 7.71% for FC alone but somewhat lower than Spain's reported 12.8% FC prevalence.[24] The persistence of constipation in older adults in Singapore likely reflects physiological factors such as age-related reductions in gut motility, medication effects (including opioids and anticholinergics), and culturally influenced dietary patterns that may be lower in fiber intake. Psychological distress was pronounced (PHQ-4 5.9 ± 3.1 for anorectal disorders) and this exceeded the Australian FC results (anxiety: 25.48%, depression: 28.03%, and somatization: 40.76%), suggesting the possibility of greater gut–brain axis dysregulation in Singapore, potentially amplified by urban stressors and culturally shaped symptom reporting and help-seeking behavior.[9]

Interpretation and Mechanisms

The higher psychological burden in Singaporean DGBI patients, particularly in constipation-related disorders, may be driven by cultural, social, and environmental factors. Singapore's multi-ethnic population (74.3% Chinese, 13.5% Malay, and 9.0% Indian) indicates heterogeneity in dietary exposure, culturally mediated symptom interpretation, and mental health stigma, factors known to influence GI symptom perception, reporting patterns, and care-seeking behavior in DGBI.[10,25] Urban stressors, including competitive work and academic culture, may exacerbate gut-brain axis dysregulation, particularly among younger adults, contributing to work absenteeism and presenteeism as seen in other RFGES countries.[9,26] This is consistent with international evidence linking perceived stress and anxiety to altered motility, visceral hypersensitivity, and autonomic imbalance.[27] In contrast, Australia's comparatively lower psychological burden among DGBI patients may reflect broader access to integrated mental health services, greater acceptance of psychological care, and public health initiatives aimed at reducing mental health stigma.[21]

Clinical, Policy, and Research Implications

Clinical implications

These findings highlight the urgent need for integrated care models in Singapore that embed psychological services within gastroenterology practice.[28] Evidence-based interventions such as cognitive–behavioral therapy and gut-directed hypnotherapy, already validated in Western cohorts, warrant local adaptation and evaluation for Singapore's multi-ethnic population. Routine use of PROMs like PHQ-12 and PROMIS-10 can facilitate early identification of psychological distress and reduced quality of life, and support personalized care planning. In Australia, the multidisciplinary approach already implemented (eg, multidisciplinary treatment for functional gastrointestinal disorders [MANTRA] trial) could serve as a model for Singapore to enhance constipation management.[29]

Policy implications

Adopting assessment tools like Rome IV Diagnostic Questionnaire in Singapore primary care could facilitate early diagnosis and reduce the burden on specialist services.[30] Public health campaigns aimed at reducing mental health stigma may encourage help-seeking, especially in Asian contexts.[31] Australia's Better Access initiative, which subsidizes psychological care, provides a potential model for similar policy development in Singapore.[23]

Research implications

Longitudinal studies in Singapore are needed to establish causality between psychological distress and DGBI progression, particularly for constipation-related subtypes. Investigating ethnicity-specific patterns could help elucidate culturally mediated differences in symptom expression and care-seeking behavior. Clinical trials evaluating culturally adapted interventions, such as mindfulness-based therapies or modified dietary approaches, are needed to establish their effectiveness and acceptability in Singapore's diverse population.[32] Additionally, microbiome research tailored to local dietary practices could provide critical insights into region-specific mechanisms underlying DGBI.[33]

Limitations and Future Directions

The cross-sectional design of our study precludes causal inferences. Generalizability may be limited by self-selection in internet surveys, which can inflate prevalence estimates due to lower social-desirability pressures when reporting bowel/anorectal symp-

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toms. The presence of recall and reporting bias can influence data accuracy, and psychological distress could influence results, particularly in Singapore, where cultural stigma may underreport mental health issues. Although the Rome Foundation protocol excluded participants with self-reported organic GI diseases or major abdominal surgery to enhance diagnostic specificity, this could result in a small negative bias. Finally, the absence of independent clinical verification, inherent to cross-sectional population studies, posed a risk of misclassification that could exclude some true DGBI cases.

A further limitation is the deviation from the predetermined age group distribution of 40/40/20, with nearly half of participants aged 18-39 (47.7%) and 40-64 (48.6%), but only 3.7% aged ≥ 65 years, limiting representativeness of older adults. Despite Singapore's high internet penetration, a recent study by Hao et al[34] found that older Singaporeans held less positive attitudes toward health technology than toward communication or entertainment tools, partly due to limited interest and discomfort with medical technology. This likely reduced senior participation and may have modestly inflated our overall prevalence estimate, given that DGBI prevalence declines with age, with the exception of fecal incontinence.[35]

Singapore's multi-ethnic composition was a key rationale for this study, but ethnicity-specific data were not collected. This limits our ability to identify cultural determinants of symptom perception or healthcare-seeking behavior. Subsequent studies should explicitly stratify by ethnicity to examine potential cultural or dietary modulators of the gut–brain axis. Future research should adopt longitudinal cohort studies to track symptom progression and psychological distress over time, incorporate ethnicity data to better understand cultural impacts, and evaluate the effectiveness of integrated biopsychosocial interventions.

In conclusion, this study, based on the RFGES database, demonstrates that nearly one-third of Singapore's adult population met Rome IV criteria for a DGBI, with constipation-related disorders representing a prominent burden, especially among women and older adults. Psychological comorbidities were strongly associated with symptom complexity, increased healthcare utilization, and reduced quality of life. The findings underscore the gut-brain axis's role and the need for culturally sensitive biopsychosocial models of care in Singapore. By establishing a robust epidemiological baseline, this study provides essential evidence to inform clinical practice, guide policy development, and support future research aimed at improving outcomes for individuals living with DGBI in Singapore and the broader region.

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Conflicts of interest
None.

Author contributions
Kewin T H Siah and Reuben K Wong contributed to conceptualization, methodology, and writing; Tamar Freud contributed to formal analysis, results interpretation, and writing; Valencia R Y Zhang contributed to writing, editing, and submission; Ami D Sperber contributed to conceptualization, supervision, and critical revision; and Olafur S Palsson and Shrikant I Bangdiwala contributed to supervision and critical revision; and all authors approved the final manuscript.

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Machine Learning Clustering Identifies Achalasia-Spectrum Phenotypes in Distal Esophageal Spasm

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Background/Aims: Distal esophageal spasm (DES) is a rare, heterogeneous esophageal motility disorder with variable treatment responses. Machine learning methods are well suited to distinguish DES phenotypes that could inform therapeutic decisions and outcomes. This exploratory study aims to identify and characterize unique DES phenotypes using unsupervised machine learning clustering.

Methods: Adults undergoing high-resolution esophageal manometry (HRM) for esophageal symptoms between 2005-2025 were included. Uniform manifold approximation and projection (UMAP) was used for dimensionality reduction of HRM metrics. DES patients were classified as within-cluster or outside-cluster according to density contours. Patient characteristics, manometric parameters, functional lumen imaging probe findings, and treatment outcomes were compared between clusters.

Results: Among 5360 patients, 42 (0.8%) had DES, 2400 (45%) normal motility, and 107 (2.0%) type III achalasia. UMAP identified 33 DES patients clustering alongside type III achalasia patients, while 9 DES patients remained outside this cluster. The primary differentiating characteristic between clusters was premature supine contraction frequency (median 8 vs 0, $P < 0.001$). Within-cluster patients demonstrated higher Esophageal Hypervigilance and Anxiety Scale scores (30 vs 20.5, $P = 0.010$). All 3 within-cluster patients who underwent peroral endoscopic myotomy achieved symptom resolution, compared to partial response in outside-cluster patients.

Conclusions: Machine learning clustering successfully identified a DES phenotype characterized by high premature contraction frequency that clusters with type III achalasia. Premature contraction frequency may serve as a clinical marker for selecting DES patients for lower esophageal sphincter-directed therapies. Given the rarity of DES, larger validation studies are needed to confirm these findings.

Keywords: Esophageal motility disorders; Esophageal spasm, diffuse; Natural orifice endoscopic surgery; Peroral endoscopic myotomy; Unsupervised machine learning

INTRODUCTION

Distal esophageal spasm (DES), defined by Chicago classification version 4.0 (CC v4.0) as ≥ 20% premature contractions without impaired esophagogastric junction (EGJ) relaxation, is a rare esophageal motor disorder with a prevalence of approximately 2% among patients undergoing high-resolution esophageal manometry (HRM) for esophageal symptoms.[1,2] DES represents a heterogeneous diagnosis that can exist along a spectrum with type III achalasia or be associated with gastroesophageal reflux disease (GERD) and opioid use.[3-6] Different phenotypes are thought to exist; however, traditional analytical methods have been insufficient to reliably identify these phenotypes.[7] Furthermore, current treatment approaches show variable success rates and identifying these phenotypes could inform which patients would benefit from invasive interventions versus alternative treatment strategies.[8-10]

The advent of machine learning has introduced novel analytical approaches for identifying patterns and unique clusters within complex datasets that might not be otherwise apparent through traditional methods. Uniform manifold approximation and projection (UMAP) is a dimensionality reduction technique that preserves local structure and relationships within complex datasets, thereby positioning it as a unique tool for identifying natural DES clusters.[11]

This single-center exploratory study aims to identify and characterize unique DES phenotypes using unsupervised machine learn-

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ing algorithms to better delineate subtypes and inform subtype-specific treatment approaches.

MATERIALS AND METHODS

Subjects

Adults ages 18-93 years presenting to our academic medical center for esophageal symptom evaluation between June 2005 and May 2025 were consecutively enrolled in an esophageal motility registry. All included subjects underwent HRM. Timed barium esophagram (TBE) and functional lumen imaging probe (FLIP) studies were performed at the treating gastroenterologist's discretion when clinically indicated. No endoscopic or surgical interventions occurred between studies. Subjects with technically limited HRM studies were excluded. Those with prior foregut surgery or pneumatic dilation were also excluded, as these can cause secondary esophageal motor abnormalities. Given the exploratory nature of this study and to distinguish clinical phenotypes, patients with reflux esophagitis, opioid use, and hiatal hernia were retained, as these conditions are associated with shortened distal latency (DL) on HRM.[3,5,6,12] The study protocol was approved by our university's Institutional Review Board (STU00224876), and retrospective analysis of coded, de-identified motility data was conducted under a waiver of informed consent.

High-resolution Esophageal Manometry

After a minimum 6-hour fast, HRM was performed using a 4.2-mm outer diameter solid-state assembly with 36 circumferential pressure sensors at 1-cm intervals (Medtronic, Shoreview, MN, USA). The catheter was placed transnasally and positioned to record from the hypopharynx to the stomach with approximately 3 intragastric sensors. Following a 2-minute baseline recording, the CC v4.0 protocol was performed: ten 5-mL liquid swallows in the supine position followed by five 5-mL liquid swallows in the upright position.[2,13,14]

While studies before 2021 were performed using older Chicago Classification protocols, CC v4.0 diagnostic criteria were retrospectively applied to all studies for consistent comparison. Investigators were blinded to clinical characteristics and other test results. Median integrated relaxation pressure (IRP) thresholds were > 15 mmHg for supine swallows and > 12 mmHg for upright swallows. Cases with inconclusive HRM findings (esophagogastric junction outlet obstruction [EGJOO] or absent contractility with borderline IRP) lacking conclusive TBE or FLIP results were excluded due to clinical uncertainty.

Timed Barium Esophagram

For inconclusive HRM findings (EGJOO or absent contractility with borderline IRP), TBE was used for further classification.[2] TBE was performed upright with 200 mL low-density barium sulfate, with anteroposterior images obtained at 1, 2, and 5 minutes.[15] A 12.5-mm barium tablet was administered if significant retention was absent at 5 minutes. Abnormal TBE was defined as barium column height > 5 cm at 5 minutes or > 5 cm at 1 minute with tablet impaction, indicating conclusive EGJ outflow disorder. Normal TBE indicated normal EGJ outflow.[16]

Functional Lumen Imaging Probe

FLIP studies were performed during sedated endoscopy using a 16-cm catheter (EndoFlip EF-322N; Medtronic).[17-19] Endoscopy was performed in left-lateral decubitus position with midazolam/fentanyl or propofol sedation at the anesthetist's discretion. Sedation choice does not significantly impact FLIP motility findings relative to HRM diagnoses.[20] After endoscope withdrawal and atmospheric pressure calibration, the FLIP catheter was placed transorally and positioned with 1-3 impedance sensors beyond the EGJ. Beginning at 40 mL, stepwise 10-mL distensions were performed to a target volume of 70 mL, with each volume maintained for 30-60 seconds.

FLIP data were analyzed using customized software (available open source at http://www.wklytics.com/nmgi) to generate FLIP Panometry plots.[21] EGJ opening analysis excluded areas with dry catheter artifact and included the EGJ distensibility index (DI) at 60 mL and maximum EGJ diameter achieved during 60-mL or 70-mL fill volumes.[21,22] Contractile response patterns were classified per the FLIP 2.0 Dallas Consensus.[23]

Evaluation of Treatments and Outcomes

Treatment, clinical course, and outcomes for all patients with a manometric diagnosis of DES were determined through manual chart review. Treatment modalities included lifestyle modifications, proton pump inhibitor therapy, nitrates, hyoscyamine, muscle relaxants/sildenafil, botulinum toxin injection, endoscopic dilation, peroral endoscopic myotomy (POEM), and surgical hernia repair.

Treatment outcomes were assessed using pre- and post-treatment patient-reported outcomes (PROs), when available. Patients undergoing motility testing were asked to complete pre-treatment questionnaires including the brief esophageal dysphagia questionnaire, Eckardt score, esophageal hypervigilance and anxiety scale (EHAS), and GERD questionnaire (GERDQ).[24-27] Post-treatment questionnaire data was collected when repeat motility testing was performed. Post-treatment Eckardt scores were also sometimes available from clinical visit documentation. When PROs were unavailable, outcomes were assessed based on clinician-documented subjec-

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tive improvement. Patients without subsequent gastroenterology visits after treatment initiation were considered lost to follow-up.

Clustering Method and Statistical Methods

UMAP was used for unsupervised dimensionality reduction, as it preserves local structure and relationships within complex datasets (Supplementary Method).[11] Prior to UMAP application, numeric predictors were normalized, variables with > 40% missing data were removed, and remaining missing data was imputed using bootstrap aggregated decision tree models (Supplementary Figure).[28] HRM data were projected into 2-dimensional space using UMAP, and DES clusters were visually identified using density contours.

HRM metrics, patient-reported outcomes, and treatment outcomes were compared between DES cases within the primary cluster versus those outside the primary cluster. Additional comparisons were made with patients diagnosed with CC v4.0 normal motility and type III achalasia, given the hypothesis that DES may exist along a spectrum with type III achalasia.[4]

Continuous variables were reported as median (interquartile range) and categorical variables as n (%). Group comparisons used chi-square tests for categorical variables and parametric tests (t-tests/ANOVA) for normally distributed continuous variables or non-parametric tests (Mann-Whitney U/Kruskal-Wallis) for non-normally distributed continuous variables. Statistical significance was set at $P < 0.05$ (two-tailed). Data analysis was conducted using R version 4.4.2, with UMAP applied using the tidymodels package.[29,30]

RESULTS

Subjects

The study cohort comprised 5360 patients with a median age of 56 years (interquartile range, 27 years), 61% of whom were females (Table 1). Within this cohort, 42 patients (0.8%) had DES, 2400 (45%) had normal motility, and 107 (2.0%) had type III acha-

Table 1. Cohort Characteristics

Characteristic	Total study cohort (n = 5360)	DES (n = 42)	Normal (n = 2400)	Achalasia III (n = 107)
Demographics				
Age (yr)	56 (IQR, 27)	61 (IQR, 21)	53 (IQR, 26)	67 (IQR, 18)
Sex, female	3255 (61.0)	26 (62.0)	1533 (64.0)	49 (46.0)
Indication				
Dysphagia	2643 (49.0)	23 (55.0)	998 (42.0)	81 (76.0)
Reflux	1004 (19.0)	6 (14.0)	564 (24.0)	3 (2.8)
Chest pain	303 (5.7)	4 (10.0)	164 (6.8)	2 (1.9)
Other	1169 (22.0)	9 (21.0)	595 (25.0)	13 (12.0)
PPI use	2130 (40.0)	16 (38.0)	979 (41.0)	21 (20.0)
Opioids use	255 (4.8)	5 (12.0)	90 (3.8)	19 (18.0)
Endoscopy				
Esophagitis (≥ LA-B)	124 (2.3)	0 (0.0)	57 (2.4)	1 (0.9)
Diverticulum	16 (0.3)	0 (0.0)	4 (0.2)	3 (2.8)
Hernia > 3 cm (n)	142 (2.6)	6 (14.0)	71 (3.0)	0 (0.0)
Manometry				
IRP (supine [mmHg])	11.0 (IQR, 10.5)	11.2 (IQR, 5.8)	9.0 (IQR, 6.0)	28.0 (IQR 15.5)
IRP (upright [mmHg])	9.0 (IQR, 11.4)	9.9 (IQR, 6.8)	7.0 (IQR, 6.2)	27.0 (IQR 16.6)
RDC IRP	4.0 (IQR, 8)	5.5 (IQR, 3.5)	2.0 (IQR, 4.9)	17.0 (IQR 15.5)
DL (supine [sec])	6.6 (IQR, 1.9)	4.1 (IQR, 0.8)	6.6 (IQR, 1.6)	3.4 (IQR 0.8)
DL (upright [sec])	6.4 (IQR, 1.8)	4.3 (IQR, 1.0)	6.4 (IQR, 1.6)	3.5 (IQR 1.2)
Premature contractions (n)	0.0 (IQR, 0.0)	6.0 (IQR, 4.8)	0.0 (IQR, 0.0)	7.0 (IQR 8.0)

DES, distal esophageal spasm; PPI, proton pump inhibitor; LA-B, Los Angeles grade B; IRP, integrated relaxation pressure; RDC, rapid drink challenge; DL, distal latency.
The total study cohort includes 2811 patients with other Chicago classification version 4.0 diagnoses that are not shown as separate column.
Continuous variables are presented as median (interquartile range [IQR]), categorical variables as n (%).

lasia. Patients with DES and type III achalasia demonstrated higher rates of opioid use (12% and 18%, respectively) compared to the remainder of the cohort.

Dysphagia was the most common indication for evaluation across all patients. However, DES patients presented more frequently with chest pain (10%) compared to type III achalasia patients (1.9%) and the overall cohort (5.7%).

Manometric parameters aligned with expected disease patterns. Type III achalasia patients demonstrated elevated median IRP in both supine and upright positions. Median DL was reduced in DES and type III achalasia patients while remaining within normal ranges for patients with normal motility. Correspondingly, the frequency of premature contractions was increased in both DES and type III achalasia groups relative to the normal motility cohort.

Cluster Comparisons

UMAP projection of HRM metrics revealed a central cluster with substantial overlap between DES and type III achalasia patients (Fig. 1). Using DES contour borders to define cluster boundaries, 33 DES, 24 normal motility, and 62 type III achalasia patients were classified as within-cluster, while 9 DES, 2376 normal motility, and 45 type III achalasia patients were classified as outside-cluster (Table 2).

The primary characteristic differentiating within- and outside-cluster observations was the number of premature supine contractions (median 8 vs 0, respectively) (Fig. 2). This difference was statistically significant among DES patients ($P < 0.001$). DL for both supine and upright swallows was also shorter in within-cluster patients. Among DES patients, no significant differences were observed between cluster groups for opioid use, large hernias (> 3 cm), or IRP, though PPI use trended higher within-cluster (42% vs 22%). Among within-cluster normal motility patients, nearly all had moderate to large hiatal hernias measured during esophageal manometry, with a mean hernia size of 3.6 cm. Within this subgroup, 25% had a hiatal hernia larger than 3 cm compared to 2.7% among outside-cluster normal motility patients. These within-cluster normal motility patients otherwise demonstrated intact peristalsis, normal IRP, normal distal contractile integral, and normal bolus transit,

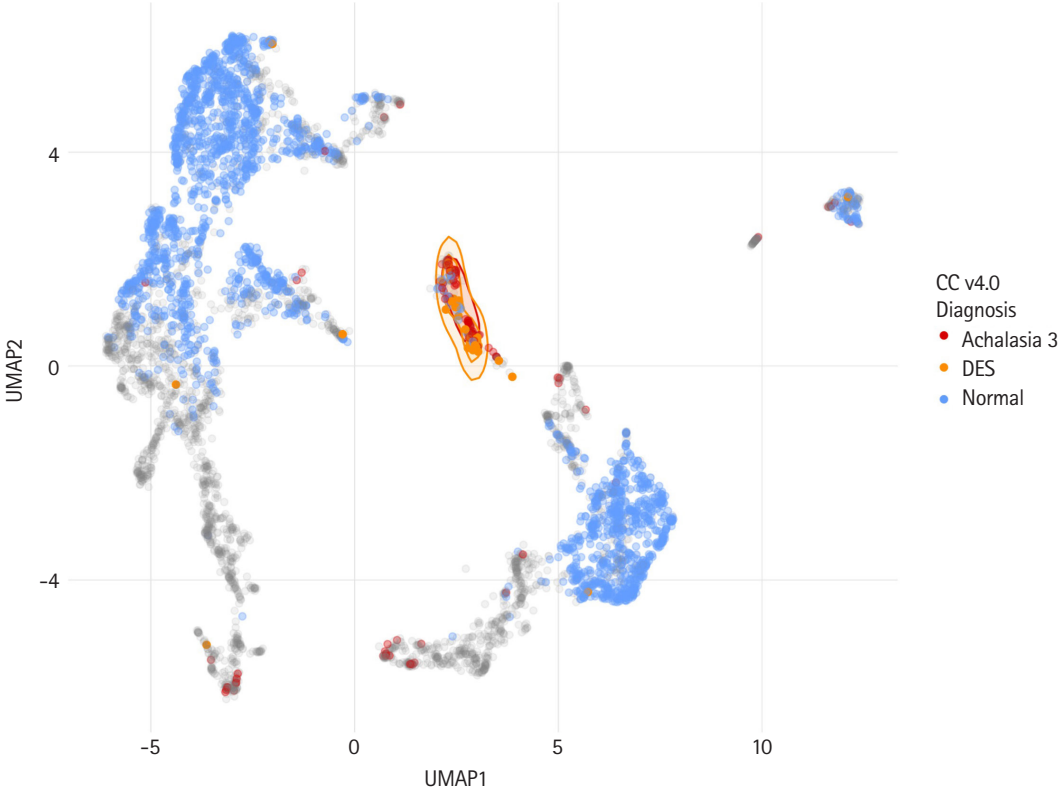

Figure 1. Uniform manifold approximation and projection (UMAP) dimensionality reduction of high-resolution esophageal manometry (HRM) data. Two-dimensional UMAP projection of HRM data showing the spatial distribution of patients with distal esophageal spasm (DES), type III achalasia, and normal motility. Gray points represent all other Chicago classification version 4.0 (CC v4.0) diagnoses. Each point represents an individual patient, with clustering performed on normalized HRM metrics after imputation of missing data. Density contours highlight the core clustering regions for DES and type III achalasia groups. DES cases demonstrate heterogeneous clustering patterns, with some cases forming a distinct cluster around type III achalasia patients, while others are distributed among different regions of the UMAP space.

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Table 2. Inter-cluster Comparison of Clinical and Demographic Variables

Characteristic	Within-cluster			Outside-cluster			*P*-value[a]
	DES (n = 33)	Normal (n = 24)	Achalasia III (n = 62)	DES (n = 9)	Normal (n = 2376)	Achalasia III (n = 45)	
Demographics							
Age (yr)	62 (IQR, 23)	66 (IQR, 16)	68.5 (IQR, 17)	60 (IQR, 14)	52 (IQR, 26)	64 (IQR, 18)	0.878
Sex (female)	22 (67.0)	17 (71.0)	29 (47.0)	4 (44.0)	1516 (64.0)	20 (44.0)	0.265
PPI use	14 (42.0)	14 (58.0)	15 (24.0)	2 (22.0)	965 (41.0)	6 (13.0)	0.578
Opioids use	4 (12.0)	4 (17.0)	10 (16.0)	1 (11.0)	86 (4.0)	9 (20.0)	0.553
Endoscopy							
Esophagitis (≥ LA-B)	0 (0.0)	3 (13.0)	0 (0.0)	0 (0.0)	54 (2.3)	1 (2.2)	> 0.999
Diverticulum	0 (0.0)	0 (0.0)	3 (4.8)	0 (0.0)	4 (0.2)	0 (0.0)	> 0.999
Hernia > 3 cm (n)	5 (15.0)	6 (25.0)	0 (0.0)	1 (11.0)	65 (2.7)	0 (0.0)	> 0.999
Manometry							
IRP (supine [mmHg])	11.0 (IQR, 6.0)	8.8 (IQR, 4.4)	27.1 (IQR, 15.0)	12.6 (IQR, 3.5)	9.0 (IQR, 6.0)	30.0 (IQR, 17.0)	0.895
IRP (upright [mmHg])	9.9 (IQR, 5.7)	11.0 (IQR, 8.0)	26.3 (IQR, 14.0)	10.9 (IQR, 8.0)	7.0 (IQR, 6.1)	25.0 (IQR, 22.0)	0.986
DCI (supine [mmHg·s·cm])	1854.0 (IQR, 2722.0)	1405.0 (IQR, 1978.0)	1920.0 (IQR, 3069.0)	1493.0 (IQR, 2480.0)	1417.0 (IQR, 1393.0)	533.0 (IQR, 2394.0)	0.631
DCI (upright [mmHg·s·cm])	1682.0 (IQR, 3031.0)	1820.0 (IQR, 1241.0)	1611.0 (IQR, 2267.0)	1842.0 (IQR, 2008.0)	1267.0 (IQR, 1365.0)	807.0 (IQR, 2638.0)	0.768
DL (supine [sec])	4.0 (IQR, 0.8)	4.4 (IQR, 0.5)	3.2 (IQR, 0.6)	4.4 (IQR, 0.4)	6.6 (IQR, 1.6)	3.7 (IQR, 1.4)	0.045
DL (upright [sec])	4.2 (IQR, 1.1)	4.3 (IQR, 0.3)	3.4 (IQR, 0.9)	5.4 (IQR, 1.0)	6.5 (IQR, 1.5)	4.2 (IQR, 1.4)	0.007
Premature contractions (n)	7.0 (IQR, 4.0)	6.0 (IQR, 5.2)	9.0 (IQR, 3.0)	4.0 (IQR, 2.0)	0.0 (IQR, 0.0)	2.0 (IQR, 5.0)	< 0.001
	DES (n = 14)	Normal (n = 4)	Achalasia III (n = 35)	DES (n = 2)	Normal (n = 422)	Achalasia III (n = 19)	
FLIP							
DI at 60 mL (mm²/mmHg)	2.1 (IQR, 1.7)	3.0 (IQR, 3.8)	0.6 (IQR, 0.6)	2.4 (IQR, 1.8)	3.4 (IQR, 3.6)	0.6 (IQR, 0.5)	0.933
EGJ diameter (maximum [mm])	15.7 (IQR, 6.3)	17.9 (IQR, 2.4)	9.6 (IQR, 4.5)	12.2 (IQR, 4.2)	18.4 (IQR, 4.9)	9.6 (IQR, 3.1)	0.440
CR							
Normal	5 (15.0)	1 (4.2)	0 (0.0)	1 (11.0)	197 (8.3)	0 (0.0)	> 0.999
Diminished	1 (3.0)	0 (0.0)	4 (6.5)	0 (0.0)	37 (1.6)	5 (11.0)	> 0.999
Disordered	2 (6.1)	0 (0.0)	13 (21.0)	1 (11.0)	40 (1.7)	15 (33.0)	0.423
Spastic	3 (9.1)	0 (0.0)	9 (15.0)	0 (0.0)	25 (1.1)	4 (8.9)	> 0.999
Absent	0 (0.0)	1 (4.2)	5 (8.1)	0 (0.0)	27 (1.1)	5 (11.0)	> 0.999

[a]*P*-value comparing DES patients within vs outside the cluster.

DES, distal esophageal spasm; PPI, proton pump inhibitor; LA-B, Los Angeles grade B; IRP, integrated relaxation pressure; DCI, distal contractile integral; DL, distal latency; FLIP, functional lumen imaging probe; DI, distensibility index; CR, contractile response.

Continuous variables are presented as median (interquartile range [IQR]), categorical variables as n (%).

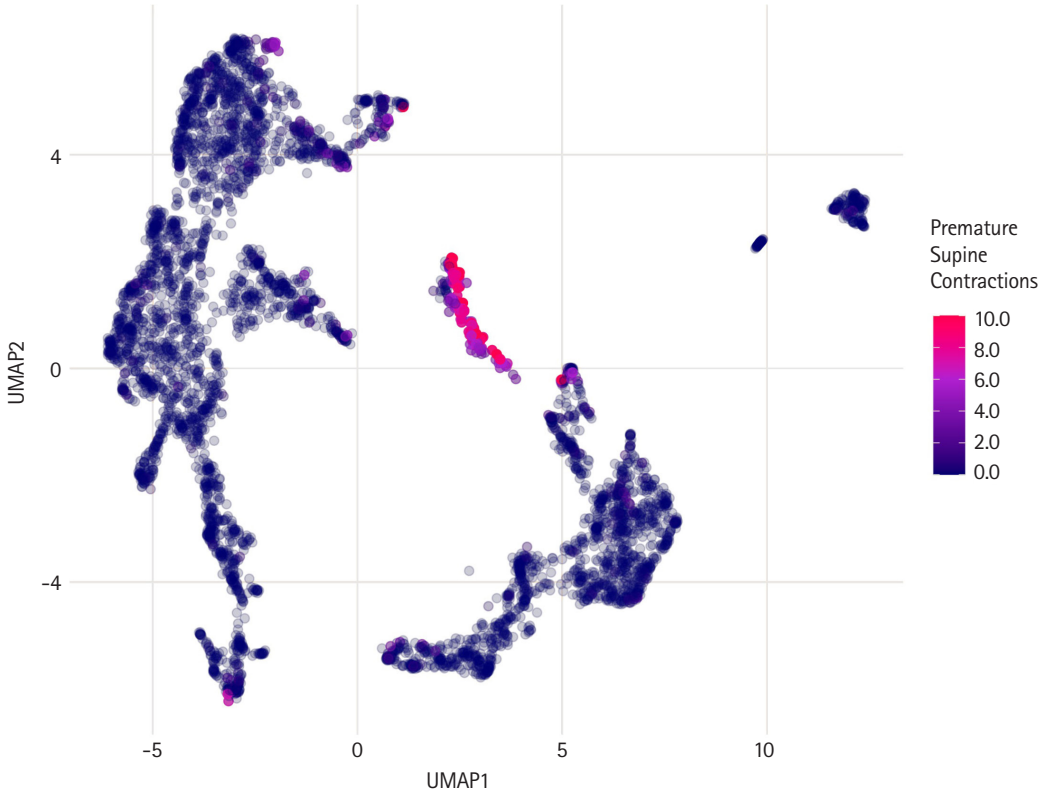

Figure 2. Uniform manifold approximation and projection (UMAP) projection colored by premature supine contraction frequency. Two-dimensional UMAP projection showing the study cohort colored by premature supine contraction frequency. Distal esophageal spasm (DES) and type III achalasia cases with higher premature contraction frequencies (red, approaching 10) cluster more closely towards the center, while cases with lower premature contraction frequencies (blue/purple, 0 to 3) tend to be distributed further from the central cluster.

findings consistent with CC v4.0 normal motility despite borderline DL values.[2]

Pre-treatment PROs showed no significant inter-cluster differences except for EHAS, where within-cluster patients demonstrated higher median total scores (30 vs 20.5; $P = 0.010$). Post-treatment PRO data were insufficient for inter-cluster comparison.

FLIP testing was completed by 14 within-cluster and 2 outside-cluster DES patients. The small outside-cluster sample precluded statistical comparison, though median values were 2.1 vs 2.4 mm^2/mmHg for distensibility index and 15.7 vs 12.2 mm for maximum EGJ diameter, respectively. Type III achalasia patients showed no inter-cluster differences in FLIP parameters. Esophagram data were insufficient for DES cluster comparisons.

Treatments

Among DES patients, PPI was the most common treatment within-cluster, while PPI and nitrates were equally common outside-cluster (Table 3). For lower esophageal sphincter-directed interventions, 5 (15%) within-cluster patients received botulinum toxin injection and 3 (9.1%) underwent POEM, compared to 2 POEM procedures in outside-cluster patients. All 3 within-cluster POEM patients reported complete dysphagia resolution. Among outside-cluster POEM patients, 1 achieved symptom resolution while the other experienced partial improvement with persistent mild chest pressure.

DISCUSSION

In this single-center exploratory study, machine learning clustering demonstrated that DES patients represent a heterogeneous population, with a subset clustering with type III achalasia based on premature contraction frequency. Despite the small sample size inherent to this rare condition, within-cluster patients exhibited higher EHAS scores and complete symptomatic response to POEM.

The clear separation between type III achalasia and normal motility patients on UMAP projection validates our clustering approach. The intermediate positioning of DES cases supports the hypothesis that it may exist along a spectrum with type III achalasia, with higher premature contraction frequencies positioning patients closer to the achalasia phenotype. A subset of normal motility patients clustered near DES and type III achalasia due to the pres-

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Table 3. Inter-cluster Distal Esophageal Spasm Treatment Differences

Treatment	DES within-cluster (n = 33)	DES outside-cluster (n = 9)	*P*-value
Treatment			
Lifestyle changes (reflux)	2 (6.1)	0 (0.0)	> 0.999
PPI	11 (33.0)	3 (33.0)	1.000
Nitrates	3 (9.1)	3 (33.0)	0.101
Sildenafil/muscle relaxants	1 (3.0)	1 (11.0)	0.387
Hyoscyamine	5 (15.0)	1 (11.0)	> 0.999
Botulinum	5 (15.0)	0 (0.0)	0.567
Dilation (not pneumatic)	4 (12.0)	0 (0.0)	0.561
POEM	3 (9.1)	2 (22.0)	0.288
Hernia repair	5 (15.0)	1 (11.0)	> 0.999
PROs (pre-treatment)			
GERDQ	8.5 (IQR 4.0)	8.0 (IQR 3.0)	0.936
BEDQ	5.0 (IQR 10.0)	7.0 (IQR 10.5)	0.802
EHAS	30.0 (IQR 15.0)	20.5 (IQR 3.0)	0.010
Eckardt	3.0 (IQR 3.0)	4.5 (IQR 2.5)	0.191

DES, distal esophageal spasm; PPI, proton pump inhibitor; POEM, peroral endoscopic myotomy; PROs, patient-reported outcomes; GERDQ, gastroesophageal reflux disease questionnaire; BEDQ, brief esophageal dysphagia questionnaire; EHAS, esophageal hypervigilance and anxiety scale; Eckardt, Eckardt score.
Continuous variables are presented as median (interquartile range [IQR]), categorical variables as n (%).

ence of moderate to large hiatal hernias, which can shorten DL by mechanically reducing effective esophageal length.[12] This anatomic effect likely positioned them closer to spastic phenotypes in UMAP space despite meeting CC v4.0 criteria for normal motility.[2] Together, these findings suggest that premature contraction frequency may serve as a marker to determine DES phenotype and guide treatment selection.

DES is recognized as a heterogeneous diagnosis with variable clinical presentations and manometric patterns.[31] DL has previously been examined as a key discriminating variable for DES, with a dichotomous threshold of less than 4.5 seconds used to identify clinically relevant premature contractions.[1] This analysis reinforces the importance of DL but differs in both methodology and interpretation. Premature contraction frequency was treated as a continuous physiologic measure within a multidimensional framework that incorporated all HRM metrics. UMAP integrates these variables holistically and identifies the features that most strongly influence local clustering structure. Within this framework, premature contraction frequency emerged as the principal factor distinguishing the DES subset that clustered with type III achalasia. Higher frequencies shifted patients toward an achalasia-like physiologic profile when evaluated alongside other motility measures.

Prior studies report lower POEM response rates in DES compared to achalasia patients.[32,33] This likely reflects DES heterogeneity, and improved characterization of achalasia-spectrum patients could yield response rates comparable to confirmed achalasia. While the 100% within-cluster POEM response rate is promising, validation in larger cohorts is needed before informing clinical practice.

Of note, several outside-cluster normal motility patients demonstrated abnormal FLIP contractile responses. This finding is not unexpected because HRM and FLIP assess related but distinct physiologic processes. HRM evaluates primary peristalsis and EGJ relaxation, whereas FLIP measures esophageal distensibility and distension-induced secondary peristalsis. As a result, normal motility on HRM can coexist with abnormal FLIP patterns, an observation described in prior studies.[34,35] Such discordance underscores the complementary roles of HRM and FLIP and highlights the need for further work to define the clinical significance of discrepant findings, particularly among patients who fall near cluster boundaries in multidimensional analyses.

The primary limitation of this study is the small DES sample size. This precluded meaningful FLIP and esophagram comparisons due to insufficient outside-cluster patients. Additionally, the retrospective study design resulted in missing data, particularly for post-treatment PROs and long-term treatment responses, with many patients lost to follow-up. Larger, multi-institutional datasets are necessary to further assess how clustering can define DES phenotypes and to overcome these data limitations.

Future research should employ advanced unsupervised clustering methods, including deep learning approaches to generate latent spaces similar to virtual disease landscapes for FLIP, which could provide probabilistic models for classifying achalasia-spectrum versus non-achalasia DES and inform candidacy for lower esophageal sphincter-directed therapies.[36] Additionally, clinically applicable algorithms incorporating premature contraction thresholds should be developed to identify optimal treatment candidates. Larger, multi-institutional datasets are necessary to externally validate these findings and better characterize DES phenotypes. Prospective studies are also needed to evaluate treatment outcomes in a standardized and comprehensive manner. The identification of an achalasia-spectrum DES subset defined by number of premature contractions on HRM represents a paradigm shift toward phenotype-driven treatment selection that could improve outcomes for patients with this challenging condition.

SUPPLEMENTARY MATERIALS

Note: To access the supplementary method and figure men-

tioned in this article, visit the online version of *Journal of Neurogastroenterology and Motility* at http://www.jnmjournal.org/, and at http://doi.org/10.5056/jnm25171.

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Conflicts of interest

John E Pandolfino: Sandhill Scientific/Diversatek (consulting and grant), Takeda (speaking), Astra Zeneca (speaking), Medtronic (speaking, consulting, patent, and license), Torax/Ethicon (speaking and consulting), EndoGastric Solutions (advisory board), and Phathom (speaking and consulting); Dustin A Carlson: Medtronic (speaking, consulting, and License). Diversatek (consulting); Braintree (consulting); Medpace (consulting); Phathom Pharmaceuticals (speaking and consulting); and Regeneron/Sanofi (speaking). Ofer Z Fass: CapsoVision (consulting), Castle Biosciences (Consulting).

Author contributions

Ofer Z Fass contributed to data analysis, visualization, interpretation, drafting of the manuscript, and approval of the final version; John E Pandolfino contributed to obtaining funding, data interpretation, editing the manuscript critically, and approval of the final version; and Dustin A Carlson contributed to study concept and design, drafting of the manuscript, obtaining funding, data analysis, data interpretation, and approval of the final version.

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SUPPLEMENTARY MATERIALS

Supplementary Method

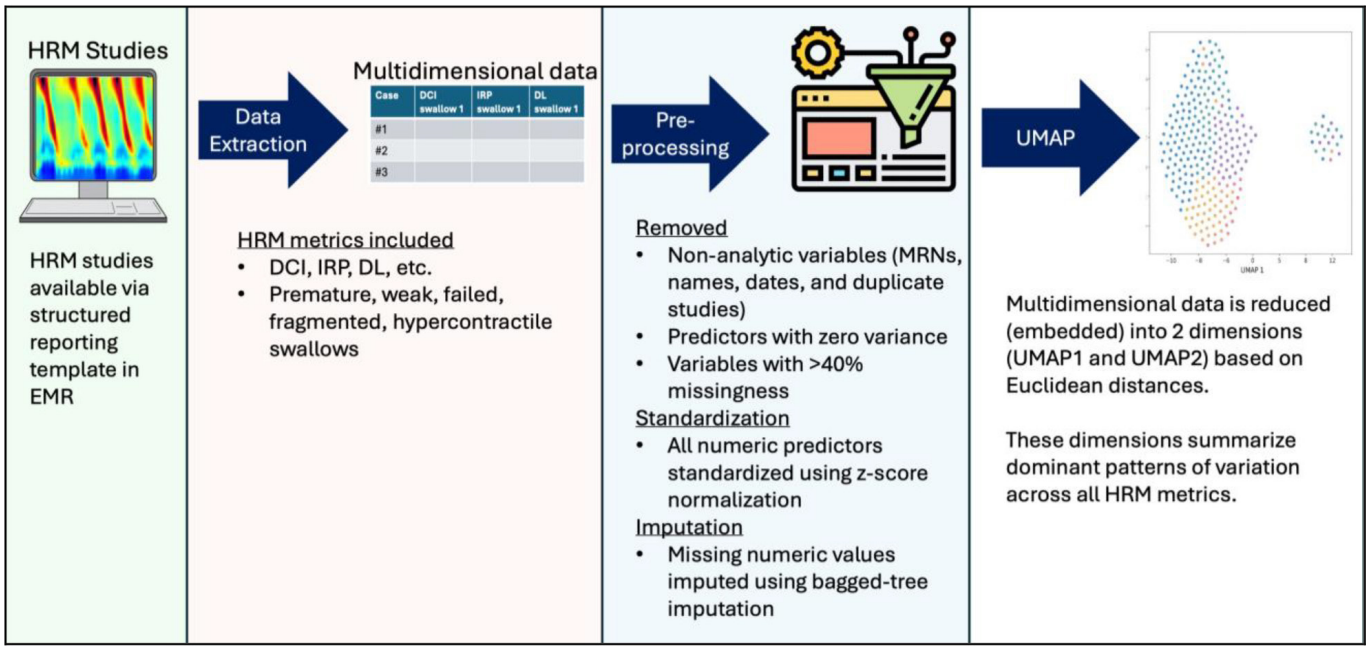
High-resolution manometry feature preprocessing and uniform manifold approximation and projection analysis

For clustering and uniform manifold approximation and projection (UMAP), we first extracted summary high-resolution esophageal manometry (HRM) metrics for each patient from the structured HRM reporting template in our electronic medical record system. Supine and upright swallows were summarized as counts of premature, weak, failed, hypercontractile, fragmented, and normal contractions, along with pressurization patterns (pan-esophageal and compartmentalized pressurization), mean distal latency, mean distal contractile integral, and integrated relaxation pressure. Provocative maneuver responses (multiple rapid swallows and rapid drink challenge) and hiatal hernia size were also included.

Preprocessing was performed using the tidymodels framework in R. Non-analytic variables (identifiers, names, dates, and duplicate studies) were removed, and predictors with zero variance were excluded. To avoid introducing noise from variables with extensive missing data, we removed predictors that were missing in more than 40 percent of patients. Remaining missing numeric values were imputed using bagged-tree imputation, and all numeric predictors were standardized using z-score normalization. Only numeric HRM metrics were used as inputs for UMAP; Chicago classification labels (normal, distal esophageal spasm [DES], type III achalasia, etc) were retained strictly as labels and were not included in the UMAP computation.

UMAP dimensionality reduction was implemented using the step_umap() function in the embed package, producing a two-dimensional representation of the standardized HRM feature space based on Euclidean distances. The resulting axes (UMAP1 and UMAP2) represent latent dimensions that capture major structure and relationships within the high-dimensional HRM data. Although not linear components like those produced by principal component analysis, they similarly summarize dominant patterns of variation in a low-dimensional space.

Inspection of HRM feature distributions across the UMAP embedding demonstrated that regions where DES and type III achalasia cases clustered were characterized by higher premature supine contraction counts and shorter distal latency values, whereas regions dominated by normal motility studies showed low premature contraction frequency and distal latency values within the normal range.



Supplementary Figure. Overview of high-resolution esophageal manometry (HRM) feature extraction, preprocessing, and uniform manifold approximation and projection (UMAP) embedding. Structured HRM studies were extracted from the electronic medical record to obtain summary metrics for supine and upright swallows. During preprocessing, non-analytic variables, predictors with > 40% missingness, and zero-variance predictors were removed. Missing numeric values were imputed using bagged-tree imputation, and all numeric features were standardized using z-score normalization. UMAP was applied to all HRM metrics to produce a two-dimensional embedding (UMAP1 and UMAP2) that captures latent structure and relationships among HRM studies. EMR, electronic medical record; DCI, distal contractile integral; IRP, integrated relaxation pressure; DL, distal latency; MRN, medical record number.

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The Fermentable Oligosaccharides, Disaccharides, Monosaccharides, and Polyols Diet Is Effective in Irritable Bowel Syndrome With Constipation: A Single Center Tertiary Care Clinical Practice Evaluation

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Background/Aims: The low fermentable oligosaccharides, disaccharides, monosaccharides, and polyols (FODMAP) diet is a well-researched treatment for diarrhea-predominant irritable bowel syndrome (IBS-D), but its value for constipation-predominant IBS (IBS-C) is uncertain. Clinicians may hesitate to recommend it for IBS-C, fearing it could worsen constipation or offer little relief. This study aims to assess the diet's effectiveness across all IBS subtypes, specifically investigating its impact on stool patterns in IBS-C and the correlation with symptom relief.

Methods: This prospective service evaluation involved 294 IBS patients (84% female, mean age 44.6) attending dietitian-led group sessions on FODMAP restriction (baseline) and reintroduction (follow-up). The primary outcome was "satisfactory relief of gut symptoms" via the global symptom question; secondary outcomes included changes in stool consistency, IBS subtype, and gastrointestinal symptoms.

Results: At baseline, 23.8% (70/294) had IBS-C and 35.7% IBS-D (105/294). All subtypes except IBS-M showed significant symptom relief at follow-up ($P < 0.001$), with no statistically significant difference between IBS-C (50.7%, 35/69) and IBS-D (44.8%, 47/105) ($P = 0.158$). After FODMAP restriction, 50.7% (35/69) IBS-C remained IBS-C, while 44.9% (31/69) shifted to unclassified IBS (IBS-U). Those who shifted (IBS-C to IBS-U) reported greater relief (74.2%, 23/31 vs 45.7% 16/35, $P = 0.019$) and reduced pain ($P = 0.006$), unlike those who remained IBS-C ($P = 0.180$).

Conclusions: IBS-C and IBS-D showed similar symptom relief after FODMAP restriction. Notably, 44.9% of IBS-C patients shifted to normal stool consistency (IBS-U), with greater symptom relief and reduced pain. FODMAP restriction does not necessarily exacerbate constipation in IBS-C and remains a viable treatment option in real world clinical practice.

Keywords: Constipation; Fermentable oligosaccharides, disaccharides, monosaccharides, and polyols; Group education; Irritable bowel syndrome; Outcome assessment health care

INTRODUCTION

Irritable bowel syndrome (IBS) is a chronic functional gut disorder affecting ~4% of the global population,[1] characterized by recurrent abdominal pain and altered bowel habits, as defined by the Rome IV criteria.[2] The pathophysiology of IBS is not fully understood but it is recognized as a disorder of gut–brain interaction influenced by biopsychosocial factors.[3]

IBS is classified into 4 subtypes; diarrhea predominant (IBS-D), constipation predominant (IBS-C), mixed stool consistency; both diarrhea and constipation predominant (IBS-M), and unclassified; neither diarrhea nor constipation predominant (IBS-U).[2] IBS-C constitutes approximately one-third of all IBS subtype cases worldwide,[1] and in the United Kingdom.[4] IBS-C is defined by more than 25% of bowel movements being Bristol stool form scale (BSFS) types 1-2 and less than 25% being BSFS types 6-7.[2]

Many IBS patients view diet as both the cause and solution to their symptoms, and they generally prefer dietary therapy over medications.[5] In contrast, patients with IBS-C often report low satisfaction with over-the-counter laxatives and fiber supplements.[6] Current IBS management guidelines recommend diet therapies, including the fermentable oligosaccharides, disaccharides, monosaccharides, and polyols (FODMAP) diet, prior to any first line pharmacological interventions in all IBS irrespective of subtype.[7,8]

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The FODMAP diet was more effective than other diets for IBS in a network meta-analysis of 13 randomized controlled trials (RCTs), but most studies focused on IBS-D, limiting the evidence for its effectiveness in IBS-C.[9] Furthermore, concerns exist that the FODMAP diet may exacerbate or fail to alleviate constipation symptoms in IBS-C.[10] The lack of evidence and unknown impact on constipation may create hesitancy among clinicians when recommending the FODMAP diet to patients with IBS-C in clinical practice, and highlights a research practice gap.

In most healthcare settings, including the United Kingdom's National Health Service, IBS-C is included in dietitian-led FODMAP treatment pathways in line with current guidelines.[7,8] Our center treats a large tertiary IBS population using a group FODMAP education program that accommodates all IBS subtypes.[11-13] Expanding research on IBS subtypes could support the development of subtype-specific guidelines, enabling more precise and high-quality nutrition care.

The aim of this study is (1) to compare the effectiveness of the restriction phase of the FODMAP diet across all IBS subtypes (IBS-D, IBS-C, IBS-M, and IBS-U), and (2) to describe the impact FODMAP restriction had on stool consistency in IBS-C and to explore the relationship between these changes and treatment outcomes.

We utilized data previously published in a FODMAP pathway comparison study in routine clinical care,[11] however this study presents new data on a different and specific topic with new insights.

MATERIALS AND METHODS

Study Design, Population, and Fermentable Oligosaccharides, Disaccharides, Monosaccharides, and Polyols Diet Advice

This study represents a sub-analysis of our previous 4-week vs 8-week FODMAP group pathway service evaluation where full details of the original study and administration of the FODMAP diet in a group setting can be found.[11] However to briefly summarise the methodology; a prospective, observational design was used to evaluate the effectiveness of group-based FODMAP diet education, within a tertiary gastroenterology service, requiring patients to follow FODMAP restriction for 4-8 weeks. Patients with IBS diagnosed by gastroenterologists using National Institute of Health and Clinical Excellence criteria[14] were referred to a dietitian for FODMAP guidance. Those suitable for group sessions participated, while those with specific needs (eg, non-English speaking) received individual appointments and were excluded from this analysis. The study took place before the COVID-19 pandemic, with all appointments conducted face-to-face. Group sessions were led by an expert dietitian in the FODMAP diet. The initial session (baseline) focused on FODMAP restriction, while the follow-up session covered FODMAP reintroduction and personalization. Patients received educational materials and had opportunities to ask questions.

Data Collection and Questionnaires

Demographic data and appointment dates were collected from electronic records while patient-reported outcomes were assessed using questionnaires (see below) completed at baseline and follow-up (after FODMAP restriction) as previously described.[11] Patients who missed the follow-up or did not complete questionnaires were excluded.

Questionnaires consisted of the following measures; global symptom relief assessed using the global symptom question (GSQ), a single question on overall symptom satisfaction.[15] Patients who responded "yes" were deemed to have achieved satisfactory relief of their IBS symptoms. Gastrointestinal symptom severity measured using the gastrointestinal symptom rating scale (GSRS), which evaluates the severity of 11 individual GI symptoms on a 4-point Likert scale with responses choices of: "none," "mild," "moderate," or "severe" symptoms.[16] IBS subtype was determined based on stool consistency assessed using the BSFS[17] and Rome IV criteria[2] at baseline and follow-up.

Dietary adherence was assessed using a 5-point Likert scale as responding to the statement "I followed the diet" (never, occasionally [a quarter of the time], half the time, often [3 quarters of the time], and always). In this study dietary adherence was classified as those responding > 50% of the time, that is, "half the time," "often," and "always."

Ethical Statement

Ethical approval was not required for this observational evaluation of routine clinical practice, in accordance with Health Research Authority guidance and its decision tool. The project was therefore registered with the Gastroenterology Medicine Clinical Effectiveness and Audit team in line with local governance procedures for service evaluations.

Statistical Methods

The primary outcome was "satisfactory relief of gut symptoms" measured by the GSQ,[15] used to compare the effectiveness of the restriction phase of the FODMAP diet across all IBS subtypes. McNemar's test was used to analyze the improvement in satisfactory relief from baseline to follow up (after FODMAP restriction) in each individual IBS subtype. To determine any differences in sat-

isfactory relief at follow up across IBS subtypes the proportion (%) of those achieving satisfactory relief in each individual subtype were compared to each other using a chi-squared test.

Our secondary outcomes included measures describing the impact of the FODMAP restriction in patients with IBS-C at baseline. Firstly, changes in stool consistency (BSFS) and related IBS-C subtype at follow-up were examined descriptively. Next, satisfactory relief (GSQ) at follow-up was compared using McNemar's test between patients who shifted from IBS-C to IBS-U (ie, from > 25% BSFS types 1-2 to > 75% BSFS types 3-5, indicating normal stool consistency[2]) and those who remained IBS-C. Secondly to analyze reductions in individual GI symptoms within each IBS subtype, GSRS scores were dichotomized: "none" and "mild" as "no moderate/severe symptoms," and "moderate" and "severe" as "presence of moderate/severe symptoms." McNemar's test was then applied to assess reductions in moderate/severe symptoms from baseline to follow-up. Thirdly to further explore the relationship between stool consistency changes and satisfactory relief, GSRS symptom changes from baseline to follow-up were compared between those who shifted from IBS-C to IBS-U and those who remained IBS-C, using McNemar's test. All statistical analyses were conducted using IBM SPSS Statistics Version 26. A *P*-value < 0.05 was considered to be statistically significant.

RESULTS

Participants

We included 294 patients with IBS, ages ranged from 18 years to 86 years (44.6 ± 15.5) and 246 (84%) were female. IBS subtypes at baseline were IBS-D (105/294, 35%), followed by IBS-U (87/294, 30%), IBS-C (70/294, 24%), and IBS-M (32/294, 11%). Ten additional patients were included in this analysis compared to the original publication (n = 284)[11] due to these patients being excluded from the original data set due to lacking data on appointment times required for the purpose of the original study. Dietary adherence was assessed in 292 of 294 patients (2 were excluded due to missing data). The overall adherence rate was 78% (230/292). These levels were consistent across all IBS subtypes, including patients who switched subtypes following FODMAP restriction (Supplementary Information).

Effectiveness of Fermentable Oligosaccharides, Disaccharides, Monosaccharides, and Polyols Restriction Diet Across All Irritable Bowel Syndrome Subtypes

Satisfactory relief in individual irritable bowel syndrome subtypes

Following FODMAP restriction, the proportion of patients reporting satisfactory symptom relief significantly increased across all IBS subtypes except IBS-M (Table 1). The highest rates of relief were observed in IBS-C, followed by IBS-U and IBS-D.

Satisfactory relief across all irritable bowel syndrome subtypes

IBS-C experienced the same level of satisfactory relief from FODMAP restriction compared to IBS-D and IBS-U. IBS-M experienced significantly less satisfactory relief than all other IBS subtypes (Table 2).

Table 1. Satisfactory Relief Assessed by the Global Symptom Question at Baseline and Follow-up in Irritable Bowel Syndrome Subtypes Following Fermentable Oligosaccharides, Disaccharides, Monosaccharides, and Polyols Restriction

IBS subtype	Baseline satisfactory reliefa number (%)	Follow-up satisfactory relief[a] number (%)	*P*-value
IBS-C	6/70 (8.6%)	39/70 (55.7%)	< 0.001
IBS-D	12/103 (11.7%)	47/103 (45.6%)	< 0.001
IBS-M	4/32 (12.5%)	8/32 (25%)	0.204
IBS-U	21/87 (24.1%)	49/87 (56.3%)	< 0.001

[a]Satisfactory relief assessed via global symptom question.
IBS, irritable bowel syndrome; IBS-C, constipation-predominant IBS; IBS-D, diarrhea-predominant IBS; IBS-M, IBS with mixed stool consistency (both diarrhea- and constipation-predominant); IBS-U, unclassified IBS (neither diarrhea- nor constipation-predominant).
P < 0.05 was considered statistically significant.

Table 2. Comparison of Satisfactory Relief at Follow-up After Low-fermentable Oligosaccharides, Disaccharides, Monosaccharides, and Polyols Restriction Diet Across Irritable Bowel Syndrome Subtypes

IBS subtype at follow-up	n	Achieved satisfactory relief[a] n (%)	Comparison vs IBS-C	Comparison vs IBS-D	Comparison vs IBS-U
IBS-C	70	39 (55.7%)	–	*P* = 0.1588	*P* = 0.9402
IBS-D	105	47 (44.8%)	*P* = 0.1588	–	*P* = 0.1136
IBS-U	87	49 (56.3%)	*P* = 0.9402	*P* = 0.1136	–
IBS-M	32	8 (25.0%)	*P* = 0.0041	*P* = 0.0463	*P* = 0.0025

[a]Satisfactory relief assessed via global symptom question.
IBS, irritable bowel syndrome; IBS-C, constipation-predominant IBS; IBS-D, diarrhea-predominant IBS; IBS-M, IBS with mixed stool consistency (both diarrhea- and constipation-predominant); IBS-U, unclassified IBS (neither diarrhea- nor constipation-predominant).
P < 0.05 was considered statistically significant.

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Impact of Fermentable Oligosaccharides, Disaccharides, Monosaccharides, and Polyols Restriction on Stool Consistency in Constipation-predominant Irritable Bowel Syndrome and Relationship to Satisfactory Relief

After FODMAP restriction, 50.7% (35/69) of patients with IBS-C remained IBS-C, while 44.9% (31/69) shifted to IBS-U (Figure). Smaller proportions changed to IBS-M (2.9%, 2/69) or IBS-D (1.4%, 1/69). At follow up satisfactory relief was achieved in 45.7% (16/35) of "IBS-C to IBS-C" but in 74.2% (23/31) of "IBS-C to IBS-U" highlighting significantly more improvement in satisfactory relief for IBS-C patients when constipation is also improved ($P = 0.019$) (Figure). Individually both IBS-C to IBS-C and IBS-C to IBS-U achieved significant improvements in satisfactory relief at follow up (both $P < 0.001$).

Gastrointestinal Symptoms

Gastrointestinal symptom response in constipation-predominant irritable bowel syndrome and constipation-predominant irritable bowel syndrome to unclassified irritable bowel syndrome

As we observed increased satisfactory relief in IBS-C patients who changed to IBS-U compared to those who remained constipated (IBS-C to IBS-C), we could assume this was due to the improvement in stool consistency. However, to understand if there were any additional factors that were different between IBS-C to IBS-C and IBS-C to IBS-U that could explain why there was more satisfactory relief, we compared GI symptoms between these 2 groups (Table 3). We did not include the IBS-C to IBS-M or IBS-C to IBS-D due to the small numbers (n = 3).

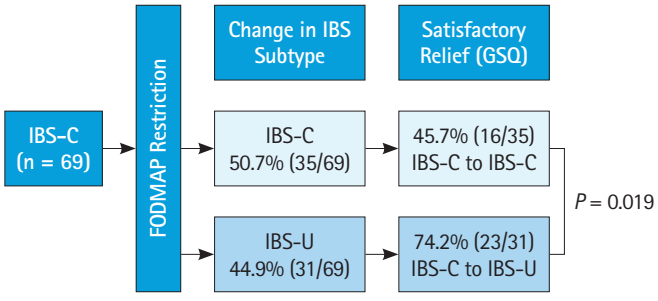

Figure. Impact of fermentable oligosaccharides, disaccharides, monosaccharides, and polyols (FODMAP) restriction on stool consistency in constipation-predominant irritable bowel syndrome (IBS-C) and relationship to satisfactory relief. Among 69 patients with IBS-C who underwent a low-FODMAP diet, 44.9% (31/69) transitioned to an unclassified IBS (IBS-U) subtype. These patients achieved significantly more satisfactory symptom relief compared to those who remained IBS-C ($P = 0.019$). GSQ, global symptom question.

When comparing a reduction in moderate/severe symptoms following FODMAP restriction using the GSRS we found a significant reduction in pain in IBS-C to IBS-U ($P = 0.006$) but not in IBS-C to IBS-C ($P = 0.180$). Bloating/distention, flatulence, borborygmi and lethargy reduced significantly in both groups ($P < 0.05$ for all), while belching, urgency, incomplete evacuation, nausea, heartburn and acid reflux did not reduce significantly in either group (Table 3). Although bloating/distention significantly reduced in both groups from baseline to follow up only 4/31 (13%) with IBS-C to IBS-U experienced bloating, while half still had bloating whom remained IBS-C (17/35, 49%). Similarly, it is worth noting that the symptom of incomplete evacuation did not change in IBS-C to IBS-C (18/35, 51% to 19/35, 54%) following FODMAP restriction, whereas in those who changed to IBS-U this symptom improved in half of patients (12/31, 39% to 6/31, 19%) (Table 3).

Gastrointestinal symptom response in all irritable bowel syndrome-subtypes

Regarding a reduction in moderate/severe GI symptoms measured using the GSRS, all subtypes experienced a reduction in bloating/distention and lethargy ($P < 0.05$ for all). All subtypes except IBS-M experienced a significant decrease in the presence of moderate/severe pain, flatulence, belching and borborygmi ($P < 0.05$ for all). All subtypes except IBS-C experienced a reduction in urgency ($P < 0.05$ for all). All subtypes except IBS-D experienced a reduction in heartburn ($P < 0.05$ for all). Only IBS-D experienced a reduction in incomplete evacuation. Finally, only IBS-U experienced a reduction in acid reflux and nausea (Table 4).

DISCUSSION

This real-world evaluation of dietitian-led group FODMAP education demonstrated that patients with both IBS-C and IBS-D experienced meaningful improvements in global symptoms. Among those with IBS-C, approximately half maintained their subtype following dietary restriction, while nearly half transitioned to a more normal stool pattern (IBS-U). Improvements in stool consistency were accompanied by greater relief of abdominal pain and overall symptoms, indicating that FODMAP restriction may improve outcomes in constipation-predominant IBS without exacerbating constipation.

Results on the FODMAP diet in IBS-C are contradictory. Halmos and colleagues[18] demonstrated the efficacy of a FODMAP diet in a cohort with more IBS-C than IBS-D, with improvements observed in both subjective and objective stool consistency. Despite these seminal results, theories evolved suggesting FODMAP restriction

Table 3. Reduction of Presence of Moderate/Severe Gastrointestinal Symptoms at Baseline Compared to After Fermentable Oligosaccharides, Disaccharides, Monosaccharides, and Polyols Restriction in Constipation-predominant Irritable Bowel Syndrome Who Changed to Unclassified Irritable Bowel Syndrome and Who Remained Constipation-predominant Irritable Bowel Syndrome

Symptom	Group	Baseline n/N (%)	Follow-up n/N (%)	*P*-value
Pain	IBS-C → IBS-C	21/35 (60%)	15/35 (43%)	0.180
	IBS-C → IBS-U	16/31 (52%)	6/31 (19%)	0.006
Bloating/distention	IBS-C → IBS-C	31/35 (89%)	17/35 (49%)	< 0.001
	IBS-C → IBS-U	23/31 (74%)	4/31 (13%)	< 0.001
Flatulence	IBS-C → IBS-C	27/35 (77%)	10/35 (29%)	< 0.001
	IBS-C → IBS-U	18/31 (58%)	5/31 (16%)	0.002
Belching	IBS-C → IBS-C	8/35 (23%)	3/35 (9%)	0.063
	IBS-C → IBS-U	6/31 (19%)	2/31 (6%)	0.219
Borborygmi	IBS-C → IBS-C	15/35 (43%)	4/35 (11%)	< 0.001
	IBS-C → IBS-U	15/31 (48%)	6/31 (19%)	0.012
Urgency	IBS-C → IBS-C	13/35 (37%)	8/35 (23%)	0.180
	IBS-C → IBS-U	10/31 (32%)	6/31 (19%)	0.289
Incomplete evacuation	IBS-C → IBS-C	18/35 (51%)	19/35 (54%)	> 0.999
	IBS-C → IBS-U	12/31 (39%)	6/31 (19%)	0.070
Nausea	IBS-C → IBS-C	10/35 (29%)	8/35 (23%)	0.774
	IBS-C → IBS-U	4/31 (13%)	1/31 (3%)	0.375
Heartburn	IBS-C → IBS-C	9/35 (26%)	3/35 (9%)	0.109
	IBS-C → IBS-U	5/31 (16%)	2/31 (6%)	0.250
Acid reflux	IBS-C → IBS-C	11/35 (31%)	5/35 (14%)	0.109
	IBS-C → IBS-U	4/31 (13%)	3/31 (10%)	> 0.999
Lethargy	IBS-C → IBS-C	26/35 (74%)	17/35 (49%)	0.004
	IBS-C → IBS-U	19/31 (61%)	8/31 (26%)	0.003

IBS-C, constipation-predominant irritable bowel syndrome (IBS); IBS-U, unclassified IBS (neither diarrhea- nor constipation-predominant).
$P < 0.05$ was considered statistically significant.

will exacerbate constipation due to reduction in small bowel water motility content and decreased colonic fermentation,[19] despite no mechanistic studies being performed that can validate this hypothesis. Although reduced fiber intake during FODMAP restriction is often cited as a risk for worsening constipation, evidence remains inconsistent.[20] A controlled feeding trial found that fiber supplementation did not improve bowel symptoms or quality of life despite increased stool water content.[21] In our group sessions, patients were advised to substitute high-FODMAP foods with balanced low-FODMAP alternatives rather than increase fiber specifically. Nonetheless, IBS-C patients improved stool consistency without targeted fiber advice, suggesting this concern may be less clinically significant. Furthermore, in the RCT conducted by Böhn et al,[22] all IBS subtypes experienced a reduction in IBS symptom severity from FODMAP restriction. While the RCT led by Pedersen et al[23] found IBS-C did not improve following FODMAP diet, however, as the authors point out with only 5 IBS-C included the lack of effect is likely due to lack of power. More recently a smartphone app that supports users to lower FODMAP intake (ie, a modified FODMAP diet) improved overall IBS symptom severity in a higher percentage of IBS-C than other IBS subtypes at both short term and long term follow up,[24] providing evidence for manipulating FODMAP intake in the primary care IBS-C patient. One study provided all low FODMAP meals for patients and found the restriction of FODMAPs showed similar benefits for IBS-C and IBS-D.[25] Furthermore in clinical practice group FODMAP education sessions improvements across all IBS subtypes have been previously reported.[12,26] Taken together, and with additional clinical practice data added from our current study, the available evidence suggests the FODMAP diet should be used in IBS-C in clinical practice to improve global symptoms.

A novel finding of this study was that patients who transitioned from IBS-C to IBS-U following FODMAP restriction reported greater satisfactory relief and pain reduction compared with those who

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Table 4. Reduction of Presence of Moderate/Severe Gastrointestinal Symptoms at Baseline Compared to After Fermentable Oligosaccharides, Disaccharides, Monosaccharides, and Polyols Restriction in All Individual Irritable Bowel Syndrome Subtypes

Symptom (GSRS)	Timepoint	IBS-C n (%)	IBS-D n (%)	IBS-M n (%)	IBS-U n (%)
Pain	Baseline	41/70 (59%)	65/102 (64%)	23/34 (68%)	46/87 (53%)
	Follow-up	24/70 (34%)	35/102 (34%)	17/34 (50%)	21/87 (24%)
	P-value	0.002	< 0.001	0.109	< 0.001
Bloating/ Distention	Baseline	58/70 (83%)	80/103 (78%)	27/34 (79%)	55/87 (63%)
	Follow-up	23/70 (33%)	43/103 (42%)	19/34 (56%)	25/87 (29%)
	P-value	< 0.001	< 0.001	0.008	< 0.001
Flatulence	Baseline	49/70 (70%)	68/104 (65%)	24/34 (71%)	59/87 (68%)
	Follow-up	17/70 (24%)	38/104 (37%)	17/34 (50%)	18/87 (21%)
	P-value	< 0.001	< 0.001	0.092	< 0.001
Belching	Baseline	16/70 (23%)	36/103 (35%)	15/33 (45%)	26/88 (29%)
	Follow-up	6/70 (9%)	16/103 (16%)	14/33 (42%)	12/88 (14%)
	P-value	0.013	< 0.001	> 0.999	0.003
Borborygmi	Baseline	31/69 (44%)	55/101 (54%)	17/33 (52%)	35/86 (41%)
	Follow-up	11/69 (16%)	29/101 (29%)	15/33 (45%)	18/86 (21%)
	P-value	< 0.001	< 0.001	0.774	0.002
Urgency	Baseline	26/70 (37%)	79/106 (74%)	26/34 (76%)	32/87 (37%)
	Follow-up	17/70 (24%)	47/106 (44%)	17/34 (50%)	20/87 (23%)
	P-value	0.064	< 0.001	0.004	0.031
Incomplete Evacuation	Baseline	33/70 (47%)	49/103 (48%)	20/32 (63%)	29/86 (34%)
	Follow-up	28/70 (40%)	35/103 (34%)	15/32 (47%)	21/86 (24%)
	P-value	0.405	0.026	0.125	0.152
Nausea	Baseline	15/69 (21%)	20/104 (19%)	15/32 (47%)	23/87 (26%)
	Follow-up	11/69 (16%)	16/104 (15%)	9/32 (28%)	12/87 (14%)
	P-value	0.481	0.503	0.070	0.013
Heartburn	Baseline	14/70 (20%)	24/103 (23%)	15/33 (45%)	14/87 (16%)
	Follow-up	5/70 (7%)	17/103 (16%)	3/33 (9%)	6/87 (7%)
	P-value	0.022	0.230	< 0.001	0.008
Acid Reflux	Baseline	15/69 (22%)	22/103 (21%)	14/32 (44%)	19/87 (21%)
	Follow-up	8/69 (12%)	15/103 (15%)	8/32 (25%)	5/87 (6%)
	P-value	0.092	0.167	0.070	< 0.001
Lethargy	Baseline	47/70 (67%)	82/105 (78%)	32/34 (94%)	53/87 (61%)
	Follow-up	28/70 (40%)	53/105 (50%)	23/34 (68%)	32/87 (37%)
	P-value	< 0.001	< 0.001	0.012	< 0.001

GSRS, gastrointestinal symptom rating scale; IBS-C, constipation-predominant irritable bowel syndrome (IBS); IBS-D, diarrhea-predominant IBS; IBS-M, IBS with mixed stool consistency (both diarrhea- and constipation-predominant); IBS-U, unclassified IBS (neither diarrhea- nor constipation-predominant). $P < 0.05$ was considered statistically significant.

remained IBS-C (Figure). Previous evidence indicates that reductions in constipation can predict decreases in abdominal pain,[27] which may explain the higher rates of satisfactory relief among patients whose bowel habits normalized. Clinically, those who shifted from IBS-C to IBS-U also showed larger, though not statistically significant, reductions in bloating, distension, and incomplete evacuation (Table 3), all factors likely contributing to overall symptom improvement. Consistent with our findings, real-world studies have similarly reported improvements in stool consistency following FODMAP restriction. Nawawi et al[28] observed normalization

from IBS-C (BSFS 1-2) to IBS-U (BSFS 3-4) in 47% of patients, while De Roest et al[29] reported improvement in 52% of those with hard stools prior to the diet. Likewise, in our cohort, 45% (31/69) of IBS-C patients transitioned to IBS-U following FODMAP restriction (Figure).

Future studies should clarify whether improvements in stool consistency, pain, or the combination of both were responsible for increased satisfactory relief. According to the U.S. Food and Drug Administration (FDA) guidance for IBS-C therapeutics, a clinical "response" is defined by concurrent improvement in stool consistency (or frequency) and at least a 30% reduction in abdominal pain.[30] Conley et al[31] have proposed applying these more rigorous FDA-defined endpoints to low FODMAP diet trials, as their adoption could enhance the assessment of efficacy and better distinguish responders from non-responders.

Despite FODMAP restriction, some patients remained IBS-C and continued to report incomplete evacuation and pain (Table 3). This is clinically relevant as it highlights a group of patients who require further support. IBS guideline management suggest introducing laxatives after dietary advice and suggest further investigation for symptoms suggestive of a defecatory disorder,[7,8] while further dietary interventions specifically aimed at constipation could be implemented.[32] Data on laxative and/or medication use and/or overlapping anorectal disorders were not collected, preventing further analysis of their potential impact on constipation symptoms.

The limitations of an observational study conducted in a single center in routine clinical practice must be acknowledged. Stool consistency measures (BSFS) to determine IBS subtype were only taken once at baseline and once at follow up. Whereas 2 weeks is suggested to define IBS subtypes in research,[33] therefore the accuracy of the IBS subtypes has limitations. How much the change we observed in subtype was due to natural fluctuations,[34] and not FODMAP restriction, is impossible to say without a control group, as is any placebo effect. Additionally, there are a range of factors that could have influenced outcomes of satisfactory relief and changes in stool consistency in IBS-C not related to the FODMAP diet. For example, differences in baseline diets, differences in FODMAP intake during the diet, fibre intake during FODMAP restriction, use/changes of medications (either laxatives to improve constipation or other medications that may have contributed to constipation), and probiotics/prebiotics, all of which were all not assessed in this study and may explain individual patient discrepancies which should be evaluated in controlled trials.

In conclusion, overall 55.7% of IBS-C achieved satisfactory relief of symptoms following FODMAP restriction in a dietitian led group education setting, with no difference compared to IBS-D. Additionally 45.7% changed stool pattern from abnormal (IBS-C) to normal (IBS-U), highlighting FODMAP restriction does not necessarily increase constipation in IBS-C. Higher rates of satisfactory relief and pain reduction were achieved in IBS-C who improved stool consistency to IBS-U when compared to IBS-C who remained constipated. This novel finding should be further explored to help understand response rates. Despite encouraging findings from our clinical practice and existing literature, controlled studies evaluating the efficacy of the FODMAP diet in IBS-C compared with IBS-D remain limited. Moreover, the long-term effectiveness and sustainability of this dietary approach are poorly understood, underscoring an important gap between research and clinical practice. For clinical practice, clinicians should not shy away from offering a dietitian led FODMAP diet to the appropriate patient with IBS-C. Monitoring the diets impact at follow up is essential, especially considering some patients may continue with hard stool consistency that may impact overall satisfactory relief and/or reduction in pain, which may require further targeted dietary or pharmacological treatment.

SUPPLEMENTARY MATERIAL

Note: To access the supplementary information mentioned in this article, visit the online version of *Journal of Neurogastroenterology and Motility* at http://www.jnmjournal.org/, and at http://doi.org/10.5056/jnm25126.

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Conflicts of interest
None.

Author contributions
Lee David Martin conceptualized the study, collected the data, and prepared the manuscript; Konstantinos C Fragkos reviewed the manuscript; and Konstantinos C Fragkos and Lee David Martin analysed the data and approved the final draft manuscript.

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SUPPLEMENTARY MATERIAL

Supplementary Information

Dietary adherence

Dietary adherence was assessed in 292 of 294 patients (2 were excluded due to missing data). The overall adherence rate was 78% (230/292).

Low Fermentable Oligosaccharides, Disaccharides, Monosaccharides, and Polyols Diet Adherence by Irritable Bowel Syndrome Subtype

IBS Subtype	Non-Adherent (n)	Non-Adherent (%)	Adherent (n)	Adherent (%)	Total (n)
IBS-C	8	11.6%	61	88.4%	69
IBS-D	27	26.5%	75	73.5%	102
IBS-M	8	23.5%	26	76.5%	34
IBS-U	20	23.0%	67	77.0%	87
Total	63	21.6%	229	78.4%	292

IBS, irritable bowel syndrome; IBS-C, constipation-predominant IBS; IBS-D, diarrhea-predominant IBS; IBS-M, IBS with mixed stool consistency (both diarrhea- and constipation-predominant); IBS-U, unclassified IBS (neither diarrhea- nor constipation-predominant).

Low Fermentable Oligosaccharides, Disaccharides, Monosaccharides, and Polyols Diet Adherence by Change in Irritable Bowel Syndrome Subtype

IBS subtype change	Non-adherent (n, %)	Adherent (n, %)	Total (n)
IBS-C → IBS-C	5 (14.7%)	29 (85.3%)	34
IBS-C → IBS-U	1 (3.3%)	29 (96.7%)	30
IBS-C → IBS-M	1 (50.0%)	1 (50.0%)	2
IBS-C → IBS-D	0 (0.0%)	1 (100.0%)	1
Total (IBS-C group)	7 (10.4%)	60 (89.6%)	67
IBS-U → IBS-U	15 (24.6%)	46 (75.4%)	61
IBS-U → IBS-C	4 (23.5%)	13 (76.5%)	17
IBS-U → IBS-M	0 (0.0%)	1 (100.0%)	1
IBS-U → IBS-D	1 (20.0%)	4 (80.0%)	5
Total (IBS-U group)	20 (23.8%)	64 (76.2%)	84

IBS, irritable bowel syndrome; IBS-C, constipation-predominant IBS; IBS-D, diarrhea-predominant IBS; IBS-M, IBS with mixed stool consistency (both diarrhea- and constipation-predominant); IBS-U, unclassified IBS (neither diarrhea- nor constipation-predominant).

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The Effects of Physical Activity on Functional Dyspepsia: A Mendelian Randomization and Randomized Controlled Study

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Background/Aims: The relationship between physical activity (PA) and functional dyspepsia (FD) remains unclear due to a lack of robust evidence. We conducted a Mendelian randomization (MR) analysis to assess potential genetic associations, followed by a randomized controlled trial (RCT) assessing the therapeutic effects of moderate-intensity exercise in FD patients.

Methods: In the MR analysis, genetic variants associated with PA traits from genome-wide association studies were analyzed. For the RCT, 260 FD patients were randomized to moderate-intensity exercise intervention or a control group. The primary endpoint was the adequate relief rate of dyspeptic symptoms.

Results: Using MR, a potential inverse association of strenuous sports activity against the development of FD was estimated (inverse variance weighte OR = 0.92, 95% CI = 0.88-0.98, $P = 0.0047$). Additionally, a suggestive inverse association was observed between other exercises (eg, swimming, cycling, keep fit, and fishing) (inverse variance weighte OR = 0.964; 95% CI, 0.937-0.993; $P = 0.013$) and FD risk. In the previous high-quality study, other exercises (eg, swimming, cycling, keep fit, and fishing) were considered moderate intensity exercise. Therefore, the MR study suggest that moderate intensity exercise may reduce the occurrence of FD. In the subsequent RCT, 114 participants in the exercise intervention group and 113 in the control group completed the study. The exercise group demonstrated a significantly higher adequate relief rate (46.9% vs 30.8%, $P = 0.008$) and greater reductions in symptom scores (all $P < 0.05$) versus controls.

Conclusions: PA, particularly moderate-intensity exercise, may reduce the risk of FD.

Keywords: Association; Functional dyspepsia; Mendelian randomization; Physical activity; Randomized controlled study

INTRODUCTION

Functional dyspepsia (FD) is a prevalent functional gastrointestinal disorder characterized by chronic or recurrent epigastric pain or burning, postprandial fullness, and early satiety.[1] With a global prevalence of approximately 16%,[2] FD imposes a substantial burden on healthcare systems and significantly impairs patients' quality of life. The chronic and relapsing nature of the disease leads to persistent health deterioration, prompting over 40% of affected individuals to seek hospital-based care, resulting in considerable socioeconomic costs.[3] Current therapeutic approaches for FD demonstrate limited efficacy, primarily offering only short-term symptomatic relief,[4] underscoring the need for more effective and sustainable treatment strategies with minimal adverse effects.

Emerging evidence suggests that physical activity (PA) confers systemic health benefits, including positive effects on cardiovas-

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cular, neurological, and respiratory systems. Previous studies have demonstrated that regular exercise can enhance gastrointestinal motility and alleviate abdominal distension in healthy individuals,[5] while improving bowel function and colonic transit time in patients with chronic constipation.[6] However, the potential therapeutic role of PA in FD remains underexplored. To date, only one small-scale randomized controlled trial (n = 72) conducted in India has evaluated exercise interventions for FD based on Rome IV criteria, demonstrating positive effects of aerobic exercise combined with conventional pharmacotherapy.[7] Given the study's limited sample size and short intervention period, further investigation is warranted to establish the efficacy of PA in FD management.

In this study, we employed a comprehensive approach to investigate the relationship between PA and FD risk. First, we performed Mendelian randomization (MR) analysis in a European population to examine potential genetic associations between PA levels and FD susceptibility. Subsequently, we conducted a randomized controlled trial to evaluate the therapeutic effects of moderate-intensity exercise on FD symptoms. We hypothesized that both genetically predicted PA levels and structured exercise interventions would demonstrate a protective association against FD development in the general population.

MATERIALS AND METHODS

Patients and Methods

Mendelian Randomization Study

Study design

The instrumental variables needed to meet the 3 assumptions: (1) genetic variants are significantly associated with physical activity, (2) they are not affected by potential confounding factors, and (3) they influence FD only through physical activity.

Genome-wide association study summary data for physical activity and functional dyspepsia

We drew summary data relative to PA from genome-wide association studies (GWAS) provided by Medical Research Council Integrative Epidemiology Unit (MRC-IEU) Consortium (n = 460 376). The genetic and phenotypic data used in this GWAS were obtained from the UK Biobank, a large-scale, population-based prospective study designed to investigate the genetic and environmental determinants of chronic diseases.[8] This resource includes extensive health and lifestyle data from over 500 000 participants, along with genomic information, enabling comprehensive analyses of complex traits. We examined 6 self-reported physical activity phenotypes, assessed via UK Biobank questionnaires. These included: (1) Heavy do-it-yourself (DIY) activities (eg, weeding, lawn mowing, carpentry, and digging, ID: ukb-b-13184); (2) Light DIY activities (eg, pruning and watering the lawn, ID: ukb-b-11495); (3) None of the heavy DIY or light DIY activities (ID: ukb-b-15869); (4) Strenuous sports (ID: ukb-b-7663); (5) Walking for pleasure (ID: ukb-b-7337); and (6) Other exercises (eg: swimming, cycling, keep fit, and bowling, ID: ukb-b-8764). The physical activity assessment was based on participant self-reports regarding their engagement in these activities over the past 4 weeks. This approach aligns with validated physical activity measurement tools.[9] The GWAS data for FD were obtained from the MRC-IEU Consortium (n = 463 010; ID: ukb-b-14814), with cases identified through ICD-10 code K30. All participants included in the MR analysis were of European ancestry. It should be noted that both the physical activity GWAS and the FD GWAS were derived from the UK Biobank cohort. This may lead to sample overlap, which can potentially introduce bias into MR estimates. The original GWAS was approved by relevant ethics committees, and informed consent was provided by all subjects (Supplementary Table).

Availability of data and materials

The data in the MR analysis could be attained from GWAS provided by MRC-IEU Consortium (Heavy DIY ID: ukb-b-13184; Light DIY ID: ukb-b-11495; None of the Heavy DIY or Light DIY ID: ukb-b-15869; Strenuous sports ID: ukb-b-7663; Walking for pleasure ID: ukb-b-7337; Other exercises eg: swimming, cycling, keep fit, bowling ID: ukb-b-8764; FD ID: ukb-b-14814). Data from the randomized controlled study section are available from the corresponding author on reasonable request.

Selection of genetic instruments

To construct the genetic instrument for physical activity, we selected single-nucleotide polymorphisms (SNPs) at the genome-wide significance level ($P \leq 5 \times 10^{-8}$). Independent SNPs were identified by applying a stringent linkage disequilibrium threshold ($r^2 < 0.001$) within a 10 000 kb sliding window. Palindromic SNPs with ambiguous strand orientation were excluded to ensure robustness. The genetic instruments need to meet the required threshold of $F > 10$ to ensure the powerful validity and the F statistic was calculated using the formula $F = (\beta/SE)$.[2,10]

Mendelian randomization analyses

The inverse variance weighted (IVW) method served as the primary analytical approach. Heterogeneity was assessed using Cochran's Q test, with a significance threshold of $P < 0.05$. In the

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absence of heterogeneity, a fixed-effects model was applied; otherwise, a random-effects model was employed.[11] Complementary analyses included MR-Egger regression, weighted median, weighted mode, and simple mode methods. A significant association was defined as $P < 0.008$ (0.05/6 exposures), while associations with *P*-values between 0.008 and 0.05 were considered suggestive.[12] A positive causal association was inferred if the IVW results were statistically significant ($P < 0.008$), even if other methods did not reach significance, provided that no substantial pleiotropy or heterogeneity was detected and the effect direction OR was consistent across all methods.[11,13] If the OR value directions of these methods were inconsistent, a stricter threshold for the *P*-value ≤ 1×10^{-8} of the instrumental variables was set.[13,14] Mendelian randomization pleiotropy residual sum and outlier (MR-PRESSO) was used to detect the existence of outliers, and the intercept obtained by MR-Egger regression determines whether there was pleiotropy, $P < 0.05$ was considered significant. The leave-one-out analysis was used to evaluate the stability of MR Results.[12] All the analyses were performed by R version 4.2.2 (R Foundation for Statistical Computing, Vienna, Austria).

Randomized Controlled Study

Study design

We conducted a single-center, randomized controlled trial involving outpatients diagnosed with FD. Eligible participants were enrolled between August 2020 and December 2021. Following screening based on predefined inclusion and exclusion criteria, patients were randomly assigned (1:1) to either the exercise intervention group or the control group using a block randomization method (block size = 4). Allocation concealment was ensured through sealed opaque envelopes. Written informed consent was obtained from all participants prior to enrollment.

Following randomization, participants in the exercise group received weekly telephone consultations from an exercise specialist to guide them in performing moderate-intensity exercise and to monitor adherence. In line with the methodology of Johannesson et al,[15] the control group also received weekly phone calls to encourage maintenance of their usual lifestyle. Neither group received any pharmacological intervention for FD. To enhance study retention, control group participants were offered post-study exercise counseling and individualized exercise prescriptions. Outcome assessors and data analysts were blinded to group allocation. The intervention duration was 3 months. Since the MR study data we used were from public databases, no additional ethical approval from the institutional review board was required. The randomized controlled study was approved by the Ethics Committee of the Second Affiliated Hospital of Xi'an Jiaotong University (Approval No. 2020031) and registered at ClinicalTrials.gov (NCT04540549; registered 29 August 2020), and all participants provided written informed consent.

Inclusion criteria

Eligible participants met the following criteria: (1) age ≥ 18 years old; (2) diagnosed with FD according to the Rome IV criteria (Supplementary Table); (3) normal findings on gastroscopy, complete blood count, liver function tests, and *Helicobacter pylori* testing within the past year; (4) no use of prokinetic agents, proton pump inhibitors, or mucoprotective drugs within the preceding 2 weeks; (5) provided written informed consent and agreed to participate in the study; and (6) capable of engaging in moderate-intensity exercise.

Exclusion criteria

Participants were excluded if they met any of the following criteria: (1) There were organic diseases that may lead to dyspeptic symptoms, such as digestive tract ulcers and tumors, pre-existing hepatobiliary, pancreatic, malignant, or metabolic diseases[16]; (2) current pregnancy, lactation, or perioperative pregnancy planning; (3) history of abdominal surgery; (4) psychiatric or psychological disorders; (5) severe cardiac, hepatic, renal or respiratory dysfunction; (6) recent use of anxiolytics, antidepressants, corticosteroids, or nonsteroidal anti-inflammatory drugs (NSAIDs); and (7) predominant symptoms suggestive of gastroesophageal reflux disease (GERD).[17]

Data collection

(1) Demographic and anthropometric characteristics: age, height, weight, sex, marriage, education level; (2) lifestyle factors: smoking and drinking; (3) symptom assessment questionnaire: weekly overall dyspeptic symptoms assessment,[18] 8 dyspeptic symptom score questionnaire, short-form leeds dyspepsia questionnaire (SF-LDQ), short-form nepean dyspepsia index questionnaire (SFNDI), hospital anxiety and depression scale (HADS); and (4) safety and follow-up: adverse events and loss to follow-up.

Moderate-intensity exercise

According to the American College of Sports Medicine (ACSM), moderate-intensity exercise is recommended for maintaining cardiopulmonary health.[19] Exercise intensity was assessed using the Borg Rating of Perceived Exertion (RPE) scale (6-20), with moderate intensity defined as fairly light (RPE 11-12) to somewhat hard (RPE 13-14).[20] The prescribed exercise regimen included: frequency: ≥ 5 sessions per week; duration: 30-60 minutes per session[19];

type: jogging or cycling; and intervention period: 3 months.

Questionnaires

Eight-item dyspepsia symptom score questionnaire. Eight-item dyspepsia symptom score questionnaire was used to assess the severity of dyspeptic symptoms, including epigastric pain, burning, postprandial fullness, early satiety, belching, bloating, nausea, and vomiting, with each symptom scoring range of 0-3 (0 = absent; 1 = mild; 2 = moderate; 3 = severe and interfering with daily activities).[21,22]

Short-form leeds dyspepsia questionnaire. Comprising 8 items, the SF-LDQ is designed to assess dyspeptic symptoms. The tool measures the frequency and severity of four manifestations: indigestion, heartburn, regurgitation, and nausea. The SF-LDQ scores ranged from 0 to 32, with higher scores corresponding to more severe dyspepsia. A score of 0 was defined as no dyspepsia, a score of 1 to 4 was defined as very mild dyspepsia, a score of 5 to 8 was defined as mild dyspepsia, a score of 9 to 15 was defined as moderate dyspepsia, and a score of 15 or above was defined as severe dyspepsia.[23,24]

Short-form nepean dyspepsia index questionnaire. SF-NDI was a 10-item questionnaire, including 5 subscales: tension/anxiety, interference with daily activities, eat/drink, knowledge/control, and work and study. Each subscale was composed of 2 items designed to evaluate the effect of dyspepsia on different aspects of patients' quality of life. Every item was rated on a scale from 1 (indicating no impact or not applicable) to 5 (representing severe impact). The scores for all items within a subscale were summed, resulting in a total SF-NDI score that could vary from a minimum of 10 points to a maximum of 100 points. A larger total score indicated a worse living standard.[25]

Hospital anxiety and depression scale. HADS included 2 subscales: hospital anxiety and hospital depression. Each subscale consisted of 7 items, totaling 14 items. Each item was scored from 0 to 3, with higher scores indicating higher levels of anxiety or depression. The total score of the 2 subscales of anxiety and depression was 21 points each, where 0-7 means no anxiety or depression, 8-10 means mild anxiety or depression, and 11-14 means moderate anxiety or depression, with a score of 15 to 21 indicating severe anxiety or depression.[15]

Study endpoints

The primary endpoint was the adequate relief rate of dyspeptic symptoms. This was assessed weekly by asking patients: "Over the past 7 days, have you had adequate relief of your dyspepsia symptoms?" with a binary (Yes/No) response. After a 3-month follow-up, adequate relief was defined as a "Yes" response in at least 50% of the weekly assessments between weeks 3 and 12 (total of 10 weeks, excluding the initial two weeks to ensure treatment stability).[18] Secondary endpoints included changes in the 8-item dyspepsia symptom score, SF-LDQ score, SF-NDI score, and the HADS score.

Sample size calculation

The sample size was determined based on the anticipated adequate relief rate of dyspeptic symptoms, using a superiority/non-inferiority clinical trial design with reference to previous studies.[18] Assuming an adequate relief rate of 40% in the control group ($P0$), we hypothesized a 20% improvement (absolute increase) in the intervention group ($P1$). With a two-sided significance level (α) of 0.05 and a power (1-β) of 90% ($\beta = 0.1$), the required sample size per group was calculated using the following formula: $N = (U\alpha+U\beta)^2 2P(1-P)/(P1-P0)^2$, $P = (P0+P1)/2$. The initial calculation yielded 108 participants per group. Accounting for an estimated 20% dropout rate, the total required sample size was 259 (rounded to 260 for balanced allocation). Thus, this study enrolled 260 participants, with 130 randomized to the moderate-intensity exercise group and 130 to the control group.

Statistical methods

All statistical analyses were performed using EmpowerStats and SPSS 20.0. Categorical variables were expressed as frequencies and percentages, and group comparisons were conducted using the chi-square test. When the expected cell count was less than 10, Fisher's exact test was applied instead. Continuous variables were presented as mean ± SD, and differences between groups were assessed using Student's t test or nonparametric tests, as appropriate. All analyses followed the intention-to-treat principle to minimize bias. The last observation carried forward method was applied to handle missing values. Graphical representations were generated using GraphPad Prism 8.0. A two-sided P-value < 0.05 was considered statistically significant.

RESULTS

Mendelian Randomization Study

Study design

In the MR analysis, we included 7662 FD cases and 455,348 normal controls.

After applying stringent SNP selection criteria, we identified the following instrument variables: Heavy DIY activity: 18 SNPs (F-statistic range: 30.6-51.7); Light DIY: 12 SNPs (F-statistic range: 29.8-84.9); No DIY (neither heavy nor light): 5 SNPs (F-statistic

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range: 30.9-39.3); Strenuous sports: 6 SNPs (F-statistic range: 31.9-49.2); Walking for pleasure: 20 SNPs (F-statistic range: 30.5-38.8); Other exercises (eg, swimming, cycling, keep fit, and bowling): 14 SNPs (F-statistic range: 30.3-61.1). All selected SNPs exhibited strong associations with their respective physical activity exposure traits (F-statistics > 10), indicating minimal risk of weak instrument bias. Detailed data sources and variant IDs are provided in Supplementary Table.

Causal effect of heavy do-it-yourself on functional dyspepsia

Q values of MR-Egger and IVW showed no significant heterogeneity between Heavy DIY and FD (P = 0.564 and 0.590, respectively) and fixed effect model was used. The IVW method indicated a suggestive inverse association (OR = 0.974; 95% CI, 0.954-0.994; P = 0.012) (Table 1). However, the effect direction of the IVW estimate was inconsistent with that of MR-Egger. To enhance robustness, we applied a stricter instrument selection threshold ($P \leq 1\times10^{-8}$), retaining 9 SNPs for analysis. Neither MR-Egger nor IVW detected significant heterogeneity ($P > 0.05$), and the fixed-effect model was again employed. In this study, no significant link was observed between Heavy DIY and FD using the IVW method (OR = 0.975; 95% CI: 0.948-1.003; P = 0.076) (Table 1 and Supplementary Fig. 1A). Horizontal pleiotropy was evaluated via MR-Egger intercept (P = 0.858) and MR-PRESSO (P = 0.560), neither of which indicated the presence of pleiotropic bias or outliers. Leave-one-out sensitivity analysis revealed that rs674094 disproportionately influenced the results; its exclusion resulted in a statistically significant association between Heavy DIY and FD (Supplementary Fig. 2A).

Causal effect of light do-it-yourself on functional dyspepsia

According to Cochran Q-test, the P-value of MR-Egger was 0.749 and that of IVW was 0.816, indicating no significant heterogeneity, and fixed effect model was used. Our MR analysis revealed no significant causal effect of Light DIY on FD (IVW OR = 0.994; 95% CI: 0.972-1.017; P = 0.600) (Table 1 and Supplementary Fig. 1B). The MR-Egger regression indicated no substantial horizontal pleiotropy (P for intercept = 0.851), and the MR-PRESSO test detected no outlier SNPs (P = 0.812). Sensitivity analysis using the leave-one-out method confirmed that no individual SNP exerted a dominant effect on the overall association (Supplementary Fig. 2B).

Causal effect of none of the heavy do-it-yourself or light do-it-yourself on functional dyspepsia

According to Cochran Q-test, the P-value of MR-Egger was 0.065 and that of IVW was 0.14, indicating no significant heterogeneity, and fixed effect model was used. We found that none of the Heavy DIY or Light DIY had no causality on FD (IVW OR = 1.059; 95% CI, 0.977-1.149; P = 0.161) (Table 1 and Supplementary Fig. 1C). MR-Egger regression detected no evidence of directional pleiotropy (P intercept = 0.773), and the MR-PRESSO test confirmed the absence of outlier SNPs (P = 0.158). Sensitivity analysis using the leave-one-out method indicated that no single genetic variant disproportionately influenced the results (Supplementary Fig. 2C).

Causal effect of strenuous sports on functional dyspepsia

The MR analysis showed no evidence of significant heterogeneity in the causal estimate of strenuous sports on FD (Cochran's Q-test: P = 0.059 for MR-Egger, P = 0.057 for IVW). Therefore, a fixed-effect model was applied. The IVW method revealed a potential inverse association of strenuous sports on FD (OR = 0.924; 95% CI: 0.875-0.976; P = 0.005) (Table 1 and Supplementary Fig. 1D), with consistent directionality across other MR methods. No significant horizontal pleiotropy was detected (MR-Egger intercept P = 0.44), and the MR-PRESSO global test (P = 0.094) confirmed the absence of outlier variants. However, leave-one-out sensitivity analysis identified rs12571549 as a highly influential variant; its exclusion nullified the association between strenuous sports and FD (Supplementary Fig. 2D), suggesting that the overall effect may be driven by this single genetic instrument.

Causal effect of walking for pleasure on functional dyspepsia

No significant heterogeneity was observed between walking for pleasure and FD, as indicated by the Q values of MR-Egger (P = 0.477) and IVW (P = 0.421), supporting the use of a fixed-effect model. The IVW method revealed a suggestive inverse association (OR = 0.975; 95% CI, 0.955-0.996; P = 0.022) (Table 1). However, the effect direction of the IVW estimate was inconsistent with that of MR-Egger. To enhance robustness, we applied a stricter instrument selection threshold ($P \leq 1\times10^{-8}$), retaining 12 SNPs for analysis. Again, no significant heterogeneity was detected (MR-Egger and IVW Q-test $P > 0.05$), and a fixed-effect model was employed. No significant association was observed for pleasure walking in relation to FD (IVW OR = 0.988; 95% CI, 0.962-1.015; P = 0.377) (Table 1 and Supplementary Figure 1E). Furthermore, the MR-Egger intercept test provided no evidence of horizontal pleiotropy (P = 0.544), and MR-PRESSO analysis confirmed the absence of outlier variants (P = 0.242). The robustness of the results was confirmed by a leave-one-out sensitivity analysis, which showed that they were not driven by any individual SNP (Supplementary Fig. 2E).

Table 1. Mendelian Randomization Results for the Relationship Between Physical Activity and Functional Dyspepsia

Exposure	Method	OR (95% CI)	*P*-value	SNPs	Heterogeneity test (*P*-value)	Pleiotropy test (*P*-value)
Heavy DIY before tightening threshold	IVW (fixed)	0.974 (0.954-0.994)	0.012	18	0.590	
	MR Egger	1.026 (0.899-1.170)	0.710	18	0.564	0.446
	Weighted median	0.966 (0.938-0.994)	0.019	18		
	Simple mode	0.965 (0.915-1.018)	0.214	18		
	Weighted mode	0.964 (0.917-1.014)	0.176	18		
Heavy DIY after tightening threshold	IVW (fixed)	0.975 (0.948-1.003)	0.076	9	0.565	
	MR Egger	1.001 (0.757-1.323)	0.995	9	0.460	0.858
	Weighted median	0.964 (0.929-1.000)	0.052	9		
	Simple mode	0.958 (0.902-1.018)	0.202	9		
	Weighted mode	0.959 (0.903-1.018)	0.206	9		
Light DIY	IVW (fixed)	0.994 (0.972-1.017)	0.600	12	0.816	
	MR Egger	0.984 (0.887-1.092)	0.769	12	0.749	0.851
	Weighted median	0.990 (0.961-1.020)	0.519	12		
	Simple mode	0.975 (0.929-1.023)	0.318	12		
	Weighted mode	0.987 (0.946-1.030)	0.567	12		
None of the Heavy DIY or Light DIY	IVW (fixed)	1.059 (0.977-1.149)	0.161	5	0.140	
	MR Egger	0.838 (0.194-3.626)	0.829	5	0.065	0.773
	Weighted median	1.037 (0.928-1.158)	0.524	5		
	Simple mode	1.015 (0.866-1.191)	0.862	5		
	Weighted mode	1.003 (0.855-1.176)	0.976	5		
Strenuous sports	IVW (fixed)	0.924 (0.875-0.976)	0.005	6	0.057	
	MR Egger	0.790 (0.544-1.146)	0.282	6	0.059	0.44
	Weighted median	0.929 (0.866-0.995)	0.037	6		
	Simple mode	0.925 (0.823-1.040)	0.248	6		
	Weighted mode	0.925 (0.833-1.026)	0.202	6		
Walking for pleasure before tightening threshold	IVW (fixed)	0.975 (0.955-0.996)	0.022	20	0.421	
	MR Egger	1.190 (0.896-1.580)	0.246	20	0.477	0.186
	Weighted median	0.979 (0.949-1.010)	0.184	20		
	Simple mode	0.994 (0.938-1.052)	0.829	20		
	Weighted mode	0.998 (0.943-1.055)	0.935	20		
Walking for pleasure after tightening threshold	IVW (fixed)	0.988 (0.962-1.015)	0.377	12	0.208	
	MR Egger	1.139 (0.730-1.776)	0.579	12	0.177	0.544
	Weighted median	1.004 (0.966-1.042)	0.854	12		
	Simple mode	1.010 (0.948-1.076)	0.761	12		
	Weighted mode	1.010 (0.949-1.073)	0.770	12		
Other exercises (eg, swimming, cycling, keep fit, and bowling)	IVW (random)	0.964 (0.937-0.993)	0.013	14	0.038	
	MR Egger	0.885 (0.709-1.104)	0.299	14	0.035	0.457
	Weighted median	0.966 (0.936-0.997)	0.031	14		
	Simple mode	0.970 (0.920-1.023)	0.284	14		
	Weighted mode	0.970 (0.926-1.017)	0.225	14		

DIY, do-it-yourself; MR, Mendelian randomization; IVW, inverse variance weighted; SNPs, single-nucleotide polymorphisms.
If there was no heterogeneity, a fixed effects model was used, and if heterogeneity was found, a random effects model was used. The tighten threshold of the instrumental variables was $P \leq 1 \times 10^{-8}$.

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Causal effect of other exercises (eg, swimming, cycling, keep fit, and bowling) on functional dyspepsia

According to Cochran Q-test, the *P*-value of MR-Egger was 0.035 and that of IVW was 0.038, indicating the existence of heterogeneity, and random effect model was used. The IVW method demonstrated a suggestive protective effect of other exercises on FD (OR = 0.964; 95% CI, 0.937-0.993; *P* = 0.013) (Table 1 and Supplementary Fig. 1F). A suggestive association was also observed using the weighted median estimator (OR = 0.966; 95% CI, 0.936-0.997; *P* = 0.031). MR-Egger regression detected no evidence of directional pleiotropy (*P* intercept = 0.457), and the MR-PRESSO method identified no outlier SNPs. Leave-one-out sensitivity analysis revealed that no SNP was found to exert a disproportionate impact on the overall results (Supplementary Fig. 2F).

Randomized Controlled Study

Baseline of patient characteristics

The MR study analysis revealed that strong sports and other exercises (eg, swimming, cycling, keep fit, and fishing) were associated with a reduced risk of FD at the genetic level. Notably, the keep fit exercise recommended by the American College of Sports Medicine refers to moderate intensity exercise.[19] Supporting this, prior high-quality research has classified other exercises (eg: swimming, cycling, keep fit, fishing) as moderate-intensity exercise.[26] This categorization aligns with previous literature that, based on Metabolic Equivalent of Task values, classified a similar composite variable including these activities as representing moderate-intensity physical activity. The evidence indicates that moderate-intensity exercise could mitigate the risk of developing FD. To further investigate this association, we conducted a RCT assessing the therapeutic effects of moderate-intensity exercise on FD symptoms.

From August 2020 to December 2021, a total of 260 patients who met the inclusion and exclusion criteria were collected. The block randomization method was used to randomly assign the patients to the moderate intensity exercise group and the control group in a 1:1 ratio with a block length of 4. During the 3-month intervention period, a total of 33 patients were lost to follow-up, including 16 patients in the exercise group and 17 patients in the control group. Finally, 114 patients (57.89% patients were female, average age was 44.36 ± 11.44 years) in the exercise group completed the study, and 113 patients (56.64% patients were female, average age was 47.35 ± 11.83 years) in the control group completed the study. The research flowchart was shown in Figure 1.

Among FD subtypes, overlapping syndrome was the most prevalent in the exercise group (39.23%, 51/130), whereas epigastric pain syndrome was most common in the control group (40.00%, 52/130). Comparative analysis of baseline characteristics—including sex, age, body mass index, location, marriage, currently smoking, currently drinking, education level, daily exercise duration, and FD subtype—showed no statistically significant differences between groups (*P* > 0.05) (Table 2). These findings confirm that the 2 groups were well-balanced at baseline.

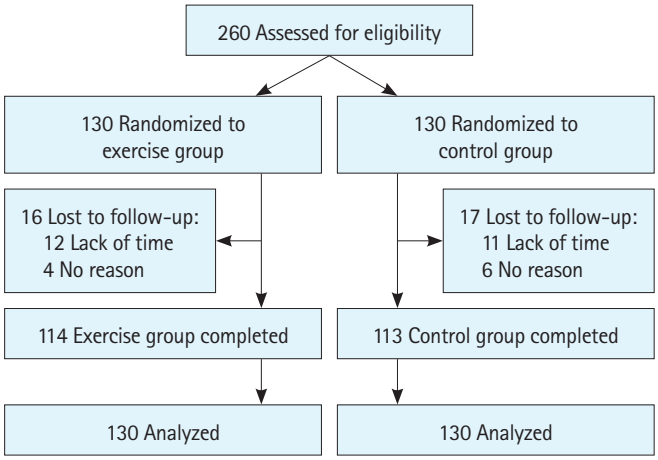

Figure. Flow chart of the study.

Adequate relief rate of dyspeptic symptoms

At the 12-week follow-up, the adequate relief rate of overall dyspeptic symptoms was significantly higher in the exercise group than in the control group (46.9% [61/130] vs 30.8% [40/130], *P* = 0.008).

Eight-item dyspepsia symptom score

At the 12-week follow-up, the exercise group demonstrated significantly greater reductions in the severity of epigastric pain, epigastric burning sensation, postprandial fullness, abdominal distension, and nausea compared to the control group (*P* < 0.05). In contrast, there was no significant difference in the decrease in severity scores of early satiety, belching, and vomiting in the exercise group compared with the control group (*P* > 0.05). Notably, the total dyspepsia symptom score decreased more markedly in the exercise group than in the control group (*P* < 0.05) (Table 3).

Short-form leeds dyspepsia questionnaire score

At the 12-week follow-up, the exercise group demonstrated significantly greater reductions in dyspeptic symptom frequency (-2.57 ± 2.35 vs -0.82 ± 1.42, *P* < 0.0001), symptom interference with lifestyle (-2.48 ± 2.28 vs -0.72 ± 1.61, *P* < 0.0001), and total SF-LDQ score (-5.05 ± 4.47 vs -1.54 ± 2.89, *P* < 0.0001) compared to the control group (Table 3). All differences were statistically significant.

Short-form nepean dyspepsia index questionnaire score

At the 12-week follow-up, the exercise group demonstrated significantly greater reductions in SF-NDI total scores compared to the control group ($P < 0.05$). Additionally, significant improvements were observed across multiple SF-NDI subdomains, including tension/anxiety, interference with daily activities, eat/drink behaviors, knowledge/control, and work/study performance (all $P < 0.05$) (Table 3).

Hospital anxiety and depression scale score

At the 12-week follow-up, intention-to-treat analysis revealed no statistically significant differences between the exercise and control groups in the reduction of anxiety scores (-0.67 ± 1.44 vs -0.78 ± 1.15, $P = 0.476$), depression scores (-0.48 ± 1.12 vs -0.42 ± 0.74, $P = 0.601$), or total HADS scores (-1.15 ± 2.32 vs -1.20 ± 1.43, $P = 0.822$) (Table 3).

Adverse reactions

Throughout the 12-week intervention period, no serious adverse events were reported in either the exercise or the control group. These findings suggest that moderate-intensity exercise over 12 weeks was well-tolerated and safe in the studied population.

DISCUSSION

Our MR analysis revealed that strong sports and other exercises (eg, swimming, cycling, keep fit, and fishing) suggested a potential inverse association with FD at the genetic level. Notably, the keep fit exercise recommended by the American College of Sports Medicine refers to moderate intensity exercise.[19] Supporting this, prior high-quality research has classified other exercises (eg, swimming, cycling, keep fit, and fishing) as moderate-intensity exercise.[26] The evidence indicates that moderate-intensity exercise could mitigate the risk of developing FD. Subsequently, our prospective RCT demonstrated that moderate-intensity exercise not only increased the remission rate of FD but also alleviated symptom severity and improved patients' quality of life. This study provides novel evidence, integrating genetic and clinical data, to support physical activity as a viable therapeutic intervention for FD. Furthermore, given its cost-effectiveness, accessibility, and lack of reliance on expensive resources, exercise represents a practical and economical treatment option for FD management.

The association between physical activity and FD remains underexplored. While some experts recommended physical activity as a primary therapy for FD,[27] robust evidence supporting its efficacy in alleviating FD symptoms is lacking. Several observational studies have reported an inverse relationship between exercise and FD. According to a study by Ford et al[28] involving 3160 participants via mailed survey, individuals with FD tended to engage in less physical exercise, and after adjusting for age, sex, body mass index, smoking, and excluding overlap with IBS, FD patients had lower frequency of moderate intensity exercise compared with control group. An internet questionnaire survey by Miwa et al[29] showed that the frequency of exercise of subjects with FD, IBS or

Table 2. The Comparison of Baseline Characteristics Between Exercise Group and Control Group

	Exercise group (n = 130)	Control group (n = 130)	*P*-value
Sex			0.912
Male	56 (43.08%)	58 (44.62%)	
Female	74 (56.92%)	72 (55.38%)	
Age (yr)	45.14 ± 11.45	46.01 ± 12.26	0.542
BMI (kg/m^2)	22.09 ± 3.05	22.12 ± 3.10	0.938
Location			0.224
Shaanxi	109 (83.85%)	101 (77.69%)	
Other	21 (16.15%)	29 (22.31%)	
Marriage			0.841
Unmarried	14 (10.77%)	13 (10.00%)	
Married	116 (89.23%)	117 (90.00%)	
Currently smoking			0.874
No	106 (81.54%)	107 (82.31%)	
Yes	24 (18.46%)	23 (17.69%)	
Currently drinking			0.620
No	108 (83.08%)	105 (80.77%)	
Yes	22 (16.92%)	25 (19.23%)	
Education level			0.785
Junior high school and below	53 (40.77%)	51 (39.23%)	
High school	24 (18.46%)	26 (20.00%)	
College and above	53 (40.77%)	53 (40.77%)	
Daily exercise time			0.184
< 1/2 hr	68 (52.31%)	58 (44.62%)	
1/2 hr < ≤ 1 hr	25 (19.23%)	28 (21.54%)	
1 hr < ≤ 2 hr	16 (12.31%)	20 (15.38%)	
> 2 hr	21 (16.15%)	24 (18.46%)	
FD subtype			0.212
Postprandial distress syndrome	31 (23.85%)	40 (30.77%)	
Epigastric pain syndrome	48 (36.92%)	52 (40.00%)	
Overlapping syndromes	51 (39.23%)	38 (29.23%)	

BMI, body mass index; FD, functional dyspepsia.
Values are expressed as the mean ± SD or n (%).

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both was significantly lower than that of the control subjects and suggested that exercise might be effective in improving the symptoms of FGID. An online survey of 2725 people in India found that uninvestigated dyspepsia may be related to lack of exercise.[30] Furukawa et al[31] conducted a cross-sectional study in a population of Japanese college students. Their research demonstrated that exercise frequency and intensity may be inversely related to the occurrence of FD. However, these studies were limited by their

Table 3. Comparison of 8-Item Dyspepsia Symptom Score, Short-form Leeds Dyspepsia Questionnaire Score, Short-form Nepean Dyspepsia Index Questionnaire Score, and Hospital Anxiety and Depression Scale Score Between Exercise Group and Control Group

	Exercise group (n = 130)			Control group (n = 130)			*P* for difference in change
	Baseline	12 wk	Change	Baseline	12 wk	Change	
Eight-item dyspepsia symptom score							
Epigastric pain	1.26 ± 1.03	0.65 ± 0.79	-0.61 ± 0.76	1.16 ± 0.95	1.05 ± 0.93	-0.11 ± 0.49	< 0.0001
Epigastric burning sensation	0.61 ± 0.95	0.28 ± 0.64	-0.33 ± 0.71	0.46 ± 0.84	0.35 ± 0.75	-0.11 ± 0.42	0.0023
Postprandial fullness	1.23 ± 1.05	0.69 ± 0.80	-0.54 ± 0.79	1.09 ± 0.96	1.01 ± 0.94	-0.08 ± 0.68	< 0.0001
Early satiety	0.34 ± 0.75	0.08 ± 0.34	-0.26 ± 0.69	0.33 ± 0.72	0.21 ± 0.59	-0.12 ± 0.48	0.0616
Belching	0.25 ± 0.66	0.12 ± 0.41	-0.13 ± 0.55	0.42 ± 0.79	0.31 ± 0.69	-0.12 ± 0.48	0.8093
Abdominal distension	1.07 ± 1.01	0.52 ± 0.77	-0.55 ± 0.85	1.03 ± 0.93	0.89 ± 0.89	-0.14 ± 0.73	< 0.0001
Nausea	0.36 ± 0.77	0.08 ± 0.37	-0.28 ± 0.70	0.28 ± 0.67	0.16 ± 0.53	-0.12 ± 0.46	0.0216
Vomiting	0.17 ± 0.54	0.05 ± 0.27	-0.12 ± 0.47	0.11 ± 0.42	0.06 ± 0.32	-0.05 ± 0.27	0.106
Total Eight-item dyspepsia symptom score	5.29 ± 2.79	2.47 ± 2.17	-2.82 ± 2.44	4.88 ± 2.43	4.05 ± 2.40	-0.84 ± 1.67	< 0.0001
SF-LDQ score							
Indigestion frequency	3.11 ± 1.02	1.98 ± 1.42	-1.14 ± 1.24	3.19 ± 0.84	2.84 ± 0.90	-0.35 ± 0.63	< 0.0001
Indigestion interference with normal activities	3.08 ± 1.05	1.95 ± 1.40	-1.13 ± 1.29	3.13 ± 0.96	2.79 ± 0.93	-0.34 ± 0.74	< 0.0001
Heartburn frequency	0.94 ± 1.52	0.52 ± 1.15	-0.42 ± 1.04	0.57 ± 1.25	0.42 ± 1.06	-0.15 ± 0.83	0.0217
Heartburn interference with normal activities	0.88 ± 1.48	0.47 ± 1.06	-0.42 ± 1.15	0.52 ± 1.21	0.41 ± 1.03	-0.11 ± 0.77	0.0118
Regurgitation frequency	1.05 ± 1.50	0.36 ± 0.92	-0.68 ± 1.26	0.88 ± 1.44	0.70 ± 1.27	-0.18 ± 0.80	0.0002
Regurgitation interference with normal activities	0.92 ± 1.45	0.28 ± 0.79	-0.65 ± 1.23	0.73 ± 1.36	0.63 ± 1.23	-0.10 ± 1.04	0.0001
Nausea frequency	0.40 ± 0.95	0.07 ± 0.40	-0.33 ± 0.87	0.38 ± 0.97	0.25 ± 0.80	-0.14 ± 0.61	0.0393
Nausea interference with normal activities	0.35 ± 0.90	0.07 ± 0.40	-0.28 ± 0.80	0.38 ± 0.97	0.22 ± 0.73	-0.17 ± 0.69	0.215
Symptom frequency score	5.50 ± 2.63	2.93 ± 2.34	-2.57 ± 2.35	5.03 ± 2.26	4.21 ± 2.15	-0.82 ± 1.42	< 0.0001
Symptom interference with lifestyle score	5.25 ± 2.49	2.77 ± 2.16	-2.48 ± 2.28	4.76 ± 2.26	4.05 ± 2.08	-0.72 ± 1.61	< 0.0001
Total SF-LDQ score	10.75 ± 5.02	5.70 ± 4.44	-5.05 ± 4.47	9.79 ± 4.43	8.25 ± 4.17	-1.54 ± 2.89	< 0.0001
SF-NDI score							
Tension/anxiety	4.32 ± 2.31	3.12 ± 1.46	-1.21 ± 1.56	4.62 ± 2.49	4.39 ± 2.38	-0.23 ± 0.70	< 0.0001
Interference	3.43 ± 2.16	2.51 ± 1.12	-0.92 ± 1.65	3.60 ± 2.19	3.45 ± 2.01	-0.15 ± 0.54	< 0.0001
Eat/drink	4.20 ± 2.09	2.99 ± 1.25	-1.21 ± 1.67	4.40 ± 2.28	4.08 ± 2.07	-0.32 ± 0.95	< 0.0001
Knowledge/control	4.62 ± 2.11	2.65 ± 0.91	-6.44 ± 4.00	4.44 ± 2.20	3.77 ± 1.76	-5.06 ± 3.42	0.0032
Work/study	3.54 ± 2.16	2.66 ± 1.21	-0.88 ± 1.54	3.59 ± 2.18	3.32 ± 1.99	-0.28 ± 0.81	0.0001
Total SF-NDI score	20.12 ± 6.00	13.93 ± 3.65	-6.18 ± 4.92	20.65 ± 7.30	19.00 ± 6.86	-1.65 ± 2.12	< 0.0001
HADS score							
Anxiety score	9.01 ± 4.08	8.34 ± 3.49	-0.67 ± 1.44	9.05 ± 3.06	8.26 ± 2.65	-0.78 ± 1.15	0.476
Depression score	9.39 ± 4.12	8.92 ± 3.80	-0.48 ± 1.12	9.53 ± 3.25	9.12 ± 3.04	-0.42 ± 0.74	0.601
Total HADS score	18.40 ± 7.79	17.25 ± 6.93	-1.15 ± 2.32	18.58 ± 5.84	17.38 ± 5.31	-1.20 ± 1.43	0.822

SF-LDQ, Short-form leeds dyspepsia questionnaire; SF-NDI, Short-form Nepean dyspepsia index questionnaire; HADS, hospital anxiety and depression scale.
Values are expressed as the mean ± SD.

reliance on the Rome III criteria and cross-sectional designs. A small RCT by Rane et al. (n = 36 per group) suggested that aerobic exercise as adjuvant therapy may improve FD symptoms,[7] but the short intervention duration and limited sample size constrain its generalizability. In contrast, our study included 130 patients per group and demonstrated through 12-week follow-up that moderate-intensity exercise exerts a measurable therapeutic effect on FD.

The role of physical activity in the pathogenesis of FD remains unclear. In patients with FD, duodenal microbiota diversity is reduced, and regular physical exercise may alter the intestinal microbiome, increasing its diversity and possibly increasing the number of beneficial bacteria.[32-34] Physical activity has been shown to increase the flow of endogenous endorphins in the blood, which can reduce the severity of symptoms.[35] Meanwhile, endorphins also interact with receptors in the brain to reduce pain perception.[36] Physical exercise triggers the release of endorphins in the brain, which aids in stress relief and improves mood.[37] Research shows that endogenous cannabinoid can cause the release of hypothalamic endorphins,[38] and physical exercise can reduce depression symptoms by increasing the activity of endogenous cannabinoid to regulate the amygdala. Previous studies have shown that the benefits of exercise are systemic. Exercise can induce the release of bioactive factors into the blood circulation, resulting in systemic beneficial effects.[39] Interestingly, while gastrointestinal symptoms and quality of life improved significantly, we did not observe a corresponding reduction in HADS. This may suggest that the therapeutic benefit of moderate-intensity exercise in FD operates primarily through mechanisms directly affecting gastrointestinal physiology (eg, motility, visceral sensitivity, and microbiota) rather than, or in addition to, a secondary effect mediated by improved mood. Alternatively, the intervention duration or intensity may have been insufficient to induce measurable changes in generalized anxiety and depression levels in this patient cohort.

There were several limitations in the study. First, the MR analysis relied on physical activity phenotypes derived from self-reported questionnaires in the UK Biobank. Measurement error and misclassification inherent to self-report may attenuate the estimated causal associations, which is a common limitation when utilizing publicly available GWAS summary statistics for exposure definition. Also, the MR analysis utilized GWAS summary statistics for both exposure and outcome derived from the UK Biobank, leading to potential sample overlap. While we employed stringent genetic instrument selection criteria, this overlap may still introduce some bias and should be considered when interpreting the MR results. Second, the MR analysis included a limited number of SNPs assessing the causal relationship between strenuous sports and FD, which may have led to an overrepresentation of rs12571549 in the leave-one-out sensitivity analysis. Third, as with most FD treatment studies, a placebo effect could not be entirely excluded. Although physical activity interventions cannot be blinded, we minimized bias by having the control group receive weekly phone consultations to reinforce habitual lifestyle adherence. Forth, the MR analysis was conducted using genetic data from individuals of European ancestry, whereas the RCT was performed in a Chinese population. Differences in genetic background, lifestyle, and environmental factors between these populations may limit the direct comparability and generalizability of the integrated findings. Future multi-ethnic studies are needed to confirm the trans-ethnic relevance of these associations. Finally, consistent with previous high-quality studies,[15] the exercise intervention was not directly supervised but instead monitored via telephone follow-up, and concomitant medication use was not restricted—reflecting real-world clinical practice. While this design enhances external validity, it may introduce variability in treatment adherence.

In conclusion, our study supports the beneficial role of physical activity in reducing the risk of FD in the general population. Strong sports and other exercises (eg, swimming, cycling, keep fit, and fishing), may be suggestively associated with a reduced risk of FD. Additionally, moderate-intensity exercise appears to alleviate gastrointestinal symptoms, enhance the rate of adequate symptom relief, and improve overall quality of life in FD patients. Further well-designed clinical trials are needed to validate these findings and elucidate the potential therapeutic benefits of physical activity for FD management.

SUPPLEMENTARY MATERIALS

Note: To access the supplementary table and figures mentioned in this article, visit the online version of *Journal of Neurogastroenterology and Motility* at http://www.jnmjournal.org/, and at http://doi.org/10.5056/jnm25159.

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Conflicts of interest
None.

Author contributions

Zhongcao Wei, Jinhai Wang, Na Liu, and Xin Xing: Conceptualization, Methodology, and Software; Zhongcao Wei and Xin Xing: data curation and writing-original draft preparation; Zhiwei Tao and Yujie Hao: visualization and investigation; Jinhai Wang and Na Liu: supervision; Zhongcao Wei: software and validation; and Zhongcao Wei and Xin Xing: writing-reviewing and editing. All authors approved the version to be published and agreed to be accountable for all aspects of the work.

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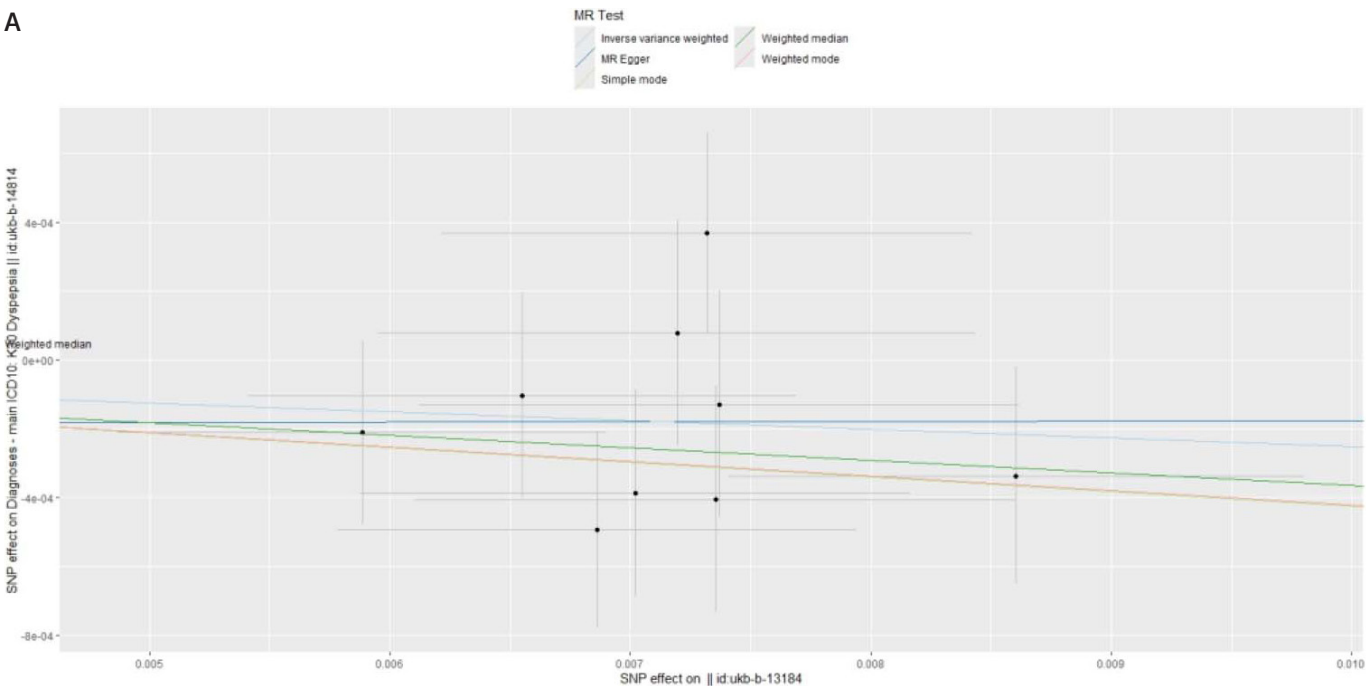
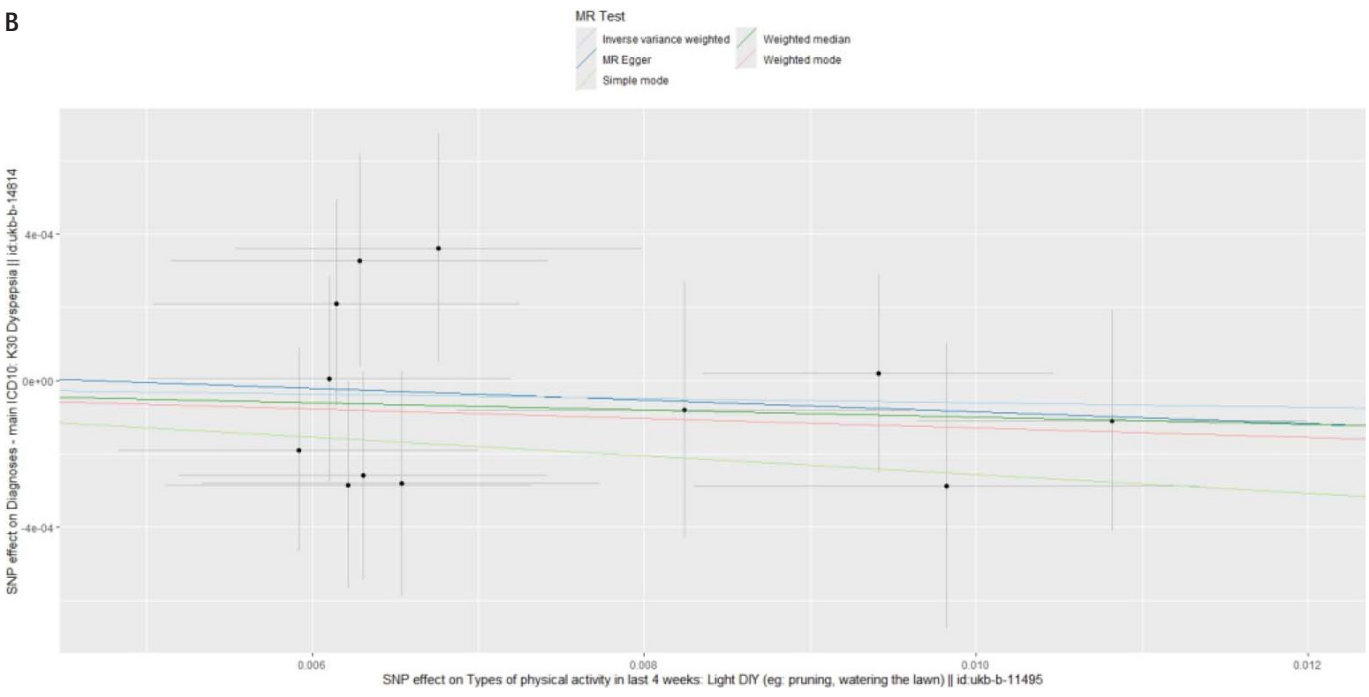
SUPPLEMENTARY MATERIALS

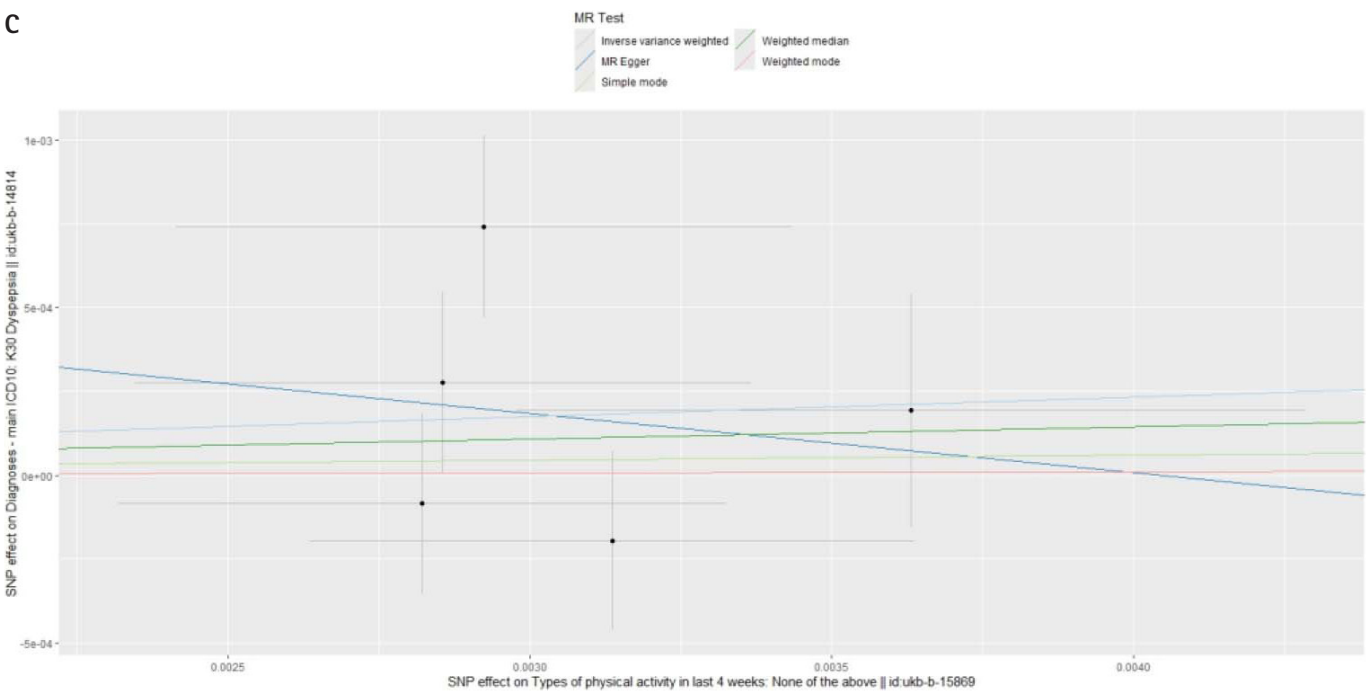
Supplementary Table. Data Sources and IDs

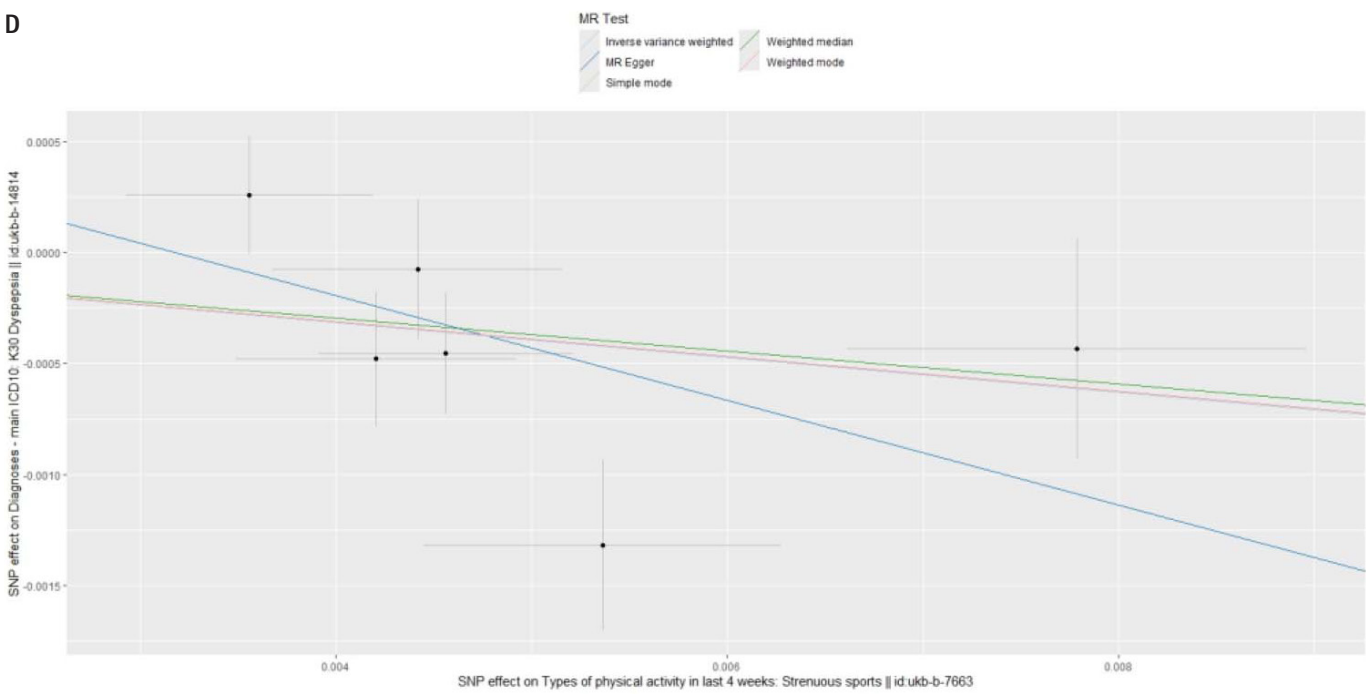
GWAS-ID	Phenotype	Year	Population	Sex	Cases	Controls	Sample size	SNPs
ukb-b-13184	Heavy DIY	2018	European	M&F	197 006	263 370	460 376	9 851 867
ukb-b-11495	Light DIY	2018	European	M&F	236 244	224 132	460 376	9 851 867
ukb-b-15869	None of the Heavy DIY or light DIY	2018	European	M&F	28 040	432 336	460 376	9 851 867
ukb-b-7663	Strenuous sports	2018	European	M&F	47 468	412 908	460 376	9 851 867
ukb-b-7337	Walking for pleasure	2018	European	M&F	329 755	130 621	460 376	9 851 867
ukb-b-8764	Other exercises	2018	European	M&F	222 470	237 906	460 376	9 851 867
ukb-b-14814	Functional dyspepsia	2018	European	M&F	7662	455 348	463 010	9 851 867

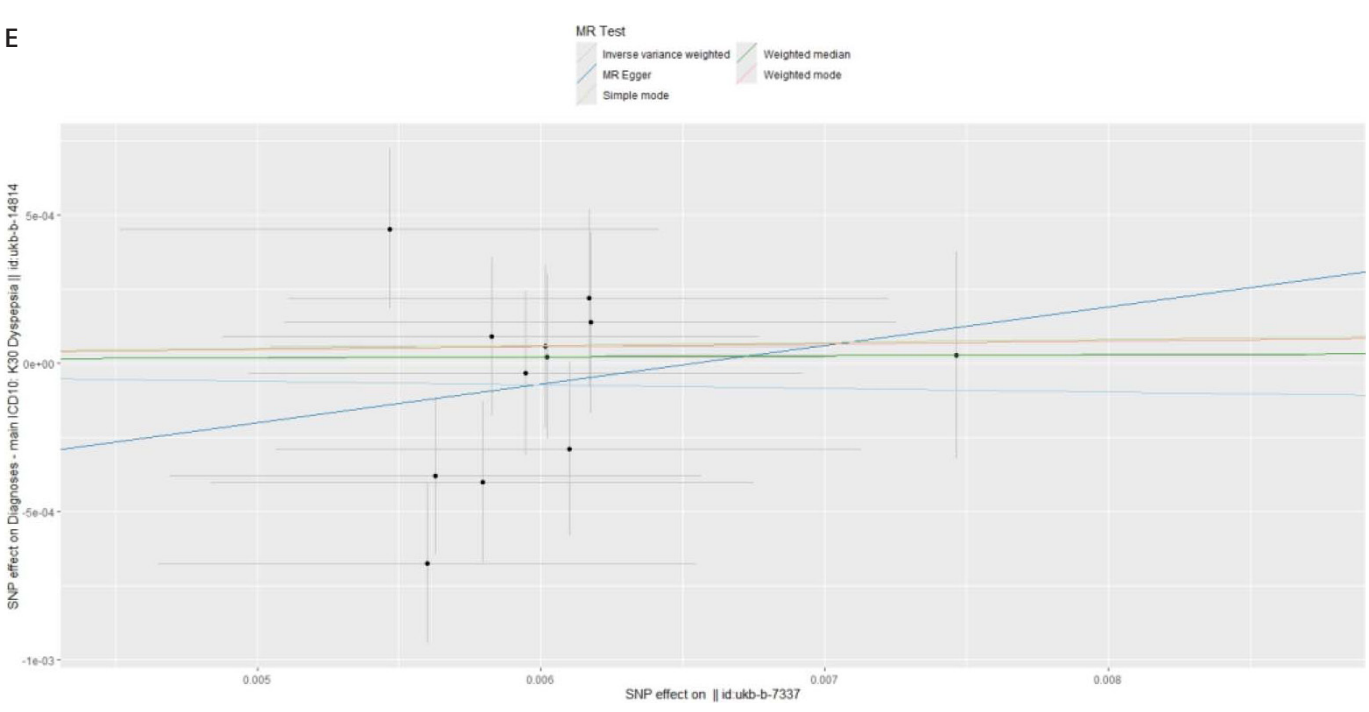
GWAS, genome-wide association studies; SNPs, single-nucleotide polymorphisms; DIY, do-it-yourself; M&F, males and females.

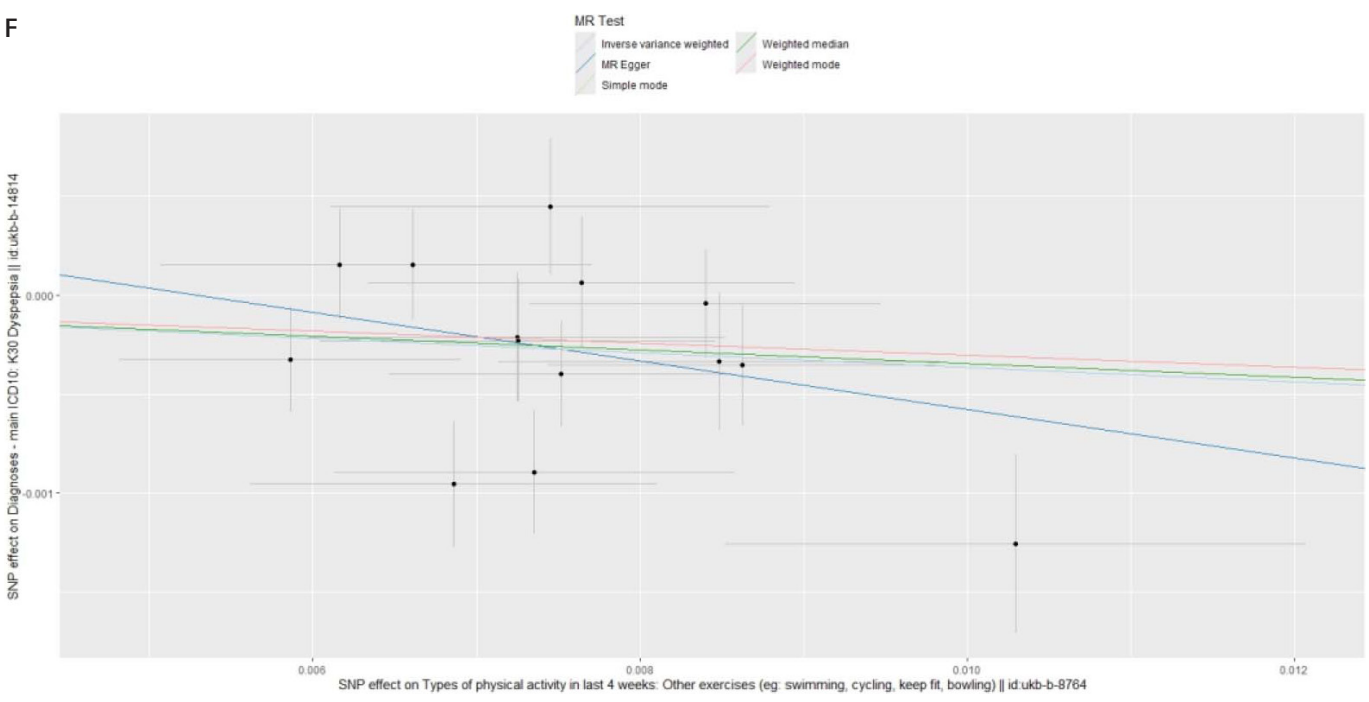




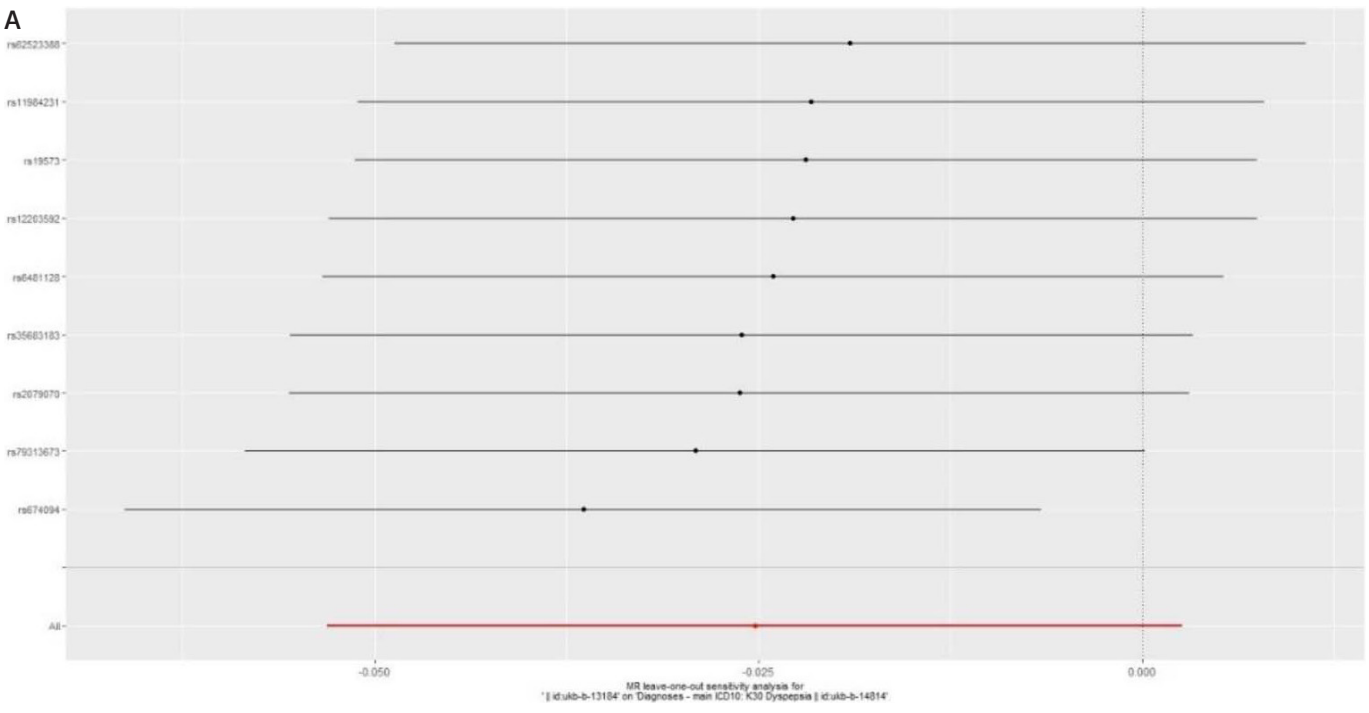


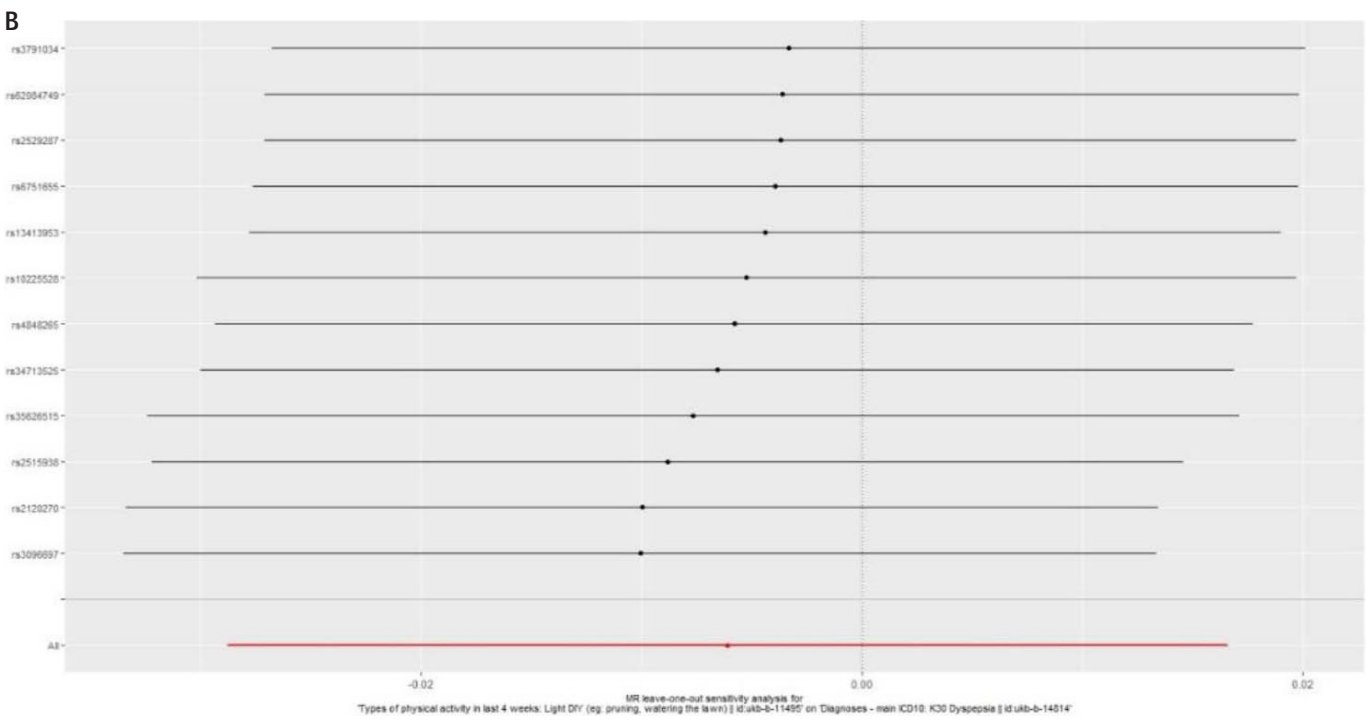


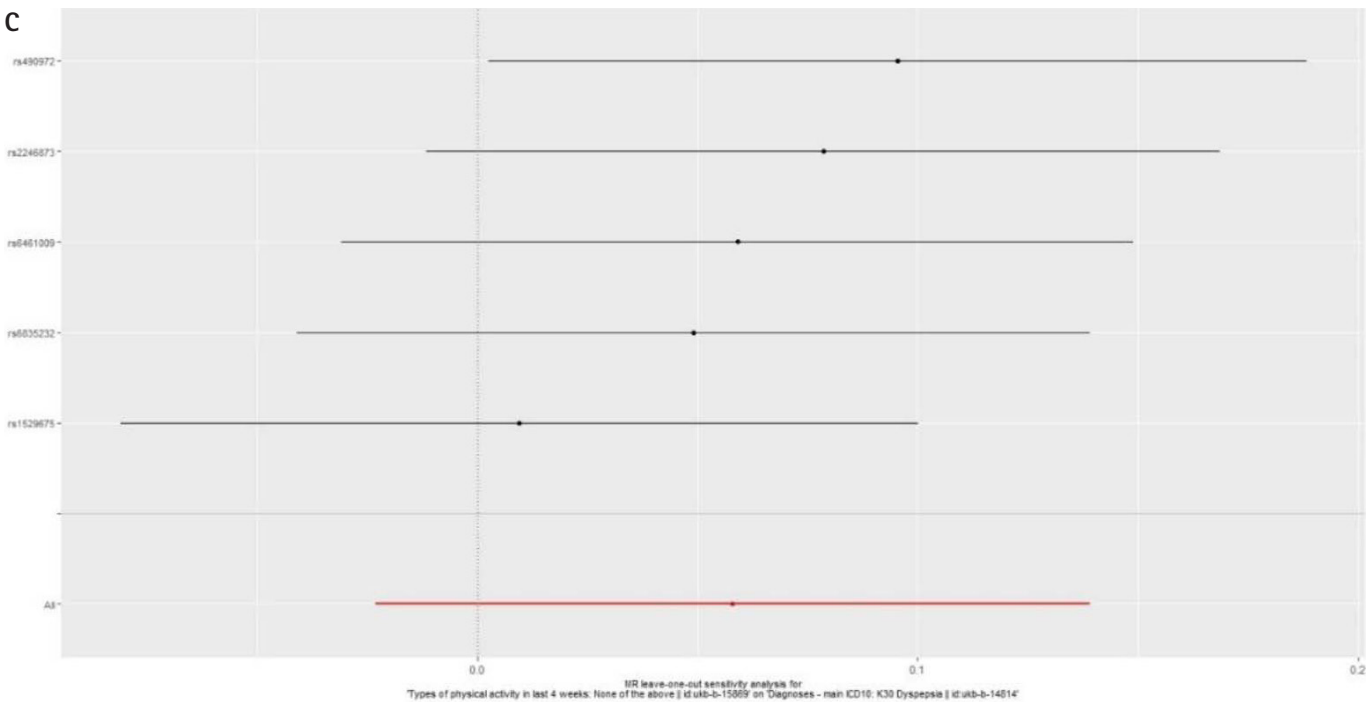


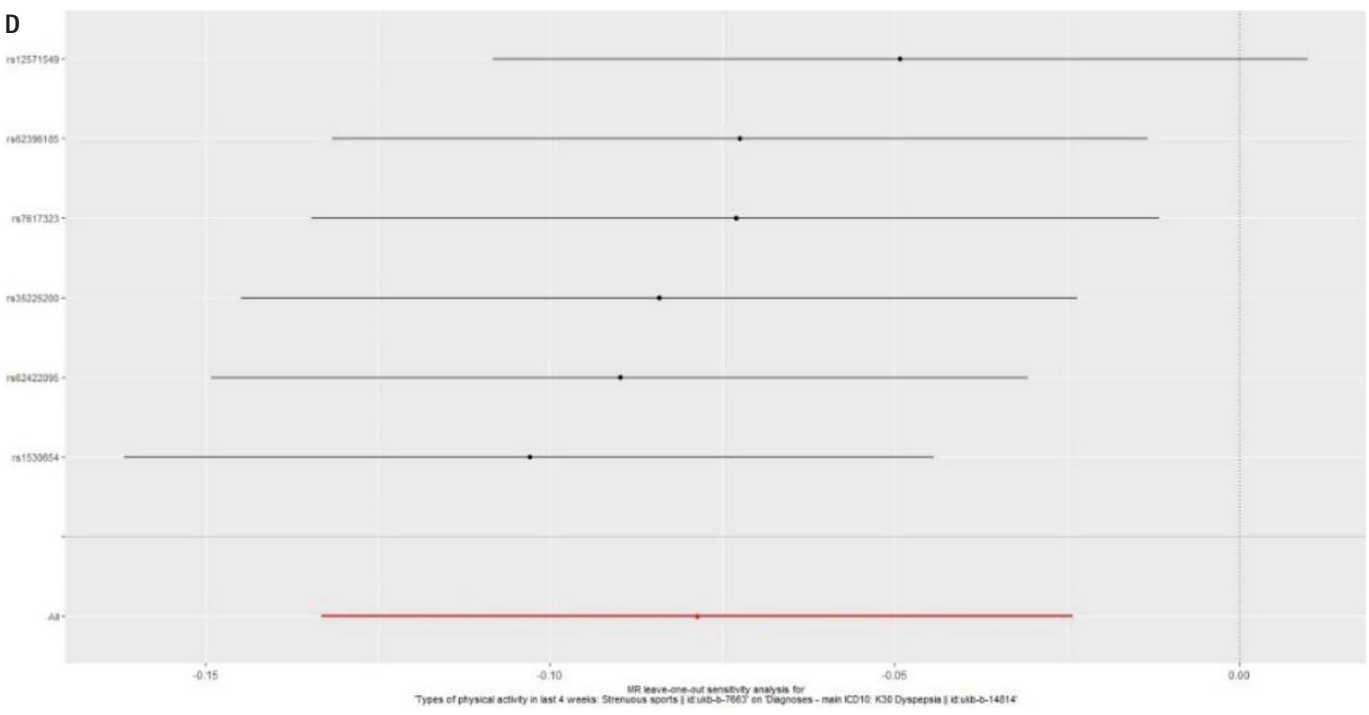


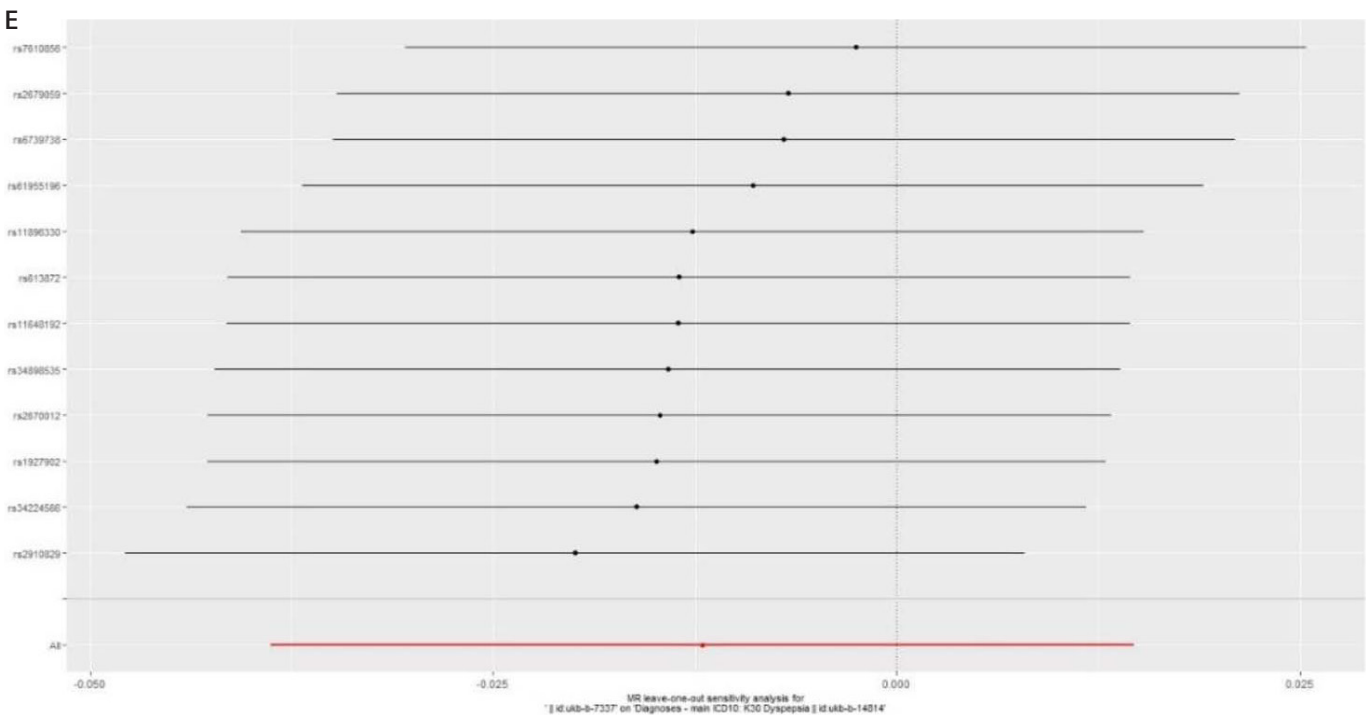
Supplementary Figure 1. Mendelian randomization (MR) scatterplots for relationship of physical activity with functional dyspepsia (FD). (A) Scatterplot of the effects of single-nucleotide polymorphisms (SNPs) on FD versus their effects on Heavy do-it-yourself (DIY). (B) Scatterplot of the effects of SNPs on FD versus their effects on Light DIY. (C) Scatterplot of the effects of SNPs on FD versus their effects on none of the Heavy DIY or Light DIY. (D) Scatterplot of the effects of SNPs on FD versus their effects on Strenuous sports. E, Scatterplot of the effects of SNPs on FD versus their effects on Walking for pleasure. (F) Scatterplot of the effects of SNPs on FD versus their effects on other exercises (eg, swimming, cycling, keep fit, and bowling).

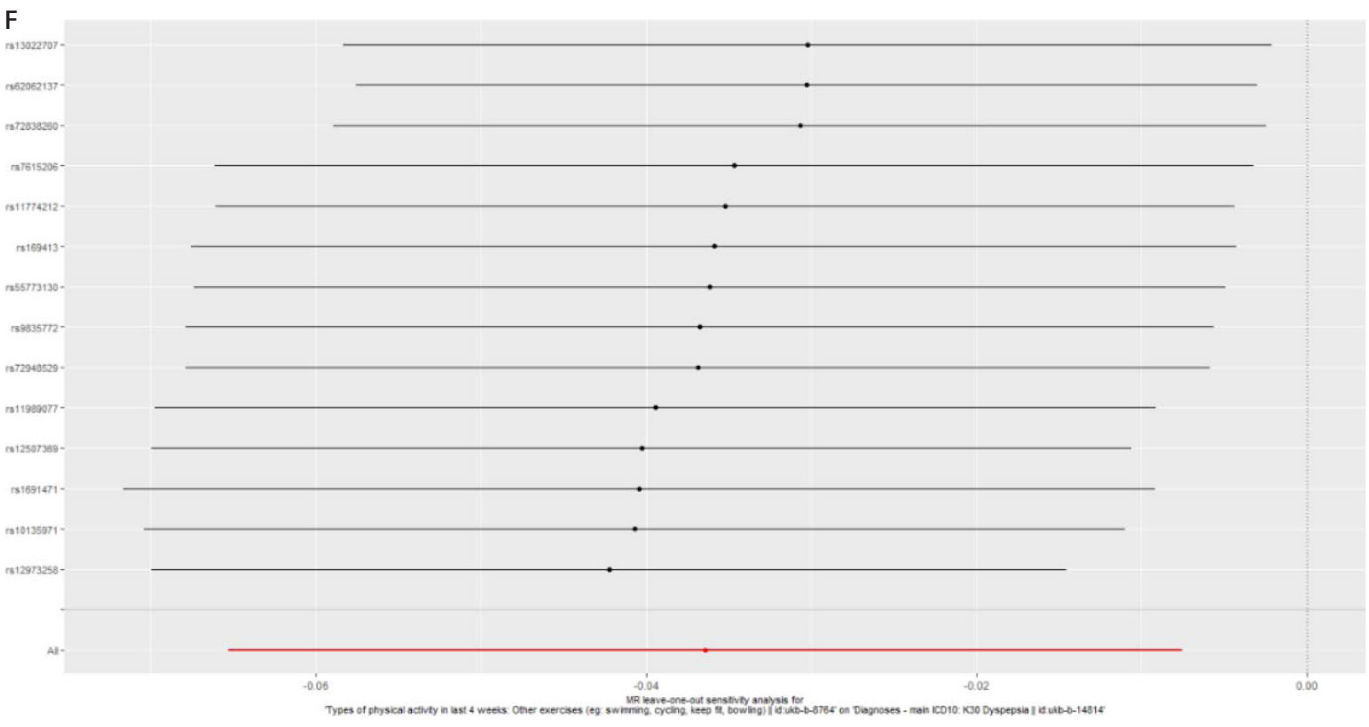












Supplementary Figure 2. Leave-one-out analyses for single-nucleotide polymorphisms (SNPs) associated with physical activity and functional dyspepsia (FD). (A) Leave-one-out analyses for SNPs associated with Heavy do-it-yourself (DIY) and FD. (B) Leave-one-out analyses for SNPs associated with Light DIY and FD. (C) Leave-one-out analyses for SNPs associated with none of the Heavy DIY or Light DIY and FD. (D) Leave-one-out analyses for SNPs associated with strenuous sports and FD. (E) Leave-one-out analyses for SNPs associated with walking for pleasure and FD. (F) Leave-one-out analyses for SNPs associated with other exercises (eg, swimming, cycling, keep fit, and bowling) and FD.

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Ineffective Esophageal Motility and Its Predictors in Gastroesophageal Reflux Disease Patients With Pathologic Acid Exposure Time

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Background/Aims: Although gastroesophageal reflux disease (GERD) and ineffective esophageal motility (IEM) often coexist, the directionality of their relationship remains uncertain. Clarifying physiological predictors of IEM in GERD patients with objectively verified pathological acid exposure may facilitate more personalized treatment strategies.

Methods: This retrospective analysis included 289 patients with typical reflux symptoms and abnormal acid exposure (acid exposure time > 6%) confirmed by 24-hour pH-impedance monitoring. Esophageal motility was assessed using high-resolution manometry, while mucosal integrity was evaluated through distal mean nocturnal baseline impedance (MNBI). Patients were categorized into normal motility and IEM groups. Clinical, endoscopic, manometric, and impedance parameters were compared between the groups.

Results: IEM was identified in 41.5% of patients. Compared with those showing normal motility, the IEM group was older (median 51.5 vs 44 years, $P < 0.001$), had reduced lower esophageal sphincter (LES) pressure (15 vs 19 mmHg, $P = 0.006$), and exhibited decreased distal MNBI (1300 vs 1780 Ω, $P = 0.005$). Multivariate regression revealed prolonged upright bolus exposure, reduced LES pressure, delayed supine distal latency, and MNBI < 1500 Ω as independent predictors of IEM. No significant associations were observed with diabetes or *Helicobacter pylori* infection.

Conclusions: In GERD patients with confirmed pathological acid exposure, IEM is common and associated with weakened LES pressure, delayed peristalsis, and reduced mucosal integrity. The link between low MNBI and IEM suggests an interaction between epithelial barrier dysfunction and esophageal motility, emphasizing the need for comprehensive functional evaluation in GERD management.

Keywords: Acid exposure; Esophageal motility disorders; Gastroesophageal reflux; Impedance; Manometry

INTRODUCTION

Gastroesophageal reflux disease (GERD) is a frequently encountered gastrointestinal condition characterized by the retrograde movement of stomach contents into the esophagus, most often manifesting as heartburn and regurgitation. Chronic exposure to refluxate may result in esophageal mucosal damage and complications, including erosive esophagitis and Barrett's esophagus. Acid exposure time (AET), measured during a 24-hour multichannel intraluminal impedance–pH (MII-pH) study, is a fundamental diagnostic parameter in GERD, representing the proportion of time the esophageal pH remains below 4. The Lyon Consensus 2.0 (2024) considers an AET exceeding 6% as definitive evidence of pathological acid reflux.[1]

Effective esophageal motility is essential for the proper clearance of refluxed gastric material. The Chicago classification version 4.0 (v4.0) characterizes ineffective esophageal motility (IEM) as the presence of over 70% ineffective swallows or a minimum of 50% failed swallows on high-resolution esophageal manometry (HREM), providing a uniform framework for assessing peristaltic function.[2] In addition to lower esophageal sphincter (LES) competence, coordinated esophageal motility and intact mucosal integrity are crucial for maintaining normal esophageal function.

Advances in impedance-based metrics, including mean nocturnal baseline impedance (MNBI) and the post-reflux swallow-induced peristaltic wave (PSPW) index, have contributed to a better understanding of GERD mechanisms.[3] MNBI provides a surrogate for esophageal mucosal integrity, whereas the PSPW index reflects the efficiency of refluxate clearance.[4] Despite their potential in classifying GERD phenotypes, the PSPW index was omitted from the 2024 Lyon Consensus 2.0 because of limited reproducibility and a lack of standardized methodology.[1]

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Although diagnostic technologies have advanced, the interplay between acid reflux and esophageal motility abnormalities remains incompletely defined. Chronic acid exposure is proposed to impair motility through inflammation-driven neuromuscular injury. Consistent with this, treatments such as anti-reflux mucosal ablation have demonstrated improvements in LES pressure and esophageal contractility along with reductions in acid exposure, suggesting that reflux-induced motility deficits may be partially reversible.[5] On the other hand, primary motility disorders, including IEM, can compromise clearance of refluxate and contribute to more severe GERD, indicating that motility dysfunction may both precede and amplify reflux pathology.[6-8]

Anatomical alterations at the esophagogastric junction (EGJ), particularly hiatal hernia (HH), may further influence the relationship between reflux and esophageal motility. Recent esophageal manometry studies have identified HH as an independent predictor of IEM, even among patients without elevated AET, suggesting that structural disruptions can directly compromise esophageal motor function.[6]

These observations highlight a potentially bidirectional and synergistic interaction between pathological acid exposure and esophageal motility dysfunction, although the underlying mechanisms remain unclear. The present study aims to identify clinical, endoscopic, manometric, and impedance-based predictors of IEM in patients with objectively confirmed pathological acid exposure (AET > 6%). By stratifying patients according to HREM findings and incorporating metrics such as MNBI, we seek to elucidate factors contributing to esophageal motility impairment within the GERD spectrum.

MATERIALS AND METHODS

Study Design and Population

This retrospective observational study was carried out at the Department of Gastroenterology Kocaeli University Faculty of Medicine, Kocaeli, Turkey, from January 2021 to January 2024. Adult patients with typical GERD symptoms persisting for at least 6 months who had undergone both HREM and MII-pH monitoring were considered for inclusion.

Approval for the study protocol was obtained from the Clinical Research Ethics Committee of Kocaeli University Faculty of Medicine (protocol no. 2025/149; dated 27 March 2025), and all procedures were performed in line with the Declaration of Helsinki and Good Clinical Practice standards. The requirement for written informed consent was waived by the institutional review board due to the retrospective nature of this study.

A total of 289 patients were included in the analysis. Participants were eligible for inclusion if they met the following criteria: (1) objectively confirmed pathological acid reflux, defined by an AET > 6% according to the Lyon Consensus 2.0;[1] (2) completion of both MII-pH and HREM studies; (3) availability of upper gastrointestinal endoscopy findings; and (4) completion of the GERD questionnaire.[9]

Exclusion criteria were as follows: achalasia (n = 6), EGJ outflow obstruction (n = 5), history of any malignancy (n = 4), MII-pH monitoring duration under 22 hours (n = 10), severe systemic illnesses such as chronic kidney disease, congestive heart failure, or cirrhosis (n = 10), prior upper gastrointestinal surgery (n = 7), and HH larger than 3 cm (n = 5). Following the application of these criteria, the final study cohort was analyzed for clinical, endoscopic, and functional parameters.

All patients underwent MII-pH and HREM after discontinuation of proton pump inhibitors (PPIs) and other acid-suppressive medications for at least 2 weeks, to ensure that measurements reflected the baseline esophageal physiology. Demographic information and symptom profiles were collected from medical records and standardized reflux questionnaires.[9] MII-pH data were analyzed following the Lyon Consensus 2.0, esophagitis severity was classified according to the Los Angeles (LA) grading system,[10] and esophageal motility disorders were defined based on the Chicago classification v4.0.[2]

Upper Gastrointestinal Endoscopy

All patients received upper gastrointestinal endoscopy due to the presence of reflux-related clinical symptoms. Procedures were performed using a FUJIFILM EG-760R gastroscope (FUJIFILM Corp, Tokyo, Japan). Endoscopic observations focused on HH, LES laxity, reflux esophagitis, and gastritis.

A hypotensive EGJ was defined as a gastroesophageal junction with a widely open or lax appearance, incomplete closure of the distal esophagus at rest, or absence of a prominent mucosal ridge. In addition, the gastroesophageal flap valve was evaluated in retroflexed view using Hill's classification (grades I-IV), which provides a standardized assessment of EGJ competence.[11] In grade I, the valve closely wraps around the endoscope, indicating normal function and effective closure; grade II demonstrates occasional opening with breathing, suggesting mild loosening; grade III shows a valve that remains largely open and does not fully close, reflecting significant dysfunction; and grade IV is characterized by the absence of a visible flap with a fully open junction, indicating severe impairment of the anti-reflux mechanism. Figure shows a representative image from a single patient demonstrating Hill grade 3. Since patients with hiatal hernias exceeding 3 cm were excluded

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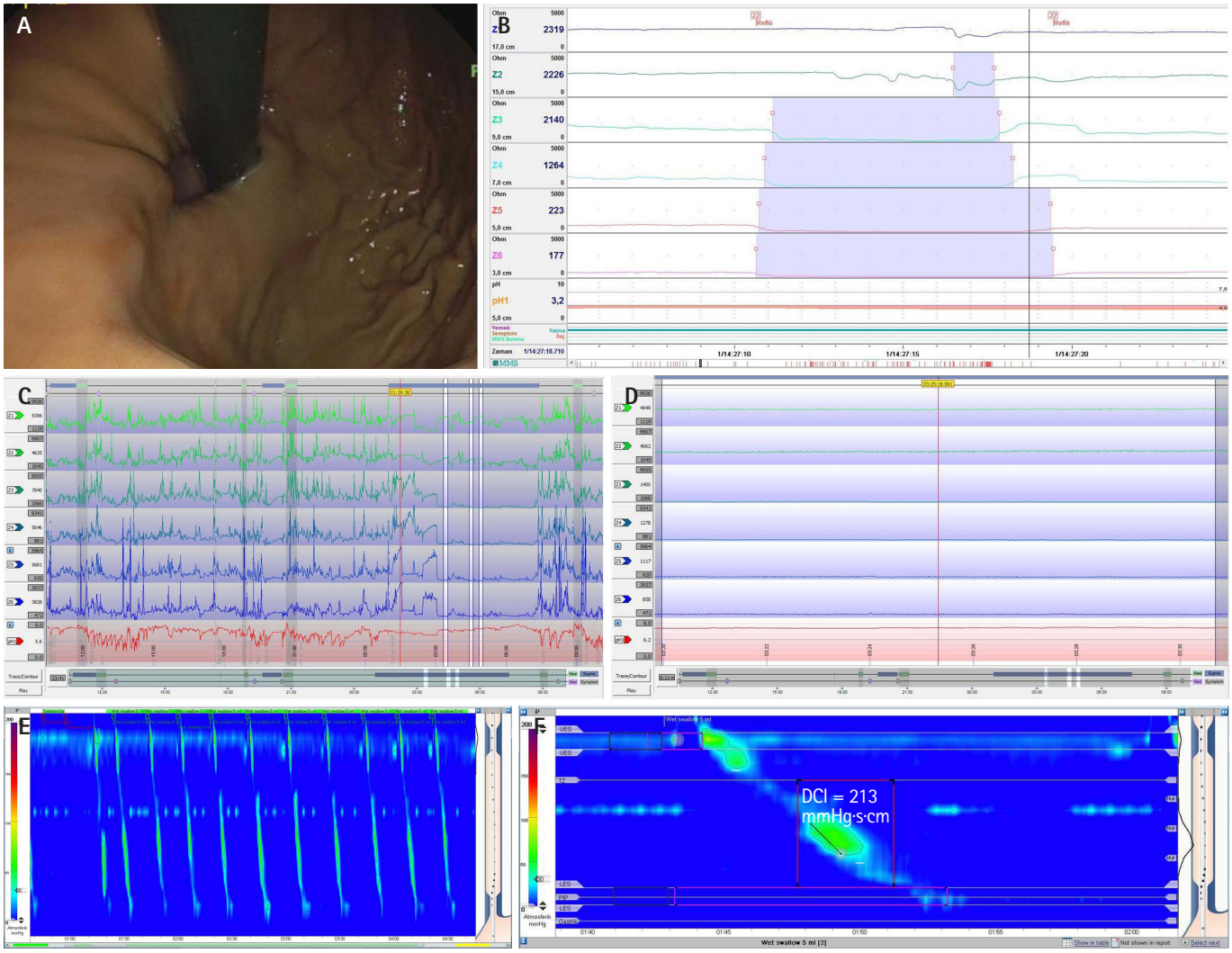

Figure. A representative case of pathological gastroesophageal reflux disease (GERD) associated with ineffective esophageal motility (IEM) is shown, integrating endoscopic, 24-hour multichannel intraluminal impedance–pH (MII-pH) monitoring, and high-resolution esophageal manometry (HREM) findings. Upper gastrointestinal endoscopy demonstrated a Hill grade 3 (A) gastroesophageal flap valve, which remained largely open and did not fully close, without evidence of erosive esophagitis. MII-pH monitoring (B) revealed pathological acid reflux, with an acid exposure time (AET) of 11.6% and 43 reflux episodes, exceeding the pathological threshold of 6%. Symptom–reflux association was significant, with a symptom index of 66.6% and a symptom association probability of 87.8%. For mean nocturnal baseline impedance (MNBI) assessment, a stable artifact-free nocturnal tracing (C) was selected, and three measurements between 1 AM and 3 AM (D) were averaged, yielding an MNBI of approximately 1274 Ω at the Z5 channel. HREM (E) demonstrated IEM according to Chicago classification version 4.0, with 10% intact, 60% weak (distal contractile integral [DCI], 100–450 mmHg·s·cm) (F), and 30% failed (DCI < 100 mmHg·s·cm) peristaltic contractions. IEM is defined by > 70% ineffective (weak) or ≥ 50% failed swallows, indicating impaired esophageal clearance and bolus transit. The average of 10 wet swallows showed a lower esophageal sphincter (LES) resting pressure of 11 mmHg, consistent with hypotension, an integrated relaxation pressure (IRP) of 9 mmHg, and a DCI of 213 mmHg·s·cm, indicative of weak peristaltic contractions, while contractile front velocity (CFV, 4.6 cm/sec) and distal latency (DL, 6 seconds) were within normal limits, suggesting impaired esophageal clearance despite preserved peristaltic timing and propagation.

from the study, cases corresponding to Hill grade IV, which reflect a completely absent gastroesophageal flap valve, were not included in the analyzed cohort.

pH–Impedance Monitoring

MII-pH monitoring was performed using a single-use Digitrapper VersaFlex-Z catheter (Medtronic, Minneapolis, MN, USA). Patients discontinued PPIs at least 2 weeks prior to the procedure and fasted for a minimum of six hours before catheter placement. The pH sensor was placed 5 cm above the LES, based on measurements from preceding high-resolution esophageal manometry. To reduce measurement variability, MNBI values were manually assessed by a single experienced physician. Automated analysis was performed using Digitrapper Reflux Reader software v6.1 (Medtron-

ic), followed by manual verification. During monitoring, patients maintained their usual daily routines and recorded meals, symptoms, and posture changes in a standardized diary.

The percentage of time that esophageal pH remained below 4.0 over a 24-hour period was used to calculate AET. An AET exceeding 6% was regarded as pathological according to the Lyon Consensus 2.0.[1]

High-resolution Esophageal Manometry

HREM was performed using a multichannel water-perfused system (MMS; Medical Measurement Systems BV, Enschede, The Netherlands, now part of Laborie Medical Technologies), integrated with Polygram Net software (version 4.01). Studies were conducted in both supine and upright positions, and data were interpreted per the Chicago classification v4.0.[2] Integrated relaxation pressure (IRP): defined as the average of the lowest 4 seconds of pressure values within the 10-second post-swallow period; it assesses lower esophageal sphincter relaxation. LES resting pressure: reflects basal tone of the LES.

Distal contractile integral (DCI) is a composite metric that quantifies the amplitude, duration, and length of esophageal contractions (expressed in mmHg·s·cm). A DCI value below 450 mmHg·s·cm is indicative of weak peristalsis, whereas a value less than 100 mmHg·s·cm denotes failed peristalsis.

Contractile front velocity: speed of peristaltic wave propagation along the esophagus. The peristaltic break length represents the extent of interruption in the 20 mmHg isobaric contour; breaks longer than 5 cm are associated with reduced efficacy of esophageal bolus clearance. DL represents the time elapsed from the onset of a swallow to the contractile deceleration point. A DL shorter than 4.5 seconds may indicate the presence of spastic esophageal contractions. Particular emphasis was placed on diagnosing IEM, defined as > 70% ineffective swallows (DCI < 450 mmHg·s·cm) or ≥ 50% failed swallows, in the absence of EGJ outflow obstruction.[2]

Data Stratification and Grouping

Only patients with pathological acid exposure (AET > 6%) were included, while those with AET ≤ 6% were excluded to ensure diagnostic consistency. The final cohort was stratified based on HREM findings into 2 groups: individuals with normal esophageal motility and those with IEM according to the Chicago classification v4.0.[2] Comparative analyses encompassed a range of variables, including demographic characteristics (age, sex, body mass index [BMI]), lifestyle factors (tobacco and alcohol use), comorbidities (hypertension, diabetes mellitus [DM]), endoscopic findings (LA classification, presence of HH, *Helicobacter pylori* infection), symptom profiles (frequency and severity of typical and atypical GERD symptoms), and MII-pH parameters (total AET, number of reflux episodes, symptom index, symptom association probability, and MNBI). Distal MNBI was determined manually, rather than through an automated system, at 3 esophageal levels (Z1, Z2, and Z5) using overnight MII-pH recordings. At each level, 3 separate 10-minute nocturnal periods—around 1 AM, 2 AM, and 3 AM—were chosen during times without swallows, reflux events, or technical artifacts. The same observer reviewed all measurements, and the mean of the three intervals at each level was calculated to provide a consistent estimate of esophageal mucosal integrity while reducing variability related to the observer.

Statistical Methods

Statistical analyses were performed using IBM SPSS Statistics version 29.0 (IBM Corp, Armonk, NY, USA). The distribution of continuous variables was examined using the Kolmogorov–Smirnov and Shapiro–Wilk tests. Normally distributed data are expressed as mean ± standard deviation, and non-normally distributed variables are summarized as median and interquartile range.

Categorical variables were summarized as frequencies and percentages. Comparisons of continuous variables between groups were performed using the Kruskal–Wallis test, followed by Dunn's post hoc analysis for pairwise comparisons when applicable. Associations between categorical variables were evaluated using the chi-square test, and Bonferroni correction was applied to adjust for multiple testing.

A two-tailed P-value of < 0.05 was considered statistically significant. Variables demonstrating significance in univariate analysis ($P < 0.05$) were subsequently entered into a multivariate logistic regression model to identify independent predictors of IEM. Adjusted odds ratios (ORs) and corresponding 95% confidence intervals (CIs) were calculated for all variables retained in the final model.

Data Availability Statement

The data presented in this study are available on request from the corresponding author.

RESULTS

The final analysis comprised 289 patients who demonstrated objectively confirmed pathological acid exposure, defined as an AET greater than 6%. The cohort consisted predominantly of male participants (60.6%, n = 175), while females comprised 39.4% (n = 114). The median age was 47 (37-57) years, and the median BMI was 26.8 (24.49-29.60) kg/m^2.

According to BMI classification, 1.0% (n = 3) were underweight

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(BMI ≤ 18.5), 27.0% (n = 78) had normal weight (18.5-24.9), 49.5% (n = 143) were overweight (25-29.9), and 22.5% (n = 65) were obese (≥ 30). These data reflect a relatively young and predominantly overweight population, consistent with typical GERD demographics.

Comorbidities and Lifestyle Factors

At least 1 comorbidity was present in 45.5% (n = 131) of patients. The most frequent conditions included hypertension (22.8%, n = 66), type 2 DM (12.5%, n = 36), allergies (9.7%), coronary artery disease (8.3%, n = 24), cardiac arrhythmia (8.3%, n = 24), asthma or chronic bronchitis (7.3%), rheumatologic disorders (6.2%, n = 18), and hyperlipidemia (4.5%, n = 13). The median comorbidity duration was 60 (24-120) months.

Among the 36 patients with type 2 DM, the median HbA1c was 6.6 (6.2-7.0). HREM revealed normal motility in 14 and IEM in 22 diabetic patients. The difference in HbA1c between these subgroups was not statistically significant (P = 0.619). Only 2 patients exhibited microvascular complications.

Regarding lifestyle habits, 55.4% (n = 160) were non-smokers, 26.0% (n = 75) were current smokers, and 18.7% (n = 54) were former smokers. Among participants who smoked or had previously smoked, the median age at initiation was 19 years (15-20), and the median cumulative smoking exposure reached 28 pack-years (8.5-42). Patients with IEM had a significantly longer duration since smoking cessation, suggesting a potential residual effect of prior tobacco exposure on esophageal motility. Alcohol consumption was reported by 10% of the cohort, with most drinking 1-2 times per week.

Symptom Profiles

Regurgitation (83%) and heartburn (73%) were the most commonly reported symptoms. Symptom duration varied widely, with a median of 36 (12-84) months. Dysphagia was reported in 42.6% (n = 72) of patients with normal motility and 33.3% (n = 40) of those with IEM, while its absence was more common in the IEM group (66.7% vs 57.4%). However, this difference did not reach statistical significance (P = 0.111).

Patients with normal motility reported significantly longer durations of regurgitation (36 [12-72] vs 24 [12-60] months, P = 0.034) and dysphagia (24 [12-36] vs 12 [9.3-57] months, P = 0.037) compared to those with IEM. These observations imply that the persistence of symptoms does not consistently correlate with the presence of esophageal motility disorders.

Helicobacter pylori Status

H. pylori status was determined by gastric biopsy. The bacterium was identified in the antrum of 28.7% (n = 83) and in the corpus of 15.6% (n = 45) of patients.

Biopsy data were unavailable in 14.2% (n = 41) of antral samples and 16.3% (n = 47) of corpus samples. Overall, 28.7% (n = 83) tested positive for *H. pylori*, 57.1% (n = 165) were negative, and 14.2% had insufficient data. No statistically significant association was found between *H. pylori* status and esophageal motility patterns (P = 0.274), suggesting limited involvement of *H. pylori* in the pathophysiology of IEM.

Endoscopic Findings

Endoscopically, EGJ appeared normal in 51.2 % (n = 148) of patients, whereas a hypotensive morphology was observed in 48.8% (n = 141). A small HH (< 3 cm) was identified in 28.4% (n = 82).

Among patients with normal manometry, 50.9% (n = 86) exhibited a normal EGJ appearance, whereas 49.1% (n = 83) demonstrated a hypotensive EGJ. In the IEM group, 51.7% (n = 62) had a normal EGJ appearance, and 48.3% (n = 58) presented with a hypotensive EGJ (P = 0.896). In the IEM group, 70.8% (n = 85) had no HH, and 29.2% (n = 35) had HH, with no significant association between manometry status and the presence of HH (P = 0.801). Concerning endoscopic valve competence, 50.3% (n = 85) of patients with normal manometry were classified as Hill grade I, 22.5% (n = 38) as grade II, and 27.2% (n = 46) as grade III. In the IEM group, 51.7% (n = 62) were Hill grade I, 20.0% (n = 24) grade II, and 28.3% (n = 34) grade III, with no significant relationship observed between manometry status and Hill grade (P = 0.878).

According to the LA classification, 76.5% (n = 221) showed no visible esophagitis. LA Grade A was observed in 14.5% (n = 42), and Grades B to D in 9.0% (n = 26). These observations demonstrate that substantial acid exposure may be present without endoscopic evidence of mucosal breaks, supporting the paradigm of non-erosive reflux disease.

pH-Impedance Findings

All patients had objectively confirmed GERD, defined by AET > 6% on MII-pH monitoring. The median upright AET was 12 (8.6-17.8), and the median DeMeester score was 37.7 (28.6-58.7), indicating substantial acid exposure across the cohort. The median number of reflux episodes was 44 (28.5-62.0).

Mucosal Integrity Values

MNBI, serving as an indicator of esophageal mucosal integrity, demonstrated median values of 1650 Ω (1033-2258) at the distal esophagus (Z5) and 2900 Ω (2046-3600) and 3042 Ω (2230-3808) at the proximal esophagus (Z2 and Z1, respectively). These values indicate varying degrees of mucosal compromise, with the most

pronounced reduction observed in the distal esophagus (Z5), consistent with chronic acid injury. The baseline demographic and clinical features of the study cohort are presented in Table 1.

Comparison of Ineffective Esophageal Motility and Normal Motility Groups

Based on HREM findings, participants were categorized into normal motility (n = 169) and IEM (n = 120) groups. IEM was observed in 41.5% of patients, indicating that esophageal motor impairment is common even in the context of uniformly pathological acid exposure (AET > 6%).

Table 1. Baseline Characteristics and Clinical Features of the Study Cohort

Variable	Value
Male	175 (60.6)
Female	114 (39.4)
Age (yr)	47.0 (37.0-57.0)
BMI (kg/m²)[a]	26.8 (24.5-29.6)
Comorbidities	131 (45.5)
Non-smokers	160 (55.4)
Current smokers	75 (26.0)
Former smokers	54 (18.7)
Alcohol consumption	29 (10.0)
Most common symptoms	
Regurgitation	240 (83.0)
Heartburn	211 (73.0)
Normal GEJ	162 (56.1)
Hypotensive GEJ	125 (43.2)
Hiatal hernia (< 3 cm)	83 (28.7)
Normal esophagus	221 (76.5)
LA-A esophagitis	42 (14.5)
LA-B or D esophagitis	26 (9.0)
AET	12.0 (8.6-17.8)
Reflux episodes	44.0 (28.5-62.0)
DeMeester score	37.7 (28.6-58.7)
Symptom-reflux association	164 (56.8)
MNBI-Z1 (Ω)	3042 (2230-3809)
MNBI-Z2 (Ω)	2900 (2046-3600)
MNBI-Z5 (Ω)	1650 (1033-2258)

[a]Body mass index (BMI) classification: underweight (≤ 18.5 kg/m²), normal weight (18.5-24.9 kg/m²), overweight (25.0-29.9 kg/m²), and obese (≥ 30.0 kg/m²). Symptom-Reflux Association: Indicates a positive correlation between symptoms and reflux events based on a positive symptom index (≥ 50%) and/or symptom association probability (≥ 95%).
IQR, interquartile range; GEJ, gastroesophageal junction; LA, Los Angeles classification; AET, acid exposure time; MNBI, mean nocturnal baseline impedance; Z1-Z5, impedance measurement channels (proximal to distal).
Values are presented as number (%) for categorical variables and median (interquartile range) for continuous variables unless otherwise specified.

Demographic and anthropometric comparisons

Sex distribution showed 39.4% females and 60.6% males overall; males were more frequent in the IEM group (65.8%) than normal motility (43.2%) but without statistical significance (P = 0.143); median age was significantly higher in the IEM group at 51.5 years (40-62) compared to 44 years (36-55) in the normal motility group (P < 0.001), while BMI was similar between groups: 26.9 (24.3-29.0) kg/m² for normal motility and 26.6 (23.8-29.0) kg/m² for IEM (P = 0.319).

Comorbidities and diabetes mellitus

Among patients without comorbidities, 63.9% had normal motility and 36.1% had IEM, whereas among those with comorbidities, 51.9% had normal motility and 48.1% had IEM (P = 0.042), suggesting systemic comorbidities may predispose to esophageal motor dysfunction; diabetes prevalence was significantly higher in the IEM group (18.3%) compared to normal motility (8.3%) (P = 0.011), with 61.1% of diabetics exhibiting IEM versus 38.9% with normal motility.

Symptom profiles

Heartburn occurred at similar rates in both groups (P = 0.894). While the median duration of symptoms tended to be longer in the IEM cohort [36 (18-60) months] compared with the normal motility group [24 (12-36) months], the difference did not reach statistical significance (P = 0.359). Severity scores were comparable, but the frequency of heartburn episodes was elevated in the IEM group (P = 0.009). Dysphagia prevalence was also similar (P = 0.114), with 64.3% of dysphagic patients showing normal motility and 35.7% IEM, though dysphagia episodes were shorter and less frequent in the IEM group (P = 0.037 and P = 0.013). Regurgitation, reported by 83% overall, was more frequent in normal motility patients (87.0%) versus IEM (77.5%), with a significantly longer median duration in the IEM group [36 (12-60) months vs 24 (12-36) months; P = 0.034]; frequency and severity were higher in the normal motility group (P = 0.020 and P = 0.018). Other GI symptoms like epigastric pain (63.3% vs 34.8%, P = 0.007) and frequent belching (52.7% vs 40.0%, P = 0.034) were more common in normal motility patients, while retrosternal pain showed no significant difference (P = 0.278). Symptom–reflux association, based on symptom index or symptom association probability, was positive in 56.8% of patients, with no significant difference between normal motility (58.0%) and IEM (55.1%) groups (P = 0.818 and P = 0.915), suggesting that symptom frequency or severity does not reliably reflect esophageal motility abnormalities.

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pH–Impedance and bolus exposure metrics

The median DeMeester score was 39.6 (33.5-58.3) in the IEM group and 31.0 (24.0-46.0) in the normal motility group, though this difference was not statistically significant ($P = 0.209$); upright AET was higher in the IEM group [15.3% (11.0-28.0) vs 13.1% (8.3-19.0); $P = 0.098$], while supine AET was similar between groups [7.4% (4.4-13.6) vs 6.1% (3.9-9.0); $P = 0.915$]; upright bolus exposure time was significantly prolonged in the IEM group [1.5% (0.7-2.5)] compared to the normal motility group [1.2% (0.6-1.8); $P = 0.024$], whereas supine bolus exposure time was minimal and similar between groups [0.1% (0.0-0.4); $P = 0.624$].

Mucosal integrity

MNBI values measured at three esophageal levels showed Z1 (proximal-most) at 3017 (2200-3766) Ω in normal motility vs 3117 (2300-3850) Ω in IEM ($P = 0.680$) and Z2 (proximal) at 2929 (2050-3600) Ω in normal motility vs 2833 (2046–3617) Ω in IEM ($P = 0.497$), indicating preserved proximal mucosal integrity regardless of motility status; however, Z5 (distal) was significantly lower in the IEM group at 1300 (850-2122) Ω compared to 1780 (1140-2315) Ω in the normal motility group ($P = 0.005$), reflecting compromised distal mucosal integrity; threshold analysis revealed values below 1500 Ω were significantly more common in the IEM group ($P = 0.006$), with stratification by MNBI categories (< 1500, 1500-2500, > 2500 Ω) showing patients with MNBI < 1500 Ω were more likely to have IEM, confirmed by pairwise post hoc Bonferroni comparisons ($P = 0.006$), and a linear-by-linear association test suggested a trend toward decreasing MNBI with worsening motility ($P = 0.060$). Collectively, these findings support the notion that distal mucosal impairment (low MNBI) is tightly linked to IEM, while proximal mucosal integrity remains relatively unaffected. Table 2 summarizes the comparison of demographic, clinical, and physiological characteristics between the normal esophageal motility and IEM groups.

Independent predictors and manometric characteristics of ineffective esophageal motility

When HREM metrics were compared, patients with IEM exhibited a lower median LES pressure (15 mmHg [9-21]) than those with normal motility (19 mmHg [12-25]; P = 0.006), reflecting compromised sphincter function potentially predisposing to increased reflux.

The median DL was 7.0 seconds in both the IEM and normal groups; however, the IEM group exhibited a significantly wider and slightly prolonged DL distribution (6.65-8.0 vs 6.1-7.9; $P = 0.021$), indicating subtle neuromuscular delay. In contrast, IRP, a marker of EGJ relaxation, did not differ significantly between groups (normal: 7 mmHg [4-12]; IEM: 6 mmHg [2.3-10.0], $P = 0.162$), indicating preserved LES relaxation in both cohorts.

In univariate analysis, several variables were significantly associated with IEM. While DM increased the odds of IEM (OR: 2.49, 95% CI: 1.21-5.08; $P = 0.013$), this effect did not remain significant after multivariate adjustment ($P = 0.126$), indicating possible confounding. Independent predictors in the multivariate model included current smoking (OR: 0.39, 95% CI: 0.18-0.83; $P = 0.016$), distal MNBI < 1500 Ω (OR: 0.50, 95% CI: 0.26-0.96; $P = 0.039$), increased upright bolus exposure (OR: 1.33, 95% CI: 1.06-1.66;

Table 2. Comparison of Demographic, Clinical, and Physiological Features Between Normal Esophageal Motility and Ineffective Esophageal Motility Groups

Parameter	Normal motility (n = 169)	IEM (n = 120)	*P*-value
Female	73 (43.2)	41 (34.2)	0.143
Male	96 (56.8)	79 (65.8)	
Age (yr)	44 (36.0-55.0)	51.5 (40.0-62.0)	< 0.001
BMI (kg/m^2)	26.9 (24.3-29.0)	26.6 (23.8-29.0)	0.319
Comorbidity	68 (51.9)	63 (48.1)	0.042
Type 2 DM	14 (8.3)	22 (18.3)	0.011
Heartburn	124 (73.4)	87 (72.5)	0.894
Regurgitation	147 (87.0)	93 (77.5)	0.050
Epigastric pain	107 (63.3)	57 (34.8)	0.007
Belching	89 (52.7)	48 (40.0)	0.034
Dysphagia	72 (42.6)	40 (33.3)	0.114
Upright AET (%)	13.1 (8.3-19.0)	15.3 (11.0-28.0)	0.098
Supine AET (%)	6.1 (3.9-9.0)	7.4 (4.4-13.6)	0.915
RE	31 (19.0-46.0)	34 (18.0-50.0)	0.865
Upright BET (%)	1.2 (0.6-1.8)	1.5 (0.7-2.5)	0.024
Supine BET (%)	0.1 (0.0-0.4)	0.1 (0.0-0.4)	0.624
SI	43 (0.0-68.0)	46 (0.0-75.0)	0.818
SAP	88 (0.0-100.0)	85 (0.0-100.0)	0.951
MNBI-Z1 (Ω)	3017 (2200-3766)	3117 (2300-3850)	0.680
MNBI-Z2 (Ω)	2929 (2050-3600)	2833 (2046-3617)	0.497
MNBI-Z5 (Ω)	1780 (1140-2315)	1300 (850-2122)	0.005
MNBI < 1500 Ω	63 (37.3)	67 (55.8)	0.006
MNBI 1500-2500 Ω	77 (45.6)	36 (30.0)	0.006
MNBI > 2500 Ω	29 (17.2)	17 (14.2)	0.130

IEM, Ineffective esophageal motility; BMI, body mass index; DM, diabetes mellitus; AET, acid exposure time; RE, reflux episodes; BET, bolus exposure time; SI, symptom index; SAP, symptom association probability; MNBI, mean nocturnal baseline impedance; Z1-Z5, impedance measurement channels (proximal to distal).
Values are presented as number (%) for categorical variables and median (interquartile range) for continuous variables unless otherwise specified.
A *P*-value of < 0.05 was considered statistically significant.

Table 3. Univariate and Multivariate Logistic Regression Analysis of Factors Associated With Ineffective Esophageal Motility

Variable	Univariate OR (95% CI)	Univariate *P*	Multivariate OR (95% CI)	Multivariate *P*
Age (yr)	1.04 (1.02-1.06)	0.001	1.02 (0.99-1.05)	0.076
Type 2 DM	2.49 (1.21-5.08)	0.013	2.14 (0.81-5.58)	0.126
Heartburn (once/wk)	1.52 (0.80-2.88)	0.200	1.61 (0.71-3.63)	0.251
Heartburn (2-3 times/week)	1.00 (0.53-1.88)	> 0.999	1.08 (0.49-2.41)	0.849
Heartburn (≥ 4 times/week)	0.47 (0.22-0.97)	0.042	0.87 (0.34-2.23)	0.765
Regurgitation (once/week)	0.54 (0.27-1.09)	0.870	0.61 (0.26-1.41)	0.246
Regurgitation (2-3 times/week)	0.68 (0.34-1.36)	0.272	0.73 (0.30-1.77)	0.489
Regurgitation (≥ 4 times/week)	0.26 (0.12-0.55)	0.001	0.49 (0.19-1.28)	0.147
Dysphagia (once/week)	1.09 (0.61-1.98)	0.763	1.82 (0.51-2.31)	0.149
Dysphagia (2-3 times/week)	0.53 (0.21-1.35)	0.184	0.61 (0.18-2.05)	0.424
Dysphagia (≥ 4 times/week)	0.24 (0.09-0.66)	0.006	0.41 (0.12-1.36)	0.149
Epigastric pain	0.52 (0.33-0.84)	0.008	0.73 (0.39-1.34)	0.307
Current smoker	0.41 (0.23-0.74)	0.003	0.39 (0.18-0.83)	0.016
Former smoker	0.67 (0.36-1.26)	0.210	0.44 (0.20-0.97)	0.041
BET (upright, %)	1.29 (1.08-1.54)	0.004	1.33 (1.06-1.66)	0.015
LES pressure (mmHg)	0.96 (0.93-0.99)	0.005	0.94 (0.91-0.98)	0.002
DL (supine, sec)	1.42 (1.13-1.80)	0.003	1.61 (1.21-2.15)	0.001
MNBI < 1500	0.44 (0.26-0.74)	0.002	0.50 (0.26-0.96)	0.039
MNBI 1500-2500	0.55 (0.28-1.10)	0.091	0.98 (0.41-2.33)	0.959

DM, diabetes mellitus; OR, odds ratio; CI, confidence interval; BET, bolus exposure time; LES, lower esophageal sphincter; DL, distal latency; MNBI, mean nocturnal baseline impedance.
This table summarizes independent risk factors for ineffective esophageal motility (IEM) identified via univariate and multivariate logistic regression analyses. Significant predictors of IEM in univariate analysis included diabetes mellitus, current smoking, reduced MNBI (< 1500 Ω), increased upright bolus exposure time, decreased LES pressure, and prolonged supine DL. In multivariate analysis, current smoking, reduced MNBI, increased upright bolus exposure time, decreased LES pressure, and prolonged supine distal latency remained independently associated with IEM. DM lost statistical significance after adjustment.

P = 0.015), decreased LES pressure (OR: 0.94, 95% CI: 0.91-0.98; *P* = 0.002), and prolonged supine distal latency (OR: 1.61, 95% CI: 1.21-2.15; *P* = 0.001). The model demonstrated good overall fit, as indicated by a statistically significant intercept (*P* < 0.001). Additionally, neither age (*P* = 0.076) nor symptom frequency—including heartburn, regurgitation, dysphagia, belching, and epigastric pain—independently predicted IEM after adjusting for covariates (Table 3). Collectively, these findings define a distinct GERD patient phenotype characterized by diminished LES tone, impaired bolus clearance, prolonged DL in the supine position, and compromised distal mucosal integrity—physiological features independently associated with IEM.

A representative patient with IEM is presented in Figure, illustrating endoscopic evaluation, HREM metrics, and MII-pH results, thereby exemplifying the physiologic interplay between IEM, impaired bolus clearance, and reflux burden.

DISCUSSION

In this well-characterized cohort of GERD patients with objectively confirmed pathological acid exposure, IEM was highly prevalent, observed in more than 40% of cases, highlighting its clinical significance. Despite comparable reflux burden, patients exhibited substantial heterogeneity in esophageal motor function and impedance-derived metrics, indicating that motor impairment contributes to GERD pathophysiology independently of acid exposure. The association of IEM with reduced LES pressure, prolonged upright bolus exposure, and decreased distal MNBI underscores the combined effect of impaired clearance and compromised mucosal integrity, supporting the concept that esophageal dysmotility represents an independent and clinically relevant pathophysiological axis in GERD rather than a simple epiphenomenon of reflux severity.

Age-related declines in esophageal neuromuscular function, including reduced peristaltic vigor and impaired secondary peristalsis, likely contribute to the increasing prevalence of IEM among

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older GERD patients.[12-20] Interestingly, symptom perception did not consistently correlate with motility impairment in our cohort: patients with preserved peristalsis often reported more prolonged regurgitation and dysphagia than those with IEM.[18,21-30] This dissociation underscores the role of altered visceral sensitivity and central sensory processing in modulating reflux symptoms independently of esophageal motor function.[19,21,22,24] Consequently, reliance on symptom burden alone may underestimate underlying motility abnormalities, highlighting the need for objective assessment using HREM and MII-pH monitoring across age groups.

Metabolic and lifestyle factors further influence esophageal motility. While obesity can increase reflux burden and mucosal injury, moderate obesity may preserve contractility, whereas morbid obesity (BMI ≥ 40) is linked to higher IEM prevalence, EGJ outflow obstruction, and elevated IRP.[31-38] In our cohort, BMI did not differ between patients with and without IEM, suggesting that motility impairment can occur independently of overall obesity. However, visceral adiposity, as assessed by waist circumference, is associated with increased GERD and mucosal compromise,[39] highlighting the importance of central adiposity in esophageal pathophysiology. Type 2 DM was linked to IEM in univariate analysis, though its effect was not independent in multivariate models, suggesting indirect mechanisms such as autonomic neuropathy or delayed gastric emptying.[40,41]

Smoking and alcohol intake can impair esophageal motility: nicotine reduces LES tone, slows clearance, and increases transient LES relaxations, whereas chronic alcohol intake can impair peristalsis and LES competence.[42-45] The inverse association between current smoking and IEM in our study may reflect residual confounding, reverse causality, or symptom-driven smoking cessation, and should not be interpreted as protective.

Structural factors, such as LES–crural diaphragm separation observed in HH, are known to correlate with increased reflux burden and abnormal MII-pH metrics.[46,47] Although HH has been linked to IEM, this relationship may be influenced by age and esophageal inflammation.[6,46] In our cohort, no significant associations were observed between esophageal motor patterns and endoscopic findings, including HH and esophagitis grade, possibly due to exclusion of patients with large HH (> 3 cm), suggesting that structural abnormalities alone are insufficient to predict motility dysfunction.

IEM was observed across the spectrum of GERD, affecting both erosive and non-erosive disease, yet its presence did not directly reflect disease severity.[47,48] Functionally, IEM impairs effective peristalsis and salivary-mediated clearance, prolonging esophageal acid contact and sustaining symptoms, particularly in patients with objectively documented pathological acid exposure.[47,49,50] In our cohort, IEM was independently associated with prolonged upright bolus exposure and reduced LES pressure, irrespective of reflux frequency or symptom correlation, suggesting clearance dysfunction that may occur independently of reflux events.[22,51-53] Endoscopic findings, including esophagitis, were similar between groups, indicating that mucosal injury primarily reflects acid exposure rather than motility impairment.[50]

The coexistence of IEM and reduced distal MNBI defines a distinct GERD phenotype characterized by impaired clearance and regionally concentrated mucosal vulnerability.[54-59] Prolonged distal latency in the supine position further supports delayed peristaltic propagation as a contributor to sustained reflux.[54,57,60]

In multivariate analysis, independent predictors of IEM included low distal MNBI (< 1500 Ω), reduced LES pressure, prolonged distal latency in the supine position, and increased upright bolus exposure, whereas neither age nor symptom burden independently predicted motility impairment. These findings reinforce the limitations of symptom-based assessment and underscore the importance of advanced physiological testing, including HREM and MII-pH monitoring, for accurate GERD phenotyping and individualized management.[61]

Clinical Implications

Our findings emphasize the critical importance of assessing esophageal motility in the diagnosis and management of GERD. The high prevalence of IEM, even in patients without severe esophagitis, highlights the limitations of relying solely on endoscopy or symptom-based evaluation. IEM may contribute to impaired bolus clearance, persistent reflux, and reduced response to PPIs. Incorporating HREM and MII-pH monitoring allows early identification of affected patients, facilitating individualized and targeted treatment planning.

Functional metrics such as distal MNBI, upright bolus exposure, and peristaltic integrity provide valuable guidance for precision-based management. The observed associations of IEM with aging, reduced LES pressure, and distal mucosal compromise underscore the importance of early detection for effective risk stratification and preoperative evaluation.

Limitations

This retrospective, single-center study design limits the generalizability of the findings and may overrepresent patients with more severe GERD. The cross-sectional nature of the study prevents causal inference, and manual interpretation of MNBI may be subject to observer variability. Additional unmeasured factors, including diet, sleep patterns, and anthropometric measures (such as abdominal and hip circumference), could have influenced the results and limit the assessment of central obesity's specific ef-

fects on esophageal motility and mucosal integrity. Furthermore, sensory testing and systematic analyses of reflux–symptom associations were not consistently available, which restricts insights into symptom perception and pathophysiological correlations.

Strengths

Despite these limitations, the study has several strengths. Strict inclusion criteria according to the Lyon Consensus 2.0 ensured a well-defined GERD cohort with pathological acid exposure (AET > 6%). Combined endoscopy, HREM, MII-pH, and impedance-derived MNBI allowed for a comprehensive assessment of esophageal motility and mucosal integrity. Conducted at a tertiary center, this cohort represents a clinically diverse, real-world population, thereby enhancing the external validity of the findings. Stratification by IEM status and multivariate analysis enabled identification of independent predictors while controlling for confounders such as age, LES pressure, MNBI, and bolus clearance. The study also examined the impact of metabolic comorbidities, including type 2 DM and obesity, on esophageal motility, further contributing to the understanding of GERD pathophysiology. Notably, many IEM patients did not exhibit endoscopic injury, highlighting the importance of functional testing for precise phenotyping.

In conclusion, IEM was frequently observed in GERD patients with pathological acid exposure and was independently associated with reduced LES pressure, impaired bolus clearance, and lower distal MNBI. Symptom severity and endoscopic findings did not consistently correspond to manometric abnormalities, underscoring the multifactorial nature of GERD. The reduction of distal MNBI in patients with IEM reflects regionally concentrated mucosal vulnerability and compromised epithelial integrity. These findings emphasize the importance of comprehensive physiological assessment using HREM and impedance measurements for accurate diagnosis and individualized management, particularly in patients with persistent or unexplained reflux. Future prospective studies are needed to elucidate the causal relationships between reflux burden and motor dysfunction and to determine whether therapies aimed at restoring motility can improve clinical outcomes and mucosal integrity.

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Conflicts of interest

None.

Author contributions

Conceptualization, methodology, validation, investigation, resources, writing—review and editing, supervision, and project administration: Ayça Eroğlu Haktanır and Altay Çelebi; software, data curation, writing—original draft preparation: Ayça Eroğlu Haktanır; formal analysis: Ayça Eroğlu Haktanır and Altay Çelebi; and All authors have read and agreed to the published version of the manuscript.

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The Association Between Usual Diet and Bowel Disorders of Gut–Brain Interaction: Analyses From the Rome Foundation Global Epidemiology Study

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Background/Aims: Food is thought to play a central role in the pathophysiology of disorders of gut–brain interaction (DGBI). Data on these associations come largely from Western countries. We evaluated associations between diet and common bowel DGBI using the database of the Rome Foundation Global Epidemiology Study (RFGES).

Methods: The RFGES database contains data collected via the Internet, personal interviews, or both from 33 countries, providing prevalence rates for common DGBI. Logistic regressions were used to analyze associations between DGBI and vegan, vegetarian, lactose-free and bread-, pasta-, and rice-predominant diets.

Results: 54 127 internet and 20 973 household survey subjects were included. Rates of adherence to a vegan or vegetarian diet were low in both surveys (0.4% to 9.8%) and differences in dietary preference were also noted between surveys with lactose-free and rice-predominant diets being more common among household survey respondents. While discrepancies between results from the 2 surveys limited the interpretation of the data, some trends were evident with a lactose-free diet linked to an increased prevalence and bread- and rice-predominant diets to lower prevalence rates of several bowel DGBI.

Conclusions: Though complicated by variations in results between the 2 survey populations, this global survey has revealed dietary factors that may impact, in the general population, on reported prevalence rates for DGBI.

Keywords: Constipation; Diet; Diarrhea; Gastrointestinal disorders, Irritable bowel syndrome

INTRODUCTION

The association between diet and gastrointestinal health has been the topic of much recent discussion and scientific inquiry. An increasing body of medical literature has underscored the role of dietary factors in the development and management of disorders of gut–brain interaction (DGBI), formerly known as functional gastrointestinal disorders, which include irritable bowel syndrome (IBS) and other functional bowel disorders (FD).[1-3] These disorders, characterized by symptom clusters and codified by the Rome criteria, are some of the most common gastrointestinal disorders, and contribute substantially to decreased health related quality of life and high medical costs.[4-7]

The pathophysiology of DGBI included under the category of bowel disorders (IBS, functional constipation, functional diarrhea, and functional bloating) is multifactorial and includes brain-gut axis disruption, changes in the gut microbiome, immune system dysregulation and visceral hypersensitivity.[8-10] The role of diet in DGBI has received recent attention as both contributing to the pathophysiology of, and informing therapeutic options for, DGBI.[1-3] Most of the data on diet and DGBI comes from intervention studies where diet is altered, and the therapeutic impact assessed. While such studies have identified dietary strategies, such as a diet low in fermentable oligosaccharides, disaccharides, monosaccharides, and

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polyols (FODMAPs), that reduce symptom burden, the relationship between an individual's usual or customary diet and the presence of DGBI has been studied less. While individual studies have variably revealed relationships between higher intake of spicy foods,[11,12] caffeine,[13,14] as well as consuming diets high in fat, protein and sugar (as in Western, "fast food" and pro-inflammatory),[12,14-17] ultra-processed foods[18,19] or FODMAPS[20] and an increase in prevalence of IBS, others have reported an association between higher intake of fiber,[13,21] and vegetables[15,22] as well as vegetarian[16,23] and Mediterranean diets[24] and a lower prevalence of IBS. These associations were not universally consistent,[25,26] however, and were performed in restricted populations (eg, by age, sex, geographic location or occupation), thereby, exposing their results to the influence of the many socio-economic and cultural factors that influence dietary choices.

The primary aim of this study is to analyze the database of the Rome Foundation Global Epidemiology Study to provide a more global assessment of relationships between diet and DGBI by evaluating associations between certain dietary patterns and the prevalence of 5 common DGBI. The secondary aim is to evaluate associations between diet as defined by starch predominance and DGBI.

MATERIALS AND METHODS

Sample

The data analyzed in this study were derived from a comprehensive survey questionnaire. To ensure consistency and accessibility the complete study questionnaire was translated by a professional company into 21 languages with linguistic validation (cognitive debriefing) and cultural adaptation. Where appropriate the translations were "localized" to dialects within a given language. All translations were reviewed by clinical monitors and the final versions were approved by native professional experts in gastrointestinal disorders. Dietary data were collected in the same manner as all other data in the questionnaire – all questionnaires were de-identified.

Adult participants from both the Internet and Household surveys of the Rome Foundation Global Epidemiology Study were studied, which has been described previously.[4,27] The surveys were conducted in 33 countries: Argentina, Australia, Bangladesh, Belgium, Brazil, Canada, China, Colombia, Egypt, France, Germany, Ghana, Holland, India, Indonesia, Iran, Israel, Italy, Japan, Malaysia, Mexico, Nigeria, Poland, Romania, Russia, Singapore, South Africa, South Korea, Spain, Sweden, Turkey, the United Kingdom and the United States.

In countries where most adults use the Internet, a secure, anonymous online survey was conducted using population samples.[27] In countries in which an Internet survey was not feasible, personal interviews were conducted in selected villages and cities without national representation.[27] The household survey countries were Bangladesh, India, Indonesia, Iran, Malaysia, and Nigeria. In China and Turkey, data were collected with both in-person and Internet surveys.

For the Internet surveys a global market research company (Qualtrics Inc, Provo, Utah, USA) enrolled subjects already registered to participate in Internet surveys. For the household surveys directions were provided in each country by a principal investigator experienced in DGBI research and face-to-face interviewing methodology. Surveys were performed according to a standardized operations manual provided by the Rome Foundation Global Epidemiology Study (RFGES). In each country, the local principal investigator selected study areas in several distinct geographic areas and balanced between urban vs rural residence, where possible. One survey participant was selected per household for surveying.[27]

In China, household and Internet surveys were conducted independently of each other. In Turkey, the Internet survey was performed first throughout the country and the household survey then designed to sample a population with the same demographic characteristics and geographic distribution throughout the country as the Internet survey.[27]

Precise details of the conduct of the survey in each of the household survey countries have been published previously.[27] To summarize, for each country a sampling frame was developed which included basic demographics and then sorted into strata to ensure appropriate representation by sex and age groups. A simple random sample of the appropriate size (proportional to the size of the stratum) was then randomly selected for a total of at least 2000 individuals. The randomly selected individual was then contacted to set up the interview.

Variables

Bowel DGBI (irritable bowel syndrome, functional constipation, functional diarrhea, and functional bloating) were defined using Rome IV criteria,[28] based on responses to the Rome IV Diagnostic Questionnaire. Subtyping of IBS according to predominant bowel habit, ie, constipation, diarrhea, mixed or unclassified, was also performed based on Rome IV guidelines.

The vegan diet was defined as not consuming milk products, meat from animals, fish and eggs; vegetarian diet as not consuming meat from animals and fish; and lactose-free diet as not consuming milk products. Bread-, pasta- and rice-predominant diets were defined as eating bread, pasta or rice 4–7 days a week, re-

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spectively, and eating the other 2 carbohydrate sources 0-3 days a week.

Statistical Methods

Data are presented separately for Internet and household surveys participants as prior studies demonstrated important differences in prevalence of DGBI between these survey methods, even when performed in the same country.[27] This may have been related to a difference in willingness to report DGBI when interviewed in person rather than when completing a survey online.

The frequency distribution of the various types of diets was initially examined separately, by mode of survey. Bivariate logistic regression examined the unadjusted association of the various types of diet with bowel DGBI. Associations between diets and bowel DGBI identified in this analysis were then further explored in multivariate analyses. Multivariable logistic regression provided the adjusted odds ratios and 95% CI for the associations between the diets and bowel DGBI, adjusting for age, sex, religion, geographical region, and living conditions. Details of age, sex, and religion distributions are presented in Table 1A. With respect to geography, sampling was performed in urban and rural areas and in larger countries in different geographic regions of the country (detailed in Sperber et al[27]). With regard to living conditions, the following parameters were included in the survey: whether living on a farm with animals, whether home had running water on tap or has electric power. Access to medical care was also a component of the questionnaire, ie, whether free, available for a small fee or at substantial cost. Here again and for the reasons stated above, all analyses were performed separately for results from the Internet and household surveys.

On all analyses differences which achieved a *P*-value ≤ 0.05 were regarded as statistically significant. No adjustments were made for multiple comparisons.

RESULTS

A total of 54 127 internet survey subjects and 20 973 household survey subjects were included.

Details of the distribution of sex, age and religion in the internet and household surveys are provided in Table 1A. While the 2 survey populations were well matched for sex and age, significant differences were noted, as expected, for religion with Christianity being the most common affiliation in the Internet survey population and Islam and Eastern/Asian religions in the household survey.

Results for dietary preferences are presented in Table 1B. For internet survey subjects, the mean age was 44 years and 49.1% were female. The majority (93.3%) did not meet criteria for either a vegan or vegetarian diet. A milk-free diet was consumed by 5.1%. The most common starch-predominant diet was bread-predominant (ie, high in bread and low in rice and pasta, consumed by 38.2%) (Table 1B).

For household survey subjects, the mean age was 44 years and 50.7% were female. Although, again, most (88.6%) subjects did not meet criteria for a vegan or vegetarian diet, more household survey respondents reported consuming vegetarian (9.8% vs 2.2%) or lactose-free (22.9% vs 5.1%) diets than those in the internet survey. In terms of starch source, consumption of a rice-predominant diet was more common (39.8% vs 11.8%) than in the Internet survey subject. In both surveys pasta-predominant diets were more common at 1.6% and 0.5%, respectively, in internet and household surveys (Table 1B).

Table 1A. Demographics (Sex, Age, and Religion) of the Internet and Household Survey Populations

Demographic	Internet survey	Household survey
Sex		
Male	50.9%	49.3%
Female	49.1%	50.7%
Age range		
18-39 yr	42.5%	41.6%
40-64 yr	41.2%	42%
≥ 65 yr	16.3%	16.4%
Religion		
Christianity	37.2%	
Islam	7.5%	37%
Judaism	3.5%	
Eastern/Asian	5.1%	36.3%
None/other	40.3%	13.7%
Missing	6.5%	13.0%

Table 1B. Prevalence of Pre-specified Dietary Patterns of Interest Among Internet and Household Survey Respondents

Diet	Internet survey respondents (n = 54 127)	Household survey respondents (n = 20 973)
Vegan	204 (0.4)	336 (1.6)
Vegetarian	1177 (2.2)	2048 (9.8)
Lactose-free	2756 (5.1)	4811 (22.9)
High in bread, low in pasta and rice	20 665 (38.2)	1607 (7.7)
High in pasta and low in bread and rice	869 (1.6)	98 (0.5)
High in rice and low in bread and pasta	6116 (11.3)	8349 (39.8)

Data are presented as n (%).

Bivariate analyses (see Table 2)

In Internet survey countries, a vegetarian diet was associated with a higher prevalence of any bowel DGBI diagnosis (odds ratio [OR] = 1.14; 95% confidence interval [CI], 1.02-1.28) and functional bloating (OR = 1.66; 95% CI, 1.27-2.18), in particular, and a lactose-free diet with IBS (OR = 1.26; 95% CI, 1.06-1.51) and diarrhea-predominant IBS (IBS-D), in particular (OR = 1.48; 95% CI, 1.09-2.00). Bread and rice predominant diets were associated with lower likelihood of any bowel DGBI (OR = 0.95; 95% CI, 0.92-0.99 and OR = 0.91; 95% CI, 0.86-0.96, respectively) and IBS (OR = 0.91; 95% CI, 0.83-0.93 and OR = 0.67; 95% CI, 0.57-0.78, respectively). A vegan diet was associated with a lower prevalence of functional constipation (OR = 0.56; 95% CI, 0.32-0.96) and a rice-predominant diet with lower odds of functional bloating (OR = 0.64; 95% CI, 0.54-0.77), constipation-predominant IBS (IBS-C) (OR = 0.55; 95% CI, 0.41-0.74) and mixed IBS (IBS-M) (OR = 0.68; 95% CI, 0.52-0.89).

In household survey countries, no associations were evident for a vegan diet while, with the exception of IBS-C and IBS-M, a vegetarian diet was associated with a lower prevalence of all bowel DGBIs, with ORs ranging from 0.2 (0.05-0.81) for IBS-D to 0.5 (0.38–0.67) for functional constipation. As in internet surveys, a lactose-free diet was associated with a higher prevalence of any bowel DGBI (OR = 1.7; 95% CI, 1.57-1.83), IBS (OR = 2.54; 95% CI, 2.03-3.17), functional constipation (OR = 1.44; 95% CI, 1.25-1.66), functional diarrhea (OR = 1.66; 95% CI, 1.27-2.17), IBS-C (OR = 2.67; 95% CI, 1.85-3.84) and IBS-M (OR = 2.42; 95% CI, 1.42-4.15), and a rice-predominant diet with IBS (OR = 1.31; 95% CI, 1.05-1.63) and IBS-D (OR = 2.05; 95% CI, 1.56-3.09). This same diet was associated with a lower prevalence of functional bloating (OR = 0.73; 95% CI, 0.56-0.95).

Multivariate analyses (see Table 3)

These analyses corrected for several factors that could influence dietary preferences: age, sex, religion, geographical region and living conditions.

In Internet survey countries, a vegan diet was associated with a lower prevalence of functional constipation (OR = 0.54; 95% CI, 0.31-0.94). In contrast, vegetarian diet was associated with a higher prevalence of functional bloating (OR = 1.39; 95% CI, 1.06-1.83), and a lactose-free diet with IBS-D (OR = 1.57; 95% CI, 1.15-2.13). Bread and rice predominant diets were associated with a lower prevalence of IBS (OR = 0.86; 95% CI, 0.78-0.95 and OR = 0.80; 95% CI, 0.67-0.95, respectively) and IBS-C (OR = 0.78; 95% CI, 0.66-0.92 and OR = 0.67; 95% CI, 0.49-0.93, respectively), in particular.

In household survey countries, in contrast to the internet survey respondents, a vegan diet was linked to a higher prevalence of any bowel DGBI (OR = 1.63; 95% CI, 1.19-2.23). Here, a vegetarian diet was linked to a lower prevalence of any bowel DGBI (OR = 0.66; 95% CI, 0.55-0.79) and functional constipation (OR = 0.62; 95% CI, 0.46-0.85) but a higher prevalence of IBS (OR = 2.44) while a lactose-free diet was associated with functional constipation (OR = 1.26; 95% CI, 1.08-1.46), IBS-C (OR = 2.43; 95% CI, 1.64-3.60) and IBS-M (OR = 2.61; 95% CI, 1.44-4.73) and a rice-predominant diet with IBS-D (OR = 1.99; 95% CI, 1.28-3.09).

DISCUSSION

In this study, we found, first, that only a small minority in both surveys selected a vegan or vegetarian diet and that the predominant carbohydrate in the diet differed substantially between internet and household respondents. The latter was not surprising given geographic and demographic differences between the 2 survey populations[4] with internet surveys performed in North and South America and Western Europe and household surveys in more developing countries, including those in sub-Saharan Africa, the Middle East and India.[4] Both vegetarian and lactose-free diets were more prevalent in household survey countries. This may reflect the influence of diets dictated by religious practices: whereas 40% of internet survey countries reported no religion, Islam and Eastern/Asian religions were common in household survey countries (37.0% and 36.3%, respectively). Second, associations between a diet and DGBI were variable, influenced by both methods of analysis and survey methodology. The differences in results between internet and household surveys are a concern. In the main survey of DGBI prevalence,[4] significant differences in DGBI prevalence were found depending on survey methodology with the overall prevalence of at least one DGBI being 20.7% in household and 40.3% in internet surveys. Furthermore, variation in country prevalence for any DGBI was much higher between household than internet survey countries.[4,27] Factors that might contribute to such a discrepancy have been explored in a prior publication.[27] While issues with the quality of data entry were more frequent in household surveys, their frequency was still too low to account for the significant differences in survey results.[27] Similarly, while socioeconomic and cultural factors may have differed between survey countries, these were accounted for in our analysis and did not impact on outcomes. Where both survey methods were employed, the results have been compared but must be interpreted with caution, as in China the nature of the populations surveyed by the 2 methods differed substantially, and in Turkey, where efforts were made to match the populations surveyed by both methods, the

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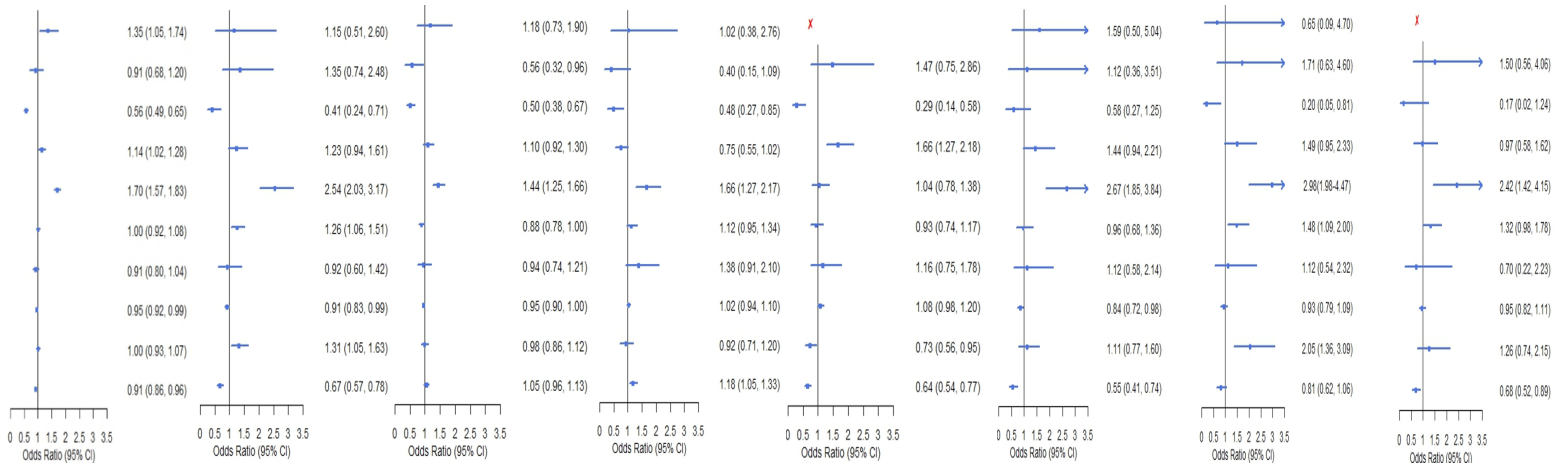
Table 2. Forest Plots of Bivariate Associations Between Dietary Patterns and Disorders of Gut–Brain Interaction in Household and Internet Surveys (Odds Ratio and 95% Confidence Interval)

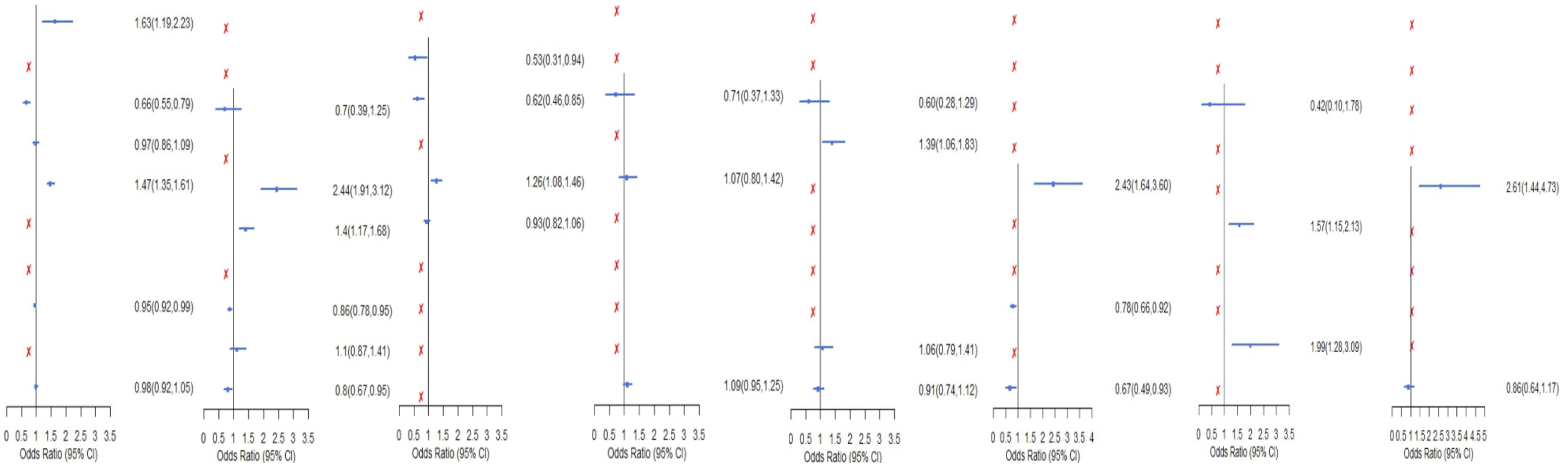
Diet	All DGBI	All IBS	Functional constipation	Functional diarrhea	Functional bloating	IBS-C	IBS-D	IBS-M
Vegan								
HHS 336	1.35 (1.05, 1.74)	1.15 (0.51, 2.60)	1.18 (0.73, 1.90)	1.02 (0.38, 2.76)	X	1.59 (0.50, 5.04)	0.65 (0.09, 4.70)	X
INT 204	0.91 (0.68, 1.20)	1.35 (0.74, 2.48)	0.56 (0.32, 0.96)	0.40 (0.15, 1.09)	1.47 (0.75, 2.86)	1.12 (0.36, 3.51)	1.71 (0.63, 4.60)	1.50 (0.56, 4.06)
Vegetarian								
HHS 2048	0.56 (0.49, 0.65)	0.41 (0.24, 0.71)	0.50 (0.38, 0.67)	0.48 (0.27, 0.85)	0.29 (0.14, 0.58)	0.58 (0.27, 1.25)	0.20 (0.05, 0.81)	0.17 (0.02, 1.24)
INT 1177	1.14 (1.02, 1.28)	1.23 (0.94, 1.61)	1.10 (0.92, 1.30)	0.75 (0.55, 1.02)	1.66 (1.27, 2.18)	1.44 (0.94, 2.21)	1.49 (0.95, 2.33)	0.97 (0.58, 1.62)
Lactose-Free								
HHS 4811	1.70 (1.57, 1.83)	2.54 (2.03, 3.17)	1.44 (1.25, 1.66)	1.66 (1.27, 2.17)	1.04 (0.78, 1.38)	2.67 (1.85, 3.84)	2.98(1.98-4.47)	2.42 (1.42, 4.15)
INT 2756	1.00 (0.92, 1.08)	1.26 (1.06, 1.51)	0.88 (0.78, 1.00)	1.12 (0.95, 1.34)	0.93 (0.74, 1.17)	0.96 (0.68, 1.36)	1.48 (1.09, 2.00)	1.32 (0.98, 1.78)
Bread-Predominant								
HHS 1607	0.91 (0.80, 1.04)	0.92 (0.60, 1.42)	0.94 (0.74, 1.21)	1.38 (0.91, 2.10)	1.16 (0.75, 1.78)	1.12 (0.58, 2.14)	1.12 (0.54, 2.32)	0.70 (0.22, 2.23)
INT 20665	0.95 (0.92, 0.99)	0.91 (0.83, 0.99)	0.95 (0.90, 1.00)	1.02 (0.94, 1.10)	1.08 (0.98, 1.20)	0.84 (0.72, 0.98)	0.93 (0.79, 1.09)	0.95 (0.82, 1.11)
Rice- Predominant								
HHS 8349	1.00 (0.93, 1.07)	1.31 (1.05, 1.63)	0.98 (0.86, 1.12)	0.92 (0.71, 1.20)	0.73 (0.56, 0.95)	1.11 (0.77, 1.60)	2.05 (1.36, 3.09)	1.26 (0.74, 2.15)
INT 6116	0.91 (0.86, 0.96)	0.67 (0.57, 0.78)	1.05 (0.96, 1.13)	1.18 (1.05, 1.33)	0.64 (0.54, 0.77)	0.55 (0.41, 0.74)	0.81 (0.62, 1.06)	0.68 (0.52, 0.89)



DGBI, disorders of gut-brain interaction; IBS, irritable bowel syndrome; IBS-C, constipation-predominant IBS; IBS-D, diarrhea-predominant IBS; IBS-M, mixed IBS; X, could not be computed.

Table 3. Forest Plots of Multivariate Associations Between Dietary Patterns and Disorders of Gut–Brain Interaction in Household and Internet Surveys (Odds Ratio and 95% Confidence Interval)

Diet	All DGBI	All IBS	Functional constipation	Functional diarrhea	Functional bloating	IBS-C	IBS-D	IBS-M
Vegan								
HHS 336	1.63(1.19,2.23)	X	X	X	X	X	X	X
INT 204	X	X	X	0.53(0.31,0.94)	X	X	X	X
Vegetarian								
HHS 2048	0.66(0.55,0.79)	0.7(0.39,1.25)	X	0.62(0.46,0.85)	0.71(0.37,1.33)	0.60(0.28,1.29)	X	X
INT 1177	0.97(0.86,1.09)	X	X	X	X	1.39(1.06,1.83)	0.42(0.10,1.78)	X
Lactose-Free								
HHS 4811	1.47(1.35,1.61)	2.44(1.91,3.12)	X	1.26(1.08,1.46)	1.07(0.80,1.42)	X	2.43(1.64,3.60)	2.61(1.44,4.73)
INT 2756	X	1.4(1.17,1.68)	X	0.93(0.82,1.06)	X	X	1.57(1.15,2.13)	X
Bread-Predominant								
HHS 1607	0.95(0.92,0.99)	0.86(0.78,0.95)	X	X	X	X	0.78(0.66,0.92)	X
INT 20665	X	1.1(0.87,1.41)	X	X	X	1.06(0.79,1.41)	1.99(1.28,3.09)	X
Rice- Predominant								
HHS 8349	0.98(0.92,1.05)	0.8(0.67,0.95)	X	X	1.09(0.95,1.25)	0.91(0.74,1.12)	0.67(0.49,0.93)	0.86(0.64,1.17)
INT 6116	X	X	X	X	X	X	X	X



DGBI, disorders of gut-brain interaction; IBS, irritable bowel syndrome; IBS-C, constipation-predominant IBS; IBS-D, diarrhea-predominant IBS; IBS-M, mixed IBS; X, could not be computed.

level of education achieved by internet respondents was higher than household respondents. It was concluded that survey methodology per se was the primary factor influencing the prevalence of DGBI[27] and while the reasons for these discrepancies cannot be fully explained, they may include a difference in response to a face-to-face, in person interview in contrast to the relative anonymity of an internet survey. In terms of the results of the dietary survey, caution must be exerted in interpreting findings such as the contrasting impacts of a vegan or vegetarian diet on DGBI prevalence between surveys as rates for vegan diets were very low in both (0.4% and 1.6%) resulting in very low numbers of responses and exposing analyses to error. Which survey method was more accurate cannot be determined with confidence. We surmise that because of a reluctance to discuss quite intimate symptoms in face-to-face interviews, the internet survey method may have provided more accurate data on symptom prevalence; while we could speculate that diet, a more "neutral" subject, might have been less affected, we have no data to support this.[27]

By correcting for several potential confounding factors, the multivariate analysis provided the most accurate assessment of diet-DGBI associations and, in general, confirmed but refined the findings from the bivariate analyses. It must be stressed that relationships between vegan and vegetarian diets and DGBI have been inconsistent in other surveys with some reporting lower rates for DGBI for those on these diets[15,23] and others not.[25,26,29-32]

There was a more consistent association between a lactose-free diet and various bowel DGBI: IBS-D in the internet survey and functional constipation, IBS-C and IBS-M in the household survey. Indeed, lactose is a common trigger for gastrointestinal symptoms such as diarrhea and abdominal bloating in IBS.[33-35] Consequently, those with DGBI may have either consciously or unconsciously avoided dairy products even though the survey results suggest that this strategy may not have proven wholly successful. Individuals with IBS are known to modify their diets to reduce symptoms of their own accord without input from health care professionals.[36]

In terms of the influence of the predominant carbohydrates in the diet, no significant associations were noted for pasta, which is likely attributable to the low numbers who reported this dietary pattern in both surveys (1.6% and 0.5% for internet and household surveys, respectively). Among internet survey respondents where bread was the predominant source of carbohydrate, a bread-predominant diet was associated with a lower prevalence for all IBS and IBS-C; no associations were noted in household survey respondents who had a much lower rate of a bread predominant diet (7.7% vs 38.2% for internet respondents). Regarding a rice-predominant diet, it too was associated with a lower prevalence for all IBS and IBS-C among internet respondents but not among household respondents (who were 4-fold more likely to consume this diet at 39.8% vs 11.3% for internet respondents), where this diet was associated with a higher prevalence of IBS-D. The reasons for such discrepant results are not apparent and may be related to the factors that may have influenced responses to household surveys. The credibility of the internet survey responses is supported by other data suggesting that a rice-based diet is associated with fewer symptoms of IBS[37] related, perhaps, to the lower impact of rice, than other carbohydrate sources, on intestinal gas.[38] A switch from a bread-predominant to a rice-predominant diet would reduce the intake of fructans which are present in a high concentration in wheat and related grains and linked with IBS symptoms.[20,39-42] It was surprising to note, therefore, that a bread-predominant diet (the main source of starch among these respondents) was associated with a lower prevalence of IBS and IBS-C in the internet survey. This observation seems credible and speaks against a major role for fructans, as well as gluten (also been implicated in the pathogenesis of IBS symptoms),[43,44] assuming that gluten-containing grains were the sources of flour for most breads. A report of lower FODMAP intake among IBS sufferers might seem to question the role of these compounds in IBS[45] but could also represent another example of dietary adjustment.

The link between diet and DGBI may be mediated by several factors, including the impact of dietary patterns on gut microbiota[46-50] and/or its immune response.[51]

The range of dietary patterns evaluated in this study was limited to those included in the survey questionnaire and judged feasible in the context of the study. Clearly other factors, not assessed, such as intakes of dietary fiber and prebiotics[52-57] or consumption of artificial sweeteners, emulsifiers and other additives might also play a role in symptom development.[58]

Strengths of our study include the rigor of its methodology, which sampled countries across the world with varying ethnicities, cultures, and dietary patterns. On the other hand, survey data can only identify associations between variables and cannot assign causation or even temporal relationships. Small numbers in relation to some of our prespecified diets further limited our ability to define associations with any DGBI.

In this large, global survey study of DGBI, we failed (albeit within the limitations of small sample sizes) to identify a consistent association between vegan or vegetarian diets and bowel DGBI. Somewhat unexpected were the associations noted between lactose-free (linked to a higher prevalence) and bread- and rice-predominant diets (linked to lower prevalence) and some bowel DGBI. Longitudinal, prospective studies are needed to elucidate the directionality of these associations. Do dietary habits cause DGBI, are they adopted in response to DGBI, or is combination of both

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involved depending on demographic, socio-economic and cultural factors?

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Conflicts of interest

None.

Author contributions

Conceptualization: Daniel Bushyhead, Ami D Sperber, and Eamonn M M Quigley; data curation: Daniel Bushyhead, Rutaba Khatun, Shrikant I Bangdiwala, Olafur Palsson, Ami D Sperber, and Eamonn M M Quigley; formal analysis: Daniel Bushyhead, Rutaba Khatun, Shrikant I Bangdiwala, and Eamonn M M Quigley; funding acquisition: Ami D Sperber; project administration: Ami D Sperber and Eamonn M M Quigley; resources: Ami D Sperber, Eamonn M M Quigley, Shrikant I Bangdiwala, and Olafur Palsson; writing–original draft: Daniel Bushyhead; and supervision: Eamonn M M Quigley and Shrikant I Bangdiwala; investigation, methodology, validation, writing–review, and editing: all authors.

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Morphological, Molecular and Motility Changes in Human Small Intestine After Loop Ileostomy

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Background/Aims: Morphological and functional changes occur after loop ileostomy surgery such as villous atrophy and decreased contractility. This study aims to examine the changes in motility of the proximal and distal limbs of the ileostomy. Morphological, molecular and functional approaches of smooth muscle are attempted to elucidate the cause of the changes in motility.
Methods: Full thickness strips of circular muscle were prepared from 25 patients (median age 64 years). Western blot was performed to quantify the expression of contractile proteins. Expression of genes was quantified using quantitative real-time polymerase chain reaction.
Results: The distal limb showed morphological atrophy and a significant reduction in the contractile response to acetylcholine and potassium chloride, as well as a reduced relaxation response to sodium nitroprusside. However, no changes in the number of platelet derived growth factor receptor-α-positive (PDGFRα$^+$) cells and interstitial cell of Cajal (ICC) were observed between the proximal and distal limbs. Western blot analysis confirmed that the expression levels of proteins associated with contraction such as myosin phosphatase targeting subunit 1, protein kinase C-potentiated phosphatase inhibitor protein-17 kDa, and Rho-associated, coiled-coil containing protein kinase 1, were significantly down-regulated in the distal limb. Despite the decrease in the expression of these proteins, neuronal cells and smooth muscle cells were preserved in the distal limb that has been diverted after ileostomy.
Conclusions: Although morphological, molecular and motility changes including down-regulation of contractile protein expression were observed in the distal loop, number of PDGFRα$^+$ cells, ICCs, and neuronal cells were preserved. These results may explain restoration of the bowel motility after ileostomy take-down with low incidence of ileus.

Keywords: Gastrointestinal motility; Ileostomy; Small intestine

INTRODUCTION

Temporary loop ileostomy is widely utilized to protect a newly constructed anastomosis in rectal cancer surgery, especially in patients undergoing sphincter-preserving procedures or receiving preoperative chemoradiotherapy, both of which increase the risk of anastomotic leakage.[1-3] By diverting the fecal stream, loop ileostomy reduces intraluminal pressure and mechanical stress on the anastomotic site, leading to significantly lower rates of leakage and reoperation. As a result, it has become a standard component of modern rectal cancer management.[4-6]

A loop ileostomy consists of 2 distinct segments: a functional proximal limb that continues to receive luminal flow and preserve normal intestinal physiology, and a nonfunctional distal limb that remains unused. The distal segment experiences prolonged deprivation of mechanical stimulation, luminal nutrients, and microbial exposure. Following confirmation of anastomotic healing, ileostomy reversal is performed. Despite restoration of luminal continuity, postoperative ileus occurs in up to 29% of patients, delaying recovery and prolonging hospitalization.[7] The physiology underlying motility recovery remains unclear. Diversion-induced structural and functional remodeling of the distal ileum is thought to contribute, but the extent to which these changes involve smooth muscle contractile pathways, neuromuscular cellular components,

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and regulatory signaling in humans has not been fully characterized. Additionally, the influence of peri-operative chemotherapy, radiation, and nutritional changes on intestinal remodeling is not well defined.

Therefore, the aim of this study is to investigate morphological, molecular, and motility changes in paired proximal and distal ileal segments collected at the time of ileostomy takedown. By analyzing full-thickness human tissues using histology, immunostaining, isometric contractility testing, and molecular profiling, we sought to provide a comprehensive and mechanistic understanding of diversion-associated alterations. This approach may clarify the physiological impact of intestinal disuse and help explain variability in motility recovery and susceptibility to postoperative ileus following ileostomy reversal.

MATERIALS AND METHODS

Tissue Acquisition

Human ileum samples were obtained from 25 patients who underwent diversion ileostomy after rectal cancer surgery and subsequently had ileostomy take-down, including 25 pairs of samples from the afferent (proximal) and efferent (distal) limbs. Following the surgery and adjuvant chemotherapy, an ileostomy takedown is conducted. This involves dissecting the exteriorized ileum, resecting it internally within the abdominal cavity, and performing an end-to-end anastomosis. For standardization, a 4-cm full-thickness segment was excised from the proximal limb (4 cm proximal to the stoma) and a corresponding 4-cm segment was excised from the distal limb (4 cm distal to the stoma). The tissues were obtained naturally as part of this surgical process, and no additional resection of the small intestine was performed. We emphasize that no unnecessary or invasive procedures were conducted on patients for research purposes. The indications for ileostomy were rectal cancer in 20 patients and rectosigmoid cancer in 5 patients. The median time from surgery to ileostomy take-down was 7 months (range 7-8 months). All 25 patients received adjuvant chemotherapy after surgery, consisting of FOLFOX (5-fluorouracil + leucovorin + oxaloplatin) in 12 patients, FL (5-fluorouracil + leucovorin) in 7 patients, and Xeloda (capecitabine) in 6 patients. Because both proximal and distal limb tissues were collected from the same patient, each paired sample served as an internal control to minimize systemic factors such as chemotherapy exposure, nutritional status, or disease-specific variability. Therefore, the observed inter-limb differences are interpreted as primarily reflecting the effects of luminal disuse, although potential contributions from chemotherapy-related changes cannot be excluded. Among 25 paired samples collected, final usable sample numbers varied depending on tissue integrity and availability of intact smooth muscle bundles: morphological analysis (n = 15 pairs), motility assessment (n = 12 pairs), Western blotting (n = 7 pairs), and quantitative real-time polymerase chain reaction (qPCR, n = 10 pairs). Exclusions occurred due to insufficient muscle layer, tissue damage during clinical handling, or inadequate RNA/protein yield. This study was conducted with the approval of the Institutional Review Board (IRB) of Dongnam Institute of Radiological and Medical Sciences (No. D-1810-016-002) and Seoul National University Hospital (No. H-0603-071-170), which ensures that the study was carried out legally and ethically in human subjects' research. The study protocol was performed in accordance with the guidelines and regulations of the Institutional Review Boards of Dongnam Institute of Radiological and Medical Sciences and Seoul National University Hospital. Written informed consent was obtained from all patients before operations.

Histological Analysis and Staining

To assess the morphological characteristics of the proximal and distal limbs, histological analysis was performed. The fixed ileum specimens were dehydrated using a series of graded alcohol concentrations and embedded in paraffin blocks. Serial sections of 4 μm thickness were obtained using a microtome and mounted on glass slides. For the identification of markers of specific cells involved in intestinal motility, immunohistochemical staining was performed on the tissue sections. The sections were deparaffinized, rehydrated, and subjected to antigen retrieval using appropriate techniques. The sections were then incubated with the primary antibody against S-100 protein, followed by appropriate secondary antibodies. The immune complex was visualized using a chromogenic substrate, and the sections were counterstained with hematoxylin for enhanced visualization. Formalin-fixed, paraffin-embedded tissue sections were stained using a Leica automatic immunostaining device (Lecia BOND I; Leica Biosystems, Nussloch, Germany). The samples were incubated with antibodies to S-100 (1:5000; DAKO, Glostrup, Denmark), C-kit (1:300; DAKO), and PDGFRα (1:100; Santa Cruz, Dallas, TX, USA). Immunostaining was performed to evaluate the presence and distribution of interstitial cells of Cajal (ICC) and platelet-derived growth factor receptor-α-positive (PDGFRα$^+$) cells within the proximal and distal limbs of the ileostomy samples. This analysis aimed to assess the cellular composition and spatial organization of these cell populations. Fixed tissue sections of the proximal and distal limbs were prepared as previously described in the Histological Analysis and Staining section. Primary antibodies (PDGFRα and C-kit) specific to PDGFRα$^+$ cell and ICC were applied to the sections, followed by incubation. The sections were then washed to remove excess pri-

mary antibodies. PDGFRα^{+} cells and ICC were manually counted within defined unit areas. The number of cells per unit area was counted for each sample. All the immunohistochemical-stained slides were reviewed by 2 experienced pathologists (Soo Young Chung and Jooyeong Song) for improved accuracy.

Isometric Tension Recordings

The tissue specimens were collected from a patient who underwent loop ileostomy for rectal cancer and subsequently had ileostomy take-down. Samples were collected from the tips of the proximal and distal limbs. The specimens were obtained in fresh condition from the operating room. After performing ileostomy take-down, ileum segments (4 × 2 cm) were removed from the resected sections. The specimens were immediately placed into oxygenated Krebs-Ringer Bicarbonate solution (KRB). The KRB contained (in mM) 120.4 NaCl, 5.9 KCl, 15.5 $NaHCO_3$, 11.5 glucose, 1.2 $MgCl_2$, 1.2 NaH_2PO_4, and 2.5 $CaCl_2$ and had a pH of 7.3-7.4 at 36.5 ± 0.5°C when bubbled to equilibrium with 97% O2 and 3% CO_2. All in situ techniques were similar to our previous studies.[8,9] The mucosal and submucosal layers were gently removed and full thickness strips (5 × 2 mm) of circular muscle were prepared from the tissue of the proximal and distal limbs. The strips were attached to an isometric force transducer (Biopac Systems, Goleta, CA, USA) and suspended in a 10-mL organ bath containing prewarmed KRB solution, while being supplied with oxygen. Muscle strips (5 × 2 mm) were mounted in 10-mL organ baths containing Krebs-Ringer bicarbonate solution (36.5 ± 0.5°C), continuously aerated with 97% O_2 and 3% CO_2. A preload tension of 10 mN was applied, and all strips were equilibrated for 120 minutes before recording spontaneous and drug-induced contractile activity. Amplitude (mN) and area under the curve (AUC, sec × mN/wave) were measured and analyzed using Graphpad Prism (version 5.0; Graphpad Software, San Diego, CA USA) and Clampfit (version 10.2; Axon Instruments, Foster City, CA, USA) software.

Western Blot Analysis

Total protein from the ileal specimens was extracted by a MagNA Lyser homogenizer (Roche, Penzberg, Germany) in protein extraction buffer (10 mM NaCl, 10 mM KCl, 10 mM EDTA, 10 mM DTT, 0.5% NP-40, and 20 mM Tris-HCl; pH 7.8) containing complete protease inhibitor (Roche, Basel, Switzerland). The homogenate was centrifuged at 12 000 rpm. for 15 minutes at 41°C and the protein concentration was measured using Bradford reagent (Bio-Rad Laboratories, Hercules, CA, USA). A total of 100 mg of protein/sample was resolved on a 10% SDS–PAGE, transferred to nitrocellulose membrane (Hybondt-c extra; Amersham Biosciences, Little Chalfont, Buckinghamshire, UK) and detected by enhanced chemiluminescence (ECL; PIERCE, Rockford, IL, USA). Quantification was performed using ImageJ software. Paired proximal–distal values were analyzed on the same blot whenever possible to reduce analytic variability. Band intensities were normalized to β-actin.

Quantitative Real-time Polymerase Chain Reaction

The mucosa and submucosa were removed by sharp dissection, and the remaining smooth muscle tissue was used for qPCR for *VAChT* as a cholinergic neuronal marker, *UCHL1* as a neuronal molecular marker and MYH11 as a molecular marker for smooth muscle cells (SMCs). Total RNA was extracted using the RNeasy Protect Mini Kit (Qiagen, Hilden, Germany). Complementary DNA was synthesized with 1 μg total RNA as the template using the SuperScript III First-Strand synthesis system according to the manufacturer's instructions (Invitrogen, Carlsbad, CA, USA). Real-time PCR was conducted using the StepOne Real-time PCR system (Thermo Fisher Scientific Inc, Waltham, MA, USA). Specific primers for *MYH11*, *UCHL1*, *VAChT*, and 18S ribosomal RNA were used.[10] The primer sequences used for qPCR were as follows: *VAChT* forward, 5'-GCTGTTTGCTTCCAAGGCTATCC-3' and reverse, 5'-GAAGGC GAACAGGACTGTAGAG-3'; *UCHL1* forward, 5'-CAGTTCAGAGGAC ACCCTGCTG-3' and reverse, 5'-CCACAGAGCATTAGGCTGCCTT-3'; *MYH11* forward, 5'-GTGAAGAACGACAACTCCTCAC-3' and reverse, 5'-TGTTGGCTCCCACGATGTAA-3. Messenger RNA expression of each gene was normalized against 18S mRNA levels. Results were calculated as ΔCt values. The relative expression levels of *MYH11*, *UCHL1*, and *VAChT* mRNA were examined in the smooth muscle layer and compared between the proximal and distal limb of ileostomy.

Drug Administration

After regular waves were detected, drugs were added to the organ bath, and changes in the tension were recorded. To evaluate the myogenic contractile and relaxation ability of the proximal and distal limbs, we applied acetylcholine (ACh, 10^{-8}–10^{-4} M), potassium chloride (KCl, 10-30 mM), and sodium nitroprusside (SNP, 10^{-8}–10^{-4} M) in the muscle strips of human ileum. All drugs were obtained from Sigma-Aldrich Chemical Co. For the analysis of amplitude, and AUC of the tension recording, recordings were analyzed for 5 minutes each before and after drug administration.

Statistical Methods

Contraction and relaxation responses induced by ACh, KCl, and SNP were normalized to each control condition (before application of each drug). Normalized response was calculated as: (Post-drug amplitude – baseline)/baseline × 100%. All data are expressed as the mean ± standard error of the mean, and statistical analysis

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was conducted using SPSS version 21.0 for Windows (IBM Corp, Armonk, NY, USA). Statistical methods now specify paired t test for normally distributed paired datasets and Wilcoxon signed-rank test when non-parametric. Statistical significance was defined as a P-value < 0.05.

RESULTS

Clinical Characteristics

The study cohort consisted of 25 patients who underwent loop ileostomy for rectal or rectosigmoid cancer followed by ileostomy reversal. The median age was 64 years (median age 64 years, range 46-77 years), and both sexes were represented. The duration of fecal diversion was relatively uniform, with a median interval of 7 months (range, 7-8 months). All patients received preoperative pelvic radiation (50.4 Gy) and adjuvant chemotherapy, including FOLFOX (n = 12), FL (n = 7), or Xeloda (capecitabine; n = 6). Nutritional status at the time of reversal was generally preserved, with a mean serum albumin level of 3.9 g/dL (range, 3.4-4.5 g/dL) and a mean body mass index of 23.5 kg/m^2 (range, 21.1-26.4 kg/m^2). Detailed patient characteristics are summarized in Table. Postoperative ileus (POI) was assessed clinically based on delayed recovery of gastrointestinal (GI) function requiring additional intervention, consistent with commonly used time-to-recovery–based definitions. In this cohort, no patient developed clinically significant POI after ileostomy reversal, and none required nasogastric decompression or additional supportive treatment. The mean time to first passage of flatus was 4.0 ± 1.3 days, indicating uneventful recovery of bowel function despite prolonged diversion and prior exposure to chemoradiotherapy.

Table. Patient Characteristics

Patient	Sex	Age (yr)	Cause of ileostomy	Duration after ileostomy (months)	Albumin (g/dL)	BMI (kg/m^2)	Radiation exposure (Gy)	Chemotherapy regimen
#1	M	68	Rectal cancer	7	3.6	21.1	50.4	FOLFOX
#2	F	77	Rectal cancer	7	3.5	22.3	50.4	FL
#3	M	72	Rectal cancer	7	4.0	25.4	50.4	FL
#4	M	76	Rectal cancer	7	4.5	24.6	50.4	FL
#5	M	73	Rectosigmoid cancer	8	3.8	23.8	50.4	FOLFOX
#6	M	56	Rectosigmoid cancer	8	3.7	25.1	50.4	FOLFOX
#7	F	62	Rectal cancer	7	4.2	24.1	50.4	FOLFOX
#8	M	65	Rectal cancer	7	4.5	25.6	50.4	FL
#9	M	72	Rectal cancer	8	4.1	23.5	50.4	FL
#10	F	61	Rectosigmoid cancer	8	3.9	22.4	50.4	FOLFOX
#11	M	68	Rectal cancer	7	3.8	23.2	50.4	FOLFOX
#12	M	63	Rectal cancer	7	4.2	26.4	50.4	FOLFOX
#13	F	61	Rectal cancer	8	3.5	21.3	50.4	FL
#14	M	52	Rectosigmoid cancer	7	3.5	24.5	50.4	FOLFOX
#15	M	64	Rectal cancer	8	3.9	21.6	50.4	Xeloda
#16	M	53	Rectal cancer	7	4.1	24.6	50.4	Xeloda
#17	F	66	Rectal cancer	7	4.3	24.5	50.4	FOLFOX
#18	M	46	Rectal cancer	7	3.7	22.4	50.4	Xeloda
#19	F	72	Rectosigmoid cancer	7	3.6	21.5	50.4	Xeloda
#20	F	54	Rectal cancer	8	4.0	25.1	50.4	FOLFOX
#21	M	69	Rectal cancer	8	3.5	23.5	50.4	Xeloda
#22	F	63	Rectal cancer	7	3.6	21.3	50.4	Xeloda
#23	M	56	Rectal cancer	8	4.2	24.3	50.4	FL
#24	F	45	Rectal cancer	8	3.4	23.7	50.4	FOLFOX
#25	F	68	Rectal cancer	8	3.5	21.3	50.4	FOLFOX

BMI, body mass index; M, male; F, female; FOLFOX, 5-fluorouracil + leucovorin + oxaloplatin; FL, 5-fluorouracil + leucovorin; Xeloda, capecitabine.

Morphological Changes Between Proximal and Distal Limb of Ileostomy

Morphologically, there was a distinct difference between the proximal and distal limbs of an ileostomy, with the distal limb clearly displaying signs of atrophy compared to the proximal limb (Fig. 1A and 1B). PDGFRα and C-kit immunohistochemistry stainings were performed on the proximal and distal limbs, respectively (Fig. 1 C-F). In 15 pairs of samples, the areas of mucosa, submucosa, and the muscle layer in the distal limb were reduced significantly compared to the proximal limb (Fig. 1G). However, no changes in the number of PDGFRα$^+$ cells and ICC (per 1 × 1 mm^2), known to be involved in intestinal motility, were observed between the proximal and distal limbs (Fig. 1H). The signals for ICC, PDGFRα$^+$ cells, and enteric neurons appeared qualitatively preserved in the distal limb, without an obvious reduction in staining patterns compared with proximal controls.

Tension Recordings of Ileal Strips

Spontaneous contractile waves of ileal strips were recorded using an isometric force transducer from the circular muscle of proximal and distal limb. Spontaneous contractions differed between the proximal and distal limbs (amplitude: 0.154 ± 0.03 mN vs 0.068 ± 0.016 mN, $P < 0.01$; AUC: 4304 ± 2201 sec × mN vs 2827 ± 1918 sec × mN, $P < 0.010$) (Fig. 2).

To measure myogenic contractility of the ileum, ACh and KCl were administered directly into the organ bath in a concentration-dependent manner. As the concentration of ACh (10^{-8} M-10^{-4} M) increased, contractions of the proximal and distal limbs became greater. However, the increase in contractions of the distal limbs was significantly lowered at high concentrations of ACh (Fig. 3; absolute tension and AUC values are provided in Supplementary Fig. 1).

The amplitudes of contractions in the distal limb were significantly lower compared to the proximal limb as ACh concentration increased to 10^{-8} M (98.4 ± 12.7% vs 105.5 ± 5.1%, $P > 0.05$), 10^{-7} M (117.8 ± 14.7% vs 116.2 ± 12.0%, $P > 0.05$), 10^{-6} M (159.5 ± 29.3% vs 123.8 ± 12.4%, $P < 0.01$), 10^{-5} (178.9 ± 37.6% vs 135.7 ± 18.9%, $P < 0.01$), and 10^{-4} M (198.9 ± 42.5% vs 144.8 ± 30.6%, $P < 0.01$) (Fig. 3C). AUCs of contractions in the distal limb were significantly lower compared to the proximal limb as ACh concentration increased to 10^{-8} M (110.2 ± 12.3% vs 102.9 ± 4.7%, $P > 0.05$), 10^{-7} M (128.0 ± 20.0% vs 118.3 ± 11.1%, $P > 0.05$), 10^{-6} M (146.2 ± 27.6% vs 120.0 ± 12.4%, $P < 0.05$), 10^{-5} (171.9 ± 30.1% vs 136.1 ± 20.6%, $P < 0.05$), and 10^{-4} M (170.8 ± 33.3% vs 130.8 ± 13.0%, $P < 0.01$) (Fig. 3D). As with ACh, contractions of the proximal and distal limbs became greater as the concentration of KCl (10-30 mM) increased. However, the increase in contractions of the distal limbs was significantly lowered at high concentrations of KCl (Fig. 4; corresponding absolute values are shown in Supplementary Fig. 2). The amplitudes of contractions in the distal limb were significantly lower compared to the proximal limb as KCl concentration increased to 10 mM (109.5 ± 11.1% vs 107.4 ± 11.6%, $P > 0.05$), 20 mM (139.1 ± 29.0% vs 121.6 ± 17.9%, $P > 0.05$), and 30 mM (177.5 ± 62.3% vs 143.7 ± 37.9%, $P < 0.05$) (Fig. 4C). AUCs of contractions in the distal limb were significantly lower compared to the proximal limb as KCl concentration increased to 10 mM (108.3 ± 7.4% vs 106.9 ± 6.7%, $P > 0.05$), 20 mM (136.7 ± 13.7% vs 124.2 ± 9.1%, $P < 0.05$), and 30 mM (159.0 ± 20.9% vs 143.9 ± 20.5%, $P < 0.05$) (Fig. 4D).

To measure myogenic relaxivity of the ileum, nitric oxide donor, SNP, was administered directly into the organ bath in a concentration-dependent manner. As the concentration of SNP (10^{-8} -10^{-4} M) increased, the amplitude and basal tone of spontaneous contractions became smaller and lower in both the proximal and distal limb. However, the decrease of contractions was less reduced in the distal limb (Fig. 5; absolute relaxation values are shown in Supplementary Fig. 3). The amplitudes of contractions in the distal limb were significantly less reduced compared to the proximal limb as SNP concentration increased to 10^{-8} M (86.8 ± 8.7% vs 90.9 ± 3.0%, $P > 0.05$), 10^{-7} M (71.1 ± 15.4% vs 85.1 ± 6.0%, $P > 0.05$), 10^{-6} M (54.8 ± 10.9% vs 77.4 ± 6.6%, $P < 0.05$), 10^{-5} (44.8 ± 13.5% vs 72.5 ± 11.3%, $P < 0.01$), and 10^{-4} M (36.9 ± 9.9% vs 62.2 ± 10.2%, $P < 0.01$) (Fig. 5C). AUCs of contractions in the distal limb were significantly less reduced compared to the proximal limb as SNP concentration increased to 10^{-8} M (90.7 ± 8.6% vs 91.9 ± 7.6%, $P > 0.05$), 10^{-7} M (76.2 ± 7.6% vs 83.4 ± 11.3%, $P > 0.05$), 10^{-6} M (68.5 ± 5.3% vs 73.6 ± 7.2%, $P > 0.05$), 10^{-5} (45.4 ± 8.2% vs 66.9 ± 8.2%, $P < 0.01$), and 10^{-4} M (32.9 ± 6.8% vs 56.8 ± 7.3%, $P < 0.01$) (Fig. 5D). The results demonstrate that myogenic contraction and relaxation in responses to ACh, KCl, and SNP are reduced in the distal limb

Expression of Contractile Proteins in Proximal and Distal Limb of Ileostomy Tissue

Ca^{2+}-sensitization mechanisms are relatively important in the contraction of GI smooth muscle and contractile proteins are involved in these mechanisms.[11,12] To investigate the difference in myogenic contraction and relaxation between the proximal and distal limb in ileostomy tissue, the expression levels of representative contractile proteins, myosin phosphatase targeting subunit 1 (MYPT1), protein kinase C-potentiated phosphatase inhibitor protein-17 kDa (CPI-17), and Rho-associated, coiled-coil containing protein kinase 1 (ROCK1), were measured.

Expression of MYPT1 and CPI-17 in distal limb decreased com-

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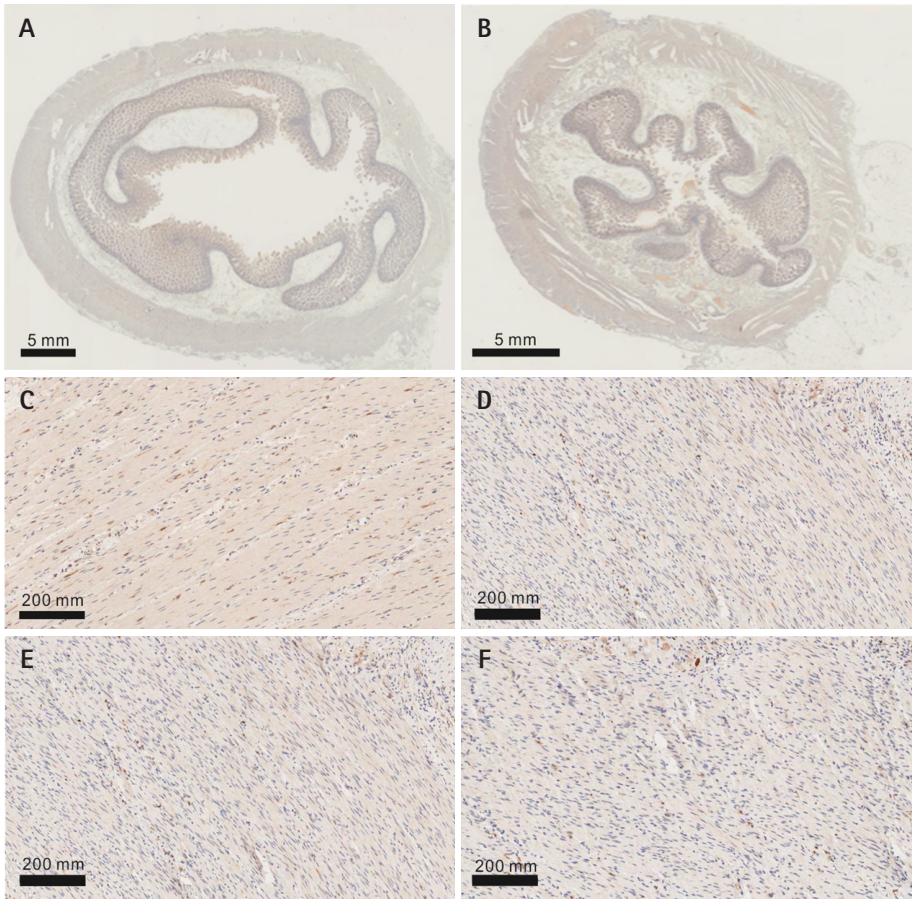

Area (mm²)	Proximal	Distal	*P*-value
Mucosa	62.1 ± 38.1	40.1 ± 26.0	< 0.0001
Submucosa	148.3 ± 107.7	86.5 ± 52.2	0.0011
Muscle layer	112.5 ± 85.8	65.6 ± 35.7	0.0034

Cell count (/mm²)	Proximal	Distal	*P*-value
PDGFRα⁺ cell	30.5 ± 5.9	28.9 ± 6.8	0.4010
ICC	26.7 ± 8.7	25.5 ± 8.9	0.7299

Figure 1. Characteristics of morphological changes in the ileostomy (n = 15). S-100 immunostaining of proximal (A) and distal (B) intestinal tissues of ileostomy. There is a significant atrophy mucosa, subserosa and muscle layer in distal limb of ileostomy. PDGFRα immunostaining of proximal (C) and distal (D) limb of ileostomy. C-kit immunostaining of proximal (E) and distal (F) limb of ileostomy. (G) Comparison of the area of the mucosa, the submucosal layer and the muscle layer. (H) Changes in the number of platelet derived growth factor receptor-α-positive (PDGFRα⁺) cells and interstitial cell of Cajal (ICC) through immunostaining. Data presented as mean ± SEM. Paired t test used for comparisons. $^{**}P < 0.01$.

pared to the proximal limb (1.15 ± 0.28 vs 0.24 ± 0.06, $P < 0.01$ in MYPT1; 1.05 ± 0.28 vs 0.26 ± 0.06, $P < 0.01$ in CPI-17). In addition, expression of ROCK1 in distal limb reduced compared to the proximal limb 1.19 ± 0.25 vs 0.51 ± 0.11, $P < 0.01$) (Fig. 6B). The results demonstrate that Ca^{2+}-sensitization mechanisms are down-regulated in the distal limb.

Expression of *VAChT*, *UCHL1*, and *MYH11* in the Proximal and Distal limb of Ileostomy Tissue

To investigate changes in nerve and SMCs in the distal limb that

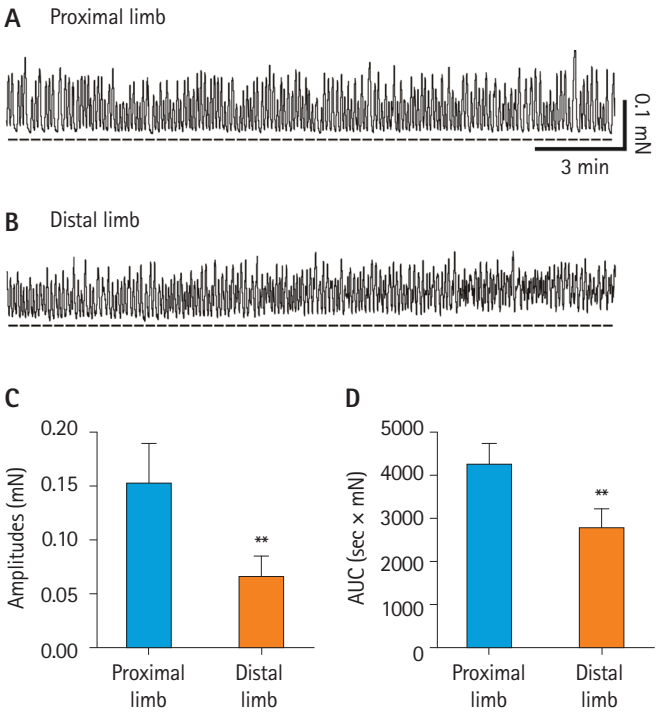

Figure 2. Spontaneous phasic activity in proximal and distal ileal smooth muscle strips (n = 12 paired samples). Raw tracing of spontaneous contractions in the proximal limb (A) and distal limbs (B) post-ileostomy without drug administration. The amplitude (C) and area under the curve (AUC) (D) of spontaneous contractions was significantly reduced in the distal limb. Data presented as mean ± SEM. Paired t test used. $^{**}P < 0.01$.

has not been used for a long time after ileostomy, the expression levels of VAChT, UCHL1, and MYH11 were measured.

Expressions of *VAChT* related with cholinergic neurotransmission and *UCHL1* related with neuronal cell in distal limb were not changed compared to the proximal limb (0.0022 ± 0.0012 vs 0.0028 ± 0.0014, $P > 0.05$ in *VAChT*; 0.051 ± 0.018 vs 0.064 ± 0.035, $P > 0.05$ in *UCHL1*). In addition, *MYH11* which is related with SMC was not different between proximal and distal limbs (2.96 ± 0.65 vs 2.76 ± 0.83, $P > 0.05$) (Fig. 7). The results suggest that no structural changes occurred in the nerve cells and SMCs in the distal limb.

DISCUSSION

This study revealed distinct morphological and functional differences between the proximal and distal limbs of the ileum following loop ileostomy. The distal limb, which remains nonfunctional due to diversion, exhibited significant villous and muscular atrophy and a marked decrease in both contractile and relaxation responses compared with the proximal limb. These findings suggest that the absence of luminal stimulation leads to structural and molecular alterations in the intestinal smooth muscle, ultimately impairing motility.[13,14] However, because all patients in this study underwent rectal cancer surgery and received peri-operative chemotherapy—with some having received radiation—these

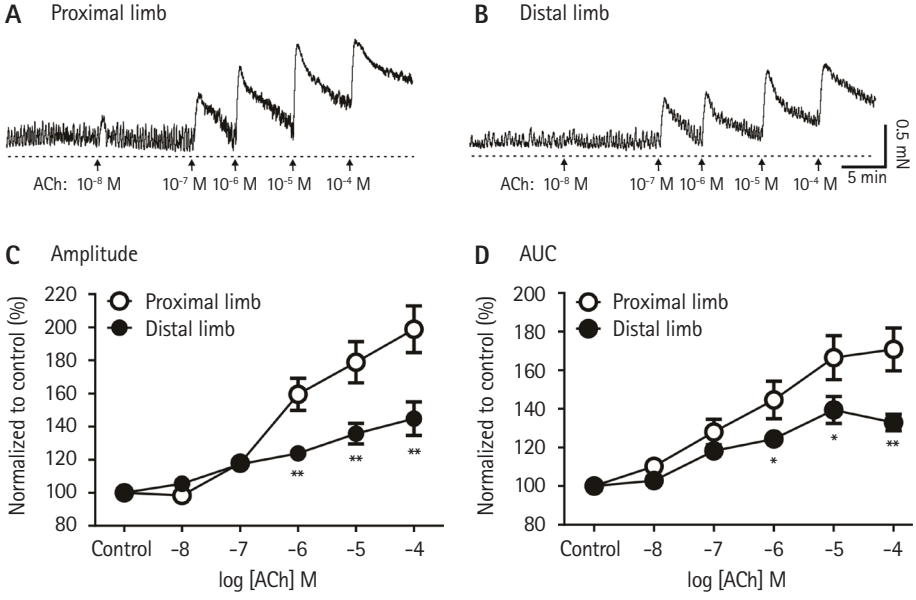

Figure 3. Acetylcholine-induced contractile responses in proximal and distal ileum (n = 12 paired samples). Raw tracing of the contractile response to acetylcholine (Ach) in the proximal limb (A) and distal limb (B) post-ileostomy. The amplitude (C) and area under the curve (AUC) (D) of contractions were significantly lower in the distal limb. The data at each concentration were normalized to control value (before administration of ACh) and expressed as a percentage. Normalized response = (post-drug amplitude − baseline)/baseline × 100%. Corresponding absolute tension values are provided in Supplementary Figure 1. Data presented as mean ± SEM. Wilcoxon signed-rank applied where appropriate. $^{*}P < 0.05$, $^{**}P < 0.01$.

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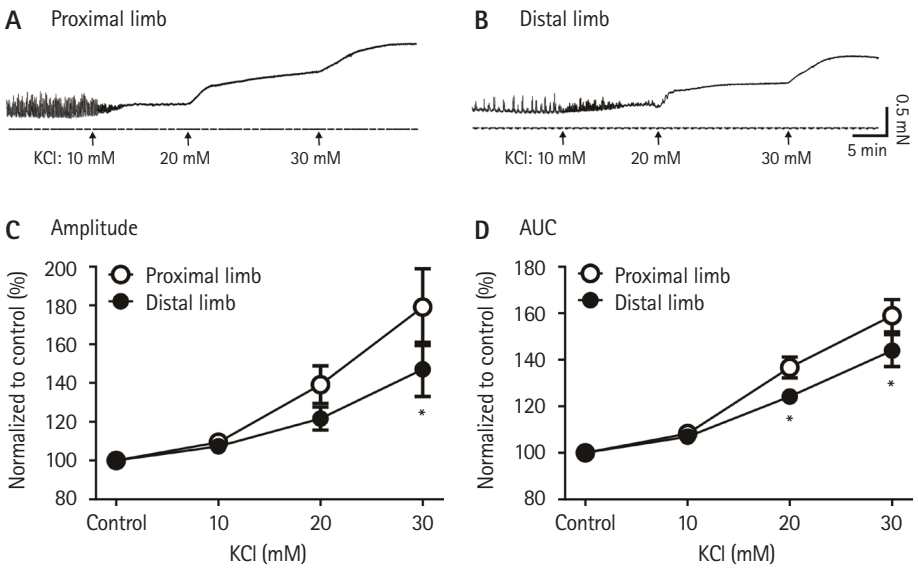

Figure 4. Potassium chloride (KCl)-induced depolarization-mediated contractions in proximal versus distal segments (n = 12 paired samples). Raw tracing of the contractile response to KCl in the proximal limb (A) and distal limb (B) post-ileostomy. The amplitude (C) and area under the curve (AUC) (D) of contractions were significantly lower in the distal limb. The data at each concentration were normalized to control value (before administration of KCl) and expressed as a percentage. Absolute values are available in Supplementary Figure 2. Data presented as mean ± SEM. Wilcoxon signed-rank test applied. *$P < 0.05$.

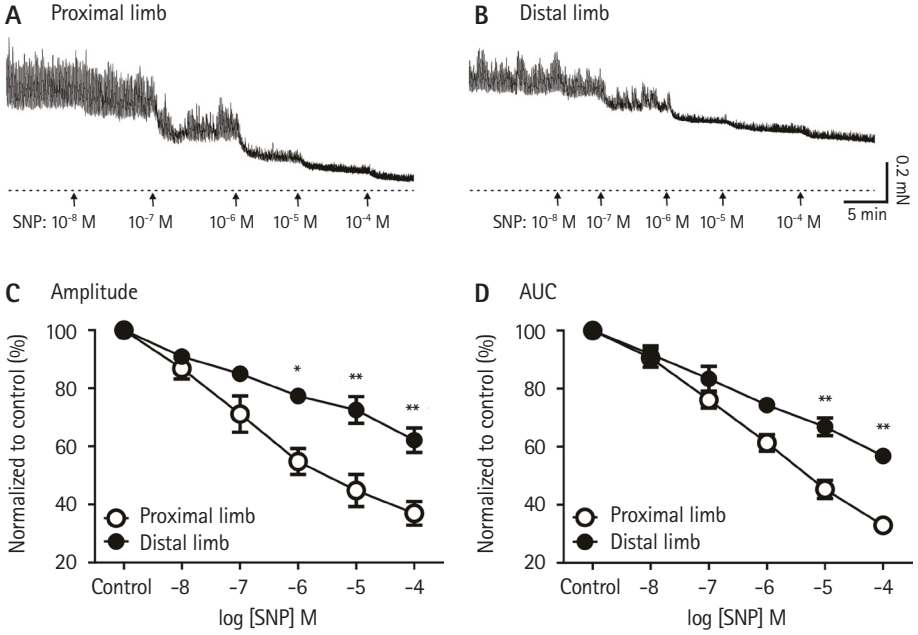

Figure 5. Nitric oxide–mediated relaxation responses to sodium nitroprusside (SNP) (n = 12 paired samples). Raw tracing of the relaxation response to SNP in the proximal limb (A) and distal limb (B) post-ileostomy. The decrease in amplitude and area under the curve (AUC) of contractions showed significant differences between the proximal (C) and distal limb (D). The data at each concentration were normalized to control value (before administration of SNP) and expressed as a percentage. Corresponding non-normalized values are available in Supplementary Figure 3. Data presented as mean ± SEM. Paired tests applied. *$P < 0.05$, **$P < 0.01$.

systemic factors may also contribute to tissue remodeling and altered neuromuscular function. Therefore, the observed alterations are likely the result of combined influences of luminal disuse and treatment-related or systemic changes, rather than disuse alone.

Atrophy involving the mucosal, submucosal, and muscular layers in the distal limb was consistent with previous animal and human studies on disuse-induced intestinal adaptation.[13-15] Such atrophic changes are likely to influence intestinal motility and ex-

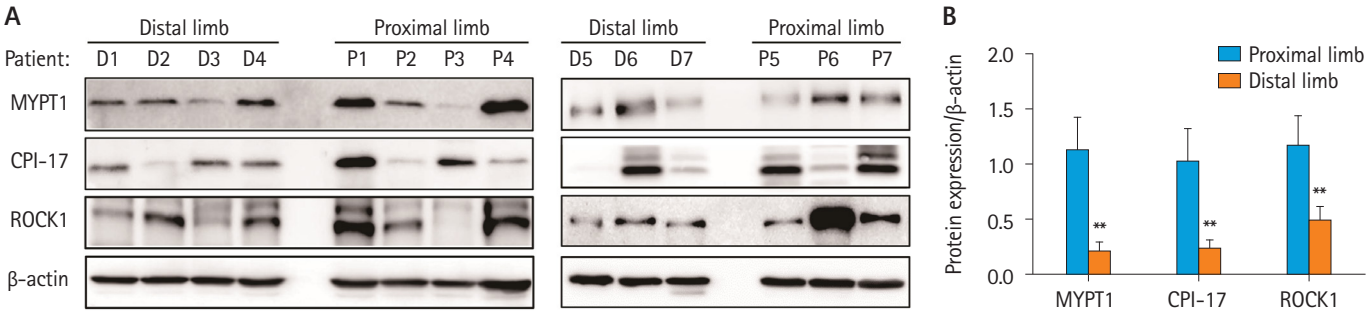

Figure 6. Expression of contractile signaling proteins in proximal and distal ileum (n = 7 paired samples). Representative Western blots of myosin phosphatase targeting subunit 1 (MYPT1), protein kinase C-potentiated phosphatase inhibitor protein-17 kDa (CPI-17), and Rho-associated, coiled-coil containing protein kinase 1 (ROCK1) in human ileum samples from proximal limb and distal limb post-ileostomy (A). The expression levels of three contractile proteins related to contraction were significantly reduced in the distal limb (B). Data presented as mean ± SEM. Paired t test used. $^{**}P < 0.01$.

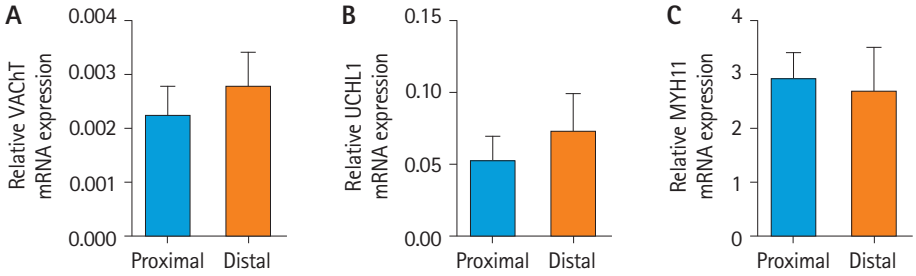

Figure 7. Gene expression profiles of neuromuscular cell markers (n = 10 paired samples). Messenger RNA expression of vesicular acetylcholine transporter (VAChT, A), ubiquitin C-terminal hydrolase L1 (UCHL1, B), and myosin heavy chain 11 (MYH11, C)using real-time quantitative polymerase chain reaction analysis in the proximal limb and distal limb post-ileostomy. Relative expression levels were not significant different between proximal limb and distal limb. Data presented as mean ± SEM. Paired t test or Wilcoxon signed-rank test applied.

plain the reduced contractile responses to ACh and KCl as well as the diminished relaxation response to SNP. Smooth muscle contraction in the GI tract is primarily regulated by intracellular calcium (Ca^{2+})-dependent activation of myosin light chain kinase, which phosphorylates myosin light chain to induce contraction. In parallel, the Ca^{2+} sensitization pathway—mediated through ROCK and CPI-17—inhibits myosin light chain phosphatase, thereby sustaining contraction.[16,17] While these findings align with mechanisms described in disuse-induced motility suppression, chemotherapy-induced mucosal damage, alterations in enteric nervous system signaling, and nutritional decline during postoperative recovery may also play a role. The reduced amplitude and AUC in distal tissue may therefore reflect an interaction between mechanical inactivity and systemic influences such as cytotoxic drug exposure, metabolic stress, or inflammatory responses rather than a single mechanistic pathway. In the distal limb, the coordinated down-regulation of MYPT1, CPI-17, and ROCK1 suggests a suppression of these Ca^{2+}-sensitization mechanisms, leading to decreased muscle contractility and impaired relaxation. These changes are likely in part a consequence of prolonged mechanical inactivity and lack of luminal contents.[18] The down-regulation of MYPT1, CPI-17, and ROCK1 in diverted segments suggests impaired Ca^{2+}-sensitization signaling, a key regulator of GI smooth muscle contractility. Although this pattern is consistent with decreased use-dependent signaling, chemotherapy and radiation are known to influence smooth muscle protein expression and could also contribute to these molecular changes. In contrast, the unchanged expression of MYH11, VAChT, and UCHL1 suggests that neuromuscular lineage identity may be preserved despite the structural atrophy observed in the distal limb. Stable mRNA expression despite histologic atrophy may reflect compensatory transcription, sampling depth differences during tissue acquisition, or dilutional effects from reduced tissue thickness rather than true preservation of functional capacity. Thus, the molecular findings may be partially attributable to disuse adaptation and partially to systemic or treatment-related effects, and the relative contributions of each factor remain to be clarified.

Despite these molecular and functional changes, the qualitative preservation of interstitial cells of Cajal (ICC), PDGFRα$^+$ cells, enteric neurons, and SMCs indicates that the basic structural framework of the SIP syncytium remains intact.[19-21] This cellular network is responsible for coordinating rhythmic intestinal motility—ICC generating pacemaker slow waves, PDGFRα$^+$ cells modulating neurotransmission, and neurons regulating excitatory and inhibitory inputs.[22,23] The maintained integrity of this network suggests

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that although contractile signaling is suppressed during diversion, the intrinsic architecture necessary for motility recovery is preserved. This may explain the clinically observed rapid restoration of bowel function after ileostomy reversal, as noted in our patients who resumed flatus passage within 4 ± 1.3 days and experienced no postoperative ileus. Although ICCs, PDGFRα^{+} cells, and enteric neuronal markers appeared qualitatively preserved in diverted segments, the absence of quantitative cell-density measurements limits definitive conclusions regarding the degree of cellular maintenance.

Although all patients in this study had rectal or rectosigmoid cancer and received adjuvant chemotherapy, paired tissue analysis within each individual minimized confounding from systemic factors such as chemotherapy, nutrition, or disease status. The observed differences therefore likely represent disuse-related changes rather than chemotherapy-induced toxicity. Nevertheless, the possibility that chemotherapy contributes to intestinal alterations cannot be excluded. The diversion period in our cohort was relatively uniform (median 7 months, range 7-8 months), but the temporal dynamics of mucosal and molecular adaptation warrant further investigation. Studies including a wider range of diversion durations and non-cancer populations could clarify time-dependent changes and generalizability.

This study is limited by its relatively small sample size and focus on short-term outcomes after ileostomy reversal. Long-term consequences of disuse and recovery, as well as broader molecular pathways regulating contractility, were not assessed. Future research incorporating transcriptomic or proteomic analyses may elucidate upstream mechanisms such as inflammatory signaling, mechanosensory regulation, or transcriptional modulation of contractile proteins.

In conclusion, this study provides insight into the morphological, molecular, and motility changes in the loop ileostomy. The decreases in the muscle layer and in the expression of contractile proteins were observed in the distal limb of the ileostomy, which may be one of the causes of ileus after ileostomy take-down. Despite these changes, there were no numerical and quantitative changes of each cell involved in motility, suggesting the possible mechanism for restoration of intestinal motility after ileostomy take-down.

SUPPLEMENTARY MATERIALS

Note: To access the supplementary figures mentioned in this article, visit the online version of Journal of Neurogastroenterology and Motility at http://www.jnmjournal.org/, and at http://doi.org/10.5056/jnm25226.

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Author contributions
Eon Chul Han, Seung-Bum Ryoo, and Tae Sik Sung performed the mechanical and electrophysiological recordings and immunohistochemistry experiments, drafted the manuscript, and contributed equally to this study; Seung-Bum Ryoo, Tae Sik Sung, and Kyu Joo Park discussed and designed the study; and Kil-yong Lee and Yoon-Hye Kwon performed the experiments and collected the data. All authors reviewed and modified the manuscript.

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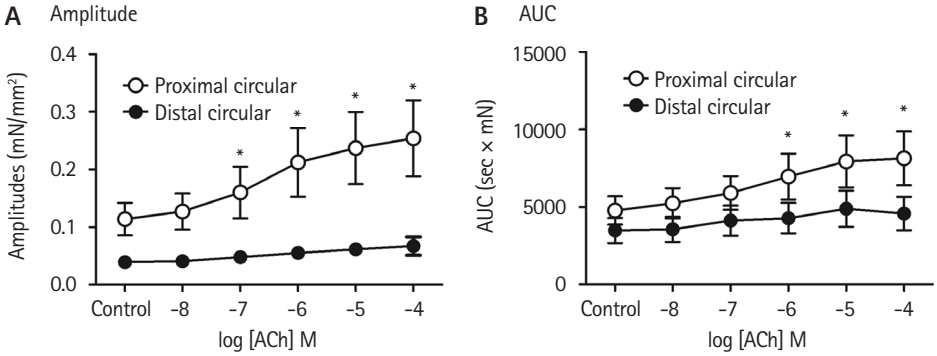
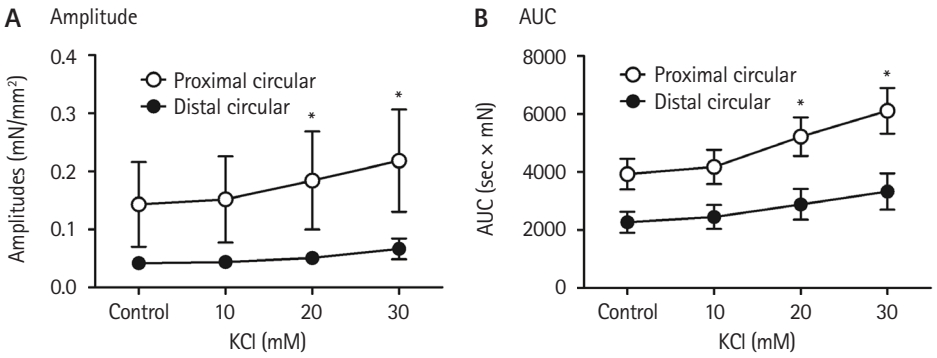
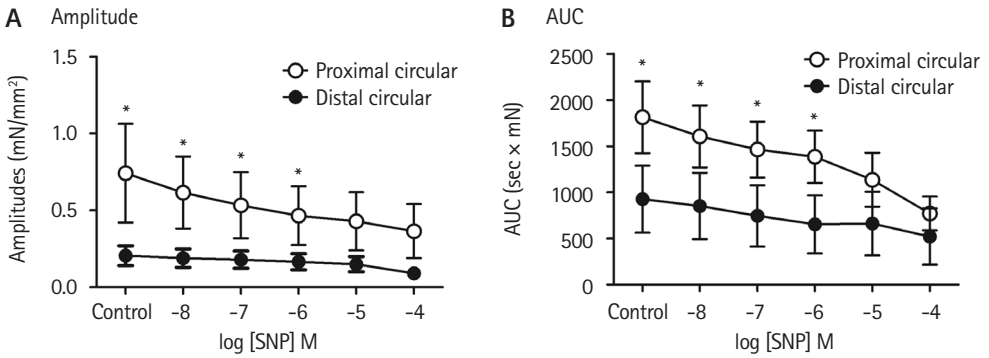
SUPPLEMENTARY MATERIALS



Supplementary Figure 1. Absolute contractile responses to acetylcholine in proximal and distal ileal circular muscle (n = 12 paired samples). Dose-response curves for amplitude (A) and AUC (B) of spontaneous contractions to cumulative acetylcholine (ACh; 10^{-8}–10^{-4} M) in proximal and distal circular muscle strips. (B) Corresponding area under the curve (AUC; g·min) for spontaneous and ACh-induced contractions. Values represent absolute tension and AUC and correspond to the normalized data shown in Figure 3 of the main text. Data are expressed as mean ± SEM. Statistical comparisons between proximal and distal limbs were performed using Wilcoxon signed-rank tests, as appropriate, $^{*}P < 0.05$ indicates a significant difference between proximal and distal segments at individual concentrations.



Supplementary Figure 2. Absolute contractile responses to potassium chloride (KCl)-induced depolarization in proximal and distal ileal circular muscle (n = 12 paired samples). Dose-response curves for amplitude (A) and area under the curve (AUC) (B) of spontaneous contractions to cumulative KCl (10, 20, and 30 mM) in proximal and distal circular strips. These data show absolute (non-normalized) tension and AUC values complementary to the normalized responses presented in Figure 4 of the main text. Data are presented as mean ± SEM. Proximal and distal segments were compared using Wilcoxon signed-rank tests. *$P < 0.05$ indicates a significant difference between proximal and distal segments at individual concentrations.



Supplementary Figure 3. Absolute relaxation responses to sodium nitroprusside in proximal and distal ileal circular muscle (n = 12 paired samples). Dose-response curves for amplitude (A) and area under the curve (AUC) (B) of spontaneous contractions to cumulative sodium nitroprusside (SNP; 10^{-8}–10^{-4} M) in proximal and distal circular muscle strips. These curves represent absolute (non-normalized) values that parallel the normalized relaxation data in Figure 5 of the main text. Data are expressed as mean ± SEM. Statistical comparisons between proximal and distal limbs were performed using Wilcoxon signed-rank tests. *$P < 0.05$ indicates a significant difference between proximal and distal segments at individual concentrations.

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Possible Mechanisms Underlying the Effects of Fecal Microbiota Transplantation in Patients With Irritable Bowel Syndrome

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Background/Aims: This study aims to identify the mechanisms underlying the effects of fecal microbiota transplantation (FMT).
Methods: This study investigated 93 patients included in our previous clinical trial: 24 received 30 g of their own feces (placebo group), while 47 and 22 received 30 g and 60 g of donor feces, respectively. The patients underwent sigmoidoscopy with biopsies, provided fecal samples, and completed questionnaires to assess irritable bowel syndrome (IBS) symptoms, fatigue, and quality of life at the baseline and at 1 month following FMT. The biopsy samples were immunostained for Musashi-1, neurogenin-3, and enteroendocrine and immune cells. The cell densities were quantified by computerized image analysis. Fecal short-chain fatty acids (SCFAs) were measured by gas chromatography, and fecal bacteria were measured by 16S ribosomal RNA polymerase chain reaction DNA amplification.
Results: The densities of stem cells, enteroendocrine cell progenitors, serotonin, glucagon-like peptide 1, and peptide YY (PYY) cells increased following FMT in the 30-g and 60-g treated groups but not in the placebo group, and they were inversely correlated with both IBS symptoms and fatigue. The densities of submucosal immune cells and mast cells decreased following FMT in the 30-g and 60-g treated groups but not in the placebo group, and they were correlated with IBS symptoms and fatigue. The fecal level of butyric acid increased in patients treated with donor feces, and this was inversely correlated with IBS symptoms and fatigue.
Conclusions: Stem cells, enteroendocrine progenitors, serotonin, and glucagon-like peptide 1 as well as low-grade inflammation may play roles in IBS symptom manifestations. The effects of FMT are probably due to the amelioration of these abnormalities.

Keywords: Glucagon-like peptide 1; Low-grade inflammation; Mast cells; Serotonin; Stem cells

INTRODUCTION

Studies of fecal microbiota transplantation (FMT) as a treatment for irritable bowel syndrome (IBS) have produced diverse results ranging from high long-term efficacy to no effect at all.[1-4] These diverse results of FMT interventions in IBS are probably due to differences in the FMT protocols used and in the included patient cohorts. The mechanisms underlying the effects of FMT on IBS symptom relief are unknown. Understanding these mechanisms may clarify the pathophysiology of IBS and lead to new therapeutic opportunities.

The densities of stem cells in the intestine and their early progeny as well as the densities of early intestinal endocrine cell progenitors are lower in IBS patients than in healthy subjects.[5] It has been speculated that abnormal clonogeny and differentiation progeny of the stem cells in the intestine and an abnormal secretory lineage differentiation progeny could play roles in the pathophysiology of IBS.[5]

Patients with IBS exhibit gastrointestinal dysmotility, visceral hypersensitivity, and abnormal secretion.[6] The enteroendocrine cells secret several hormones that control gastrointestinal motility, visceral sensitivity, and gastrointestinal secretion.[6] The density of enteroendocrine cells is reportedly lower in patients with IBS than in healthy subjects.[6]

Intestinal mucosal low-grade inflammation is considered to be one of the factors contributing to the pathophysiology of IBS.[7,8] Among the immune cells, an increase in submucosal mast cells appears to be associated with the activation of IBS symptoms.[7,8]

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Short-chain fatty acids (SCFAs) are the main metabolites of the bacterial fermentation of undigested and unabsorbed fiber and carbohydrates in the intestine.[9,10] SCFAs affect intestinal contractility, visceral pain, epithelial proliferation, epithelial barrier function, and host immunity.[11]

In our previous randomized, double-blind, placebo-controlled study we found that FMT decreased the symptoms and increased the quality of life (QoL) of IBS patients.[1] The present study involved a large proportion of the IBS patients included in that study, and was designed to establish whether the improvement following FMT is associated with changes in the densities of intestinal stem cells and their early progenitors, secretory lineage differentiation progenitors, enteroendocrine cells, submucosal immune cells, and fecal SCFAs, as well as the composition of fecal bacteria.

MATERIALS AND METHODS

Study Design and Randomization of Patients

The design of this study has been described in detail previously (Supplementary Protocol).[1] Briefly, the patients were randomized to receiving 30 g of their own feces (placebo group) or 30 g or 60 g of feces from a donor (30-g and 60-g groups, respectively) at a ratio of 1:1:1. Patients underwent sigmoidoscopy, and completed 3 questionnaires to assess their symptoms and QoL at the baseline and at 1 month after FMT. During sigmoidoscopy, 5 biopsy samples were obtained at different sites of the sigmoid colon about 30 cm to 35 cm from the anus.

Patients

This study included 93 of the 164 patients who participated in our previous clinical trial and agreed to undergo sigmoidoscopy.[1] When the study was completed and opened, there were 24 patients in the placebo group, 47 patients in the 30-g group, and 22 patients in the 60-g group. The detailed data of these patients are given in Table 1. Detailed patient characteristics have been reported previously.[1]

Fecal Sample Collection, Preparation, and Administration

The procedure for performing FMT has been described in detail earlier.[1] Shortly, donor fecal material was frozen immediately and was kept at −80°C, and thawed at 4°C for 2 days before transplantation. The fecal samples were mixed with 40 mL of sterile saline (0.9% NaCl) and filtered. The transplant was administered into the duodenum via a gastroscope.

Symptom and Quality of Life Measurements

IBS symptoms and fatigue were measured by the IBS Severity Scoring System (IBS-SSS) and Fatigue Assessment Scale (FAS) questionnaires.[12,13] QoL was measured using the IBS-QoL questionnaire.[14] Response to FMT was defined as a decrease in IBS-SSS total score of ≥ 50 points.

Table 1. Baseline Characteristics of the Irritable Bowel Syndrome Patients Included in the Study

Characteristic	Overall (n = 93)	Placebo FMT (n = 24)	30-g FMT (n = 47)	60-g FMT (n = 22)	*P*
Age (yr)	40.0 ± 13.5	45.0 ± 15.8	38.7 ± 12.7	37.9 ± 11.6	0.193
Sex, female/male	73/20	19/5	34/13	20/2	0.216
IBS-D	32	5	18	9	
IBS-C	39	12	17	10	0.422
IBS-M	22	7	12	3	
IBS duration, years	15.5 ± 7.9	14.5 ± 8.0	17.2 ± 9.3	14.7 ± 5.7	0.322
IBS-SSS total score	310.8 ± 73.9	292.7 ± 83.0	319.5 ± 73.0	312.0 ± 64.6	0.402
FAS total score	31.5 ± 5.0	30.6 ± 4.4	31.4 ± 5.3	31.4 ± 5.1	0.976
PPI medication	32 (34.4)	8 (33.3)	16 (34.0)	8 (36.4)	0.974
Birth-control medication	64 (68.8)	16 (66.7)	33 (70.2)	15 (68.2)	0.952
Antimigraine medication	5 (5.4)	1 (4.2)	4 (8.5)	0 (0.0)	0.329
Medication against asthma/allergies	10 (10.8)	3 (12.5)	6 (12.8)	1 (4.5)	0.560
Medication with levothyroxine	1 (1.1)	0 (0.0)	1 (2.1)	0 (0.0)	0.610
Medication with heart/vascular drugs	3 (3.2)	1 (4.2)	2 (4.3)	0 (0.0)	0.619

Data are presented as mean ± SD, n/n, n, or n (%) values.
FMT, fecal microbiota transplantation; IBS, irritable bowel syndrome; IBS-D, diarrhea-predominant IBS; IBS-C, constipation-predominant IBS; IBS-M, mixed IBS; IBS-SSS, IBS Severity Scoring System; FAS, Fatigue Assessment Scale; PPI, proton-pump inhibitor.

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Immunocytochemistry and Computerized Image Analysis

After fixation of the biopsy samples in 4% buffered paraformaldehyde, they were embedded in paraffin, and cut into 5-μm-thick sections. Immunocytochemistry was done using the avidin-biotin-complex (ABC) method with the Vectastain Elite ABC Plus kit and peroxidase (horse anti-mouse/anti-rabbit IgG) (Vector Laboratories, Burlingame, CA, USA). The primary antibodies used are summarized in Table 2.

Morphometric measurements were performed using a microscope with a camera linked to a computer. Olympus cellSens imaging software was used. The objective used in the analysis was ×40. Using this objective, each field on the monitor represented a tissue area of 0.14 mm^2. Immunostained sections were coded, and the co-authors that performed measurements were not aware of identities of the slides. Measurements were made in 10 randomly chosen fields. The method used has been validated earlier.[15]

The numbers of Musashi-1 and neurogenin-3 cells were measured by manually counting the number of immunoreactive cells and the number of crypts by pointing and clicking the computer mouse in each field. The density of Musashi-1 cells was expressed as the number of cells per crypt, and that of neurogenin-3 as the number of cells per 10 fields. The density of enteroendocrine cells was determined from the number of enteroendocrine cells and the area of the epithelial cells. The density of enteroendocrine cells was expressed as the number of endocrine cells/mm^2 of epithelium. The number of immune cells was determined by counting the number of immune cells in the lamina propria manually in each field. The density of immune cells was expressed as the number of immune cells per field of view.

Determination of Fecal Short-chain Fatty Acid Levels

Each fecal sample was weighed, homogenized with 2.5 mL of 3 mmol/L 2-ethylbutyric acid/0.5 mmol/L H_2SO_4, and distilled in vacuum. The SCFAs level and the levels of acetic, propionic, iso-butyric, butyric, iso-valeric, valeric, iso-capronic, and capronic acids were measured using gas chromatography (Agilent 7890A; Agilent, Santa Clara, CA, USA).[16] The levels of SCFAs are expressed in mmol/kg wet weight.

Table 2. Characteristics of Antibodies Used

Antibodies	Supplier	Dilution	Detect
Polyclonal rabbit anti-synthetic peptide conjugated to keyhole limpet hemocyanin derived from within residues 1-100 of human Musashi-1	Code ab21628, Abcam, Cambridge, UK	1:100	Stem cells and in their early progeny
Monoclonal mouse anti-protein expressed in 293T cells transfected with human neurogenin-3 expression vector	Code ab87108, Abcam	1:100	Early endocrine cell progenitors
Monoclonal mouse anti-human chromogranin	Clone LK2H10, Invitrogen, Kissimmee, FL, USA	1:500	General marker for enteroendocrine cells
Monoclonal mouse anti-serotonin	Clone 5HT-H209, Invitrogen	1:300	Serotonin producing cells
Polyclonal rabbit anti- human PYY	PAS-95608, Invitrogen	1:300	PYY producing cells
Monoclonal mouse-anti- GLP-1	EPR 4043, Abcam	1:500	GLP-1 producing cells
Polyclonal rabbit anti-PP	Diagnostic BioSystems, Pleasanton, CA, USA	1:100	PP producing cells
Polyclonal rabbit anti-somatostatin	A0556, Lot 10048798, Dako, Carpinteria, CA, USA	1:100	Somatostatin producing cells
Monoclonal mouse anti-human CD45	Code no. M0701, Dako	1:600	CD45 is a common antigen expressed by cells in the hematopoietic system and their progenitors
Monoclonal mouse antihuman CD5	Code IS082, Dako	1:500	CD5 is a marker of both B and T lymphocytes
Monoclonal mouse anti-human CD57	Code no. I5647, Dako	1:200	CD57 is expressed by subsets of NK cells and $CD8^+$ lymphocytes, and by a small percentage of $CD4^+$/$CD45RO^+$ T lymphocytes
Monoclonal mouse anti-human CD23	Code IS781, Dako	1:400	CD23 is a common marker for B lymphocytes
Monoclonal mouse anti-human CD68	Code no. M0814, Dako	1:100	CD68 is localized in human monocytes, macrophages, and myeloid cells.
Monoclonal mouse anti-human mast-cell tryptase	Code no. M7052, Dako	1:800	Mast cells

PYY, peptide YY; GLP-1, glucagon-like peptide 1; PP, pancreatic polypeptide.

Bacterial Analyses

Fecal bacterial contents were analyzed as described in detail previously by 16S ribosomal RNA polymerase chain reaction DNA amplification covering variable regions V3-V9, and by DNA probe hybridization of 48 bacterial markers.[17]

Statistical Methods

To test differences between the placebo, 30-g, and 60-g groups in response, sex, IBS subtypes, and medication Fisher's exact test was used. The Kruskal-Wallis test with a Dunn's multiple-comparisons post-hoc test with adjusted *P*-values were used to assess differences between placebo, 30-g, and 60-g groups at the baseline in IBS duration, IBS-SSS, and FAS total scores. The nonparametric Mann-Whitney U test was used to analyze differences between the baseline and at 1 month following FMT in the placebo, 30-g, 60-g, responder, and nonresponder groups. Correlations were analyzed using the nonparametric Spearman test, and *P*-values were adjusted for multiple correlations using Bonferroni correction. Differences between IBS subtypes regarding IBS-SSS, FAS, and IBS-QoL scores and the densities of Musashi-1, neurogenin-3, enteroendocrine cells, and immune cells were analyzed using the Kruskal-Wallis test with a Dunn's multiple-comparisons post hoc test, where adjusted *P*-values were recorded. Cohen's d was used to determine the effect size of FMT on IBS-SSS, FAS, and IBS-QoL total scores, and the densities of Musashi-1, neurogenin-3, enteroendocrine cells, and immune cells as well as the SCFAs levels.

Ethics

The study was approved by the West Regional Committee for Medical and Health Research Ethics, Bergen, Norway (Approval No. 2017/1197/REK vest). This study was registered at www.clinicaltrials.gov (NCT03822299).

RESULTS

Responses, Symptoms, and Quality of Life

Four patients in the placebo group responded to FMT (17%), while 39 patients (83%) and 19 patients (86%) responded in the 30-g and 60-g groups, respectively. FMT effect size as measured by Cohen's d on different parameters is given in Table 3.

The IBS-SSS total score, the scores for its 4 items, and the FAS total score were significantly decreased in the 30-g, 60-g, and responder groups, but not in the placebo or nonresponder groups (Supplementary Tables 1 and 2). The IBS-QoL total score and the scores for its domains other than domain 7 (sexual function) increased in the 30-g and responder groups, but not in the placebo, 60-g, or nonresponder groups (Supplementary Table 3).

Densities of Stem Cells, Early Endocrine Cell Progenitors, and Enteroendocrine Cells

The densities of Musashi-1, neurogenin-3, chromogranin A, serotonin, PYY, and glucagon-like peptide 1 (GLP-1) cells increased at 1 month after FMT in the 30-g, 60-g, and responder groups,

Table 3. The Fecal Microbiota Transplantation Effect Size as Measured by Cohen's d on Different Parameters Measured in the Study

Parameters	Effect size		Interpretation	
	30-g	60-g	30-g	60-g
IBS-SSS total scores	0.9	1.2	Large effect size	Large effect size
FAS total scores	0.9	0.8	Large effect size	Large effect size
IBS-QoL total scores	0.5	0.4	Medium effect	Medium effect size
Density of Musashi-1 cells	1.4	1.4	Large effect size	Large effect size
Density of neurogenin-3 cells	3.1	2.7	Large effect size	Large effect size
Density of chromogranin A cells	1.6	2.0	Large effect size	Large effect size
Density of GLP-1 cells	2.0	1.9	Large effect size	Large effect size
Density of serotonin cells	0.6	0.9	Medium effect size	Large effect size
Density of PYY cells	0.5	0.8	Medium effect size	Large effect size
Density of total immune cells	1.6	1.7	Medium effect size	Large effect size
Density of mast cells	0.7	0.6	Medium effect size	Medium effect size
Fecal SCFAs levels	0.7	0.8	Medium effect size	Large effect size
Fecal butyric acid levels	0.4	0.7	Small effect size	Medium effect size

IBS-SSS, Irritable Bowel Syndrome Severity Scoring System; FAS, Fatigue Assessment Scale; IBS-QoL, Irritable Bowel Syndrome Quality of Life questionnaire; GLP-1, glucagon-like peptide 1; PYY, peptide YY; SCFAs, fecal short-chain fatty acids.

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but not in the placebo or nonresponder groups (Fig. 1 and 2). Few cells expressed pancreatic polypeptide and somatostatin in the analysed biopsy tissues to allow a reliable comparison.

Densities of Immune Cells and Mast Cells

The densities of immune cells and mast cells decreased significantly after FMT in the 30-g, 60-g, and responder groups, but not

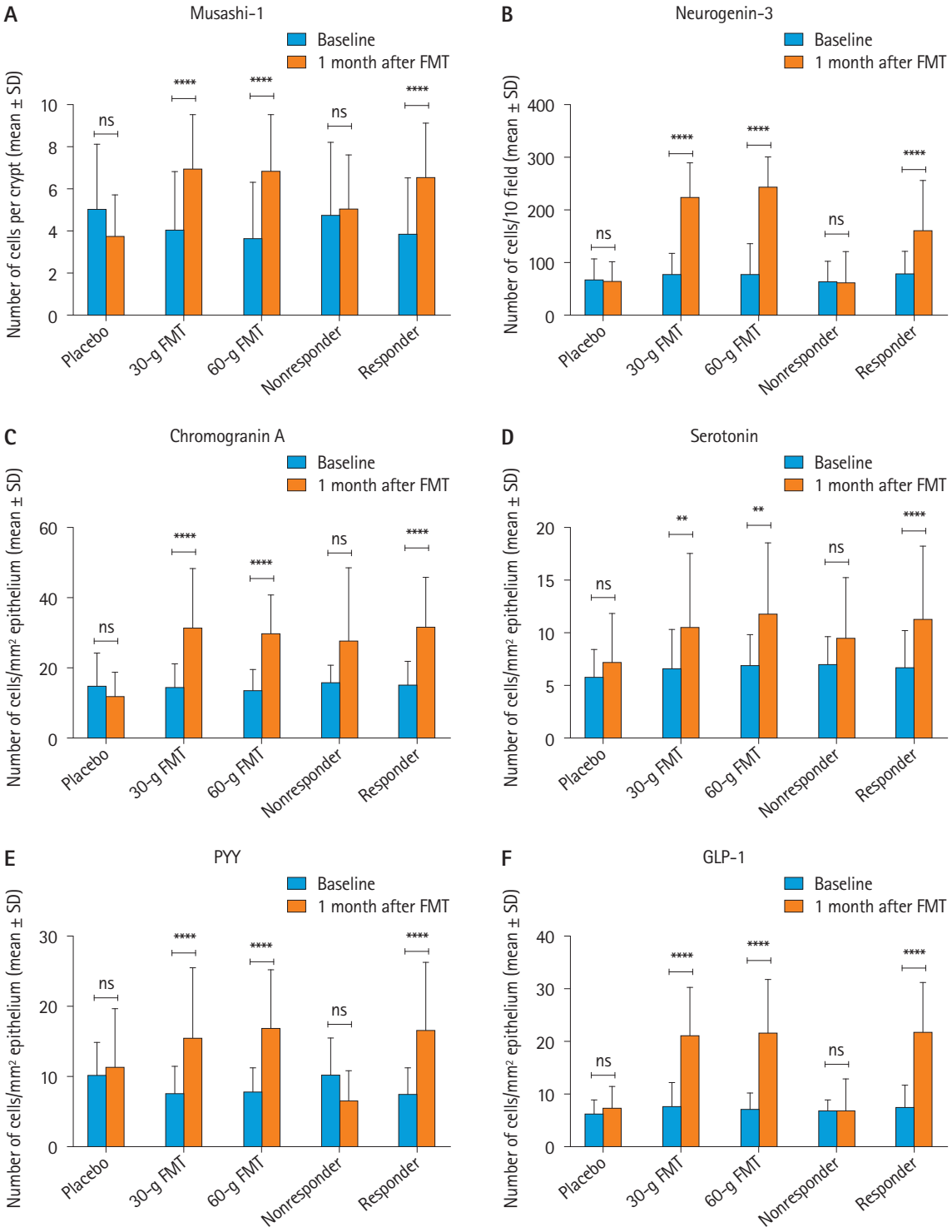

Figure 1. Densities of cells expressing Musashi-1 (A), neurogenin-3 (B), chromogranin A (C), serotonin (D), peptide YY (PYY) (E), and glucagon-like peptide 1 (GLP-1) (F) in the placebo, 30-g, 60-g, nonresponder, and responder groups at the baseline and at 1 month after fecal microbiota transplantation (FMT). *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$; ****$P < 0.0001$.

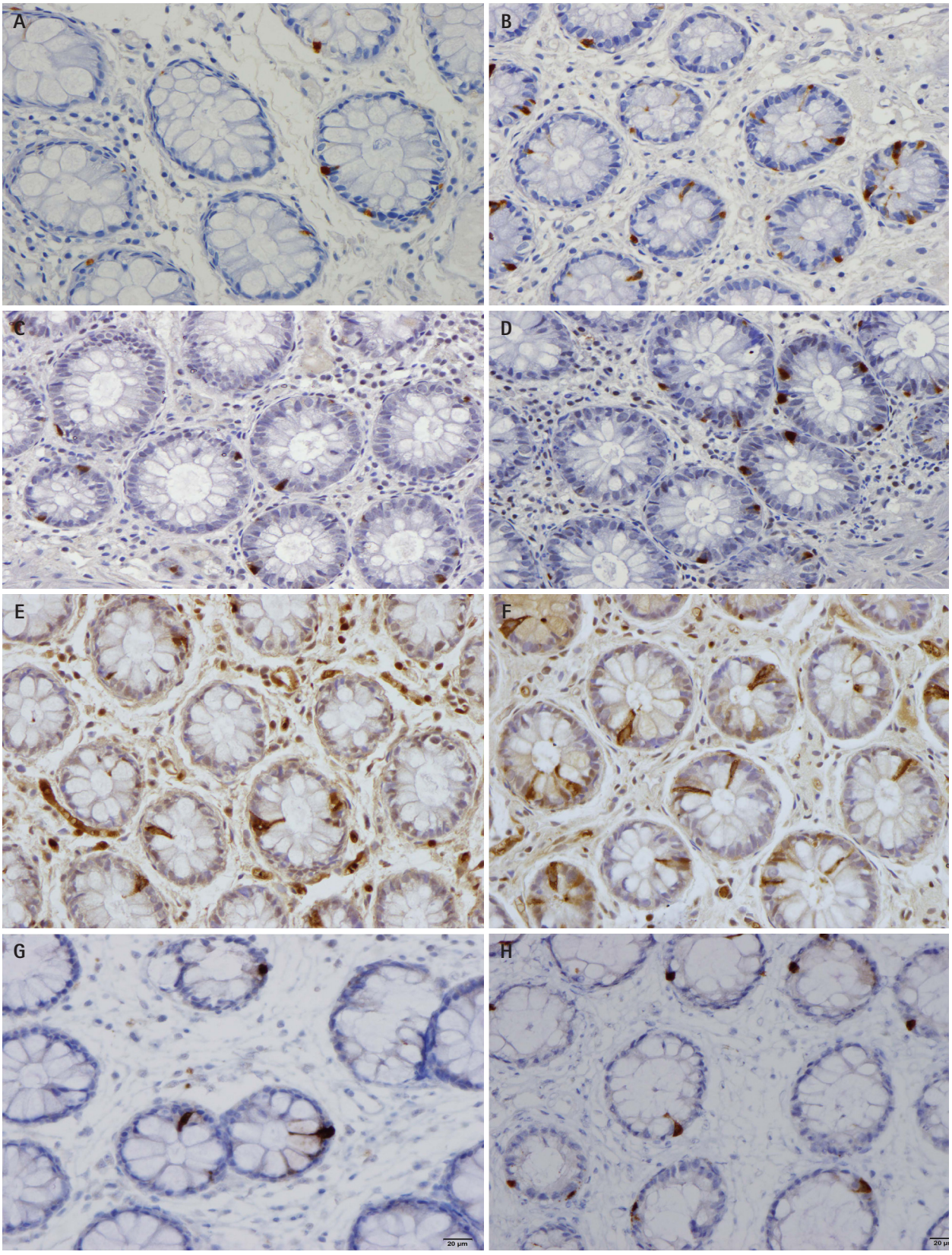

Figure 2. Enteroendocrine cells in a patient from the 30-g group. Chromogranin-A-expressing cells at the baseline (A) and 1 month after fecal microbiota transplantation (FMT) (B). Serotonin-expressing cells at the baseline (C) and 1 month after FMT (D). Peptide YY (PYY)-expressing cells at the baseline (E) and 1 month after FMT (F). Glucagon-like peptide 1 (GLP-1)-expressing cells before FMT (G) and 1 month after FMT (H).

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in the placebo or nonresponder groups (Fig. 3). There were few other immune cell types in the analyzed biopsy tissues to allow a reliable analysis.

Fecal Short-chain Fatty Acid Levels

The total SCFA level increased significantly at 1 month after FMT in the 30-g, 60-g, and responder groups, but did not change in the placebo or nonresponder groups (Supplementary Table 4). The level of butyric acid also increased at 1 month after FMT in the 30-g, 60-g, and responder groups, but not in the placebo or nonresponder groups (Supplementary Table 4). The levels of acetic, propionic, iso-butyric, isovaleric, valeric, iso-capronic, and capronic acids did not change in either the placebo or donor-treated groups at 1 month after FMT.

Bacterial Analyses

The DI decreased from 3.1 ± 0.1 (mean ± SD) at the baseline to 2.1 ± 0.8 at 1 month following FMT in the 30-g group ($P < 0.0001$),

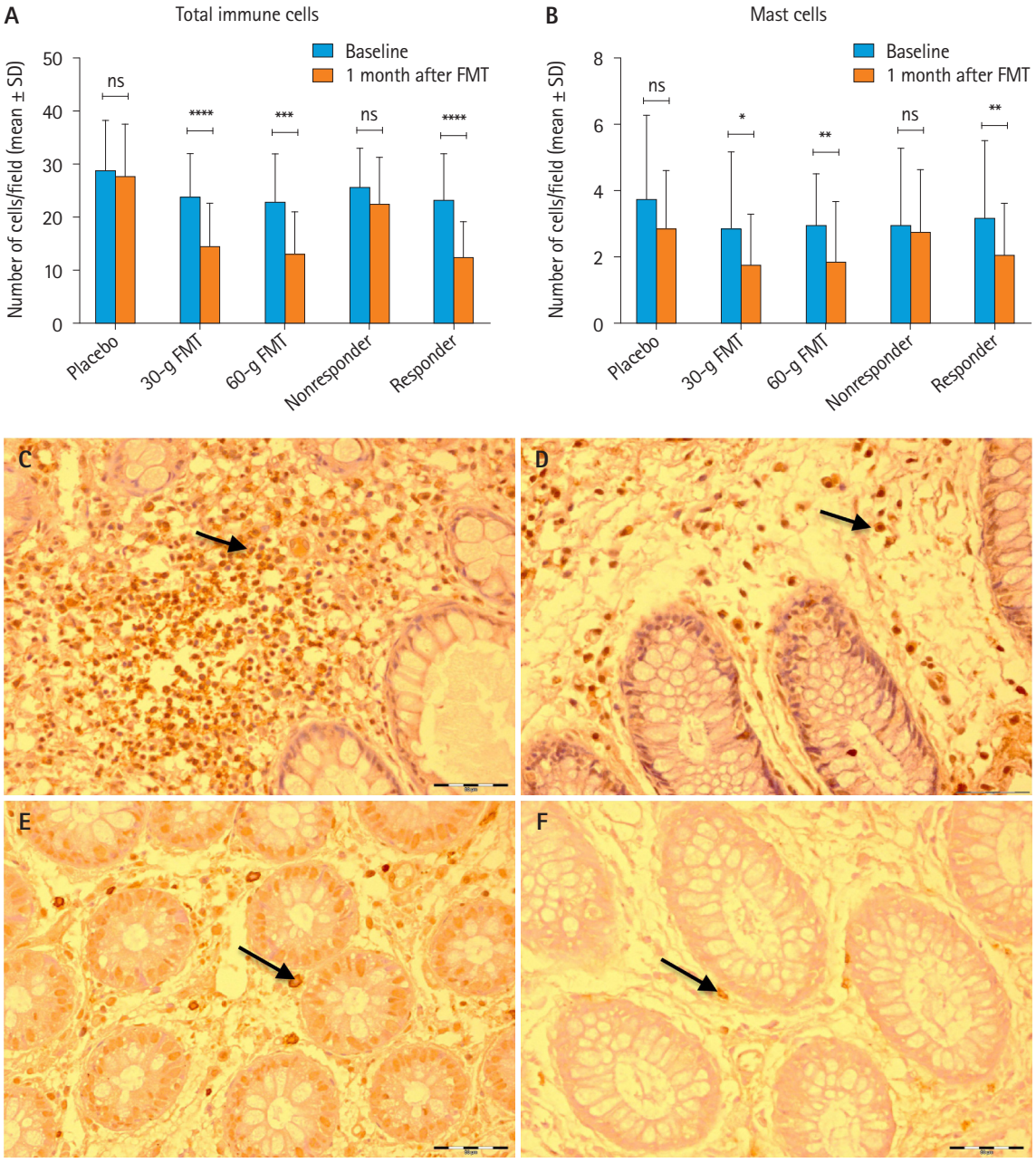

Figure 3. Immune cells as detected by CD45 (A) and mast cells (B) at the baseline and at 1 month after fecal microbiota transplantation (FMT) in the placebo, 30-g, 60-g, nonresponder, and responder groups. Photomicrographs of immune cells (arrows) before FMT (C) and at 1 month after FMT (D), and mast cells (arrows) before FMT (E) and at 1 month after FMT (F) in a patient from the 60-g group. *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$; ****$P < 0.0001$.

and from 2.9 ± 0.9 to 1.9 ± 0.8 in the 60-g group (P = 0.0005). The DI in the placebo group was 2.8 ± 0.9 at the baseline and 2.7 ± 1.0 at 1 month after FMT (P = 0.800). Abundances of different bacteria is given in Supplementary Table 5.

Correlations

Densities of stem cells, early endocrine cell progenitors, enteroendocrine cells, and immune cells

The density of cells expressing Musashi-1 was significantly inversely correlated with the IBS-SSS total score, and the FAS total score (Supplementary Fig. 1 and 2). The density of neurogenin-3-expressing cells was significantly inversely correlated with the IBS-SSS total scores and FAS total scores (Supplementary Fig. 3 and 4).

The densities of chromogranin-A, serotonin, and GLP-1 cells were inversely correlated with the IBS-SSS total scores and FAS total scores (Fig. 4 and 5). The density of these cells was also not correlated with the FAS total score or the scores for its items (Supplementary Fig. 5, 6, and 8). The density of PYY cells was not correlated with the IBS-SSS total score or FAS total scores (Fig. 5 and Supplementary Fig. 7). The density of was inversely correlated with the IBS-SSS total score and the scores for its items as well as with the FAS total score and the scores for its items (Fig. 5 and Supplementary Fig. 8).

The density of total immune cells and mast cells were positively correlated with the IBS-SSS total score, FAS total score, and the scores for its 2 items (Fig. 6 and Supplementary Fig. 9 and 10).

Fecal short-chain fatty acid levels

The butyric acid level was inversely correlated with the IBS-SSS total score and the scores for its 4 items as well as with the FAS total score and the scores for its 2 items (Supplementary Fig. 11 and 12).

Differences Between Irritable Bowel Syndrome Subtypes

There was no difference in the response to FMT between IBS subtypes (P = 0.900). There was no difference between IBS subtypes regarding IBS-SSS, FAS, and IBS-QoL scores either at the baseline or at 1 month after FMT (Supplementary Fig. 13). There was also no difference between the IBS subtypes regarding the densities of Musashi-1, neurogenin-3, chromogranin A, serotonin, PYY, and GLP-1 cells at the baseline and at 1 month after FMT (Supplementary Fig. 14). There was no difference between the IBS subtypes regarding total immune cells or mast cell densities, or the fecal level of butyric acid either at the baseline or at 1 month after FMT (Supplementary Fig. 15).

DISCUSSION

The density of stem cells is lower in IBS patients than in healthy subjects, and it has been speculated that the low density of enteroendocrine cells in IBS patients could be caused by abnormal clonogeny and differentiation progeny of the intestinal stem cells.[5] In the present study, the density of intestinal Musashi-1-expressing cells increased in IBS patients who received donor FMT and in the FMT responder group, but not in the placebo or nonresponder groups. Furthermore, the density of Musashi-1-expressing cells was inversely correlated with the IBS-SSS total score and the scores for all of its items. These findings indicate that the increases in the densities of stem cells and their early progenitors play roles in the symptom improvement seen after FMT. The density of Musashi-1-expressing cells was also inversely correlated with the FAS total score and the scores for its items, suggesting that the increases in the densities of stem cells contribute to the improvement in fatigue seen after FMT. The present observations suggest that transplantation of epithelial organoids containing stem cells could be a useful therapeutic approach for IBS in the near future.[18-20]

The density of neurogenin-3 is lower in IBS patients than that in healthy subjects.[5] The present study found that density of neurogenin-3-expressing cells increased in donor-treated IBS patients and in the FMT responder group, but not in the placebo or nonresponder groups. Although the density of these cells was inversely correlated with the IBS-SSS total score, it was only inversely correlated with the score for the item of dissatisfaction with bowel habits. This suggests that this increase in the density of neurogenin-3-expressing cells only affects intestinal motility. The density of these cells was not correlated with the FAS total score or the scores for its items, indicating that the changes in neurogenin-3 cells after FMT do not affect fatigue in IBS patients. Mutation in neurogenin-3 results in a life-threatening condition called enteric anendocrinosis, which is associated with severe malabsorption and diabetes.[21] This condition is accompanied by the almost total loss of small- and large-intestine enteroendocrine cells.[21] In experimental animals, changes in intestinal neurogenin-3 expression result in a shift in the secretory lineages, with more goblet cells and fewer enteroendocrine cells.[21] Moreover, a low neurogenin-3 expression is associated with the low density of enteroendocrine cells seen in autosomal recessive disorder, congenital malabsorptive diarrhea, small-intestine allograft rejection, and NEUROG3-knockout mice.[22-24]

The density of cells expressing the common marker for enteroendocrine cells (chromogranin) in the colon has been reported to be lower in IBS patients than in healthy subjects.[6] In the present study, the density of these cells increased after FMT in the donor-treated

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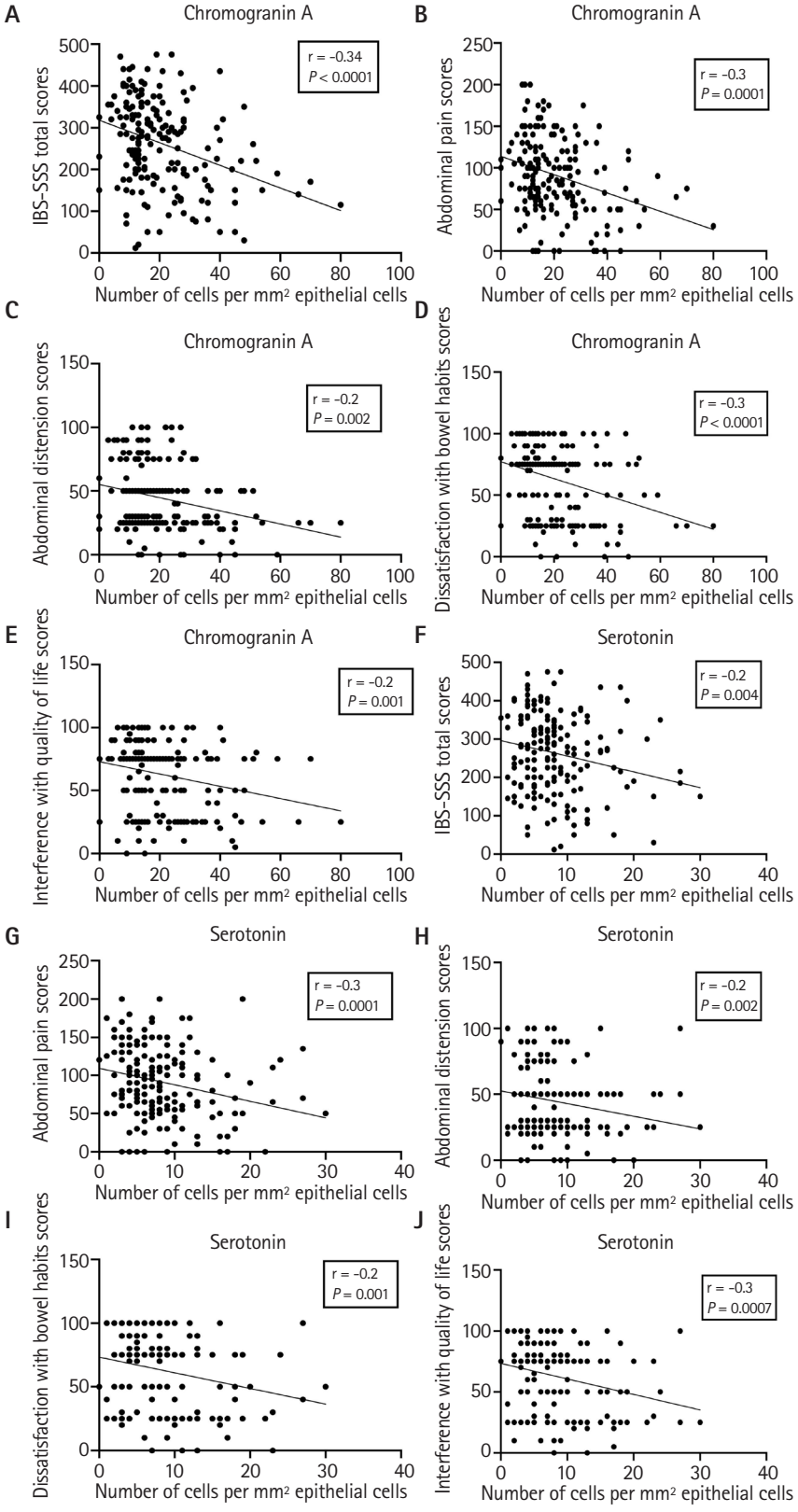

Figure 4. Correlations of the density of chromogranin-A-expressing cells with the Irritable Bowel Syndrome Severity Scoring System (IBS-SSS) total score (A) and the scores for abdominal pain (B), abdominal distension (C), dissatisfaction with bowel habits (D), and interference with quality of life (E). Correlations of the density of serotonin-expressing cells with the IBS-SSS total score (F) and the scores for abdominal pain (G), abdominal distension (H), dissatisfaction with bowel habits (I), and interference with quality of life (J).

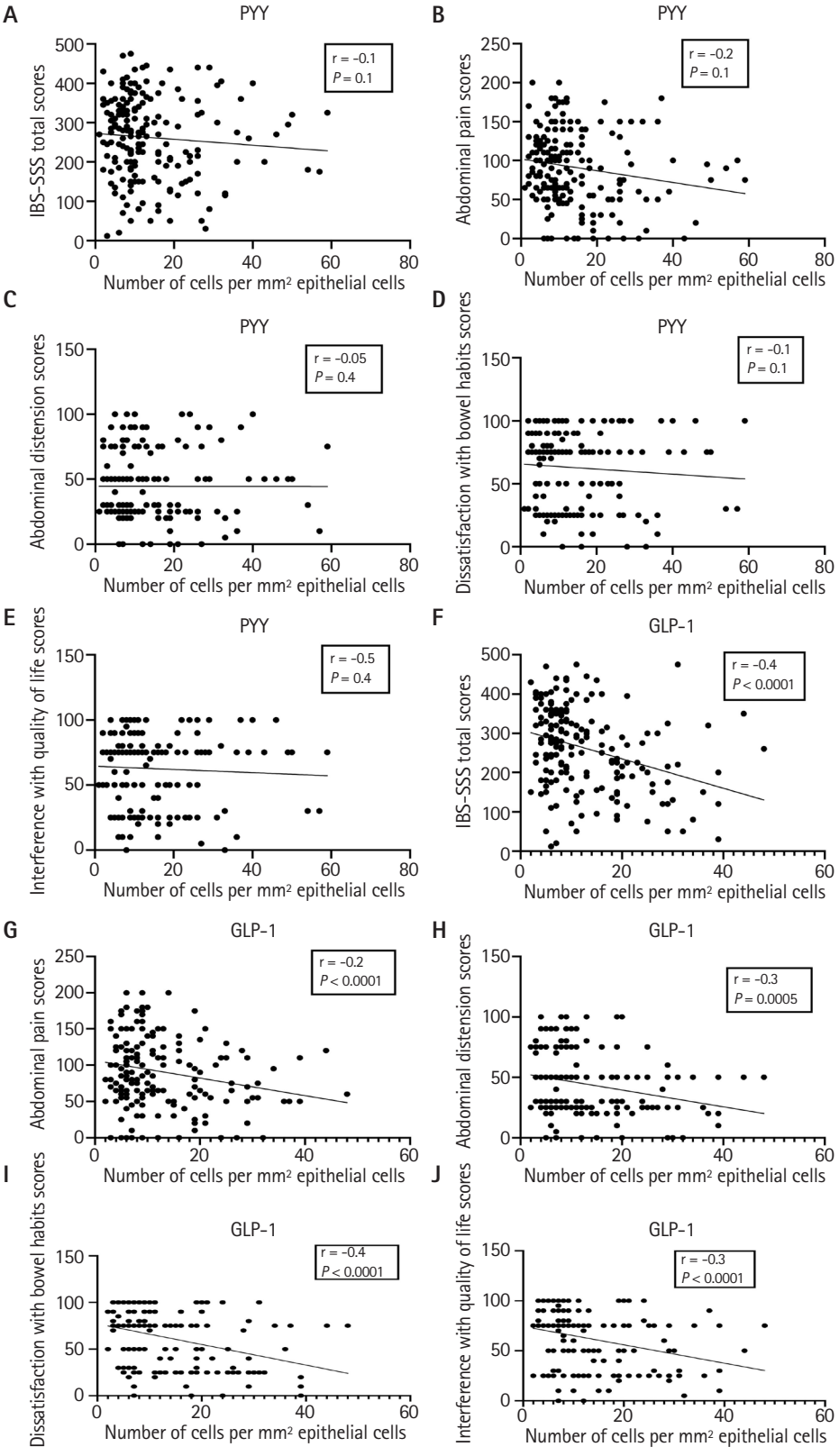

Figure 5. Correlations of the density of peptide YY (PYY)-expressing cells with the Irritable Bowel Syndrome Severity Scoring System (IBS-SSS) total score (A) and the scores for abdominal pain (B), abdominal distension (C), dissatisfaction with bowel habits (D), and interference with quality of life (E). Correlations of the density of glucagon-like peptide 1 (GLP-1)-expressing cells with the IBS-SSS total score (F) and the scores for abdominal pain (G), abdominal distension (H), dissatisfaction with bowel habits (I), and interference with quality of life (J).

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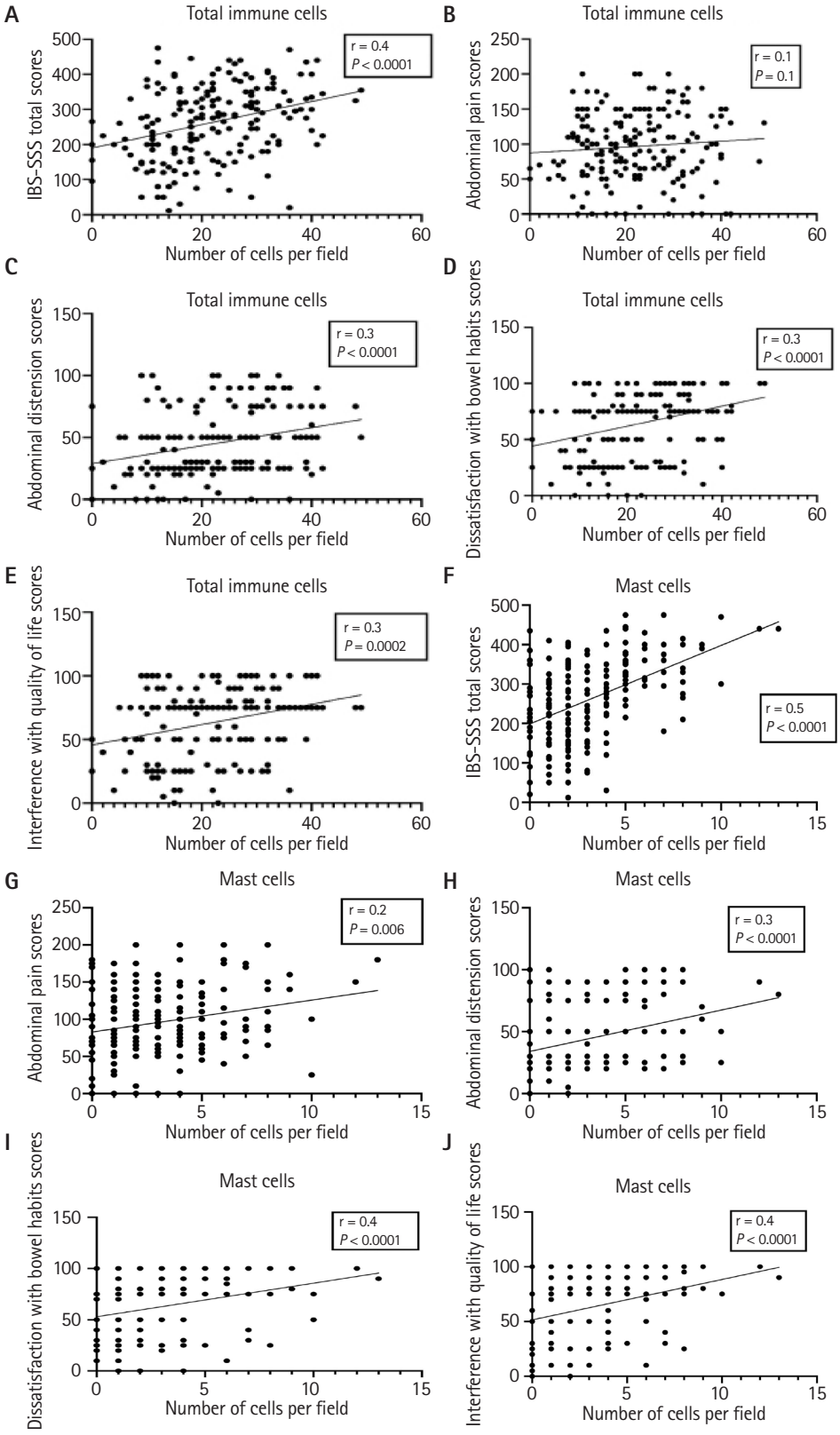

Figure 6. Correlations of the density of submucosal immune cells with the Irritable Bowel Syndrome Severity Scoring System (IBS-SSS) total score (A) and the scores for abdominal pain (B), abdominal distension (C), dissatisfaction with bowel habits (D), and interference with quality of life (E). Correlations of the density of submucosal mast cells with the IBS-SSS total score (F) and the scores for abdominal pain (G), abdominal distension (H), dissatisfaction with bowel habits (I), and interference with quality of life (J).

groups and in responder group, but not in the nonresponder or placebo groups. The density of chromogranin-A-expressing cells was inversely correlated with the IBS-SSS total score. These observations indicate that FMT probably exerts its effects by affecting the density of enteroendocrine cells that regulate gastrointestinal motility, visceral sensitivity, and secretion. Moreover, the density of chromogranin A was inversely correlated with the FAS total score and the scores for its items, indicating that enteroendocrine cells may also play a role in the improvement of fatigue recorded after FMT.

This study found that the density of serotonin cells increased in the IBS patients who received donor FMT and in responder group, but not in the placebo or nonresponder groups. The density of these cells was inversely correlated with the IBS-SSS total score and the scores for its 4 items. The density of serotonin cells in the large intestine has been reported to be lower in IBS patients than in healthy subjects.[6] However, the density of these cells was not correlated with the FAS total score, suggesting that serotonin does not affect fatigue in IBS patients. Serotonin plays an important role in visceral sensitivity by activating the submucosal branch of the enteric nervous system.[6] Serotonin may exert its effects on the central nervous system through an endocrine or a synaptic pathway from neuropods (axon-like long basal processes extending to the lamina propria) to neurons. Serotonin is believed to stimulate 5-hydroxytryptamine 1-7 receptors in circumventricular brain nuclei (hypothalamus and area postrema) via an endocrinal mechanism.[21] Furthermore, serotonin stimulates colonic motility, accelerates small- and large-intestine transit, and inhibits gastric emptying.[6] The importance of serotonin in manifesting IBS symptoms is already recognized, and led to the use of serotonin receptor antagonists and selective serotonin-reuptake inhibitors as symptomatic pharmacological treatments for IBS.[25]

The density of PYY cells in the large intestine has been found to be lower in IBS patients than in healthy subjects.[6] The density of PYY cells increased significantly at 1 month after FMT in the 30-g, 60-g, and responder groups, but not in the placebo or nonresponder groups. The density of these cells was inversely correlated with the abdominal pain score, but not with the IBS-SSS total score or the scores for its other 3 items. PYY modulates serotonin release from colonic enterochromaffin cells via the endogenous NK2/NK3 cascade system.[26] This could explain the inverse correlation between the density of cells expressing PYY and abdominal pain observed in the present study. The density of PYY-expressing cells was not correlated with the FAS total score or the scores for its items in the present study.

The density of GLP-1-cells increased in the IBS patients treated with donor FMT and in responder group in this study, but not in the placebo or nonresponder groups. Moreover, the density of these cells was inversely correlated with the IBS-SSS total score and the scores for its items, indicating an association between IBS symptoms and GLP-1. The density of GLP-1-expressing cells was also inversely correlated with the FAS total score, suggesting a role for GLP-1 in fatigue observed in IBS patients. GLP-1 inhibits gastric and pancreatic secretion, delays gastric emptying, is a major mediator of the ileal brake, and exerts some incretin effects.[6,27,28] GLP-1 acts through a hormonal and synaptic neuropod mode of action on GLP-1 receptors located in the hypothalamus, nucleus of the solitary tract, and area postrema.[21] GLP-1 analogue ROSE-010 have been shown to reduce abdominal pain in IBS patients, to delay gastric emptying and to accelerate colonic transit.[29,30] It has been suggested that ROSE-010 has an optimal effect on IBS-C and IBS-M.[31] The present study showed that there was no difference in the increased density of GLP-1 expressing cells between the 3 major IBS subtypes, namely diarrhea-predominant IBS (IBS-D), constipation-predominant IBS (IBS-C), and mixed IBS (IBS-M). This observation advocated that GLP-1 analogue can also be used in treating IBS-D. The GLP-1 analogues are used currently for treating diabetes and for weight loss.[21] The present observations confirm the previously published results that GLP-1 analogues could be of value in treating IBS symptoms and probably the associated fatigue.

The density of total immune cells decreased after FMT in the lamina propria in the donor-treated groups and responder group, but not in the placebo or nonresponder groups. The density of submucosal immune cells was correlated with the IBS-SSS total score and with the scores for abdominal distension and dissatisfaction with bowel habits, but not with the score for abdominal pain. These findings support the suggestion by Barbara and colleagues that low-grade inflammation plays a significant role in the pathogenesis of IBS.[8] The density of immune cells was also correlated with the FAS total score and the scores for its sub-items, indicating that submucosal immune cells play a role in fatigue associated with IBS. This finding suggests that low-dose 5-aminosalicylic acid or budesonide could be useful in treating IBS.

The density of submucosal mast cells decreased in the IBS patients who received donor FMT and in responder group, but not in the placebo or nonresponder groups. The density of these cells was correlated with the IBS-SSS total score and the scores for its items including abdominal pain and dissatisfaction with bowel habits scores. Mast cells are capable of producing serotonin, and possess serotonin receptors.[32] Submucosal mast cells lie close to enteric nerves, and by releasing inflammatory mediators they affect the enteric nervous system and muscle function.[8] The present observations underline the importance of mast cells in IBS symp-

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tom manifestation and support the hypothesis of Barbara and colleagues that mast cell–nerve interactions could be important for IBS pathophysiology.[33] Barbara and colleagues proposed controlling mast-cell functions using pharmacological drugs such as anti-IgE antibodies, inhibitors of the intracellular protein tyrosine kinase Syk, second-generation antihistamines, and proteinase-activated receptor antagonists.[33]

The present observations that the fecal level of butyric acid increased in IBS patients who received 30-g or 60-g of the donor feces and in the responder group, but not in the placebo or non-responder groups are in agreement with previously published results.[34] Furthermore, the present findings that the level of butyric acid was inversely correlated with the IBS-SSS total score and the scores for its items are also in agreement with previous published observations.[34] Butyric acid provides colonic epithelial cells with energy and regulates gastrointestinal motility, contractility, visceral sensitivity, epithelial proliferation, epithelial barrier function, and local immunity.[35] The important role played by butyric acid in manifesting IBS symptoms is supported by previous observations of the butyric acid levels being inversely correlated with intestinal transit time in healthy subjects, and inversely correlated with colon transit time and positively correlated with stool frequency in IBS patients.[35,36] Furthermore, treating IBS patients with microencapsulated sodium butyrate improved their abdominal pain, diarrhea, and constipation. Butyric acid regulates serotonin expression by intestinal enterochromaffin cells and also regulates the expression of PYY and GLP-1 by activating the free fatty acid receptor 2 (FFAR2), FFAR3, G protein-coupled receptor 43 (GPR43), and GPR41 receptors.[10,37-42] In the present study, the level of butyric acid was inversely correlated with the FAS total score and the scores for its items. The possible mechanisms underlying these effects remain to be investigated.

The bacteria whose abundance was increased in IBS patients who received FMT included *Alistipes* spp., *Faecalibacterium prausnitzii*, *Eubacterium biforme*, and *Lactobacillus* spp. *Alistipes* spp. hydrolyses the serotonin precursor tryptophan into indole.[43] They also produce the anti-inflammatory acetic, succinic, and propionic SCFAs.[43,44] *Faecalibacterium prausnitzii* and *Eubacterium biforme* are the main producers of butyric acid in the large intestine.[45-51] *Faecalibacterium prausnitzii* also produces an anti-inflammatory single 15-kDa protein that blocks the nuclear factor kappa B pathway, and increases the expression of serotonin by affecting major genes regulating serotonin expression.[52,53] *Lactobacillus* spp. regulate epithelial barrier function and modulate immune responses by increasing the expression of mucin by intestinal cells, and blocking the adherence of bacteria, fungi, and parasites.[54,55] These findings raise the question whether probiotics containing *Faecalibacterium prausnitzii*, *Eubacterium biforme*, *Alistipes*, and *Lactobacillus* spp. could be useful in treating IBS and fatigue.

The main strengths of this study were that it included IBS patients belonging to the 3 main IBS subtypes (ie, IBS-D, IBS-C, and IBS-M) as well as both female and male patients with IBS. The main limitations of the study were that it did not include the fourth IBS subtype, it was dominated by female IBS patients, and it showed only short-term changes and not the long-term changes as well as the sample size of the patients' cohort did not allow analysis of the difference between IBS-subtypes.

In conclusion, the densities of stem cells, enteroendocrine cell progenitors, serotonin, GLP-1, and PYY cells increased after FMT in IBS patients treated donor´s transplant. The densities of these cells were inversely correlated with IBS symptoms and fatigue, indicating their roles in manifesting IBS symptom and fatigue. The abundance of bacteria that produce butyric acid increased after FMT in IBS patients who received donor feces and hence the level of butyric acid also increased. Moreover, the densities of submucosal immune cells and especially, mast cells decreased after FMT in IBS patients who received donor's transplant. The densities of these cells were positively correlated with IBS symptoms and fatigue. Although these observations showed association rather than causation or mechanistic proof, they may suggest new treatment approaches for IBS. Furthermore, these parameters can be used as predictive biomarkers for FMT response in IBS and eventually durable effects.

SUPPLEMENTARY MATERIALS

Note: To access the supplementary protocol, tables and figures mentioned in this article, visit the online version of *Journal of Neurogastroenterology and Motility* at http://www.jnmjournal.org/, and at http://doi.org/10.5056/jnm26028.

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Conflicts of interest

None.

Author contributions

Magdy El-Salhy designed the study, obtained the funding, administered the study, recruited the patients, performed FMT, performed sigmoidoscopy, collected, analyzed, interpreted the data, and drafted the manuscript; Tora Skarvatun performed immunostaining of colonic sections, quantified the densities of stem cells, stem cell progenitors, enteroendocrine and immune cells, and critically revised the manuscript for important intellectual content; and Dag Arne Lihaug Hoff, Johan Lunding, Elisabeth Kjelsvik Steinsvik, Odd Helge Gilja, and Jan Gunnar Hatlebakk contributed to the design of the study and to the analysis and interpretation of the data, and critically revised the manuscript for important intellectual content.

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SUPPLEMENTARY MATERIALS

Supplementary Table 1. IBS-SSS total scores and scores for its four items in the placebo, 30-g, 60-g, responders, and nonresponders groups at the baseline and at 1 month after FMT.

Time	Group	Total score	1	2	3	4
Baseline	Placebo FMT	292.7±83.0	93.1±49.2	53.5±25.9	73.8±25.0	73.3±19.0
	30-g FMT	319.5±73.0	111.1±43.5	52.0±26.4	77.0±18.1	75.6±18.6
	60-g FMT	312.0±64.6	109.5±39.6	46.1±22.6	73.5±17.5	70.1±26.0
	Responders	313.9±65.9	109.4±42.2	52.8±24.3	78.4±15.1	76.2±18.3
	Non-responders	305.2±87.5	100.0±48.0	49.2±26.2	72.3±24.1	70.9±22.3
1 month after FMT	Placebo FMT	262.3±56.3	81.9±56.3	47.9±30.8	61.7±31.2	58.8±29.8
	30-g FMT	201.2±108.9****	78.9±47.9**	35.9±23.4***	49.1±30.0****	51.2±29.3****
	60-g FMT	188.9±87.4****	65.0±40.5***	29.8±19.3**	40.9±26.8****	45.5±24.5**
	Responders	160.4±47.2****	66.8±47.2****	31.8±23.6****	44.0±29.3****	46.2±28.4****
	Non-responders	311.7±89.7	93.9±46.8	48.0±25.3	60.6±29.4	60.6±26.9

IBS-SSS items: 1, abdominal pain; 2, abdominal distension; 3, dissatisfaction with bowel habits; 4, interference with quality of life.
Data are mean±SD values. **, p<0.01; ***, p<0.001; ****, p<0.0001 relative to placebo FMT.

Supplementary Table 2. FAS total scores and scores for its two items in the placebo, 30-g, 60-g, responders, and non-responders groups at the baseline and at 1 month after FMT.

Time	Group	Total score	Physical fatigue	Mental health issues
Baseline	Placebo FMT	31.6±4.4	16.4±2.4	15.2±2.5
	30-g FMT	30.9±5.4	16.0±3.0	15.5±2.8
	60-g FMT	31.6±5.1	15.7±2.8	15.9±2.8
	Responders	31.3±5.3	15.9±2.9	15.8±2.7
	Nonresponders	31.0±4.6	16.4±2.7	15.1±2.7
1 month after FMT	Placebo FMT	30.3±5.8	16.0±2.7	14.3±3.3
	30-g FMT	27.2±7.0*	14.3±3.9*	13.3±3.1**
	60-g FMT	27.6±5.9*	13.2±2.9**	13.2±3.1**
	Responders	27.0±6.2***	14.0±3.2**	12.9±2.7****
	Nonresponders	30.0±6.7	15.9±3.6	14.7±3.5

Data are mean±SD values. *, p<0.05; **, p<0.01; ***, p<0.001; ****, p<0.0001 relative to baseline.

Supplementary Table 3. IBS-QoL total scores and scores in its eight domains in the placebo, 30-g, 60-g, responders, and non-responders groups at the baseline and 1 month after FMT.

Time	Group	Total score	1	2	3	4	5	6	7	8
Baseline	Placebo FMT	115.6±20.3	26.3±6.4	23.0±5.6	11.9±2.8	10.6±2.5	6.9±2.8	14.4±3.2	7.5±1.9	11.0±2.6
	30-g FMT	106.7±22.3	25.3±6.6	19.3±5.6	11.0±3.4	10.0±2.7	5.8±2.7	13.3±3.4	7.4±2.0	10.5±2.5
	60-g FMT	112.7±51.3	27.1±4.9	20.6±4.8	12.1±2.4	10.3±2.4	7.1±3.3	14.1±3.4	8.0±1.1	10.6±2.6
	Responders	109.6±22.1	25.9±6.3	20.0±5.3	11.7±3.1	10.0±2.6	6.3±3.1	13.6±3.5	7.7±1.6	10.7±2.5
	Nonresponders	111.9±21.3	26.2±6.7	21.6±6.0	11.1±2.9	10.6±2.6	6.6±2.5	14.0±3.2	7.3±2.1	±
1 month after FMT	Placebo FMT	123.3±28.5	29.4±7.6	23.3±5.7	12.9±4.4	12.3±3.6	8.2±3.0	14.8±3.6	7.3±2.3	11.3±3.1
	30-g FMT	158.0±86.0***	39.0±24.0***	30.0±22.0**	23.0±16.0***	15.0±10.0***	13.0±10.0**	24.0±16.0***	10.0±6.0	15.0±10.0*
	60-g FMT	131.7±18.1**	30.4±6.8*	23.4±4.9*	13.9±3.2*	12.8±2.5*	9.3±3.0*	16.0±3.9*	8.2±1.5	12.6±2.6*
	Responders	126.7±22.7****	30.9±6.4****	22.5±4.3**	14.2±3.3***	12.1±2.2****	8.2±3.0***	15.6±3.2***	8.2±1.4	11.7±2.4*
	Nonresponders	118.0±27.0	28.4±7.3	21.8±6.1	12.2±3.9	11.7±2.7	7.3±2.7	15.0±3.3	7.4±2.1	11.2±2.9

IBS-QoL domains: 1, dysphoria; 2, interference with daily activities; 3, body image; 4, health worries; 5, food avoidance; 6, social reaction; 7, sexual function; 8, impact on relationships.
Data are mean±SD values. *, p<0.05; **, p<0.01; ***, p<0.001; ****, p<0.0001 relative to the baseline.

Supplementary Table 4. Faecal SCFA levels in the placebo, 30-g, 60-g, responders, and non-responders groups at the baseline and at 1 month after FMT.

Time	Group	Total SCFAs	Acetic acid	Propionic acid	Isobutyric acid	Butyric acid	Isovaleric acid	Valeric acid	Isocapronic acid	Capronic acid
Baseline	Placebo FMT	85±46	46±20	14±10	1.5±0.9	10±2	2.9±3.1	2.0±2.1	0.0±0.0	0.6±0.7
	30-g FMT	79±40	45±21	14±10	1.4±1.0	11±8	2.1±1.4	1.9±2.0	0.0±0.0	0.7±1.3
	60-g FMT	68±27	42±15	12±6	1.3±0.7	12.2±8.1	1.9±1.1	0.8±0.2	0.0±0.0	0.6±0.8
	Responders	81±44	44±20	13±9	1.4±0.9	10.4±7.9	2.1±1.3	1.7±1.8	0.0±0.0	0.6±1.0
	Non-responders	77±42	42±19	12±8	1.7±1.8	13.8±8.5	1.9±1.8	1.4±0.8	0.0±0.0	0.5±0.7
1 month after FMT	Placebo FMT	70±24	40±14	12±5	1.6±0.9	7±2	2.3±1.5	1.5±0.8	0.0±0.0	0.5±0.7
	30-g FMT	98±48*	40±15	11±7	1.3±0.6	16±10**	1.9±0.9	1.6±0.7	0.0±0.0	0.5±0.9
	60-g FMT	98±48*	45±27	14±10	2.2±1.9	19±12*	2.4±1.7	0.9±0.2	0.0±0.0	0.5±0.6
	Responders	104±52**	43±19	12±7	1.5±0.7	15.7±9.6****	2.2±1.8	1.6±0.8	0.0±00	0.5±0.8
	Non-responders	68±22	40±14	11±5	1.4±0.8	12.0±5.9	1.4±0.7	1.5±0.6	0.0±0.0	0.4±0.7

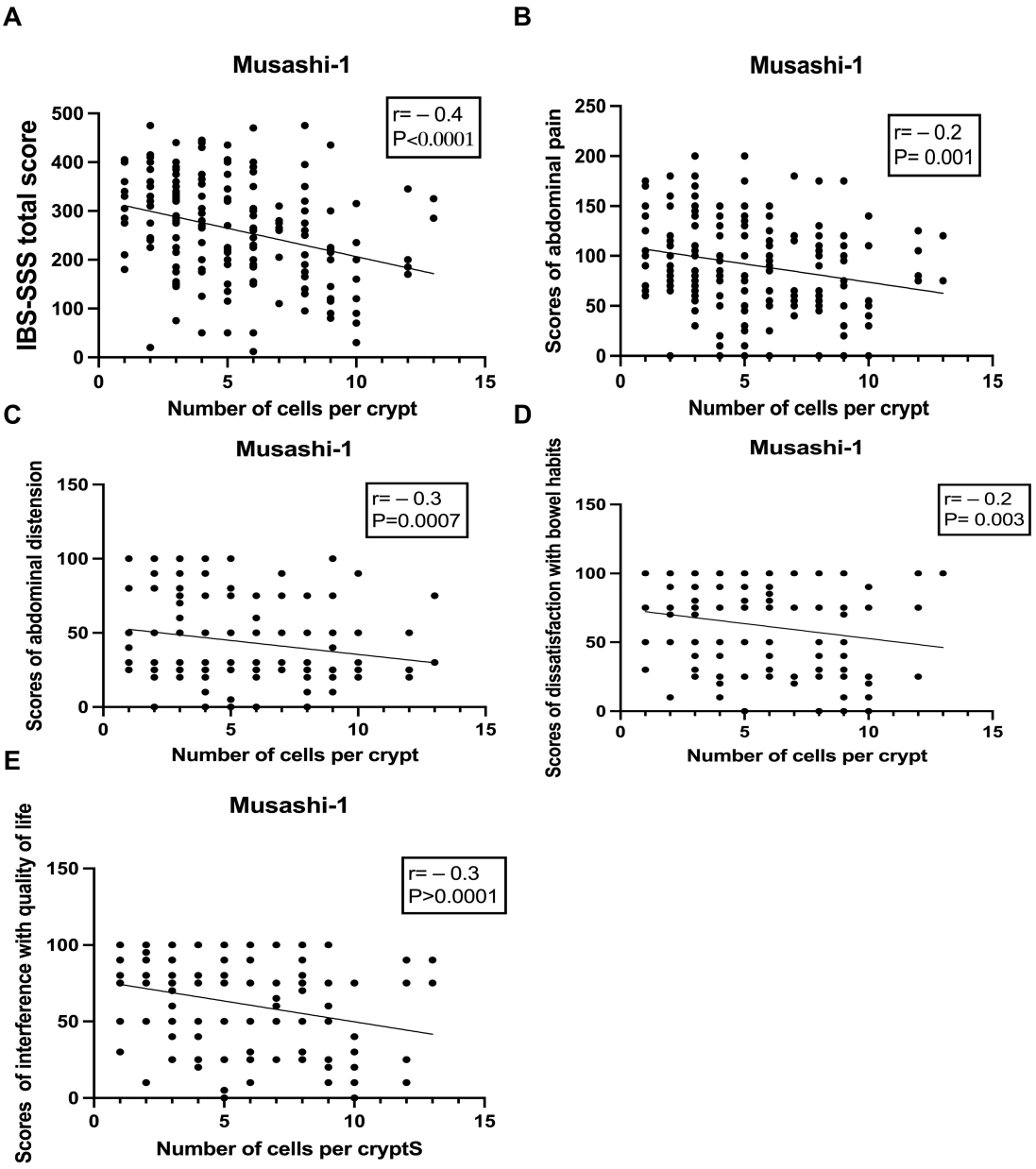
Data are mean±SD values in mmol/kg wet weight. *, $p<0.05$; **, $p<0.01$; ****, $p<0.0001$ relative to baseline.

Supplementary Table 5. Bacteria whose abundance differed significantly between the baseline and at 1 month after FMT in the placebo and donor-treated groups.

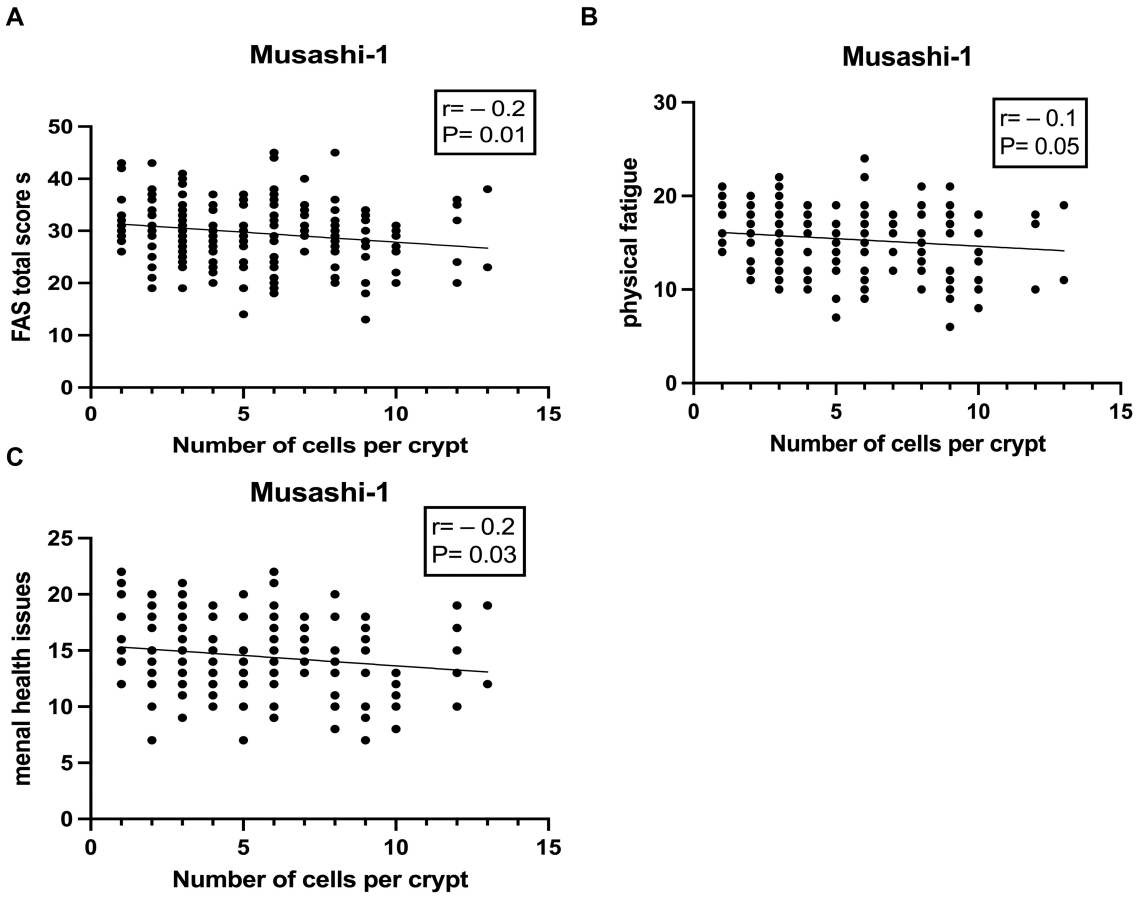
Bacteria	Placebo group		Donor-treated groups	
	Baseline	1 month after FMT	Baseline	1 month after FMT
Actinobacteria spp.	165.3±106.5	173.1±271.3	18.5±36	34.4±52.5***
Actinomycetales spp.	10.8±8.0	21.9±18.4**	14.9±28.2	33.7±46.2****
Bifidobacterium spp.	187.3±128.5	129.3±206.5*	154.7±131.1	146.3±154.8
Alistipes spp.	55.0±35.0	99.3±129.1	65.3±50.7	241.4±156.9****
Alistipes onderdonkii	118.0±28.5	23.2±18.3****	112.6±39.4	54.2±79.2****
Bacteroides fragilis	36.3±140.8	37.4±45.6*	15.3±27.6	21.5±26.4*
Bacteroides pectinophilus	245.5±6.2	14.2±8.6****	238.3±43.4	20.9±12.7****
Bacteroides spp.	132.6±13.9	44.0±134.1****	144.9±24.7	29.4±54.5****
Bacteroides and *Prevotella* spp.	326.0±216.4	427.3±274.7	400.3±192.0	853.4±295.3****
Bacteroides stercoris	34.5±79.2	24.0±38.0	22.6±14.5	35.5±37.4
Bacteroides zoogleoformans	28.6±18.9	11.9±24.9***	30.3±18.9	75.1±93.3
Parabacteroides johnsonii	7.0±5.8	8.0±6.9	7.3±7.9	28.8±35.4****
Parabacteroides spp.	24.8±26.7	33.8±23.9*	32.4±56.3	111.6±116.6****
Firmicutes spp.	495.9±151.6	464.7±246.7	689.8±236.5	532.4±166.3***
Bacilli spp.	216.5±191.5	298.7±287.5	216.4±155.8	300.6±171.9*
Catenibacterium mitsuokai	56.8±34.6	14.7±43.8****	73.5±95.9	60.9±183.7****
Clostridia spp.	388.1±6.9	437.6±27.0	384.9±73.3	441.7±84.7**
Clostridium methylpentosum	72.8±6.9	25.5±27.0****	82.7±50.1	20.6±12.5****
Clostridium spp.	8.4±8.8	11.8±28.5	7.6±13.6	4.6±9.2**
Coprobacillus cateniformis	42.2±2.5	6.8±6.0	40.7±7.1	7.3±8.3****
Dialister invisus	94.9±145.7	179.2±186.4*	100.0±130.8	199.3±271.0
Dialister invisus and *Megasphaera micronuciformis*	45.7±72.8	142.5±171.2*	50.4±76.4	134.4±236.1
Dorea spp.	41.8±28.2	96.5±104.1	41.1±31.3	46.9±75.1
Eubacterium biforme	65.±114.5	72.3±153.4	38.7±94.9	102.6±121.0****
Eubacterium hallii	106.5±72.4	140.8±183.6	89.7±48.3	107.7±75.1
Eubacterium rectale	47.8±42.4	85.8±144.2	45.2±40.1	112.1±105.3*
Eubacterium siraeum	62.7±47.6	50.5±64.4	62.3±67.9	88.8±115.8
Faecalibacterium prausnitzii	248.5±91.6	337.4±215.8	228.6±110.9	402.1±163.5****
Lachnospiraceae spp.	604.0±136.6	787.8±201.0	641.6±158.2	878.1±133.6****
Lactobacillus ruminis and *Pediococcus acidilactici*	5.8±20.9	8.0±8.6	12.1±74.8	12.6±30.2
Lactobacillus spp.	80.8±94.8	142.9±206.6	37.1±78.0	153.2±25.5****
Lactobacillus **ssp. 2**	11.2±19.4	5.8±8.3	9.7±18.8	18.7±25.1
Phascolarctobacterium spp.	17.4±23.6	17.2±16.2	27.2±45.8	53.8±109.7
Ruminococcus albus and *Ruminococcus bromii*	93.5±89.7	82.3±97.3	82.8±76.3	114.4±104.4
Ruminococcus gnavus	36.1±76.2	86.6±99.7*	44.8±87.6	91.5±15.2**
Streptococcus agalactiae and *Eubacterium rectale*	41.0±34.1	46.5±109.3	35.2±33.2	56.5±9.4***
Streptococcus salivarius ssp. *thermophilus* and *Streptococcus sanguinis*	47.2±78.1	77.5±133.0	51.8±71.8	42.3±67.8
Streptococcus salivarius ssp. *thermophilus*	100.9±76.5	143.8±206.3	111.2±78.0	89.3±113.9***
Streptococcus spp.	19.4±25.1	46.5±109.3****	22.2±28.5	25.3±41.1
Streptococcus **ssp. 2**	62.6±19.7	60.7±145.8	63.9±16.7	29.2±39.8****
Veillonella spp.	289.4±66.8	523.9±185.0****	318.0±92.6	631.6±180.2****
Firmicutes **spp.**	202.6±141.3	195.9±234.2	274.9±198.2	196.3±124.8
Proteobacteria spp.	65.5±46.9	131.0±259.7	71.9±65.3	54.6±63.9**
Acinetobacter junii	199.3±1.4	1.8±3.4****	194.6±26.9	2.0±3.7****

Enterobacteriaceae spp.	36.8±1.1	3.5±5.1****	41.8±29.2	3.6±5.6****
Shigella and *Escherichia* spp.	41.3±85.3	135.9±262.8	69.5±129.2	45.1±96.7
Mycoplasma hominis	70.9±53.7	102.2±115.4	115.9±126.9	69.2±58.5
Akkermansia muciniphila	108.2±136	114.4±226.0	87.3±126.6	79.9±110.5

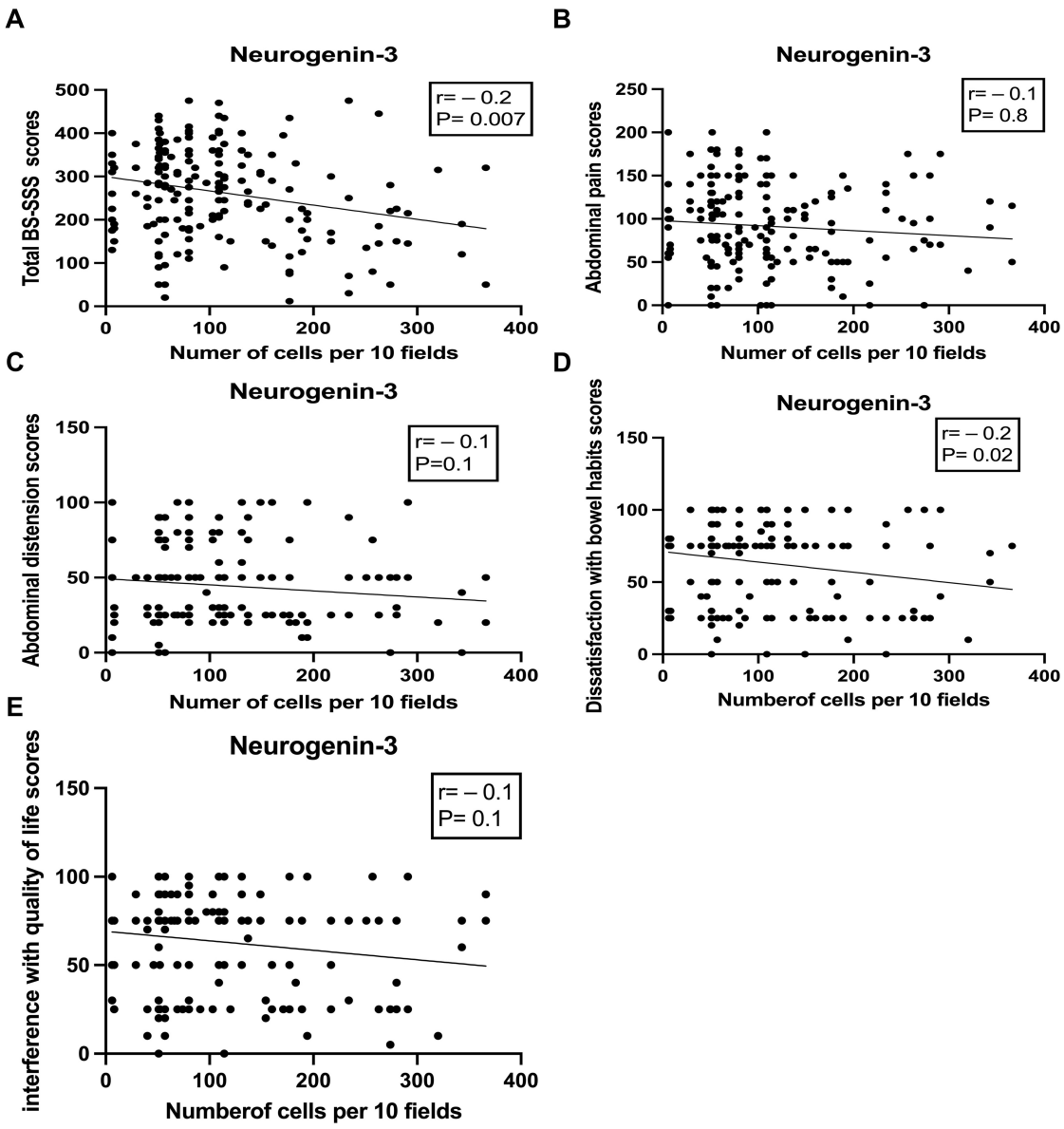
Data mean±SD values. *, $p<0.05$; **, $p<0.01$; ***, $p<0.001$; ****, $p<0.0001$ relative to baseline.
The shaded bacteria markers are those that changed in the donor-treated groups but not in the placebo group.



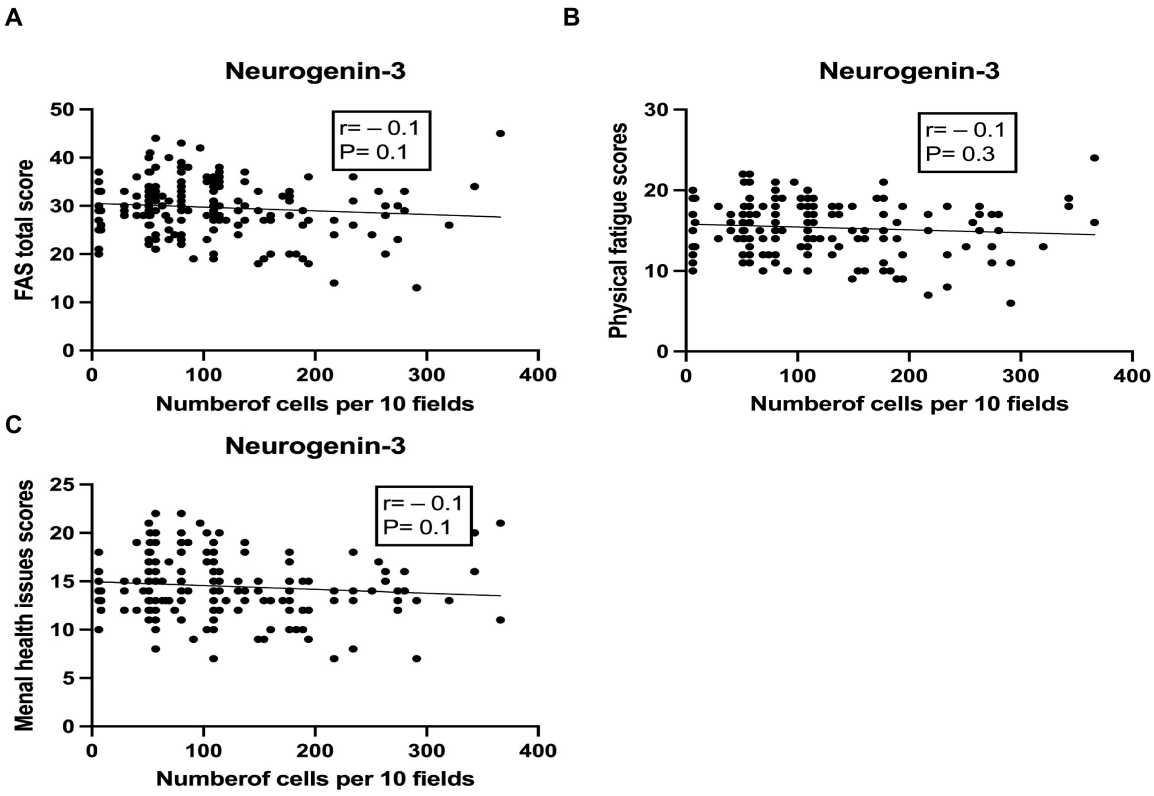
Supplementary Figure 1. Correlations of the density of Musashi-1-expressing cells with the IBS-SSS total score (A) and the scores for abdominal pain (B), abdominal distension (C), dissatisfaction with bowel habits (D), and interference with quality of life (E).



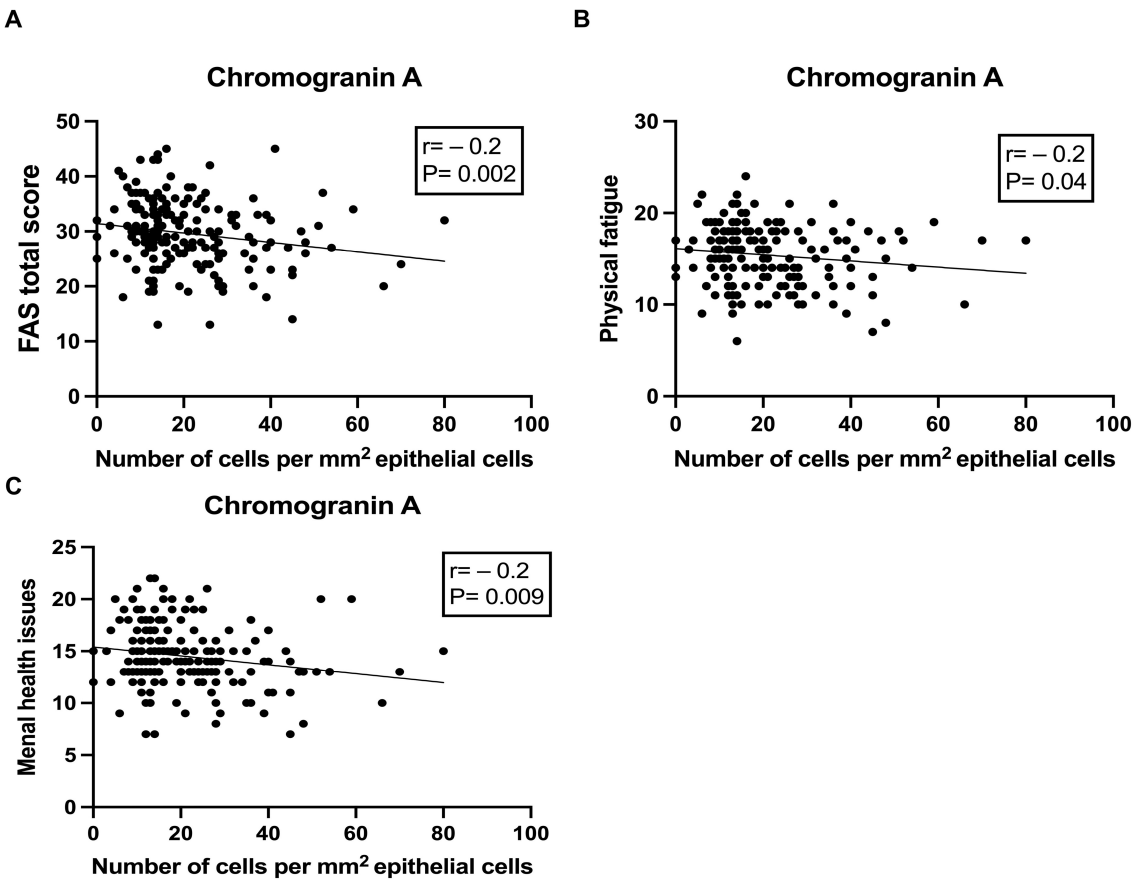
Supplementary Figure 2. Correlations of the density of Musashi-1-expressing cells with the FAS total score (A) and the scores for physical fatigue (B) and mental health issues (C).



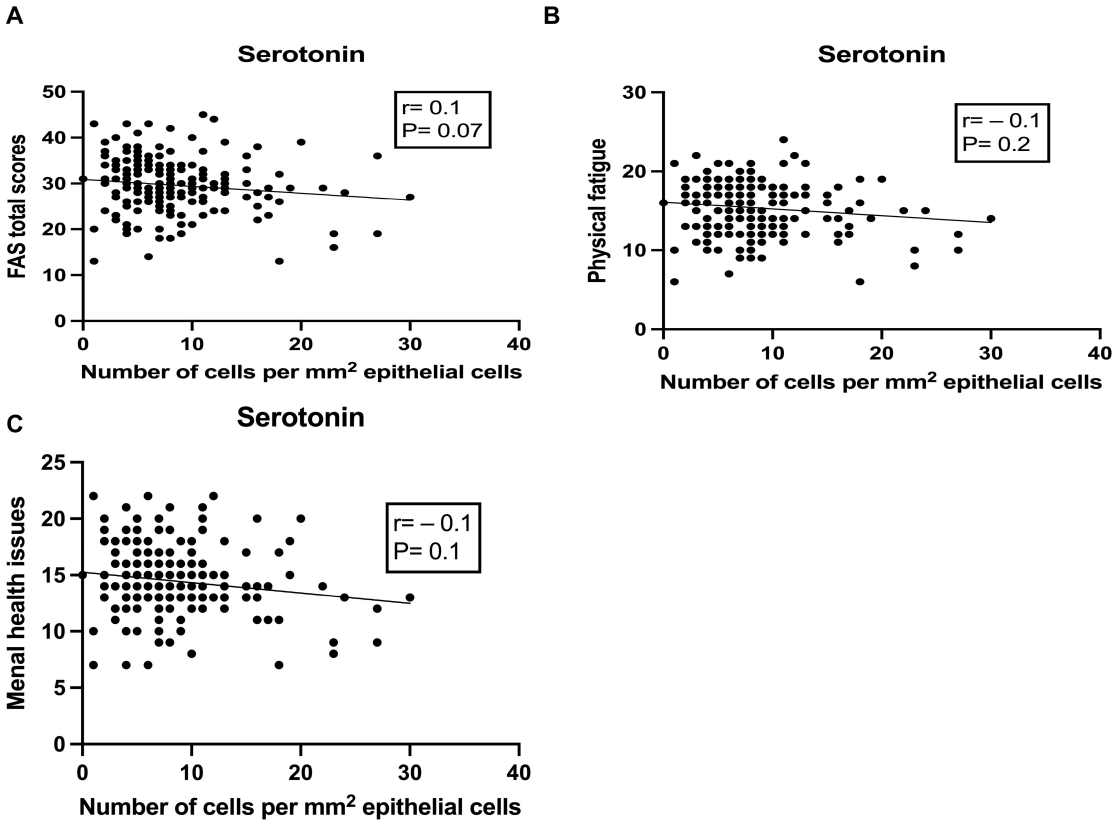
Supplementary Figure 3. Correlations of the density of neurogenin-3-expressing cells with the IBS-SSS total score (A) and the scores for its four items (B, C, D, E).



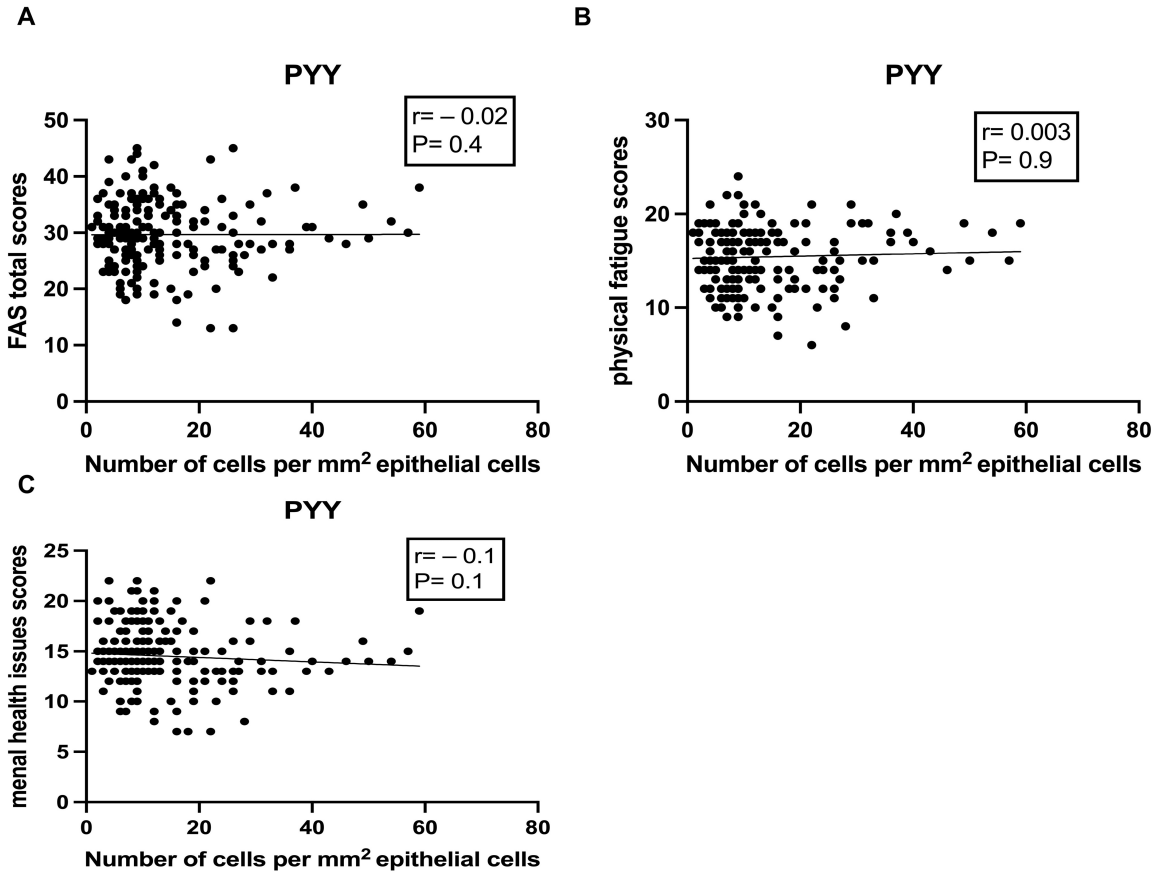
Supplementary Figure 4. Correlations of the density of neurogenin-3-expressing cells with the FAS total score (A) and the scores for its two items (B, C).



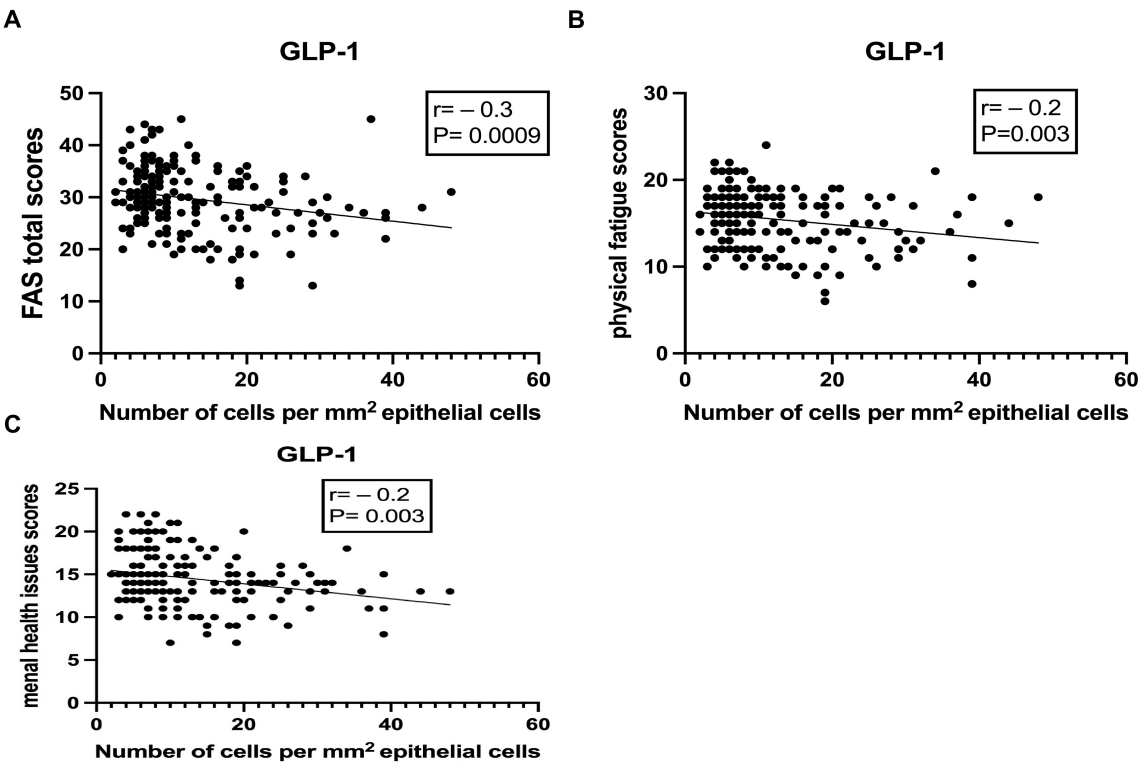
Supplementary Figure 5. Correlations of the density of chromogranin-A-expressing cells with the FAS total score (A) and the scores for its two items (B, C).



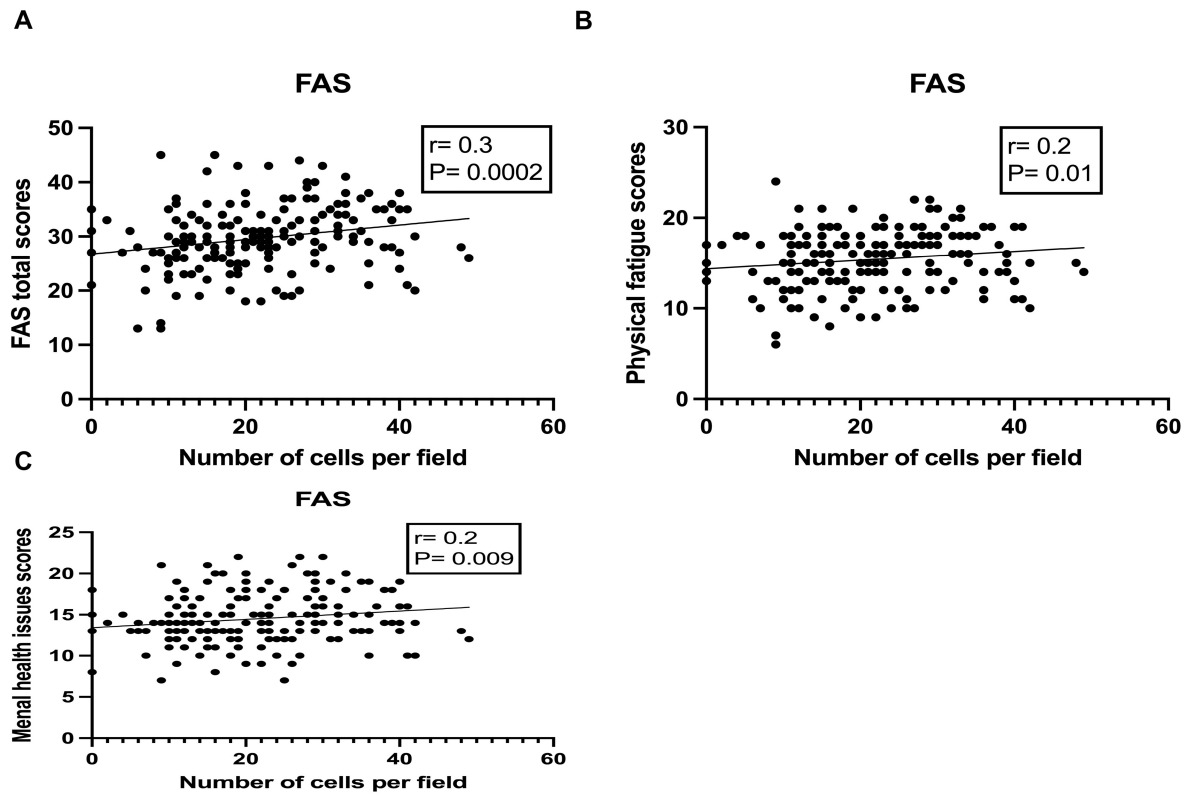
Supplementary Figure 6. Correlations of the density of serotonin-expressing cells with the FAS total score (A) and the scores for its two items (B, C).



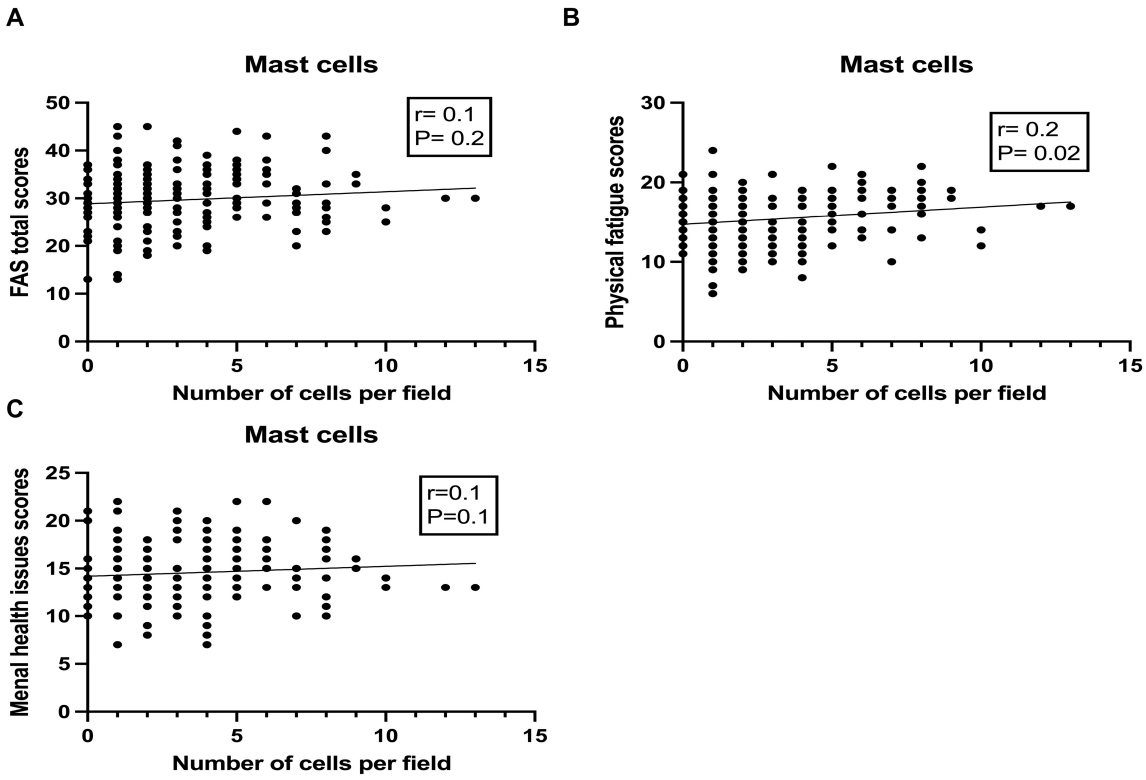
Supplementary Figure 7. Correlations of the density of PYY-expressing cells with the FAS total score (A) and the scores for its two items (B, C).



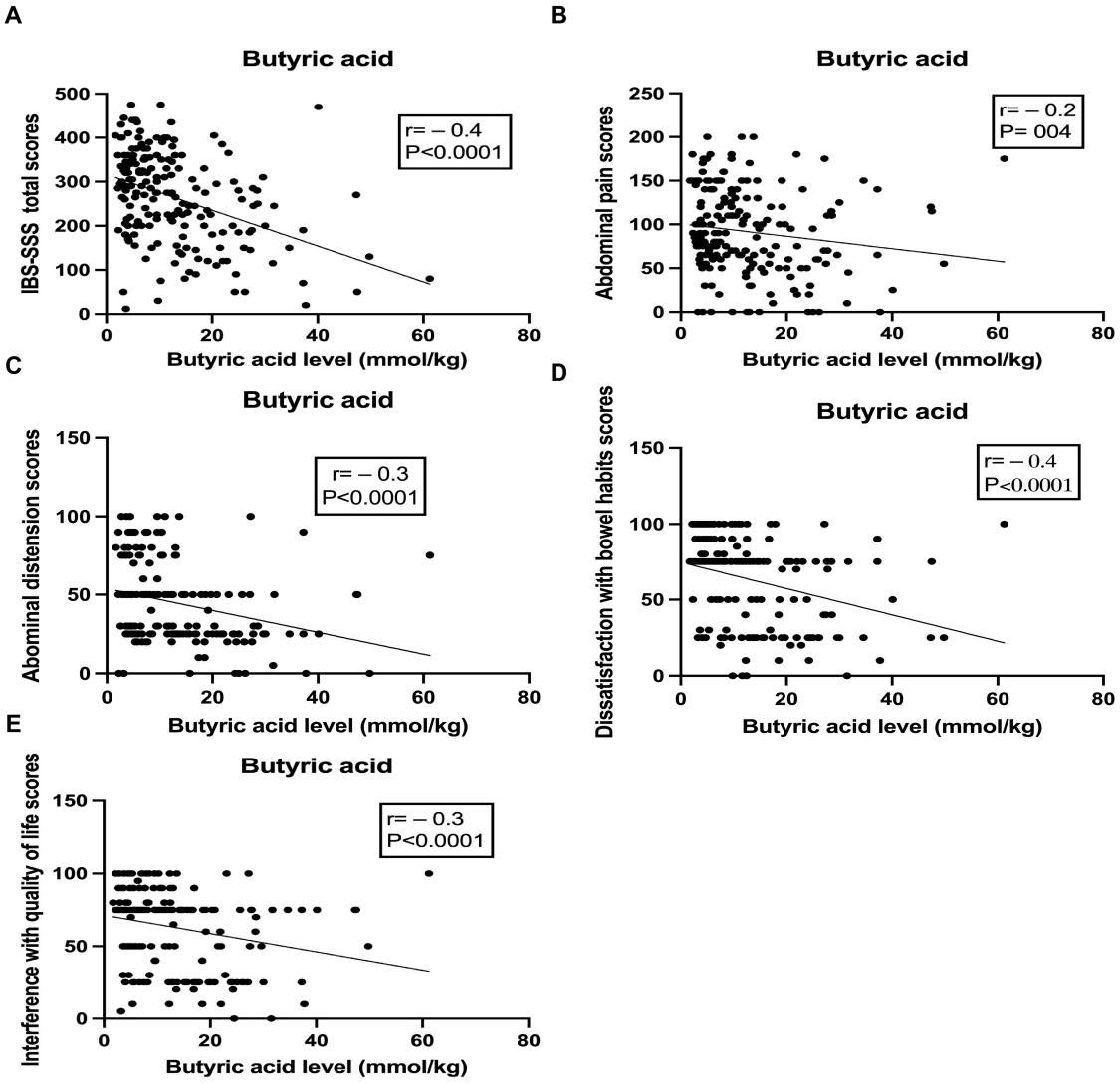
Supplementary Figure 8. Correlations of the density of GLP-1-expressing cells with the FAS total score (A) and the scores for its two items (B, C).



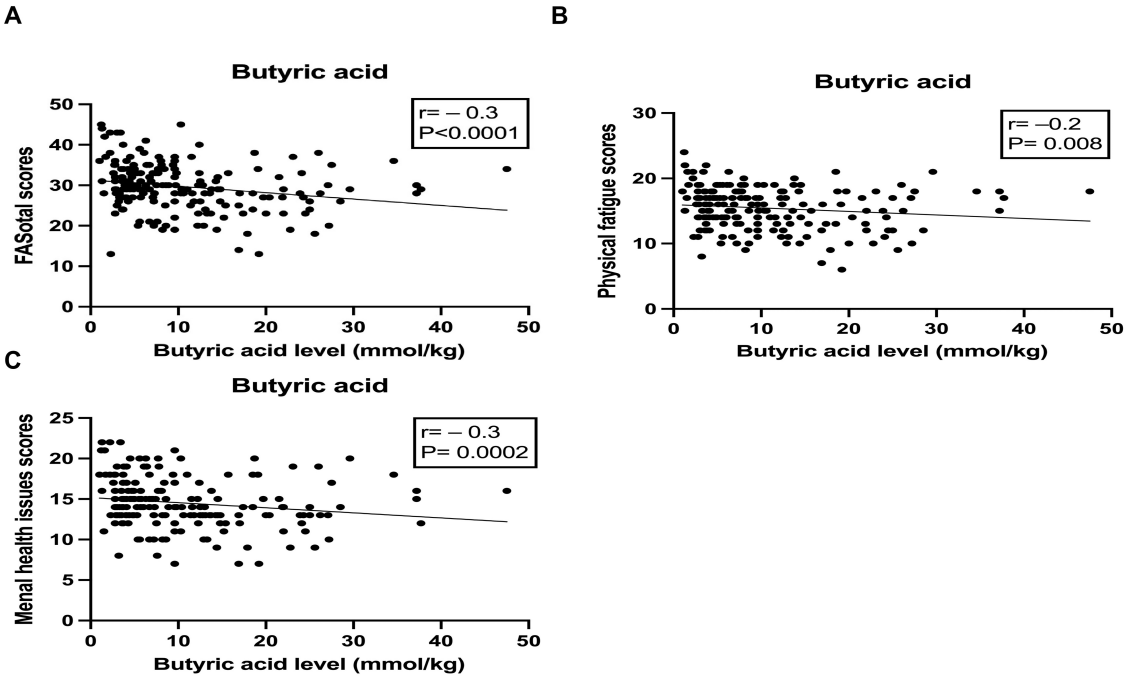
Supplementary Figure 9. The density of cells expressing the common marker for immune cells (CD45) was positively correlated with the FAS total score and the scores for its two subitems.



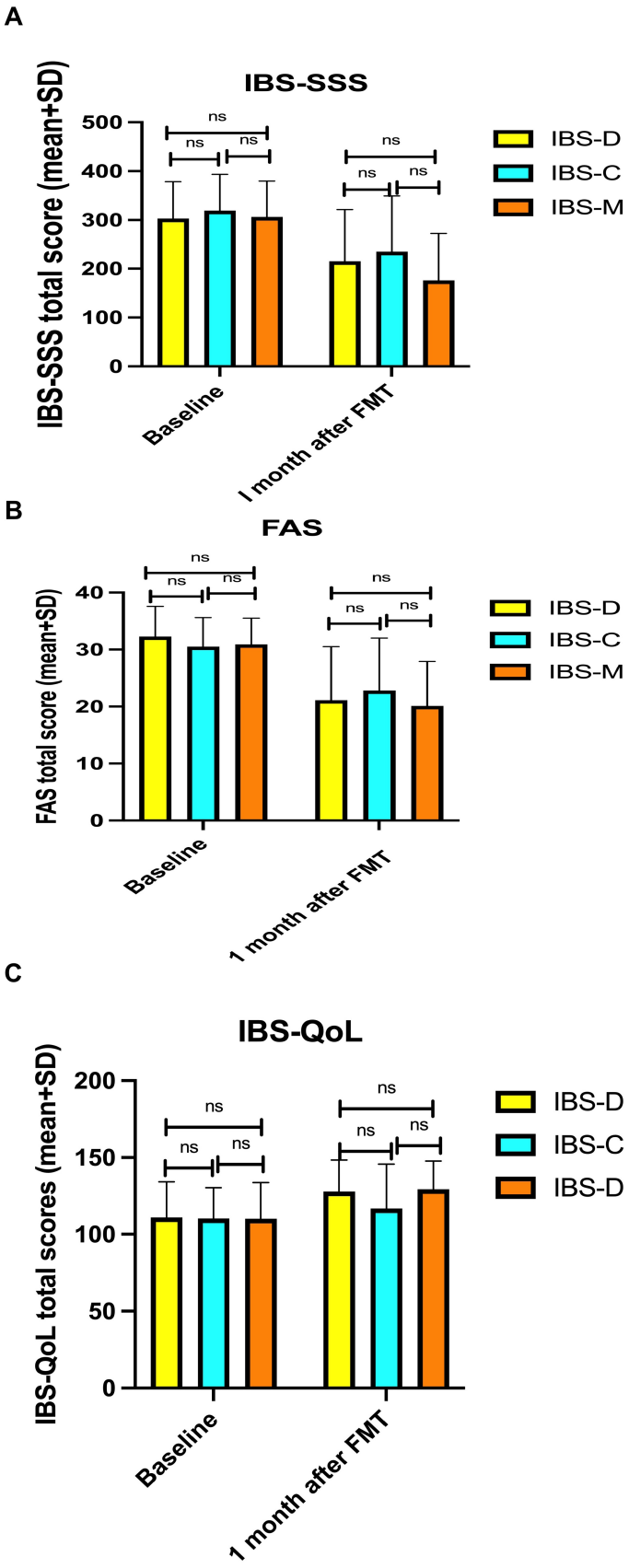
Supplementary Figure 10. Correlations of the density of mast cells with the FAS total score (A) and the scores for physical fatigue (B) and mental health issues (C).



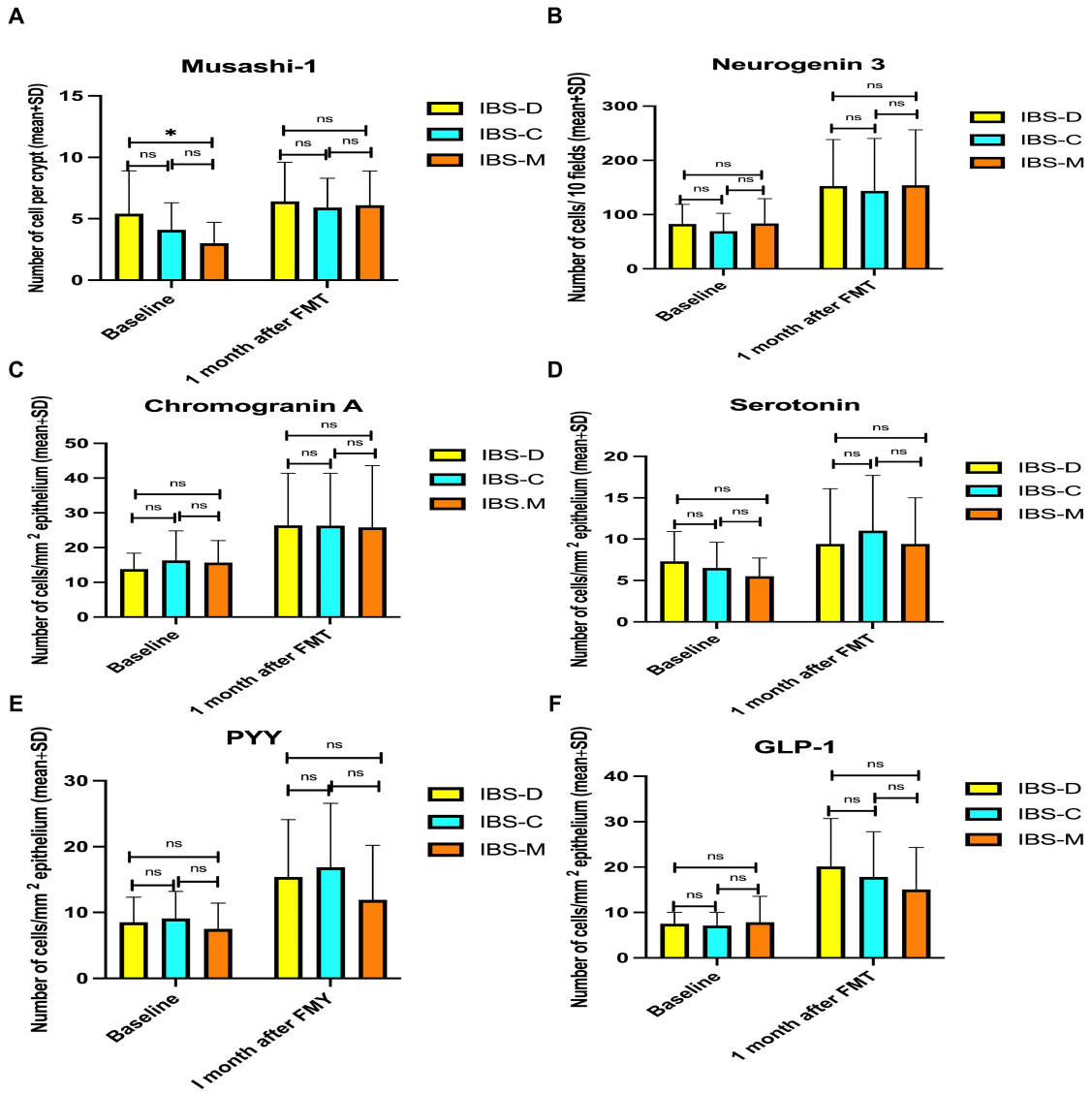
Supplementary Figure 11. Correlations of fecal butyric acid levels with the IBS-SSS total score (A) and the scores for abdominal pain (B), abdominal distension (C), dissatisfaction with bowel habits (D), and interference with quality of life (E).



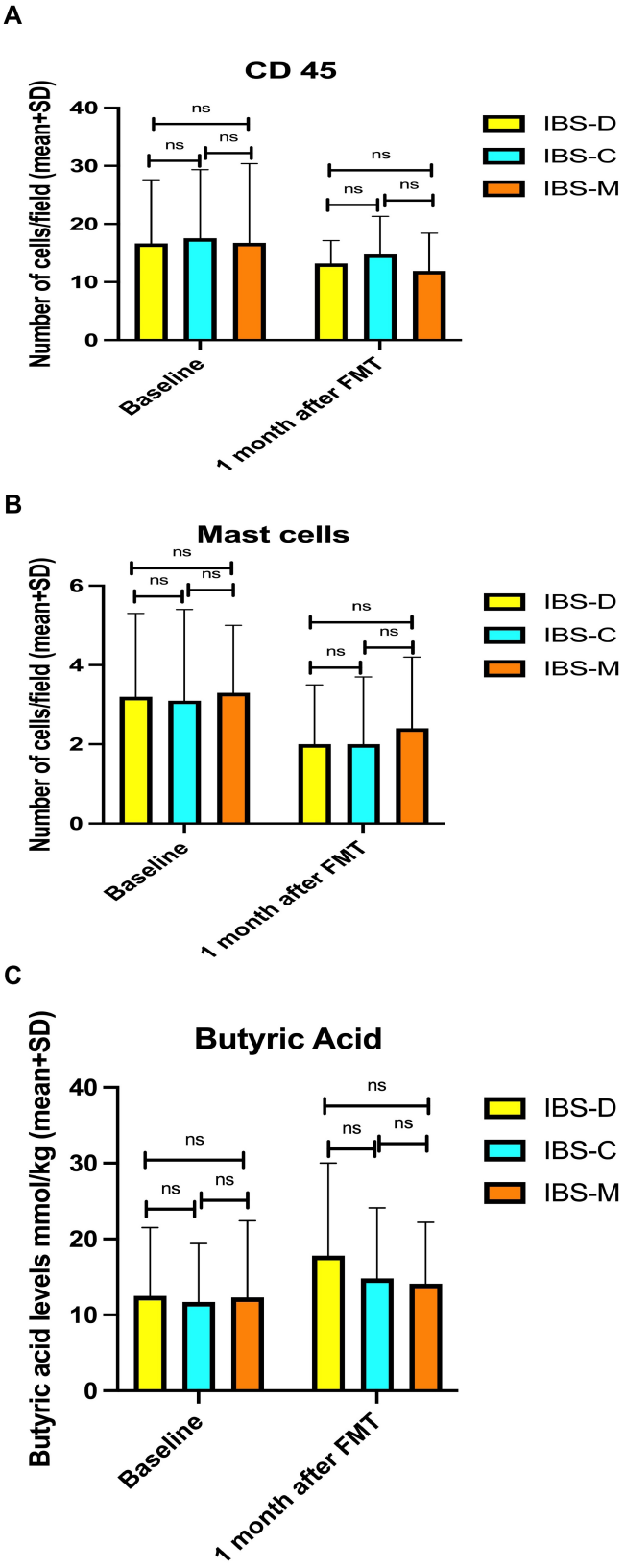
Supplementary Figure 12. Correlations of fecal butyric acid levels with the FAS total score (A) and the scores for physical fatigue (B) and mental health issues **(C)**.



Supplementary Figure 13. Differences between IBS subtypes in the IBS-SSS total score (A), FAS total score (B), and IBS-QoL total score (C). ns, nonsignificant.



Supplementary Figure 14. Densities of Musashi-1-expressing cells (A), neurogenin-3-expressing cells (B), chromogranin-A-expressing cells (C), serotonin-expressing cells (D), PYY-expressing cells (E), and GLP-1-expressing cells in patients with IBS-D, IBS-C, and IBS-M. ns, nonsignificant.



Supplementary Figure 15. Densities of immune cells (A) and mast cells (B), and faecal butyric acid levels **(C)** in patients with IBS-D, IBS-C, and IBS-M. ns, nonsignificant.

Supplementary Protocol

Brief summary

Irritable bowel syndrome (IBS) is a common chronic gastrointestinal disorder that affects 10–20% of the worldwide population. The prevalence of IBS in Norway is reportedly between 8% and 25%. The pathophysiology of IBS is incompletely understood, and there is no effective treatment for this condition. Imbalance (dysbiosis) of the gut microbiome has been found in patients with IBS. In the absence of an effective method to restore the dysbiosis, transplantation of a microbiome from healthy individuals with a well-functioning gut (so-called faecal microbiota transplantation [FMT] to those with IBS has been performed. Two randomized controlled trials (RCTs) have been published recently: one found that FMT reduced the symptoms and improved the quality of life, whereas it had no effect in the other study. In order to clarify these contradictory results, a new RCT enrolling a larger number of patients is required. In this study, we intend to recruit 165IBS patients from those attending the outpatient clinic at Stord Hospital in a randomized, double-blind placebo trial. A single healthy donor with a well-characterized microbiome is going to be used. The effects on symptoms, quality of life, fatigue as well as dysbiosis before and after FMT will be investigated. The patients will be randomized to placebo (own faeces) or to 30 or 60 g of the donor faeces in the ratio 1:1:1.

Study design

Patients

Two hundred patients who conform with the inclusion and exclusion criteria listed below will be included. In addition, the patients will be examined physically, with blood tests performed to exclude inflammation and liver, kidney, and thyroid diseases. They will undergo further gastroscopy with duodenal biopsies to exclude celiac disease, and colonoscopy to exclude malignancy and inflammatory bowel disease. Microscopic colitis will be excluded by examining tissue obtained by colonoscopy with segmental biopsy sampling.

Inclusion criteria

The following inclusion criteria will be applied:

1. Adult patients between 18 and 85 years who fulfil Rome IV criteria for the diagnosis of IBS.
2. No other gastrointestinal organic cause(s) for their symptoms.
3. Moderate to severe IBS symptoms (IBS Symptom Severity Scale [IBS-SSS] score ≥175).
4. Having followed the Stord Hospital program for at least 3 months, which includes information about IBS delivered by a gastroenterologist in a 60-minute consultation, adherence to a modified NICE diet, intake of 25–35 g psyllium husk fibres daily, and regular exercise.
5. Having attended the IBS school course at Haukeland Hospital or Stord Hospital, which includes a 2-day course (12 hours) with information delivered by a gastroenterologist, specialized nurse, psychiatrist, dietitians, physiotherapist, and a representative of the patient's organization.

Inclusion criteria

The following exclusion criteria will be applied:

1. Pregnant, planning pregnancy, or lactating.
2. Taking antibiotics or probiotics within 1 month prior to FMT.

3. Having undergone any abdominal surgery, with the exception of appendectomy, cholecystectomy, caesarean section, or hysterectomy.
4. Comorbidity such as kidney failure or chronic heart disease.
5. Organic gastrointestinal disease or systemic disease, with the exception of reflux esophagitis.
6. Alcohol or drug abuse.
7. Severe psychiatric disorders.

Donor selection and screening
A single donor will be selected and screened according to the European and international guidelines. The donor should not be a first-degree relative of any of the patients, since the intestinal microbiota is affected by the genetic composition, and a close relationship could result in similarity of the faecal microbiota between the donor and recipient.

Protocol
The protocol and design of the study are illustrated in Figure 1.
Faeces collection, preparation, and administration
Faeces collected from the donors and recipients will be stored at –80°C. Frozen faeces (30 or 60 g) from the donor or patients (placebo) are then thawed at 5°C and dissolved in 40 mL of 0.9% sterile saline per 30 g of faeces. The dissolved stool is administered to the patient after an overnight fast via the working channel of gastroduodenoscope in pars descendens duodenum distal to the papilla of Vater. The working channel of the gastroscope is then flashed with another 40 mL of sterile saline.

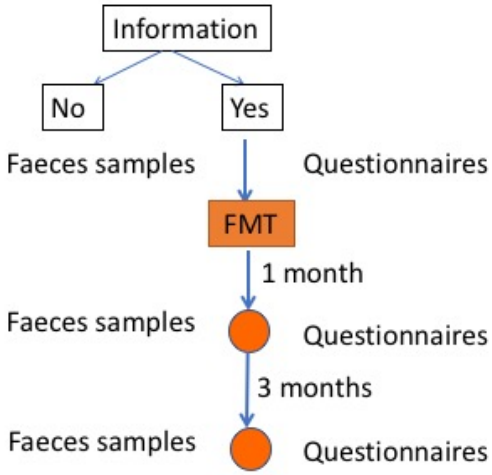


Figure 1. Schematic presentation of the study protocol. Sigmoidoscopy at the baseline and at 1 month after FMT was performed in patients who agreed to undergo sigmoidoscopy.

Methods
Questionnaires
The following questionnaires will be utilized in the assessments:
1. IBS-SSS.
2. Birmingham Symptom Scale.

3. IBS-Quality of Life Questionnaire, attachment 3.
4. SF-NDI (Short-Form Nepean Dyspepsia Index).
5. FAS (Fatigue Assessment Scale).

Microbiome analysis

The gut microbiota will be analysed using the GA-map™ Dysbiosis test (Genetic Analysis AS, Oslo, Norway) by algorithmically assessing the faecal bacterial abundance and profile (quantified as the dysbiosis index [DI]), and potential deviation in the microbiome from normobiosis (6). The GA-map test is based on faecal homogenization, mechanical bacterial cell disruption, and automatically extracting total bacterial genomic DNA using magnetic beads. The DI is based on 54 DNA probes targeting more than 300 bacterial strains based on their 16S rRNA sequence in 7 variable regions (V3–V9). Twenty-six bacteria probes are species specific, 19 detect bacteria at the genus level, and 9 probes detect bacteria at higher taxonomic levels. Probe labelling will be performed by single-nucleotide extension and hybridization to complementary probes coupled to magnetic beads, and signals will be detected using the BioCode 1000A 128-Plex analyser (Applied BioCode, Santa Fe Springs, CA, USA). A DI of >2 shows a microbiota profile that differs from that of the normobiotic reference collection (DI=1–2, nondysbiosis; DI**=2–4**, moderate dysbiosis; DI=4–5, severe dysbiosis).

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Symptomatic Inlet Patch as a Cause of Chronic Cough and Dysphagia: A Case Report

INTRODUCTION

The inlet patch or heterotopic gastric mucosa of the proximal esophagus is a benign lesion characterized by the presence of ectopic gastric tissue, likely of embryological origin.[1] Its prevalence ranges between 1.1% and 10%,[2] and it is usually asymptomatic, being incidentally detected during gastroscopy. In symptomatic cases, it may cause globus sensation, dysphagia, chronic cough, or laryngopharyngeal reflux, especially when acid secretion is present.[1] Rarely, it is associated with serious complications such as ulcers, fistulas, or adenocarcinoma. When a typical endoscopic appearance is observed, histological confirmation and further surveillance are not required.[3]

Published studies have emphasized that acid-related mechanisms can play a significant role in chronic cough and other extraesophageal symptoms, even in the absence of pathological distal reflux. This supports the hypothesis that localized acid exposure, such as that produced by a heterotopic gastric inlet patch, may elicit chronic laryngopharyngeal symptoms through direct mucosal irritation or by triggering a cough reflex hypersensitivity.[4]

CASE REPORT

A 53-year-old male presented with chronic cough and intermittent high dysphagia for solids, with 2 episodes of food impaction that resolved spontaneously. He denied heartburn, chest pain, regurgitation, or associated constitutional symptoms. An upper endoscopy revealed a gastric mucosal island (inlet patch) in the upper third of the esophagus (Fig. 1), with no other abnormalities. The work-up was completed with high-resolution esophageal manometry, which showed hypotonia of the lower esophageal sphincter with transient relaxations and signs of sliding, findings that may favor gastroesophageal reflux. Esophageal body motility was normal. Baseline dual-channel esophageal pH-metry without proton pump inhibitors (Fig. 2) showed a markedly elevated acid exposure time (AET) at the proximal level (total AET, 38.1%), without a corresponding increase in distal acid exposure (total AET, 0.4%). Therefore, gastroesophageal reflux is within physiological limits. The localized acid exposure observed in the proximal esophagus may be directly attributable to the heterotopic gastric mucosa identified on endoscopy. Correct probe placement was confirmed by chest X-ray, with one sensor located in the distal esophagus and the other in the proximal esophagus.

DISCUSSION

The findings of this clinical case support the hypothesis that heterotopic gastric mucosa may be clinically relevant in symptomatic patients. The 24-hour esophageal pH-metry showed episodes of acid exposure in the proximal esophagus, corresponding to the location of the inlet patch. In this case, the presence of symptoms such as chronic cough and dysphagia, in the absence of other abnormal test results, suggests that the inlet patch may play an active symptomatic role, likely through acid secretion by parietal cells in the heterotopic mucosa. In clinical practice, these findings may influence treatment decisions. In selected patients, treatment of symptomatic inlet patch with proton pump inhibitors, mucosal protectants such as sucralfate, and/or alginate-based preparations has been proposed, or even endoscopic ablation with argon plasma.[2]

Although the overall efficacy of acid suppression in chronic cough remains controversial, its selective success in cases like this reinforces the pathogenic role of focal acid secretion from heterotopic gastric mucosa.[5]

In our case, treatment with lansoprazole 30 mg once daily led to a partial but clinically significant improvement in the patient's dysphagia and chronic cough. At one-year follow-up, this partial improvement persisted, indicating sustained symptomatic benefit despite the absence of complete resolution.

This clinical improvement reinforces the need to consider the inlet patch in the differential diagnosis of patients with atypical gastroesophageal reflux symptoms or persistent otolaryngological manifestations, especially when conventional studies are negative.

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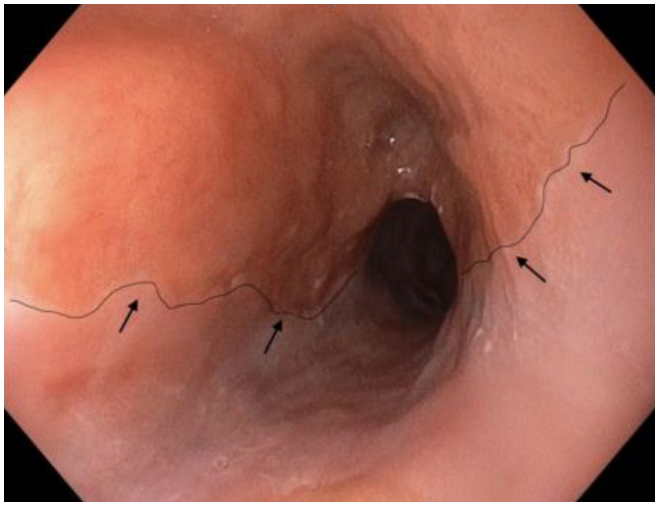

Figure 1. Upper gastrointestinal endoscopy showing a well-defined, oval, salmon-colored lesion (black arrows) approximately 3 cm in its longest axis is identified in the proximal esophagus; compatible with heterotopic gastric mucosa or inlet patch.

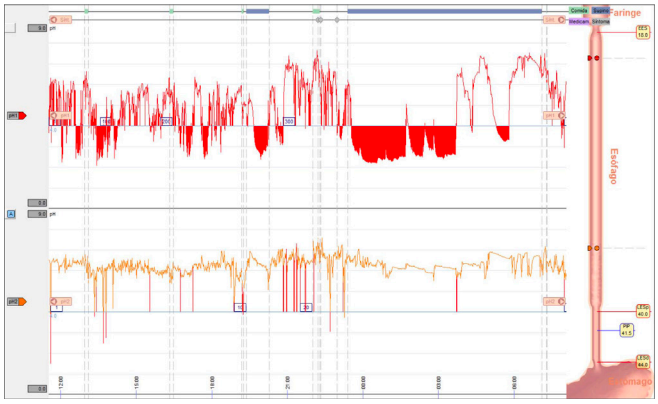

Figure 2. Baseline dual-channel esophageal pH-metry performed without proton pump inhibitors shows absence of pathological acid gastroesophageal reflux in the distal sensor, with a total acid exposure time (pH < 4.0) of 0.4% (normal < 4.5%). The proximal channel shows a markedly elevated acid exposure time of 38.1% (normal < 0.9%).

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Author contributions: Elena Aznar Ruiz, Ana Zatarín Vallés, Marta Aparicio Cabezudo, Laura García Pravia, and Constanza Ciriza de los Ríos, consultant gastroenterologists, contributed to the conception and design of the study and to the interpretation of the data. Marina Arroyo Cadenas, specialized gastrointestinal motility nurse, performed the esophageal motility study. Elena Aznar Ruiz drafted the manuscript. All authors critically revised the manuscript for important intellectual content and approved the final version of the manuscript.

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Functional Dyspepsia and Gastric Emptying Scintigraphy: Why 4-Hour Imaging and a 255-kcal Meal Matter

TO THE EDITOR: Park et al[1] examined functional dyspepsia (FD) using technetium-99m-phytate gastric emptying scintigraphy and reported no between-group difference in gastric emptying half-life, yet retention rates in the FD group were consistently lower than those in controls across time points.[1] We commend the authors but note several protocol features that may have contributed to an appearance of relatively faster emptying and attenuate group differences.

First, the test meal was ~180 kcal. We acknowledge that Park et al[1] described cultural and practical considerations for this meal choice; however, lower-calorie/low-fat meals are known to shorten gastric residence and may reduce detection of delayed emptying relative to the validated 255-kcal low-fat egg-white standard. Lower-calorie, low-fat meals shorten gastric residence and can reclassify delayed emptying as normal or rapid. Consensus statements recommend a standardized low-fat egg-white meal (~255 kcal) with fixed timing at 0, 1, 2, and 4 hours.[2,3]

Second, 4-hour imaging was conditional on the 2-hour result. Routine 4-hour acquisition detects more abnormalities than 2-hour protocols and is the basis of widely used reference values; therefore, we suggest obtaining 4-hour images in all subjects to align with widely used consensus protocols and reference values, not only those exceeding a 2-hour threshold.[4-6]

Third, baseline imbalances (higher rates of current smoking, higher body mass index, and diabetes in FD) may confound retention curves and group comparisons.[1] Covariate-adjusted models (eg, mixed-effects analyses of retention over time with prespecified stratification by diabetes/body mass index) and sensitivity analyses using complete 0-4–hour data and 4-hour–only classification would clarify whether the "faster" pattern persists after addressing protocol-related bias.

In summary, the combination of a sub-255 kcal meal, non-mandatory 4-hour imaging, and metabolic confounding may partly account for the observed retention pattern in FD. We respectfully encourage the authors to provide the above sensitivity analyses to strengthen inference for clinicians and readers.

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Preoperative Esophageal Physiologic Testing in Bariatric Surgery: How Far Should We Go?

TO THE EDITOR: We read with great interest the article by Leelakiatpaiboon et al,[1] who compared high-resolution esophageal manometry (HREM) with the upper gastrointestinal (UGI) series for diagnosing sliding hiatal hernia (HH) in morbidly obese patients undergoing bariatric surgery. Their paired design, the use of intraoperative findings as the reference standard, and the focus on clinically relevant hernias (≥ 2 cm) provide valuable data for preoperative assessment.[1]

The authors report excellent diagnostic performance of HREM (sensitivity 92.3%, specificity 93.7%) and substantially better inter-observer agreement than the UGI series. These results are consistent with previous reports in which HREM more accurately detects and grades sliding HH than endoscopy or radiology when surgical findings are used as the gold standard.[2] We agree that HREM can refine risk stratification in high-volume bariatric centers with access to experienced motility specialists.

We would, however, like to place these findings in the context of current guideline recommendations. For classic antireflux surgery, consensus statements such as ICARUS and other expert panels regard preoperative esophageal manometry and reflux monitoring as mandatory to document gastroesophageal reflux disease and exclude major motility disorders.[3,4] In contrast, bariatric surgery guidelines from major societies do not mandate routine manometry or pH-impedance testing in all bariatric candidates; rather, objective reflux testing is generally reserved for selected patients with significant reflux symptoms or diagnostic uncertainty, and many programs still rely mainly on symptoms, UGI series, and esophagogastroduodenoscopy for preoperative evaluation.[5,6]

Against this background, the algorithm proposed by Leelakiatpaiboon et al,[1] which incorporates systematic HREM and 24-hour multichannel intraluminal impedance–pH monitoring for all morbidly obese patients being considered for bariatric surgery, may be difficult to implement in centers with limited resources or motility expertise.[1] Comparative studies that evaluate stepwise strategies—reserving HREM and reflux monitoring for patients with troublesome reflux symptoms, discordant UGI or endoscopic findings, or planned concomitant antireflux procedures—would help determine the most cost-effective and pragmatic approach.

Nevertheless, the authors are to be commended for providing robust prospective data on HREM for HH detection in the bariatric setting. Their work highlights the potential role of HREM in clarifying esophagogastric anatomy and reflux burden and will stimulate further discussion on how best to integrate advanced physiologic testing into bariatric surgery pathways.

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Reinterpreting Amitriptyline in Extra-esophageal Gastroesophageal Reflux Disease Through the GABAergic Inhibitory Axis

TO THE EDITOR: In a single-center, single-blind, randomized crossover pilot trial, Woo et al[1] evaluated the efficacy of low-dose amitriptyline (10 mg nightly) added to standard proton pump inhibitor (PPI) therapy for patients with extra-esophageal manifestations of gastroesophageal reflux disease (GERD) refractory to ≥ 4 weeks of PPI treatment. In this study of 40 patients, both the amitriptyline plus PPI arm and the PPI-only arm demonstrated substantial symptom improvement and quality-of-life gains; however, the addition of amitriptyline did not confer a statistically significant advantage over PPI monotherapy (65% vs 85% response rate, $P = 0.273$).[1]

This neutral result highlights important methodological considerations. First, participants were defined as "PPI-refractory" based on 4 weeks of acid suppression without objective physiologic testing (eg, pH-impedance and reflux symptom correlation), which may result in inclusion of patients with acid-responsive disease, functional heartburn, or visceral hypersensitivity that are indistinguishable by symptom assessment alone. Second, amitriptyline's neuromodulatory effects—operating through enhanced descending inhibitory pathways—typically manifest over longer periods than the 4-week treatment window used in this trial. Third, the crossover design lacked a washout period, potentially blunting detection of cumulative neuromodulatory effects and introducing carryover in central sensitization. Collectively, these factors limit inference regarding amitriptyline's potential efficacy among physiologically enriched subpopulations.

Comparative literature in functional esophageal disorders offers

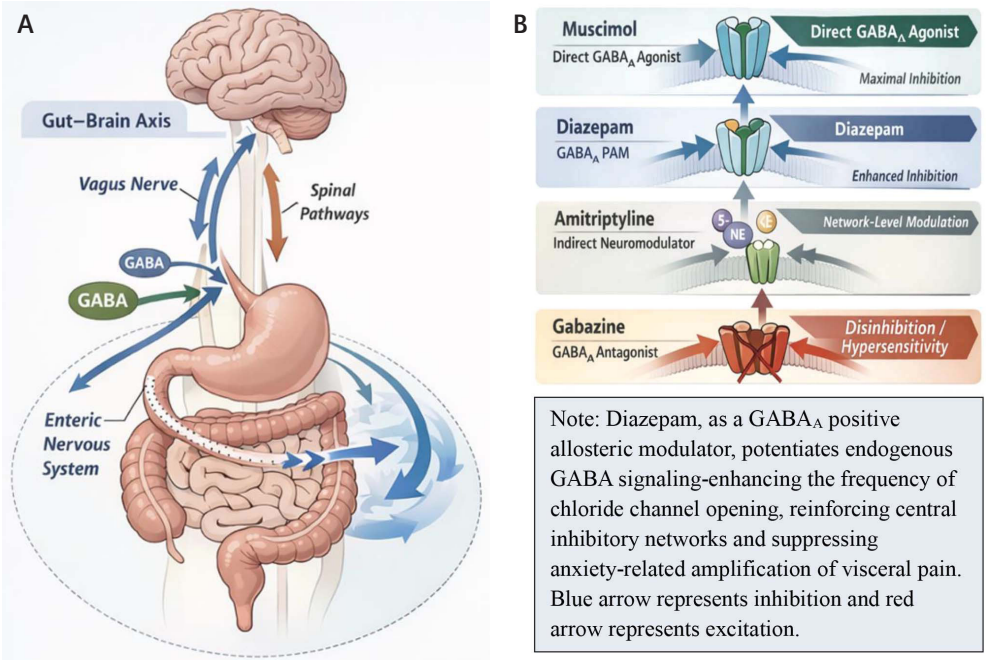

Figure 1. Hierarchical engagement of central inhibitory control along the gut–brain axis. Schematic illustration of graded inhibitory modulation of esophageal sensory input within the gut–brain axis. (A) This figure illustrates the inhibitory control involving vagal (brainstem descending modulation), spinal (segmental nociceptive gating), and enteric (local neuromodulation within ENS) pathways for visceral hypersensitivity. (B) This figure illustrates the direct gamma-aminobutyric acid type A ($GABA_A$) receptor activation, ie, Muscimol produces immediate suppression of visceral afferent signaling, whereas positive allosteric modulation (diazepam) enhances endogenous inhibitory tone (attention to note in [B]). Amitriptyline acts indirectly by strengthening descending monoaminergic pathways that converge on GABAergic interneurons, resulting in slower, network-level neuromodulation. In contrast, $GABA_A$ antagonism (gabazine) removes inhibitory restraint and models visceral hypersensitivity. This framework highlights the importance of mechanistic alignment between neuromodulator properties, patient phenotype, and trial design.

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Table. Positioning of Neuromodulatory Agents Along the GABAergic Inhibitory Axis

Agent	Primary pharmacologic action	GABA system interaction	Conceptual role in neurogastroenterology
Muscimol	GABA-agonist	Direct receptor activation	Maximal experimental inhibition of visceral input
Diazepam	GABA-positive allosteric modulator	Amplifies endogenous GABA	Suppression of reflex hyperexcitability
Amitriptyline	Tricyclic antidepressant	Inderect GABA facilitation	Network-level neuromodulation
Gabazine	GABA-antagonist	Blocks inhibitory signaling	Experimental model of hypersensitivity

GABA, gamma-aminobutyric acid.

additional context. In a randomized placebo-controlled trial of nortriptyline in non-erosive reflux disease, 21 days of treatment significantly reduced acid-induced brain responses on functional magnetic resonance imaging, though clinical symptom differences were not significant, suggesting central modulation without detectable short-term clinical benefit.[2] Similarly, imipramine did not outperform placebo in patients with esophageal hypersensitivity or functional heartburn, although modest quality-of-life improvements were reported.[3] Meta-analyses indicate that while central neuromodulators may reduce symptom severity in conditions such as functional chest pain and globus, evidence for benefit in reflux hypersensitivity and functional heartburn remains inconsistent.[4] Furthermore, a Bayesian network meta-analysis of antidepressants in GERD suggested that certain agents (eg, combined flupentixol-melitracen and selective serotonin reuptake inhibitors) may achieve higher symptom remission rates compared to placebo, whereas traditional tricyclic antidepressants did not show superiority.[5]

These findings underscore 2 important points. First, antidepressant neuromodulators modulate visceral sensation and central pain processing, as reflected in increased pain thresholds and altered brain activity in experimental paradigms, even when clinical endpoints lag.[6] Second, heterogeneity in pathophysiology matters: neuromodulators may benefit subsets of patients characterized by visceral hypersensitivity or central sensitization rather than those whose symptoms are driven predominantly by acid exposure or peripheral mechanisms.

Building on this nuanced evidence base, we propose a hierarchical mechanistic framework (Figure) in which amitriptyline is conceptualized as an anxiolytic-based, network-level neuromodulator that exerts indirect effects on GABAergic inhibitory circuits. In contrast to direct $GABA_A$ receptor agonists or positive allosteric modulators (Table), amitriptyline's pharmacodynamics involve enhancement of descending monoaminergic inhibition and require sufficient treatment duration and appropriate patient phenotyping to reveal clinical efficacy. This perspective may help reconcile mixed clinical results and guide future trial designs in extra-esophageal GERD toward mechanism-enriched cohorts, extended treatment windows, and objective physiological subtyping.

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Erratum

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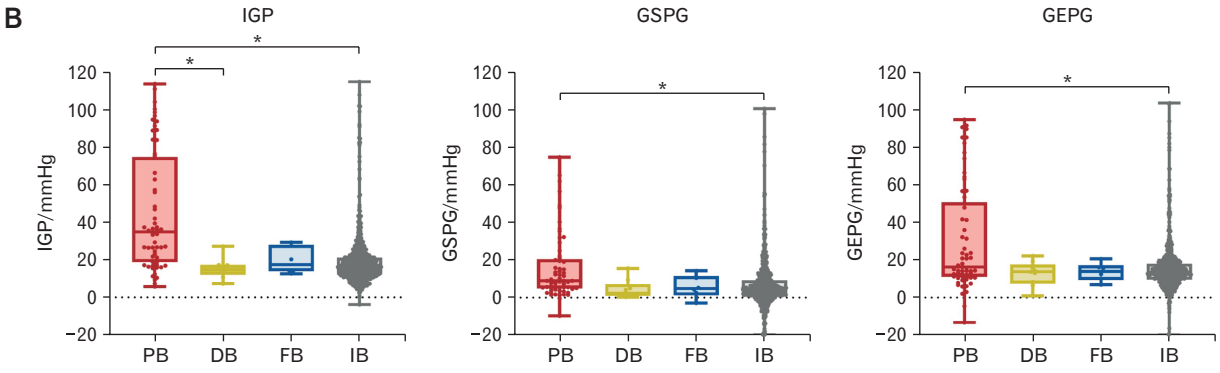
Belching and Reflux: The Relationship and the Underlying Mechanism

Songfeng Chen, Xingyu Jia, Dianxuan Jiang, Mengyu Zhang, Niandi Tan, Qianjun Zhuang, and Yinglian Xiao

(J Neurogastroenterol Motil 2026;32(2):254-266; https://doi.org/10.5056/jnm25189)

The original version of this article contains 1 mistake.
The x-axis label in Figure 4B should have read as "PSGB, DSGB, FSGB, ISGB."



Previous Figure 4B

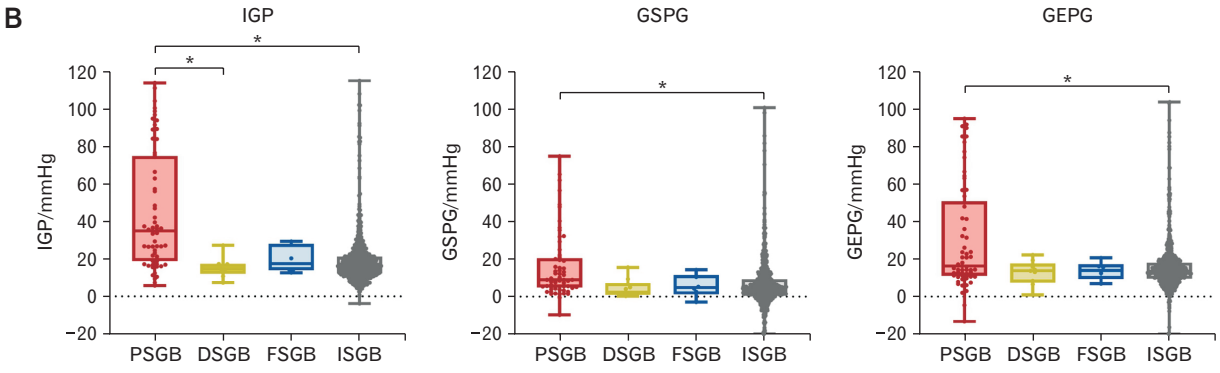


Figure 4B after correction

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ETHICAL POLICIES OF JNM THE JOURNAL OF NEUROGASTROENTEROLOGY AND MOTILITY

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Journal of Neurogastroenterology and Motility

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The purpose of these regulations is to establish and observe research ethics for research papers submitted to the Journal of Neurogastroenterology and Motility.

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Article 6

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ACTH	adrenocorticotropic hormone (adrenocorticotropin)
ADP	adenosine diphosphate
ADP	ase adenosine diphosphatase
ALT	alanine aminotransferase
AMP	adenosine monophosphate
anti-HAV	antibody to hepatitis A virus
anti-HBc	antibody to hepatitis B core antigen
anti-HBe	antibody to hepatitis B e antigen
anti-HBs	antibody to hepatitis B surface antigen
anti-HDV	antibody to hepatitis D (delta) virus
AST	aspartate aminotransferase
ATP	adenosine triphosphate
ATP	ase adenosine triphosphatase
BUN	blood urea nitrogen
cAMP	adenosine 3',5'-cyclic monophosphate
cDNA	complementary DNA
cGMP	guanosine 3',5'-cyclic monophosphate
CNS	central nervous system
CT	computed tomography
DNA	deoxyribonucleic acid (deoxyribonucleate)
EDTA	ethylenediaminetetraacetic acid
HAV	hepatitis A virus
HBcAg	hepatitis B core antigen
HBeAg	hepatitis B e antigen
HBsAg	hepatitis B surface antigen
HBV	hepatitis B virus
HCV	hepatitis C virus
HDV	hepatitis D (delta) virus
H&E	hematoxylin and eosin
HIV	human immunodeficiency virus
HLA	human leukocyte antigen
Ig	immunoglobulin
IL	interleukin
IFN	interferon
Km	Michaelis constant
MHC	major histocompatibility complex
MRI	magnetic resonance imaging
mRNA	messenger RNA
NANB	non-A, non-B (hepatitis)
NSAID	nonsteroidal anti-inflammatory drug
PG	prostaglandin
PGI	prostacyclin
PNS	peripheral nervous system
RNA	ribonucleic acid
TGF	transforming growth factor
TNF	tumor necrosis factor
TXA2	thromboxane A2
UDCA	ursodeoxycholic acid
VIP	vasoactive intestinal polypeptide
vol	volume
vs	versus
wt	weight

Measurements and Units of Measure

Combining Prefixes

T	tera- (10^{12})
G	giga- (10^{9})
M	mega- (10^{6})
k	kilo- (10^{3})
h	hecto- (10^{2})
da	deca- (10^{1})
d	deci- (10^{-1})
c	centi- (10^{-2})
m	milli- (10^{-3})
μ	micro- (10^{-6})
n	nano- (10^{-9})
p	pico- (10^{-12})
f	femto- (10^{-15})
a	atto- (10^{-18})

Units

A	ampere(s)
Å	angstrom(s)
cal	calorie(s)
CFU	colony-forming unit(s)
C	coulomb(s)
cpm	counts per minute
cps	counts per second
cm^3	cubic centimeter(s) (not cc)
Ci	Curie(s)
cycle/min	cycles per minute
cycle/sec	cycles per second
Da	dalton(s)
day(s)	day(s) (do not abbreviate)
dL	deciliter
°C	degree(s) Celsius
d	density
dpm	disintegrations per minute
dps	disintegrations per second
eV	electron volt(s)
Eq	equivalent(s)
°F	degree(s) Fahrenheit

F	farad
ft	foot
G	gauss
g	gram(s)
$t_{1/2}$	half-life
H	henry(ies)
Hz	hertz
hr	hour(s)
in	inch(es)
IU	international unit(s)
J	joule(s)
K	kelvin
kcal	kilocalorie(s)
kDa	kilodalton(s)
kg	kilogram(s)
L	liter(s)
×	magnification
m	meter(s)
mg	microgram(s)
mL	microliter(s)
mm	micrometers (do not use microns)
msec	microsecond(s)
mL	milliliter(s)
mmHg	millimeter(s) of mercury
min	minute(s)
mo	month(s)
mol/L	molar
mol	mole(s)
msec	millisecond(s)
Ω	ohm(s)
osm	osmole(s)
oz	ounces(s)
Pa	pascal(s)
lb	pound(s)
psi	pound-force per square inch
rpm	revolutions per minute
rps	revolutions per second
sec	second(s)
U	unit(s)
V	volts(s)
W	watt(s)
wk	week(s)
yr	year(s)

Statistical Terms

ANOVA	analysis of variance
CI	confidence interval
r	correlation coefficient
df	degrees of freedom
$\bar{x}$	mean
NS	not significant
n	number of observations
OR	odds ratio
p	probability
SD	standard deviation
SE	standard error
SEM	standard error of the mean
F	variance ratio

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Acknowledgements

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- **Books**

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